DURABLE RESEARCH CONCEPT

20 Watts Was Enough

A biologically inspired R&D blueprint for sparse, grounded, continual, energy-efficient AI.

EDITION

Full concept book

CONTENTS

20 canonical documents + 1 generated appendix

LENGTH

82,099 words

SOURCE

Generated from the private Git repository

NAVIGATION

Contents

GENERATED RESEARCH STATUS

Scientific claims are not runnable tests

Evidence status describes support for a source-domain claim. Test readiness describes whether this project can execute a frozen experiment. The two axes are deliberately reported separately.

CLAIM DENOMINATOR: 1,375

Written protocols versus executable evidence

1,262 of 1,375

Protocol-covered claims

At least one linked candidate or fixture contains all eight required written protocol facets. This is design coverage, not an experimental result.

0 of 1,375

Workstation-executable claims

A linked artifact must have a checked workstation-ready manifest and a runnable scientific harness. Smoke plumbing cannot promote a claim.

ARTIFACT DENOMINATOR: 31

Experiment implementation lane

31

COMPLETE WRITTEN PROTOCOLS

1

VALIDATED SMOKE HARNESS

0

WORKSTATION-READY EXPERIMENTS

INDEPENDENT AXES

Evidence status by highest test tier

HIGHEST TEST TIER	ESTABLISHED	PLAUSIBLE	SPECULATIVE	DISPUTED	UNKNOWN	TOTAL
Ledger only	93	14	2	4	0	113
Linked description	0	0	0	0	0	0
Protocol complete	966	193	57	46	0	1,262
Workstation executable	0	0	0	0	0	0

"Established" means the cited observation is established within its stated scope. It does not mean the proposed AI translation has passed this project's experiment.

NOT MISSING AT RANDOM

113 ledger-only claims

Evidence inputs	81
Source reproductions	24
New experiment needed	8

The engineering gaps collapse into 5 proposed experiment families. Evidence inputs and source reproductions are not falsely counted as project tests.

Open the proposed backlog

NEAREST EXECUTABLE TARGET

candidate-010: smoke-ready

Candidate 010: reset-coupled staged verification

6/9 machine-checkable structural promotion gates currently pass

- ✓ **Explicit execution-track claim scope**
  execution track is limited to C-170
- ✓ **Frozen full profile**
  data.full_profile identity and SHA-256 match the frozen repository file
- ✓ **Full experiment-path tests**
  implementation.full_tests is part of the workstation suite executed by npm test
- ✓ **Corruption-evident output ledger**
  hash-chain output is wired through the tested checkpoint ledger
- ✓ **Deterministic resume**
  runner imports the declared, tested checkpoint/resume implementation
- ✓ **Measured-energy provider**
  external-meter provider capability, exclusions, test, and frozen config hash match
- ○ **Frozen confirmation seeds**
  seeds.confirmation does not disclose a frozen non-empty seed list
- ○ **Frozen held-out seeds**
  seeds.held_out does not disclose a frozen non-empty seed list
- ○ **Strict recomputed promotion evidence**
  strict promotion evidence is pending and no claim-eligible bundle exists

Open the experiment contract **Open the harness notes**

The private site contains the complete artifact table and live generated coverage report.

20 Watts Was Enough

> *A biologically inspired R&D blueprint for sparse, grounded, continual, energy-efficient AI.*

The name is a joke with a serious target: metabolic accounting places the adult human brain's whole-organ power budget at roughly 17–20 watts, while contemporary AI systems often buy capability by activating and moving far more state than a task needs. The number names the constraint; comparisons with silicon require a shared task, quality envelope, and system boundary. See C-001 and the energy-model chapter.

Central thesis

The brain is evidence that useful, adaptive intelligence can exist under a strict power and communication budget. This project asks which *computational constraints* behind that fact can become engineering requirements:

- separate total capacity from active capacity;
- ground representations in temporally aligned perception, action, and outcome;
- allocate computation according to uncertainty and task demand;
- separate rapid episodic learning from slow structural learning;
- consolidate before pruning and hardening;
- store mutable facts in inspectable memory instead of forcing all knowledge into weights; and
- measure energy, data movement, quality, and uncertainty together.

Project status

Stage: concept and evidence framework. There is no model implementation yet.

The repository is the canonical source. The original Google Doc and Gemini discussions are preserved under `sources/` as historical, non-authoritative inputs.

The default normative context is the European Union and Germany. Legal, standards, and conformity claims remain applicability-, jurisdiction-, version-, and date-qualified under the `normative baseline`; foreign material is preserved as comparative unless a concrete project hook makes it applicable.

The same files are also rendered as a private research site. The site is a generated reading surface—not a second document store. Saving a Markdown, equation, bibliography, or Mermaid file updates the local preview through hot reload; publishing creates an owner-only online edition from a committed Git state. The site also provides a downloadable A4 book containing this README and all canonical concept chapters, generated from the same files and checked for staleness during the build. See decision 0005.

Concept map

CHAPTER	QUESTION
Thesis and principles	What is the project's central engineering hypothesis?
Working architecture	How do runtime, adaptation, reconstructive generation, maintenance, and resource control form one system?
Biology is a launchpad	Which biological constraints transfer, and which substrate limits should engineering escape?
Cross-domain convergence	How do different sciences collapse into shared problem–solution principles without losing causal differences?
Structural growth and routing	How does a persistent capability gap create, test, specialize, merge, protect, or retire conditional modules?
Sensorimotor grounding	Which event, clock, opportunity, action, intervention, history, language, uncertainty, and provenance contracts ground a world model?
Representative adaptive performance	How are learning, transfer, fatigue, risk, recovery, coordination, and selection compared under the conditions in which action is actually possible?
Active acoustic inference	How do calibrated sound, timing, propagation, masking, spatial action, active emission, separation, and exposure form one operator-qualified inference loop?
Operator-qualified sensing	What could a physical measurement resolve, what remains prior-dependent, when is another observation worth its cost, and which substrate should execute the transform?
Active chemical sensing	How do reaction, transport, mixtures, active sampling, plume motion, adaptation, drift, analytical confirmation, exposure, and cost qualify a chemical inference?
Reliability under mission profiles	How should variable physical components be characterized, operated, protected, repaired, repurposed, or retired across real stress histories?
Physical computation boundaries	Which fundamental, device, circuit, workload, facility, and lifecycle boundary supports an energy claim?
Sparse predictive compute	How do event, context, and resource loops price the next computation or observation?
Memory and consolidation	How does an episode become a retained skill, external fact, weakened trace, or deletion?
Maturity and structural consolidation	When should a structure be protected, reopened, compacted, or retired?
Hardening and factual memory	Which qualified path becomes compiled, remains a reusable skill, enters versioned factual memory, or escalates?
System synthesis	How do runtime, task, resource, adaptation, and maintenance control planes fit together?
Energy evaluation contract	How are lifecycle boundaries, equal budgets, break-even, uncertainty, and null models enforced?
Research roadmap	What evidence and experiments are required before building the full system?

Supporting material:

- `research/claims.md` — stable claim IDs and evidence status
- `research/references.bib` — primary-source bibliography
- `research/adoption-matrix.md` — what to use, test, explore, or watch
- `research/principle-registry.md` — canonical deduplicated problem–solution invariants
- `research/field-coverage.md` — generated OECD/DFG field census, taxonomy blind spots, and breadth queue
- `research/domain-inventory.md` — audited, partial, and queued scientific fields
- `research/discovery-policy.md` — open-world search, extraction, deduplication, and promotion rules
- `research/normative-baseline.md` — EU/Germany default, source-role hierarchy, and applicability record
- `research/audits/` — dated primary-source research passes and engineering null-model audits
- `research/neuroscience-opportunity-map.md` — underused neural mechanisms and falsifiable translations
- `research/comparative-biology.md` — candidates from animals, plants, immune systems, and adaptive networks
- `research/source-crosswalk.md` — imported ideas mapped into evidence and principle bundles

- `research/open-questions.md` — unresolved decisions
- `math/` — notation, boundaries, and derivations
- `math/visual-models.md` — interpretable plots of the current efficiency equations and break-even boundaries
- `assets/` — editable diagram and future figure sources
- `decisions/` — durable project decisions
- `experiments/candidates/` — falsifiable, equal-budget experiment contracts
- `experiments/fixtures/` — reusable cross-candidate stress benchmarks that add no architecture by themselves
- `experiments/test-coverage.md` — generated claim-to-protocol coverage and workstation execution readiness
- `experiments/test-readiness-summary.json` — compact machine-readable readiness surface used by the site and book
- `CHANGELOG.md` — human-readable evolution

Editing rule

Do not regenerate the project from a prompt. Make an incremental change to a chapter, claim, equation, diagram, or decision; update links and the changelog; then run:

```
pwsh -File scripts/validate-docs.ps1
npm run validate:math
```

For the live rendered edition, install the locked dependencies once and start the watcher:

```
npm ci
npm run dev
```

The preview is served at `http://localhost:3000`. Internal Markdown links, GitHub-style tables, LaTeX equations, the bibliography, and editable Mermaid sources are rendered in the same searchable reader.

Regenerate the full concept book after changing the README, a concept chapter, diagram renderer, plot, or book stylesheet:

```
npm run generate:book-pdf
npm run validate:book-pdf
```

The tracked artifact is `public/downloads/20-watts-was-enough-full-concept-book.pdf`; the private site offers it directly for download, while `/book` remains a printable HTML edition. Raw source captures, audit ledgers, and experiment fixtures remain in the searchable private site rather than being duplicated into the reading edition.

See `CONTRIBUTING.md` for the full workflow.

License

Thesis and design principles

Scope

This chapter states the project's argument, its engineering requirements, and the limits that keep a biological analogy from becoming mythology.

Thesis

The brain demonstrates that adaptive behavior, continual learning, perception, memory, and action can coexist under a tight energy and communication budget (C-001). Contemporary AI demonstrates a different strength: highly parallel optimization over vast data and parameter spaces.

The project hypothesis is that a capable artificial system should combine the search capacity of modern learning with constraints that biology cannot evade:

1. only a small, relevant fraction of capacity should be active for an event;
2. communication and memory movement must be priced, not hidden behind FLOPs;
3. learning must begin from aligned perception, action, and consequence rather than language alone;
4. rapid acquisition must not directly rewrite stable long-term structure;
5. consolidation must precede destructive compression; and
6. stable skills, mutable facts, and active reasoning should use different storage and execution paths.

The intended result is a system whose cost scales primarily with the information and uncertainty relevant to a task, rather than with its full stored capacity.

Biological observation

Neural signaling is metabolically costly, cortical activity is constrained, and biological learning spans multiple timescales. Sensory and motor experience precedes mature language. These observations motivate design constraints.

They do **not** establish a single brain algorithm, an equivalent number of digital operations, or a guarantee that every biological mechanism is efficient on silicon.

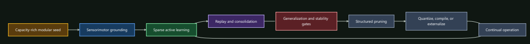
Proposed AI translation

The blueprint combines four separations:

- **capacity versus activity:** sparse routers select modules, depth, memory, and precision;
- **prediction versus correction:** predictable state follows a cheap path; residual uncertainty receives more computation;
- **experience versus structure:** fast memory captures episodes while slow learning integrates regularities under regression tests; and
- **reasoning versus knowledge:** stable skills may be compiled or quantized, while mutable facts remain retrievable and attributable.

Development is a controlled lifecycle:

Wide diagram · fit to page



Editable source: `../assets/diagrams/developmental-pipeline.mmd`.

Efficiency mechanism

The optimization target is constrained quality per unit of total system energy, with latency and uncertainty limits. A model is not efficient merely because it has sparse weights; the runtime must avoid loading and operating on inactive state, and routing overhead must remain smaller than the work skipped.

The top-level objective is developed in the energy model:

$$\max \frac{Q}{E_{\text{facility}}} \quad \text{subject to} \quad L_{p95} \le L_{\max},\ R \le R_{\max},\ Q \ge Q_{\min}.$$

Here Q is task quality, E_{facility} is measured facility energy, L_{p95} is tail latency, and R is a defined risk or calibration metric.

Evidence status

- Metabolic constraints on neural activity are established within the scope of C-001.
- Conditional computation is established in narrower engineered systems under C-003 and C-004.
- The complete developmental pipeline is a speculative integration. No cited paper validates it as one system.

Speculative extensions

- Learn a global energy price that modules must "bid" against when requesting compute.
- Let stable routing motifs become physically colocated or compiled for lower data-movement cost.

- Allow controlled structural growth when no existing module can absorb a new regime without interference.

Failure modes

- Anthropomorphic labels conceal incompatible mechanisms.
- Sparsity saves theoretical FLOPs but increases real communication or latency.
- Energy optimization suppresses rare but important computation.
- Early specialization prevents transfer and produces brittle modality silos.
- "Hardening" turns uncertain or mutable behavior into an uncorrectable path.

Measurable predictions

The integrated hypothesis survives only if, at matched quality:

- active parameters and memory traffic grow more slowly than total capacity;
- average energy falls without unacceptable tail-risk or calibration loss;
- sequential learning retains prior capability better than a matched single-timescale baseline; and
- grounded intervention tasks improve beyond gains explained by additional data or parameters alone.

Working architecture: three coupled loops

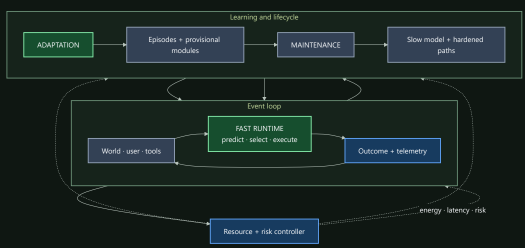
> *A capable system should execute, adapt, and consolidate on different timescales while one controller prices energy, latency, and risk.*

Scope

This is the shortest complete description of the proposed system. It joins the project's individual mechanisms into one operating architecture and shows where information moves, where learning happens, and which decisions remain reversible.

Architecture at a glance

Wide diagram · fit to page



Editable source: `../assets/diagrams/three-coupled-loops.mmd`.

The three loops solve different problems:

LOOP	TIMESCALE	PRIMARY DECISION	NORMAL OUTPUT
Fast runtime	event	What should run now?	action, answer, or escalation
Adaptation	episode to session	What did this outcome teach us?	attributable episode, hypothesis, or provisional module
Maintenance	replay to release cycle	What should become stable, move, weaken, or disappear?	consolidated memory and restructured execution

The resource controller crosses all three. It can restrict an action because of energy, latency, communication, or risk, but it cannot silently replace the task objective. Measured outcomes return to the controller so estimated costs can be corrected.

Biological observation

Living systems repeatedly separate processes by timescale and locality:

- many components are available, while only a small subset is strongly active together (C-001, C-003);
- local circuits resolve routine events while larger systems coordinate exceptions (C-017, C-024);
- fast traces change behavior before slow structure is rewritten (C-008, C-019);
- replay, forgetting, and structural maintenance are selective processes (C-036–C-042);
- stable structure can be protected and later reopened under bounded intervention (C-043–C-045); and
- local demand can recruit adjacent resource supply (C-049–C-051).

The shared principle is separation of concerns: execution, adaptation, memory, structure, and physical supply change at different speeds and respond to different signals. Digital systems add exact copying, external stores, checkpoints, shadow evaluation, and rollback. Those capabilities make the separations easier to test and safer to revise.

Proposed AI translation

1. Fast runtime

For each event, the runtime forms a predictive state containing expected observations, uncertainty, current goals, and relevant recent context. A hierarchical selector then chooses the cheapest admissible path:

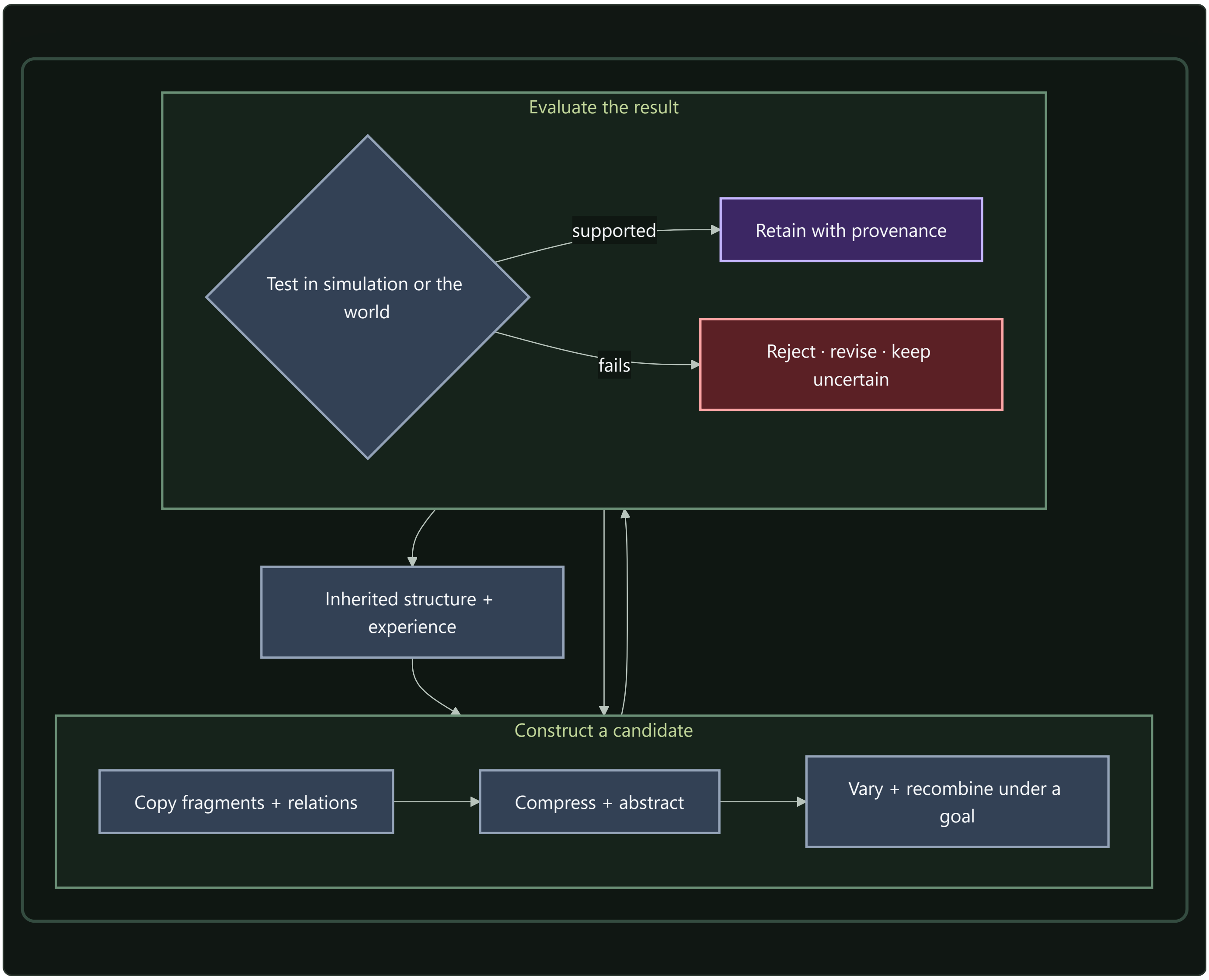
- a hardened low-cost path for a familiar, low-risk event;
- one or more conditionally active experts;
- another layer or reasoning step;
- episodic or factual retrieval;
- a sensor, tool, or environment intervention; or
- escalation to a human or a more capable controller.

The output is more than an answer. It includes measured latency, energy, communication, calibration, and any observed consequence. That telemetry is the input to later learning and resource decisions.

2. Adaptation

An outcome is captured with provenance before it can alter slow parameters. It may update temporary state, create a provisional adapter, change a local connection, or simply become a replay candidate. A surprising event is not automatically a lesson; it may be noise, attack, contradiction, or evidence of a new regime.

There is no ex-nihilo generator in this design. Every proposal is built from inherited structure and experience: complete patterns, fragments, relations, or abstractions are copied and transformed. The generative cycle is therefore explicit:



Editable source: `../assets/diagrams/generative-recombination.mmd`.

Stochasticity can alter which variants are tried. Usefulness comes from the entire cycle: decomposition, recombination, intervention, comparison, and selection. The supporting evidence and unresolved distinctions are tracked in the endogenous generation audit and is falsified first through Candidate 004.

RECONSTRUCTIVE ADAPTATION THROUGH EXTERNAL STATE

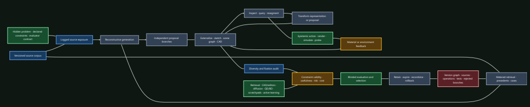
Visual art and design cognition make the cycle more operational. Proposal generation is reconstructive: learned categories, exemplars, relations, styles, constraints, tools, and prior transformations shape what can be proposed (C-842–C-860). That does not reduce the process to nearest-neighbor retrieval. A system can externalize a partial state, inspect consequences that were not explicit in its preceding plan, change representation, probe the world, and use the result to create a new branch. The novelty is relative to a declared history; the validity comes from constraints and observed outcomes.

The adaptation loop therefore distinguishes seven event types:

1. **Exposure:** which sources, examples, styles, and constraints were available before the task;
2. **Retrieval:** which versioned sources were deliberately requested during the task and at what cost;
3. **Reconstruction:** which fragments, relations, or abstractions formed a proposal branch;
4. **Externalization:** which editable representation—text, sketch, scene graph, CAD state, program, simulation, or prototype—made the proposal inspectable;

5. **Epistemic action:** which render, measurement, simulation, query, or material intervention was selected to reduce a decision-relevant uncertainty;
6. **Evaluation and selection:** which hard constraints, qualified evaluators, risk limits, and selection rule admitted or rejected a version; and
7. **Retention:** which source, parent, operation, test, rejected branch, dependency, expiry, and rollback records were preserved.

Wide diagram · fit to page



Editable source: versioned-reconstructive-design.mmd.

Several separations are non-negotiable. More proposals are not necessarily better proposals; proposal diversity, constraint validity, selection regret, and realized outcome are different measurements. Similarity to an example can transfer a useful structural relation or cause fixation and negative transfer, so similarity and task effect are both reported. A provenance graph can record derivation without proving intention, authenticity, correctness, or originality. Aesthetic response is observer-, context-, expertise-, and history-qualified rather than a universal scalar.

Parallel branches are useful only if they remain genuinely independent before critique. A shared canvas can coordinate work and simultaneously propagate an early fixation. The comparison therefore includes private branches, delayed reveal, source substitution, example removal, evaluator swap, material swap, and representation conversion. Every alternative pays for retrieval, proposals, tools, simulations, human critique, physical material, waste, retention, and recovery.

Fixture F-002 tests this complete loop against a mature composition of retrieval, CAD and editors, generative sampling, explicit search and quality-diversity, scratchpads and shared workspaces, active learning, simulation, version control, provenance, and independent reranking. The mathematical contract keeps exposure, retrieval, relative novelty, reinterpretation, information gain, fixation, negative transfer, constraint margins, evaluator qualification, selection regret, lineage, reconstructability, and lifecycle cost separate. If that ordinary stack ties, the additional architecture is removed while the fixture remains as a negative result.

3. Maintenance

Maintenance receives a budgeted set of possible actions. For any memory, module, or route it may:

- defer while collecting more evidence;
- replay against related and conflicting cases;
- merge redundant records while retaining provenance;
- externalize mutable propositions to an inspectable store;
- reduce influence or plasticity;
- delete state after retention and safety checks;
- reopen mature structure in a reversible branch;
- change logical topology or physical placement; or

- promote stable computation into a cheaper representation.

The formal action model is defined in `../math/memory-lifecycle.md`. Structural changes remain shadowed and reversible until regression, calibration, and energy tests pass.

4. One event end to end

1. A sensor, user, or tool changes the current state.
2. The runtime predicts what matters and estimates uncertainty.
3. The selector chooses depth, modules, memory, precision, and possible intervention under the current budget.
4. Execution produces an outcome and physical telemetry.
5. The adaptation loop stores an attributable episode and may construct a bounded hypothesis or provisional module.
6. The maintenance loop later replays the episode against related and conflicting evidence.
7. A validated regularity may be consolidated, compiled, quantized, externalized, relocated, or used to change future routing.
8. Regression or fragility tests can reject the change and restore the previous state.

5. State ownership

STATE	PRIMARY OWNER	NORMAL WRITE PATH	REASON FOR SEPARATION
Current predictive state	runtime	every event	cheap, transient, task-specific
Attributable episodes	adaptation	observed outcome	preserves evidence before abstraction
Provisional hypotheses	adaptation	bounded generation and trials	permits novelty without global drift
Reusable skills and representations	slow model	validated consolidation	stable transfer across events
Mutable propositions	factual memory	sourced, versioned update	correction, attribution, and conflict
Hardened execution paths	compiled store	promotion pipeline	lower repeated interpretation cost
Lifecycle and fragility state	maintenance	replay, probes, regressions	repair remains separate from execution
Resource and risk policy	controller	calibrated policy update	local mechanisms cannot hide physical cost

Efficiency mechanism

Efficiency is an event-level constrained decision. At time t, the selector chooses an admissible action a from $\mathcal{A}_t$:

$$a_t^* = \arg\max_{a \in \mathcal{A}_t} \mathbb{E}[\Delta V(a) \mid s_t]$$

subject to

$$E(a) \le B_t^E, \qquad L(a) \le B_t^L, \qquad R(a) \le \epsilon_t^R.$$

Here s_t is the predictive state; $\Delta V(a)$ is expected task-value improvement; $E(a)$ is energy in joules; $L(a)$ is latency in seconds; $R(a)$ is the declared risk measure; and B_t^E, B_t^L, and ϵ_t^R are event-specific limits. Measured cost after execution recalibrates the estimator.

Lifecycle comparison includes the work usually hidden outside inference:

$$E_{\text{life}} = E_{\text{run}} + E_{\text{memory}} + E_{\text{network}} + E_{\text{adapt}} + E_{\text{maint}} + E_{\text{recovery}}.$$

An architecture advances only if it improves the joint frontier of quality, risk, latency, and lifecycle energy against its strongest conventional baseline. Skipped FLOPs alone are not an efficiency result.

Evidence status

COMPONENT	CURRENT EVIDENCE	TRANSLATION STATUS
Conditional experts and early exit	C-003, C-004	implemented mechanisms; physical benefit remains system-specific
Predictive residual allocation	C-005–C-007, C-022	plausible composition; grounded experiment required
Fast/slow memory and replay	C-008–C-010, C-036–C-042	scoped evidence; lifecycle controller experimental
Structural maturation and reopening	C-012, C-043–C-045	biological interventions established; digital state machine speculative
External factual memory	C-014	usable mechanism; conflict and energy policy unresolved
Local resource and context control	C-046–C-051	biological observations established; proposed control planes experimental
Endogenous proposal generation	C-061–C-066	constituent observations established; integrated curriculum speculative
Versioned reconstructive design	C-842–C-860	constituent observations scoped; complete externalize–inspect–transform–evaluate loop unvalidated
Complete three-loop system	none	unvalidated synthesis

The engineering analogue audit maps the architecture to feedback control, value of information, caches, schedulers, change detection, adaptive routing, replicated state, and system identification. Each experiment must compare against the strongest relevant version of those methods.

Speculative extensions

- Receiver-specific fast and slow context channels under a hard bit budget.
- Modules bidding for compute by expected value improvement while a global controller enforces physical and safety constraints.
- Active fragility probes on shadows or replicas before topology changes.
- Capability-gap repair that admits complementary modules instead of restoring arbitrary capacity.
- Joint learning of logical routing, memory placement, and hardware locality.

Each remains outside the core architecture until its isolated contract beats the appropriate engineering baseline.

Failure modes

- **Dense control disguised as sparsity:** routing, broadcast, or monitoring moves nearly as much state as dense execution.
- **Controller conflict:** runtime, adaptation, maintenance, and resource loops oscillate or optimize incompatible signals.
- **False maturity:** a shortcut appears stable and is hardened before compositional and intervention tests.
- **Memory laundering:** unsupported episodes become slow parameters without provenance or contradiction checks.
- **Maintenance dominates cost:** replay, probes, migrations, and regression testing consume the savings of conditional execution.

- **Fixation propagates through shared state:** early examples or collaborators collapse independent branches before alternatives are tested.
- **Generation impersonates evaluation:** the proposal model's own score is reported as independent evidence of validity or usefulness.
- **Lineage impersonates truth:** a complete derivation record is treated as proof of correctness, intention, authenticity, or originality.
- **Renaming without mechanism:** a familiar cache, controller, or scheduler receives a biological label without a different causal operation.

Measurable predictions

The architecture earns implementation only if isolated experiments support the following predictions:

1. Conditional execution reduces measured data movement and wall energy at matched quality, calibration, and tail risk.
2. Fast attributable memory improves adaptation latency while slow-model regression remains bounded.
3. Reversible maturity gates reduce catastrophic drift without blocking necessary relearning or newcomer admission.
4. Structured recombination plus targeted intervention produces more valid, useful novelty than matched-budget stochastic sampling alone.
5. Editable externalization, inspection, representation change, and epistemic action improve hidden-constraint validity, selection regret, and transfer beyond the complete retrieval/editor/generation/search/tool stack without increasing fixation or lifecycle cost.
6. Maintenance improves retention and adaptability after replay, monitoring, migration, and recovery costs are counted.
7. At least one proposed mechanism beats the strongest ordinary controller, scheduler, cache, router, or system-identification baseline.

Until those tests pass, this is a working architecture: precise enough to reject, modular enough to simplify, and explicitly incomplete.

Biology is a launchpad, not a ceiling

Scope

This chapter defines how the project may borrow from living systems without turning a biological implementation into a specification. The brain is an existence proof for efficient adaptive intelligence, but neither evolution nor neural tissue optimized the same objective, hardware, scale, or reliability contract as an artificial system.

The working rule is:

> *Reproduce the useful constraint or computation, not the substrate accident.*

Biological observation

Biological intelligence is shaped by slow, noisy, failure-prone components; metabolically expensive communication; physical growth; local chemical signals; and an inability to checkpoint, clone, or roll back a whole organism. Brains compensate with local processing, sparse activity, redundancy, specialized cell types, multiple timescales, and continual maintenance.

Those compensations are informative, but they are not automatically optimal on silicon. Digital systems offer different capabilities: fast switching and interconnect, exact copying, explicit addressing, external storage, reversible experiments, global synchronization when it is worth its cost, and precision that can be selected per operation. At the same time, digital systems still pay heavily for data movement, memory access, communication, cooling, and idle capacity. The relevant question is which constraints and useful computations survive a change of substrate.

The speed comparison must stay qualitative until a shared task and latency boundary exist. Axonal conduction, synaptic integration, transistor switching, GPU kernels, and cluster collectives are different operations. Selecting the fastest number from each domain would recreate the invalid comparison rejected by C-016.

Proposed AI translation

For each biological candidate, record five transformations:

1. **Observed function:** what the organism demonstrably achieves.
2. **Biological constraint:** which physical or evolutionary limit shaped it.
3. **Candidate invariant:** the computation that may survive a substrate change.
4. **Silicon implementation:** the least literal engineered mechanism that tests the invariant.
5. **Escape hatch:** the biological limitation that engineering should not inherit.

BIOLOGICAL PATTERN	CANDIDATE INVARIANT	SILICON-NATIVE ESCAPE HATCH
Mostly local signaling	Price communication and keep repeated work near its state	Permit fast global exchange when its measured value exceeds its traffic cost
Slow, noisy spikes	Event-driven, uncertainty-sensitive updates	Use dense vector arithmetic or exact digital state where it is cheaper
Synaptic and dendritic computation	Compute near stored state; route locally before global aggregation	Implement fused kernels, hierarchical memory, or programmable modules rather than literal morphology
Sleep and replay	Separate acquisition from protected integration	Consolidate asynchronously, continuously, or from exact checkpoints
Development and pruning	Explore with reversible capacity, then commit after evidence	Grow, clone, roll back, and reallocate modules without waiting for physical development
Multiple memory systems	Match update rate and provenance to information lifetime	Use databases, caches, logs, tools, and versioned weights unavailable to animals
Homeostasis and repair	Treat stability and maintenance as active control loops	Use telemetry, deterministic tests, redundancy, and replacement hardware

This transformation is a project decision, not evidence that any row will produce a gain. The candidate inventory is maintained in the neuroscience opportunity map and comparative-biology map. Its relationship to existing AI is tracked in the adoption matrix.

Efficiency mechanism

The approach preserves useful functions in the cheapest available form and uses digital capabilities—including exact copies, reversible branches, and direct addressing—where they improve the measured system.

For a biological candidate m, the engineering experiment compares at least three systems: a conventional baseline B, a literal or close biological translation L_m when meaningful, and a silicon-native abstraction S_m. Using the measurement contract from the energy model, the candidate is interesting only if S_m improves the quality–risk–energy frontier:

$$(Q, R, E, L)_{S_m} \succ (Q, R, E, L)_B,$$

where Q is task quality, R is the declared risk metric, E is energy within the declared boundary, and L is latency. The symbol $\succ$ means Pareto dominance under pre-registered tolerances; it does not collapse unlike units into a decorative score.

Evidence status

- Neural energy constraints are established only within the scope of C-001.
- Nonlinear dendritic subunits, homeostatic scaling, neuromodulated plasticity, and specialized inhibitory control are biological observations under C-017 through C-020; their proposed artificial abstractions are not thereby validated.
- Performance of the complete cross-substrate system remains speculative until matched engineering tests establish it.

Speculative extensions

- Learn when global communication is worth buying instead of banning it.
- Compile frequently reused local circuits into fast deterministic paths while retaining a slower plastic path for exceptions.
- Search jointly over algorithm, memory hierarchy, interconnect, precision, and physical embodiment rather than treating hardware as a final deployment detail.

- Import organizational principles from organisms without centralized brains, including plants, immune systems, cephalopod limbs, and adaptive transport networks.

Failure modes

- A metaphor is mistaken for a mechanism.
- “Biological” becomes an unearned synonym for efficient or intelligent.
- “Silicon is faster” is used to ignore memory movement, synchronization, or thermal limits.
- A negative result for one implementation is treated as falsifying the underlying biological abstraction—or vice versa.
- The project claims an idea is absent from AI after finding only that it is absent from mainstream foundation models.
- Evolutionary fitness is confused with task accuracy, truthfulness, or human values.

Measurable predictions

This framing earns its place if, across isolated mechanism experiments:

- silicon-native abstractions equal or outperform literal translations at matched quality and system boundary;
- retained biological constraints predict where energy or interference is saved;
- removing a proposed invariant removes the benefit even when the biological surface form remains; and
- at least one hybrid uses a capability unavailable to biology—such as exact rollback or external versioned memory—to improve continual learning without increasing the declared risk metric.

Cross-domain convergence and principle deduplication

> *Different sciences often describe the same constrained operation in different nouns. The architecture should pay for the operation once.*

Scope

This chapter turns the project's open-world research policy into an architectural method. Findings may originate in neural tissue, plants, immune systems, animal collectives, ecosystems, control theory, databases, or materials. They enter the design only after their causal operation has been normalized and compared with the principles already present.

The canonical unit is therefore neither a paper nor an organism. It is a versioned problem–solution invariant with a stable `P-` identifier, scoped evidence, an engineering null model, and an experiment that can reject its AI translation. The detailed registry remains in the principle ledger; this chapter explains how its bundles compose into one system.

Biological observation

The current evidence corpus repeatedly encounters five pressures.

1. **Scarcity:** more states, candidates, or possible actions exist than can be active at once. Examples include constrained cortical signaling (C-001), sparse insect odor codes (C-025), immune selection (C-028), and congestion-triggered reserve routes (C-035).
2. **Locality:** a complete central description is too slow or expensive. Dendritic branches integrate locally (C-017); cephalopod limbs retain peripheral control (C-024); and collective systems can coordinate through sparse influence or shared environmental state (C-031, C-054).
3. **Reversible change:** recent evidence must affect behavior before it deserves permanent structure. Eligibility traces (C-019), plant priming (C-026), selective replay (C-036), and retrieval-sensitive memory (C-039) all separate a temporary state from a later commitment decision.
4. **Stability:** selection and learning create positive feedback that must be bounded. Synaptic scaling (C-018), feedback inhibition (C-025), reconstruction around a target organization (C-033), and ecological recovery dynamics (C-058, C-059) expose different parts of that control problem.
5. **Repeated work:** a stable solution should migrate from expensive general search into topology, placement, representation, or material. Relevant observations include body–controller co-development (C-029), structured pruning (C-012), lower-precision representation (C-013), and activity-dependent resource placement (C-050).

These recurrences do not make the underlying mechanisms identical. They reveal where multiple fields expose the same engineering pressure and where one shared experiment can replace several renamed proposals.

Proposed AI translation

From domain language to a mechanism record

Every retained observation is rewritten as the following tuple:

$$M = \langle p, s, b, x, f, G, \tau, \rho, \phi \rangle,$$

where:

SYMBOL	FIELD	REQUIRED CONTENT
p	problem	the failure or objective faced by the observed system
s	source organization and scale	form/material, process/organism, ecosystem/collective, or an explicit cross-scale interaction
b	constrained budget	energy, bandwidth, material, time, risk, or capacity with declared units
x	sensed state	what the mechanism can actually observe
f	causal operation	the intervention-supported state transformation
G	information topology	local, hierarchical, broadcast, pairwise, or environment-mediated flow
τ	timescale	event cadence or duration in declared steps or seconds
ρ	reversibility	what can decay, reopen, roll back, or be reconstructed
ϕ	failure boundary	conditions under which the effect disappears, reverses, or becomes harmful

The tuple is a structured research record, not a numerical embedding. Its fields retain units and provenance. Two findings are candidates for one principle only when their problem, causal operation, information topology, and timescale agree at the abstraction needed by an experiment.

Source organization and scale are not optional metadata. A material structure, an organism-level control loop, and an ecosystem-level interaction can deliver a similar function through different causal paths, authority, and timescales; they do not merge until those differences are shown irrelevant to the discriminating experiment.

Biomimetic transfer is a search method, not an evidence grade

Biomimetics gives this project useful process language, but it does not create a shortcut through the evidence ledger. The search runs in both directions:

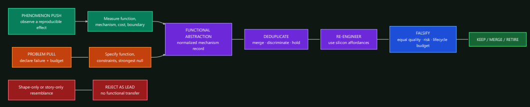
1. **Phenomenon push** begins with a measured effect in a living system and asks which artificial problem shares its function, constraints, causal operation, and failure boundary.
2. **Problem pull** begins with a measured artificial-system failure and asks which scientific fields contain a mechanism that solves the normalized problem under a comparable budget.

`Phenomenon push` broadens the established biology-push/solution-driven route; `problem pull` broadens technology-pull/problem-driven design. The broader names matter because this project searches physical, formal, social, and engineering sciences as well as biology.

Both routes converge on M, then pass through deduplication, silicon-native redesign, a strong conventional null, and a falsifiable equal-budget test. A visual likeness without transferred function is rejected; a transferred function without a causal and cost model remains only a lead;

and a validated source mechanism does not establish that its artificial translation works.

Wide diagram · fit to page



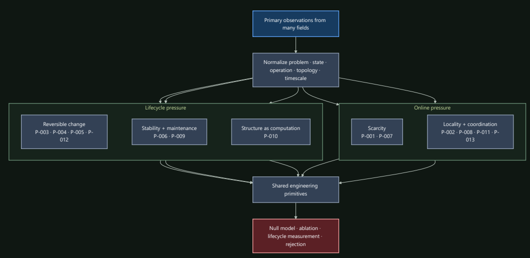
Editable source: biomimetic-bidirectional-transfer.mmd.

The public descriptions of biomimicry are useful for discovering biological strategies and unifying themes, while ISO 18458:2015 and VDI 6220 Part 2 provide more precise terminology and process frames. None is evidence that a particular mechanism transfers or improves efficiency. The dated method audit records the source hierarchy and the parts adopted here.

Five solution families

The thirteen current `P-` principles remain distinct, but they compose into five navigational families:

Wide diagram · fit to page



Editable source: `../assets/diagrams/recurring-solution-families.mmd`.

FAMILY	INCLUDED PRINCIPLES	SHARED ARTIFICIAL PRIMITIVE	IMPORTANT SEPARATION INSIDE THE FAMILY
scarcity	P-001, P-007	budgeted selection plus the option to buy more evidence	selecting a candidate is different from deciding whether uncertainty warrants more work
locality and coordination	P-002, P-008, P-011, P-013	stateful local modules with typed, priced communication paths	escalation, temporal binding, and shared workspaces solve different coordination failures
reversible change	P-003, P-004, P-005, P-012	versioned candidates, traces, memories, and topology with explicit promotion or decay	delayed credit, episodic content, candidate diversity, and graph mutation require different state
stability and maintenance	P-006, P-009	a slower controller that observes aggregate state and can throttle, repair, replay, or reopen	regulating a variable online is different from scheduling lifecycle work
structure as computation	P-010	migrate mature recurring work into topology, placement, precision, or compiled paths	each target has different invalidation, migration, and recovery costs

Deduplicate before architecture expansion

A new finding passes through six decisions:

1. **Scope the evidence.** Record the observed system, intervention, result, uncertainty, and boundary as a `C-` claim.
2. **Normalize the mechanism.** Fill every field of M without organism-themed names standing in for operations.
3. **Search the registry.** Compare against all existing `P-` records, including held candidates and negative evidence.
4. **Merge or discriminate.** Merge when the same control loop would be tested; otherwise name the smallest experiment that distinguishes the mechanisms.
5. **Name the strongest null.** A scheduler, cache, controller, estimator, database, routing rule, or standard learning method gets the same interface and budget.
6. **Change one canonical object.** Update a principle, primitive, equation, diagram, or experiment instead of adding another themed architecture.

This creates a many-to-one evidence graph:

```
many observations → scoped claims → fewer principles → shared primitives → decisive experiments
```

Composition without double counting

Principles can cooperate while remaining separately testable. For one event:

1. prediction-error allocation decides whether more evidence is valuable;
2. selective allocation chooses a module within the active budget;
3. local autonomy runs that module near its state;
4. a temporary trace records unresolved credit;
5. the maintenance plane later decides whether to replay, protect, merge, or forget it; and
6. structural offloading compiles only the recurring path that survives those tests.

Calling the entire chain “sparsity,” “memory,” or “homeostasis” would hide which operation produced a gain. Each experiment therefore removes one principle at a time while holding the other interfaces constant.

Passive self-organization is a mandatory null

Efficient-looking structure does not establish sensing, represented goals, or counterfactual action. Fracture sets, drainage networks, and river avulsions can arise from local stress, gravity, flow, conservation, thresholds, and stored geometry (C-232–C-242). The project calls a mechanism adaptive control only when it specifies:

1. a service variable external to the adaptation law;
2. observations with spatial/temporal support and latency;
3. a decision that could choose differently under different evidence;
4. authority and an actuator;
5. movement, reserve, monitoring, and recovery budgets;
6. persistent state and reset cost; and
7. a guarantee or tested failure envelope.

This criterion makes passive physics a stronger baseline. If local flow–structure feedback produces the same topology and service without a controller, the controller must justify its sensing, decision, switching, and maintenance cost. Connectivity is not throughput (C-234); material removed by “pruning” must appear as transport, storage, or output elsewhere (C-239); and a topology change can reassign or destroy service rather than improve it (C-242).

Transported fields are not messages

Microbial signals, metabolites, fungal nutrients, plant hormones, process streams, queues, and network packets can all create a local field that later receivers use. The common systems object is not "communication in nature." It is a transported, transformed, delayed, and locally decoded quantity whose physical path may also change future capacity (C-563, C-578, C-580).

For transported type k, use the local balance

$$\frac{\partial c_k}{\partial t} = D_k \nabla^2 c_k - \mathbf{u} \cdot \nabla c_k + p_k - q_k - r_k c_k,$$

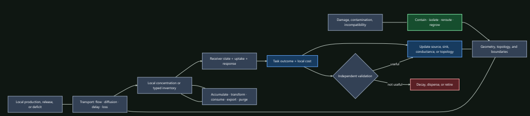
where concentration c_k is mol/m³, diffusion coefficient D_k is m²/s, velocity $\mathbf{u}$ is m/s, production and uptake rates p_k, q_k are mol/(m³ s), and first-order loss r_k is 1/s. Digital fields use an analogous typed byte or item ledger; they do not inherit molecular units or diffusion laws. Geometry, boundary conditions, sampling support, and receiver state are part of the contract.

Several fields can be superposed at the same endpoints. If the observed outcome is

$$y = \sum_{g=1}^{G} h_g(c_g, \theta_g) + b + \varepsilon,$$

where G is the number of typed channels, g indexes one channel, c_g is its transported concentration or inventory, h_g is its receiver-dependent effect, θ_g is channel state, b is background, and ε is measurement error. Observing y or cutting every channel together does not identify any one h_g. Each claimed channel needs an independently targetable intervention or a justified identification model. Rillig et al.'s concurrent-fungal-network perspective makes this problem explicit for fungal guilds sharing plant endpoints; the project adopts it as an engineering inference, not as evidence that such guild combinations have a demonstrated beneficial function (DOI, C-217, C-584).

Wide diagram · fit to page



Editable source: transported-field-contract.mmd.

The diagram deliberately spans several fields while retaining their differences:

- microbial quorum output depends on production, confinement, flow, loss, and receptor state; it is not a vote count (C-563, C-575);
- cross-feeding can help, harm, or create a weakest-link dependency, so local transfer is not system utility (C-571);
- fungal flow can both deliver material and remodel conductance while rapid containment remains distinct from later bypass (C-577–C-581); and
- distinct symbiotic partners can exchange unequally under conflicting objectives (C-584); and

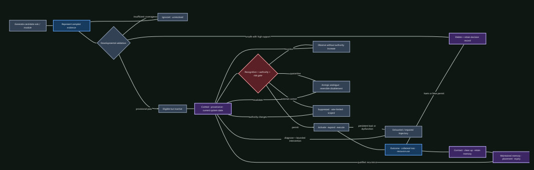
- concurrent typed channels sharing endpoints remain causally unresolved when the experiment observes or disables only their aggregate.

The held fixture belongs jointly to P-011/P-013 and existing Candidates 001 and 013 (C-585). It must beat typed pub/sub or queues, fixed-graph adaptive routing, backpressure/primal–dual allocation, and make-before-break state transfer. Charge production, transport, storage, cleanup, contamination, topology edits, failed delivery, reserve, and measurement. If geometry and cleanup do not affect the task, use the simpler digital primitive.

Inactive is not one state

The immune evidence in C-727–C-747 does not add another architecture principle. It contributes a stricter state contract. Representation, recognition, authentication, authorization, activation, suppression, deletion, impairment, contraction, memory, and recovery answer different questions and have different reversal costs.

Wide diagram · fit to page



Editable source: typed-tolerance-lifecycle.mmd.

An activation bit cannot distinguish an absent module from an unobserved one, a policy-suppressed module from a damaged one, or post-response contraction from retained memory. That ambiguity makes reactivation unsafe and makes resource accounting wrong. Each transition therefore records evidence support, authority, expected useful effect, collateral-loss terms, resource budget, expiry, recovery target, and observed outcome.

Recognition supplies evidence; it does not grant authority. Deletion requires a higher support threshold than reversible monitoring or quarantine because a rare useful capability can be lost. Local copies pay maintenance, refresh, invalidation, and inconsistency costs. Recovery ends only when useful service and reserve are both restored.

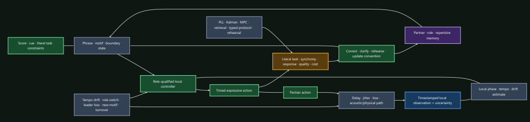
The typed lifecycle mathematics compares this contract against a complete ordinary stack: typed state machines, calibrated risk and abstention, least privilege, constrained control, anomaly detection, evolutionary or ensemble search, replay, placement, and resource-aware scheduling. If the ordinary stack reproduces the decisions and frontier, the biological vocabulary is removed and the state contract remains.

Coordination without a privileged clock

Music cognition provides a combined stress regime for mechanisms that already exist elsewhere in the project: multiscale prediction, local timing control, transformation-aware memory, constrained generation, role protocols, repair, and cumulative transmission. The evidence in C-748–C-783 does not justify a music-specific principle.

Its useful residue is a benchmark: agents must preserve literal phrase constraints and bounded relative timing while tempo, expressive microtiming, delay, roles, motifs, leaders, and partners change, without reading a privileged shared clock.

Wide diagram · fit to page



Editable source: shared-clock-free-coadaptation.mmd.

The fixture refuses several easy substitutions:

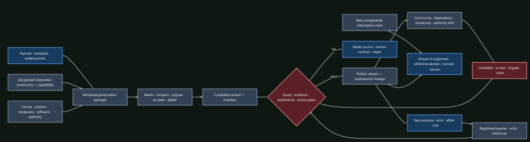
- similar mean tempo is not pairwise synchronization;
- stimulus-frequency neural power is not evidence of endogenous prediction;
- low onset error is not the correct phrase or role;
- familiar-partner performance is not co-adaptation when retrieval explains it;
- novelty is not validity, responsiveness, or usefulness; and
- within-group convention compression is not cross-group task success.

Fixture F-001 compares the full composition with fixed correction, PLL/adaptive oscillator, Kalman identification, distributed MPC, a paid centralized conductor, transformation-aware retrieval, and typed role/version protocols. The mathematical contract keeps local clock error, phase, phrase prediction, role safety, message bytes, latency, rehearsal, recovery, and joules separate. If online estimation plus retrieval and explicit protocol state ties, the residual is retired.

Preserved bits are not preserved meaning

Library and archival science adds a dependency that storage abstractions often hide: information remains useful only for a declared interpreter community with the formats, schemas, vocabularies, identity mappings, authority evidence, software, units, permissions, and practiced capability needed to use it. The audit evidence will occupy C-784–C-803.

Wide diagram · fit to page



Editable source: query-registered-preservation.mmd.

This separates several failure modes that are otherwise mistaken for one success:

- fixity detects bit change; it does not establish truth or authenticity;
- provenance records lineage; it does not validate the assertion;
- a valid ontology constrains representation; it does not prove instance data;
- a citation edge is not endorsement or comparable impact;

- findability and interoperability do not establish correctness or safe reuse; and
- retaining records does not retain the tacit capability needed to interpret or act on them.

The query-registered preservation mathematics defines finite native-unit tolerances, evidence reachability, interpreter and dependency validity, migration regression, unregistered-query abstention, and complete lifecycle cost. The Candidate 017 track must beat ordinary OAIS/PREMIS-style packaging, versioned schema/vocabulary, retained-source recovery, and query-regression tests. If it does not, the ordinary preservation stack remains the implementation.

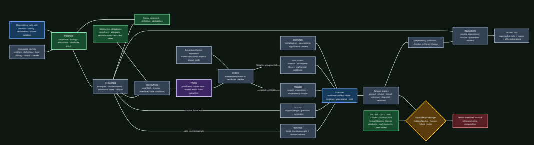
Candidate production is not acceptance

Mathematical practice supplies the sharpest version of a distinction already needed throughout this project: producing a candidate and earning acceptance are different operations. Conjectures, analogies, abstractions, proof sketches, retrieved lemmas, numerical patterns, solver traces, and learned tactic choices can all direct search. None of those origins establishes the resulting claim. The evidence and formal-system boundaries are tracked in C-861–C-880.

The transferable lifecycle is typed:

1. **Propose.** Bind a candidate to the exact problem, definitions, logic, axioms, library, corpus, proposal ancestry, and accessible evidence.
2. **Challenge.** Search examples, counterexamples, countermodels, boundary cases, definition changes, and adversarial encodings before commitment.
3. **Decompose.** Turn the target into a dependency DAG whose child artifacts reconstruct the parent goal through named interfaces and side conditions.
4. **Prove or refute.** Produce a proof term, checked finite reduction, solver certificate, admissible model, or exact counterexample—not only a verdict.
5. **Check.** Bind the artifact to the exact instance and library version, then run a small independent checker while publishing shared trust roots.
6. **Publish.** Retain `proved`, `refuted`, `tested`, `unknown`, `disputed`, and `retracted` as different states with evidence, provenance, and cost.
7. **Invalidate.** A definition, axiom, checker, dependency, or library change quarantines its reverse-dependency closure until rebuild or recheck.

Wide diagram · fit to page



Editable source: versioned-proof-discovery-lifecycle.mmd.

This lifecycle prevents five common substitutions:

- finite agreement is evidence for a tested range, not a universal result;
- a counterexample rejects the encoded universal claim but does not choose its unique repair;
- individually plausible lemmas do not close a parent goal unless their proof artifacts reconstruct it;

- a checker establishes the encoded proposition relative to its logic, definitions, axioms, libraries, preprocessing, and trusted implementation; it does not validate the informal intent or importance; and
- participant or model count does not establish independent checking when sources, parsers, libraries, prompts, or failure modes are shared.

Fixture F-004 compares the joined lifecycle with complete interactive and automated theorem provers, formal libraries, CDCL SAT, DPLL(T) SMT, constraint and model finding, CEGAR/CEGIS, learned premise/tactic guidance, exact numerical and exhaustive methods, and ordinary expert review. The mathematical contract defines immutable identities, leakage closures, abstraction obligations, proof-DAG reconstruction, certificate cost, typed states, invalidation, human effort, and joules. If the mature stack ties, the composition is retired; the state distinctions and negative result remain.

Efficiency mechanism

Deduplication saves two different resources.

First, it reduces research duplication. If plant priming, eligibility traces, and cache admission all motivate a temporary state before commitment, the project maintains one promotion interface and tests the domain-specific differences as variants. Papers and claims remain separate; architecture and instrumentation are reused.

Second, the resulting families define where the runtime should avoid repeated work:

- scarcity families reduce unnecessary activation and acquisition;
- locality families reduce movement and synchronization;
- reversible-change families prevent every event from rewriting durable state;
- maintenance families move repair and integration off the critical path; and
- structural offloading reduces the recurring cost of mature behavior.

For principle j, lifecycle acceptance uses the energy contract from chapter 80:

$$\Delta E_j(N) = N\,(E_{B,\text{event}} - E_{j,\text{event}}) - E_{j,\text{introduce}} - E_{j,\text{maintain}} - \mathbb{E}[E_{j,\text{recover}}],$$

where N is a dimensionless count of qualified events, event terms are joules per qualified event, and introduction, maintenance, and expected recovery are joules over the same observation horizon. A positive $\Delta E_j(N)$ is only an energy result; quality, risk, calibration, latency, and resilience must also remain inside their declared envelopes.

Evidence status

PROPOSITION	STATUS	BASIS
several scientific domains expose recurring scarcity, locality, memory, stability, and structural pressures	established within the current scoped corpus	claims ledger and domain inventory
candidate production and scoped formal acceptance are different artifact states	established or plausible within the audited formal systems and experiments	C-861–C-879; the joined lifecycle in C-880 remains plausible
the thirteen current principles are the correct deduplication	plausible working taxonomy	principle registry; boundaries remain revisionable
five families provide a useful navigation layer without erasing mechanism differences	proposed synthesis	must improve retrieval, experimental reuse, and reviewer agreement
recurrence across less-related domains predicts a useful artificial primitive	speculative	requires prospective tests against matched null models
the complete composition improves lifecycle efficiency	speculative	isolated and composed experiments have not yet established it

Speculative extensions

- Represent claims, principles, primitives, experiments, null models, and failures as a queryable versioned graph while keeping Markdown canonical.
- Track negative results as first-class edges so a rejected translation is not repeatedly rediscovered under another domain name.
- Estimate reviewer agreement on mechanism tuples before allowing a new principle ID.
- Search specifically for counterexamples: fields where the same pressure produces a different stable solution or where the recurring solution fails.
- Use contradictions between domains to generate new experiment regimes rather than averaging the difference away.
- Maintain silicon-native escape routes beside every principle: exact copying, direct addressing, typed storage, rollback, high-speed communication, and variable precision can change which operation is cheapest.

Failure modes

- **Naming duplication:** the same feedback loop appears repeatedly as a new organism-inspired component.
- **False convergence:** similar diagrams hide different sensed variables, causal operations, or timescales.
- **Overcompression:** a family becomes so broad that no ablation can isolate its mechanism.
- **Evidence laundering:** recurrence is treated as proof of independence, optimality, or transfer to AI.
- **Null-model neglect:** a familiar cache, scheduler, controller, or estimator is omitted because the biological story sounds novel.
- **Interface drift:** two variants share a principle ID while receiving different inputs, budgets, or evaluation envelopes.
- **Positive-result bias:** failed translations disappear, so the same proposal returns with a new metaphor.
- **Double-counted savings:** two principles claim the same avoided operation or compare against different baselines.
- **Verifier laundering:** checker acceptance is reported as truth about an informal statement without exposing formalization, axioms, dependencies, or the trusted implementation.
- **Finite-to-universal leakage:** a sampled numerical or test range is silently promoted from `tested` to `proved`.
- **Correlated checking:** producer and verifier share the parser, library, preprocessing, or defect that matters while being counted as independent.
- **Taxonomy lock-in:** stable IDs are mistaken for immutable scientific truth.

Measurable predictions

1. Independent reviewers given only normalized mechanism records agree on merge-versus-separate decisions more often than reviewers given titles and domain descriptions alone.
2. As domain coverage grows, the number of supporting claims per accepted principle rises faster than the number of principles; a near one-to-one ratio signals failed deduplication.

3. At least one proposed organism-specific mechanism is experimentally indistinguishable from an existing primitive at matched interface and cost and is removed rather than renamed.
4. Experiments built around shared principles reuse telemetry, null models, and failure regimes across more than one source domain.
5. Per-principle ablations attribute lifecycle savings to distinct avoided operations; overlapping savings disappear when all variants use one common baseline and boundary.
6. A principle promoted from recurrent evidence survives at least one regime derived from a domain outside the one that originally motivated its AI translation.
7. A typed propose–challenge–prove–check lifecycle reduces bad acceptance, localizes repair, and improves dependency invalidation beyond a generator plus checker alone after search, certificates, libraries, human effort, and joules are charged; otherwise Fixture F-004 retires the composition.

Structural growth, specialization, and conditional routing

Scope

Define how a modular system acquires new capacity without running, training, or retaining every possible module for every event. Growth is admitted only for a measured capability gap, new capacity earns traffic in probation, and mature capacity remains eligible for merge, reopening, or retirement.

The central object is a **capacity lifecycle**, not a continuously expanding pool. Birth, routing, specialization, placement, consolidation, and removal are separate decisions with separate costs.

Biological observation

Developing nervous systems generate and reorganize more cells and connections than remain in mature circuits. In the studied mouse retinogeniculate system, relative activity and complement signaling changed microglial engulfment and retention of developing inputs (C-043). This supports activity-sensitive structural refinement by a slower maintenance process, without specifying one general pruning rule.

Protection is not necessarily permanent. Targeted extracellular and receptor interventions reopened specific forms of adult visual-cortex plasticity (C-044, C-045). The resulting engineering states are candidate, consolidating, protected, reopened, and retiring. A module can be stable without becoming impossible to revise.

Other biological systems expose control operations that recur at different scales:

- germinal-center affinity maturation combines variation, selection, expansion, and later protection of useful lineages (C-028);
- *Physarum* and fungal networks couple use-dependent reinforcement, exploration, fusion, and contraction to changing flow (C-027, C-034);
- ants can open reserve routes under crowding before throughput falls (C-035); and
- activity can recruit local energy production, alter resource placement, and change local vascular supply in scoped neural preparations (C-049–C-051).

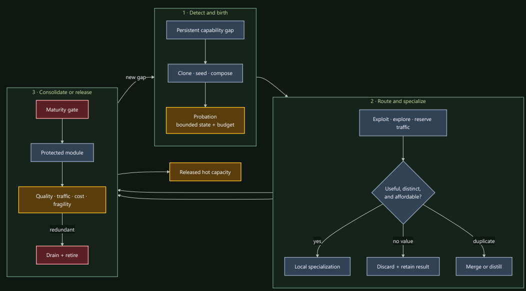
These mechanisms are not interchangeable. Together they impose a useful system constraint: variation and reserve capacity consume resources; selection needs an independent test; frequently used structure may stabilize; unused or duplicated structure must be able to leave the hot path.

A defined microbial-community experiment adds a practical design tactic: adding candidates selected for missing functions repaired the tested community better than merely restoring organism count (C-056). The inverse problem also matters. Functional redundancy was associated with poor newcomer engraftment in two small reanalyzed human microbiome datasets (C-057). Mature capacity can resist both harmful and beneficial entrants, so an artificial newcomer needs protected evaluation traffic rather than permission from incumbent routing logits alone.

Proposed AI translation

The capacity lifecycle

Wide diagram · fit to page



Editable source: `../assets/diagrams/structural-growth-routing.mmd`.

The initial system contains modality encoders, predictive shared state, conditionally addressable experts, hierarchical routers, episodic and factual memory interfaces, and a declared reserve. Total addressable capacity may be large, but each event receives only a bounded route. Reserve capacity is stored, placed, and periodically tested; it is not free merely because it is inactive.

1. Detect a capability gap

Growth begins with a gap record, not a global loss spike. A valid record groups attributable episodes that existing routes fail in a consistent way and asks whether the failure is better explained by:

- missing capability;
- insufficient active compute or depth;
- missing context or memory;
- router error or capacity congestion;
- interference inside an existing module;
- distribution drift; or
- corrupted data, tools, or feedback.

The maintenance plane attempts the cheaper explanations first. New capacity is eligible only when the gap persists across resampling or recurrence, existing modules cannot absorb it inside their interference and resource bounds, and a candidate has a declared validation contract. This ordering prevents every hard example from becoming an expert.

2. Choose a birth operation

The proposal names both the new structure and what it is expected to repair:

BIRTH OPERATION	BEST INITIAL CONDITION	PRINCIPAL COST
Clone and diverge	one incumbent is close but suffers interference	copied parameters, optimizer state, later deduplication
Activate a seed	the gap is genuinely outside active coverage	stored reserve, cold-start training, placement
Compose a module	existing primitives are adequate but repeatedly coordinated	router depth, boundary traffic, compilation work
Reopen a protected module	a previously valid skill must change	regression risk, branch validation, rollback

Every birth creates a versioned provisional module. It cannot write the slow model, replace an incumbent, or claim permanent memory during probation. Initialization source, training episodes, expected role, resource ceiling, and discard path are recorded before it receives traffic.

3. Give the candidate probation traffic

The router divides admitted work into three explicit budgets:

- **exploit traffic** goes to the strongest validated route;
- **exploration traffic** compares plausible candidates on informative events; and
- **reserve traffic** preserves failover and tests paths that would otherwise decay unnoticed.

A newcomer receives a capped share of gap-relevant episodes plus matched control episodes. The control traffic reveals whether it learned a capability or merely a narrow identifier for the failure cluster. Incumbents cannot reduce the evaluation share through their own confidence, but the maintenance plane can stop the trial for quality, risk, latency, memory, or energy violations.

For event x, use a dimensionless routing objective

$$\mathcal{L}_{\text{route}}(x) = \mathcal{L}_{\text{task}}(x) + \lambda_E \frac{\widehat{E}(x)}{E_0} + \lambda_B \frac{\widehat{B}(x)}{B_0} + \lambda_{\text{bal}} \mathcal{L}_{\text{balance}}(x) + \lambda_{\text{churn}} \mathcal{L}_{\text{churn}}(x),$$

where $\widehat{E}(x)$ is estimated joules/event, $\widehat{B}(x)$ is estimated bytes/event across named memory and network boundaries, and E_0 and B_0 are declared reference scales with the same units. All λ coefficients and losses are dimensionless. The physical joule and byte measurements remain separate reported outcomes; normalization does not turn them into task quality.

Balance keeps one expert from taking all traffic. Churn is applied only after a route has accumulated evidence of stable specialization; penalizing early movement would protect arbitrary initialization.

When module reports become strategic

Ordinary routing remains the default. Shadow-price feedback already represents scarcity under declared controller conditions (C-133), and auction or matching language adds nothing when the router can observe costs and every module shares the system objective. A market-like mechanism is in scope only when a persistent module holds decision-relevant private information, can improve its future traffic by misreporting, and faces a real opportunity consequence it cannot reset or evade.

That regime creates specific failures. Selection pressure on a visible metric can damage poorly measured substitute tasks (C-139); proper scoring needs an independently verified outcome and does not establish competence or causal contribution (C-140); peer agreement can reward shared error or collusion (C-141); and fixed-agent truthfulness does not solve false identities or a biased allocator (C-143).

Candidate 008 therefore begins with a cooperative applicability control that it should not beat. Only then does it introduce hidden costs, adaptive metric gaming, protected outcomes, entrants, identity resets, collusion, and allocator deviation. Withheld audits, lineage-bound consequences, protected entrant traffic, and replayable commitments remain only if they improve external task, risk, energy, and latency outcomes after their evaluation and storage costs.

4. Measure specialization rather than naming it

A candidate becomes useful when it improves a defined region of behavior while remaining distinguishable from existing modules. Evidence includes:

- causal improvement when the candidate is admitted and regression when it is ablated;
- reduced interference on incumbents or protected history;
- consistent advantage on held-out gap and recurrence episodes;
- a stable but non-exclusive routing region;
- calibration and rare-case behavior inside its declared envelope; and
- physical cost that remains inside its allocation.

Low routing entropy alone is not specialization: a router can collapse onto a module for the wrong reason. High activation diversity alone is not useful: a pool can fragment one capability across many expensive duplicates. The test is complementary causal contribution at a measured lifecycle cost.

5. Hand off to merge, protection, or retirement

At the end of probation, the candidate has three normal outcomes:

1. **Discard.** It adds no reliable capability. Preserve the negative result so the same proposal is not regenerated indefinitely.
2. **Merge or distill.** It reproduces an incumbent or several modules have converged on one operation. Build a compact branch, rerun intervention and recurrence tests, then drain duplicates.
3. **Consolidate.** It contributes a distinct reusable capability. Pass it to the maturity lifecycle for protection, reopening rules, structured pruning, and rollback.

A protected module remains monitored for traffic, unique contribution, physical placement, recovery, and newcomer exclusion. Persistent redundancy returns it to a merge-or-retire gate. Logical topology changes can be evaluated with Candidate 001, which charges reconfiguration, migration, reserve, controller, and recovery costs.

Conventional null models

The growth controller must beat ordinary ways of allocating or restructuring capacity, not only a frozen weak model:

ID	NULL MODEL	WHAT IT TESTS
N0	Capacity-matched dense monolith	whether modular growth is needed at all
N1	Fixed-capacity sparse MoE with tuned load balancing	whether conditional routing alone explains the gain
N2	Fixed modules plus adapters or low-rank updates	whether new structure beats ordinary parameter-efficient adaptation
N3	Periodic global architecture or topology optimization	whether a standard batch redesign explains local lifecycle control
N4	Usage- or magnitude-prune/retrain cycle	whether causal merge/retire gates add value
N5	Random valid birth, merge, and retirement under equal budgets	whether the lifecycle signals carry information
N6	Trace-aware oracle with future gap labels	unattainable ceiling; never a superiority baseline

Equalize initial capacity, maximum stored capacity, active work, optimizer updates, training examples, router information, tuning trials, migration bytes, validation work, and wall-clock opportunity. Charge candidate failures and discarded births. Otherwise growth buys more search while the null models are asked to solve the task in place.

A topology change carries state

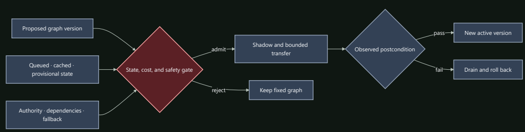
Changing an edge label is not the same as transferring a working system. The process-engineering audit makes the missing state visible: an installed path may hold inventory, energy, contamination, pending work, actuator authority, wear, calibration, maintenance obligations, and shared protection dependencies (C-501, C-510). Two endpoint configurations can both be feasible while the path between them is unsafe (C-512).

For digital modules, use a typed transition inventory rather than pretending that bytes obey material conservation. For state class k,

$$I_k(t_1) - I_k(t_0) = A_k - D_k - X_k + R_k,$$

where I_k is bytes present at a named boundary, A_k is admitted bytes, D_k is deliberately deleted bytes, X_k is exported bytes, and R_k is internally replicated bytes over $[t_0, t_1]$. Each term is bytes and carries a provenance, validity, and ownership version. This is an accounting contract, not a claim that information is physically conserved (C-517).

```
Wide diagram · fit to page
```



Editable source: conservation-qualified-reconfiguration.mmd.

Candidate 001 therefore includes a physical stress track. It must beat fixed-graph adaptive control, multi-mode supervisory control, and offline redesign after installed reserve, transition state, flushing, downtime, maintenance, and rollback are charged. If those nulls tie it, “adaptive topology” describes an implementation choice rather than an efficiency mechanism (C-516).

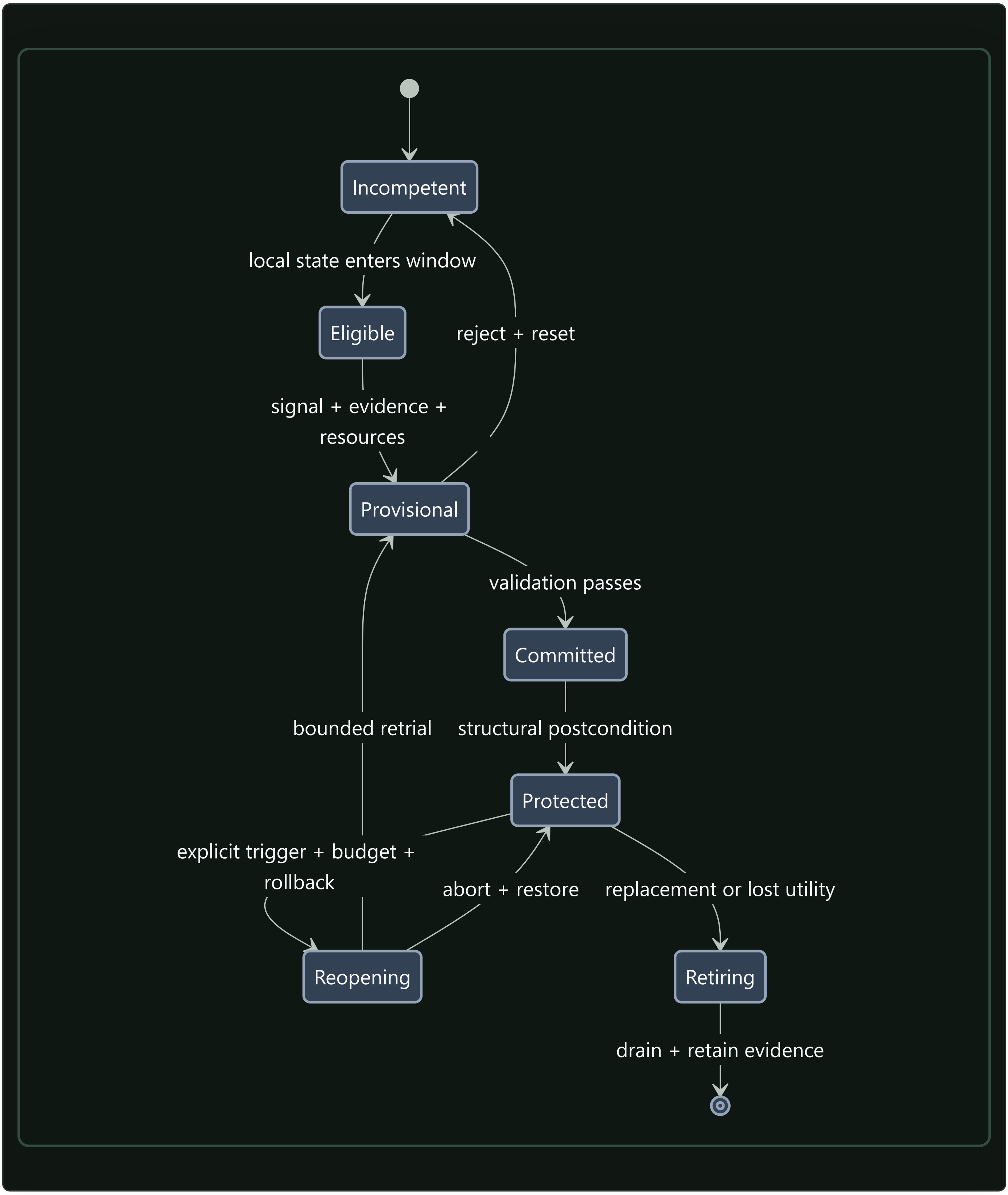
Competence-gated structural transition

A signal is not a complete command. Developmental experiments show that concentration, duration, position, stage, and receiver state can change the response to the same extracellular input (C-539–C-549). Commitment can later be redirected, but reopening is an intervention with selection, resource, integrity, and safety costs rather than free reversal (C-550, C-551).

For module m and transition-contract version v, define the dimensionless admission predicate

$$G_{m,v}(t) = \mathbf{1}\left[q_m(t) \in \mathcal{C}_v, t \in W_v, e_m(t) \ge \theta_v, \mathbf{r}_m(t) \succeq \mathbf{r}_v^{\min}, R_{m,v}(t) = 1\right].$$

$q_m(t)$ is typed local competence state; $\mathcal{C}_v$ is the permitted state region; t is seconds; W_v is a declared time interval in seconds; e_m and evidence threshold θ_v are dimensionless; resource vectors $\mathbf{r}_m$ and $\mathbf{r}_v^{\min}$ are compared componentwise in their native units; and $R_{m,v}$ is a dimensionless predicate that a tested rollback path exists. The signal is eligible to create a provisional transition only when $G_{m,v} = 1$.



Editable source: competence-gated-transition.mmd.

The fixture splits responsibilities instead of adding another candidate:

- Candidate 002 tests low-bandwidth signal decoding by versioned receivers;
- Candidate 009 owns admissibility, authority, dependencies, and invalidation;
- Candidate 010 owns provisional evidence before commitment and reset;
- Candidate 006 owns structural write, health, shadow, and physical reopening; and

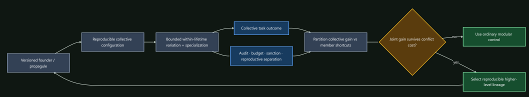
- Candidate 014 owns support, stage, vintage, and the observations defining competence.

Reject the composition if an ordinary gate, schedule, staged verifier, and versioned migration procedure matches quality, interference, rollback, availability, bytes, latency, and joules. The fixture is useful only if the interaction among signal, receiver history, finite window, commitment, and reopening predicts failures those separate controls miss (C-562).

When the unit of adaptation changes

A cooperating set of modules is not automatically a higher-level unit. The claim becomes testable only after defining its boundary, child-configuration event, inherited state, within-lifetime state, descendant relation, collective performance, member-level incentives, and conflict-control cost. This separates aggregation from individuality and a demographic bottleneck from a reproducible founder boundary (C-282–C-295).

Wide diagram · fit to page



Editable source: conflict-bounded-unit-transition.mmd.

For collective lineage k, let Z_k be task-native collective performance and W_k its dimensionless admitted-descendant weight. The selection accounting is

$$\Delta \bar{Z} = \frac{\mathrm{Cov}(W_k, Z_k)}{\bar{W}} + \frac{\mathbb{E}[W_k \Delta Z_k]}{\bar{W}}.$$

The first term is change among declared collectives and the second is transmission change. A nested partition separately reports member-level shortcuts. The identity is accounting, not causal proof; group and descendant definitions are preregistered. Candidate 016 must beat typed modular systems, permissions and tests, clean versioning, external evaluation, routed experts, and ensemble/population selection after enforcement, false-sanction, interface, founder, reserve, and recovery costs.

Efficiency mechanism

For experts $i = 1 \dots n$, let $g_i(x) \in \{0, 1\}$ be the dimensionless event gate, $C_i(x)$ be executed operations/event under a declared precision, and $C_{\text{router}}(x)$ use the same convention:

$$C_{\text{active}}(x) = C_{\text{router}}(x) + \sum_{i=1}^{n} g_i(x) C_i(x).$$

This separates addressable parameter capacity from executed work, as sparse mixture-of-experts systems demonstrate in specific implementations (C-003). It does not price parameter reads, dispatch, all-to-all communication, imbalance, cold starts, or maintenance.

The lifecycle energy per served event is therefore

$$\bar{E}_{\text{capacity}} = E_{\text{route}} + \sum_{i=1}^{n} g_i E_i + E_{\text{comm}} + \frac{E_{\text{birth}} + E_{\text{train}} + E_{\text{place}} + E_{\text{validate}} + E_{\text{merge/retire}} + \mathbb{E}[E_{\text{recovery}}]}{N_{\text{served}}},$$

where every E term is in joules, E_i is expert execution energy/event, and N_{served} is the number of events over the comparison horizon. Report stored parameter and optimizer bytes, bytes moved/event, latency, utilization, quality, calibration, and risk alongside energy.

Growth is efficient only if conditional execution saves more than candidate search, idle reserve, placement, validation, and later contraction consume. Lottery-ticket results show that competitive sparse subnetworks can exist in tested settings (C-012); they do not establish that this lifecycle discovers them or realizes energy savings on a target system.

Evidence status

ELEMENT	STATUS	ROLE IN THIS CHAPTER
conditional expert routing (C-003)	established in published systems	capacity and active compute can be separated in suitable implementations
competitive sparse subnetworks (C-012)	established in tested settings	staged selection and pruning are viable operators
use-dependent biological topology (C-027, C-034)	established in scoped organisms/models	motivates reinforcement, decay, exploration, and contraction tests
diversity, selection, and protection (C-028)	established in the cited immune experiment/model	motivates a bounded candidate lifecycle
congestion-triggered reserve use (C-035)	established in the cited ant setup/model	reserve paths should be priced and tested before overload
developmental refinement and reopening (C-043–C-045)	established in scoped neural preparations	supports distinct candidate, protected, and reopened states
local resource demand and placement (C-049–C-051)	established in scoped neural preparations	physical supply and movement belong in routing cost
shadow prices, matching, proper scoring, metric pressure, identity, and allocator credibility (C-133–C-143)	established under scoped economic models and experiments	ordinary routing remains the null; contestable allocation is conditional on a measured strategic-information problem
audit-backed contestable allocation (C-144)	speculative systems composition	Candidate 008 must lose its distinction when modules are cooperative and directly observable
capability-gap repair (C-056)	established for the defined mouse community and challenge	motivates selecting additions by missing function
functional redundancy and engraftment (C-057)	plausible association	motivates protected newcomer evaluation and a lock-in test
higher-level heredity and conflict accounting (C-282–C-295)	scoped population-genetic and evolutionary results; artificial composition speculative	Candidate 016 tests whether a collective becomes a useful adaptation unit beyond ordinary modular lifecycle controls
complete grow–route–specialize lifecycle	speculative synthesis	requires comparison with N0–N6

Speculative extensions

- Let modules request a birth trial with a compact capability-gap certificate; maintenance allocates the trial, not the requesting module.
- Maintain seed modules at several parameter and precision scales so a new role need not start from the largest available structure.
- Learn placement jointly with specialization only after migration cost and rollback are measurable.
- Use recurrence-aware cold storage: retire a module from hot execution while retaining enough checkpoint and routing evidence to restore it if its regime returns.
- Allow two candidates to share an encoder or memory interface while keeping their update authority and resource accounts separate.
- Test local birth/retirement against periodic global architecture optimization under recurrent rather than one-way task sequences.

Failure modes

FAILURE	OBSERVABLE SIGNATURE	REQUIRED RESPONSE OR ABLATION
Router collapse	one module takes most traffic; queue tails or overflow rise	fixed-capacity MoE and stronger load-balancing baseline
Candidate inflation	birth rate and stored bytes rise without held-out gap closure	cap trials; compare no-growth and random-birth nulls
Fragmentation	many modules show overlapping ablation effects and high boundary traffic	merge/distill branch with causal coverage tests
Incumbent lock-in	a superior newcomer cannot acquire evaluation traffic	reserved probation share; compare router-logit admission
Premature localization	modality-specific routes lose cross-modal transfer	shared-module and monolithic controls on compositional tests
Reserve starvation	no path remains for faults or new regimes	price and enforce declared reserve capacity
Reconfiguration thrash	repeated births, moves, merges, or retirements dominate cost	hysteresis and slower maintenance epochs
Stranded capacity	cold modules occupy memory but never serve, fail over, or restore	cold-storage, deletion, and restore-value comparison
Cosmetic sparsity	active gates fall while loaded bytes, communication, or joules do not	hardware trace and dense-kernel ablation
Maintenance inversion	search, validation, migration, and rollback cost exceeds runtime saving	full lifecycle equation and fixed-structure nulls
Rare-role deletion	average quality holds while rare or safety-critical cases regress	protected recurrence suite and reconstructable checkpoint
False higher-level unit	aggregate reward rises but collective inheritance is transient or member shortcuts dominate	preregister descendant relation; selection partition; ordinary modular and external-evaluator nulls

Measurable predictions

1. Capability-gap-driven births close held-out failure clusters with fewer admitted candidates and lower lifecycle energy than random birth, periodic fixed growth, and capacity-matched adapter baselines.
2. A protected probation share lets genuinely better newcomers establish causal value faster than incumbent-logit admission without increasing harmful promotions at the same validation budget.
3. Successful specialization produces a stable, non-exclusive routing region and positive unique ablation value; routing entropy alone predicts promotion less reliably.
4. Conditional growth reduces executed operations and bytes moved/event relative to a capacity-matched monolith while preserving quality, calibration, and rare-case performance.
5. Causal merge/retire gates preserve recurring and intervention capability better than usage- or magnitude-only pruning at matched hot capacity.
6. Reserve capacity improves recovery after faults or returning regimes enough to justify its stored bytes, idle energy, and periodic test traffic.
7. Joint routing and placement lowers communication energy only after migration, cold-start, and rollback work are included.
8. A collective lifecycle advances only when cost-adjusted between-collective selection and inherited capability persist under member shortcuts and turnover beyond ordinary lifecycle governance.
9. The complete lifecycle advances only if it improves the quality–risk–latency– energy–adaptability frontier over N0–N5; a parameter-count or FLOP reduction alone does not satisfy the prediction.

Multimodal sensorimotor grounding

Scope

Grounding means that a model's state is constrained by temporally ordered observation, action, and consequence—not that every concept must be reducible to pixels or motor commands. This chapter defines the trajectory record from which such state can be learned, how language attaches to it, and the tests that separate action-conditioned structure from passive multimodal correlation.

The central unit is a versioned episode containing asynchronous sensor events, commanded and realized interventions, exogenous events, and language with provenance. The model must predict and act under missing modalities, timing error, partial observability, and finite sensing energy. A representation is treated as grounded only to the extent that it supports held-out prediction, intervention, control, and reference at a declared uncertainty.

Biological observation

Biological perception is embedded in a closed loop. During active whisker exploration, measured contact mechanics predicted primary sensory-neuron firing better than whisker angle alone in the scoped mouse experiment (C-022). The result establishes that self-generated sensing mechanics can matter to the sensory code; it does not make embodiment necessary for every capability.

Other evidence constrains parts of the translation:

- sparse coding can learn localized structure from natural images in the model studied under C-002;
- latent target prediction can learn semantic image representations without pixel reconstruction under C-006;
- children in a controlled toy study targeted actions toward unresolved causal structure under C-062;
- pedagogical demonstration narrowed subsequent exploration in the scoped toy experiments under C-063; and
- causal transparency changed copying of irrelevant demonstrated actions in a comparative puzzle-box experiment under C-064.

Together these findings motivate aligned experience, active evidence collection, and separation of imitation from outcome learning. They do not yet establish the project-level hypothesis that embodied multimodal training yields more general physical concepts than comparable text-centric training. That claim remains speculative under C-007.

Proposed AI translation

Canonical trajectory record

One episode is

$$\tau = (\nu, \mathcal{O}, \mathcal{A}, \mathcal{X}, \mathcal{Y}),$$

where τ is an episode record and all five components are immutable after publication; corrections create a new version.

SYMBOL	MEANING	UNIT OR TYPE
ν	episode metadata	identifiers and versioned configuration
$\mathcal{O}$	asynchronous observation-event set	timestamped records
$\mathcal{A}$	commanded and realized action-event set	timestamped records
$\mathcal{X}$	exogenous event set	timestamped records
$\mathcal{Y}$	language-event set	timestamped text or token records

Metadata ν contains episode and environment identifiers; train, development, or test split; actor or policy version; simulator or device version; clock domains; calibration versions; random seed when applicable; and parent episode for a counterfactual branch. Frames from one episode never cross data splits independently.

An observation event from modality r is

$$o_{r,k} = (r, t^{\mathrm{cap}}_{r,k}, t^{\mathrm{recv}}_{r,k}, v_{r,k}, m_{r,k}, q_{r,k}, p_{r,k}),$$

where:

- r is a modality identifier such as vision, audio, touch, proprioception, force, temperature, or depth;
- k is an event index;
- $t^{\mathrm{cap}}_{r,k}$ is physical capture time in seconds;
- $t^{\mathrm{recv}}_{r,k}$ is the time in seconds at which the learner could first access the event;
- $v_{r,k}$ is the sensor value with the sensor's declared unit, shape, encoding, and quantization;
- $m_{r,k}$ is a dimensionless missingness/status code;
- $q_{r,k}$ is a quality record with named quantities and units, such as exposure time in seconds or signal-to-noise ratio in decibels; and
- $p_{r,k}$ is a provenance pointer to source, calibration, transformation, and checksum records.

Capture and receipt time are both required. Offline alignment uses capture time; a deployable causal policy may use an event at time t only when $t^{\mathrm{recv}}_{r,k} \le t$. This prevents a delayed sensor packet from becoming accidental future information.

An action event is

$$\alpha_j = (t^{\mathrm{cmd}}_j, t^{\mathrm{on}}_j, t^{\mathrm{off}}_j, a_j, \tilde{a}_j, \pi_j, p_j),$$

where t_j^{cmd}, t_j^{on}, and t_j^{off} are command, realized onset, and realized offset times in seconds; a_j is the commanded action; $\tilde{a}_j$ is the measured realized action; π_j is the behavior-policy probability for a discrete action (dimensionless) or probability density for a continuous action (in reciprocal action-volume units) when it is known; and p_j is its provenance pointer. Each action component declares its physical unit—for example newtons, newton-metres, metres/second, radians, or a discrete tool identifier. Command and realization are never silently substituted for one another.

An exogenous event $x_l \in \mathcal{X}$ records event and receipt timestamps t_l^{event} and t_l^{recv} in seconds, event type, measured magnitude and unit, and provenance. Examples include another agent's action, an uncommanded collision, lighting change, or simulator reset. Known simulator state s_t^* may be stored for evaluation in its declared physical units, but it is oracle information and is withheld from learned systems unless a baseline explicitly receives it.

A language event is

$$y_n = (t_n^{\mathrm{start}}, t_n^{\mathrm{end}}, w_n, c_n, r_n, p_n),$$

where the two timestamps are seconds, w_n is text or a token sequence, c_n identifies speaker and communicative role, r_n is a set of referent pointers to objects, events, trajectory intervals, or external records, and p_n is source provenance. A retrospective caption, an instruction available before action, a question, a report from another agent, and model-generated text have different roles even when their words match.

Observation is a versioned contract

A sensor record is not the latent state. Propagation, foreground, exposure, instrument response, background, calibration, reconstruction, threshold, and selection intervene before a downstream model sees it (C-218–C-221). For latent target θ, nuisance state η, measurement y, and response version H_v,

$$y = H_v(\theta, \eta) + \epsilon,$$

where y and H_v have matching declared units and ϵ follows a declared noise and background model. If the analyzed record was selected by event $S = 1$, inference is conditional on that event:

$$p(\theta, \eta \mid y, S = 1, v) \propto p(y, S = 1 \mid \theta, \eta, v) p(\theta, \eta \mid v).$$

The trajectory record therefore carries response and calibration version, exposure, selection state, detection power for non-detections, reconstruction choice, and data vintage. Cross-modal fusion also retains association uncertainty and shared dependencies; a different sensor type is not evidence of conditional independence (C-224).

Candidate 014 tests whether propagating this contract with each claim adds value beyond a complete typed, calibrated, lineage-aware, selection-aware, and simulation-checked evidence stack. The candidate must abstain on response null spaces and exact degeneracies rather than buy more observations that cannot identify the missing direction (C-229).

Missingness is observed state, not a zero tensor

For modality r and decision time t, define $M_t^{(r)} \in \{0, 1, 2, 3, 4\}$ as a dimensionless status:

CODE	MEANING
0	present and inside calibration range
1	absent by experimental design

CODE	MEANING
2	unavailable because of sensor or transport failure
3	present but occluded, saturated, clipped, or outside calibration
4	status unknown

The raw value is not imputed before status is retained. If an imputed value is needed, the model receives both the imputation and $M_t^{(r)}$. Training dropout is labeled as synthetic missingness and is sampled in contiguous outages as well as isolated frames. Confirmatory tests include natural missingness, failure, corruption, and combinations not seen during training.

Missingness may itself be informative—for example a tactile sensor becomes available only after contact. Evaluation therefore distinguishes performance gained from legitimate availability structure from shortcut prediction of a label or environment identifier.

Temporal alignment contract

Each modality has a calibrated clock offset δ_r in seconds and residual jitter scale σ_r in seconds. The corrected capture time is

$$\bar{t}^{\mathrm{cap}}_{r,k} = t^{\mathrm{cap}}_{r,k} - \delta_r.$$

Alignment does not mean forcing all modalities onto one frame rate. Encoders consume timestamped events or aggregate them inside declared causal windows. For a decision at time t, the available history is

$$\mathcal{H}_t = \{o_{r,k} : t^{\mathrm{recv}}_{r,k} \leq t\} \cup \{\alpha_j^{\leq t} : t_j^{\mathrm{cmd}} \leq t\} \cup \{x_l : t_l^{\mathrm{recv}} \leq t\} \cup \{y_n : t_n^{\mathrm{end}} \leq t\},$$

where $\mathcal{H}_t$ is a set of records, not a unit-bearing scalar, and $\alpha_j^{\leq t}$ contains only action fields measured by time t; a future realized offset or actuator trace is not revealed with the earlier command. The equation admits a language event after its end time; a streaming system may instead add explicitly timestamped token prefixes. Outcome targets may use later events, but the state used to choose an action may not.

Alignment robustness is tested by adding known per-modality offsets and jitter, dropping clock-synchronization messages, and withholding one calibration version. A mechanism that works only at exact simulator step boundaries has not learned a robust temporal relation.

Observation, intervention, and consequence

An action token is useful only when its consequence can be distinguished from background change. Training retains four separable cases:

1. passive observation with no controlled action;
2. policy-chosen action with known or estimated behavior probability π_j;
3. randomized or scripted intervention with known assignment; and
4. paired simulator branch from the same saved initial state with one changed action.

Only the third and fourth directly support an interventional comparison. Policy-chosen logs are confounded by the policy's state and require a declared system-identification, propensity, or model-based assumption. Exogenous events remain separate from realized action so the model cannot credit itself for an outside cause.

For a horizon $\Delta > 0$ seconds, an action-conditioned predictor estimates

$$p_\theta\left(z_{t+\Delta} \mid z_t, \tilde{a}_{[t,t+\Delta)}, M_{\le t}\right),$$

where $z_t \in \mathbb{R}^d$ is a dimensionless latent state of width d, θ is the parameter vector, $\tilde{a}_{[t,t+\Delta)}$ is the realized action sequence over the interval with per-component physical units, and $M_{\le t}$ is missingness history. The predictor is not called causal merely because action is an input. Causal interpretation depends on the data case and evaluation intervention.

This realized-action form is used for training and retrospective scoring. A prospective rollout conditions on a candidate command a and integrates over the calibrated distribution of actuator delay and realized action; it does not assume that a command is executed exactly.

Predictive state and action loop

The observation encoder forms

$$z_t = E_\phi(\mathcal{H}_t),$$

where E_ϕ is an encoder with dimensionless parameters ϕ and z_t is dimensionless. Separate heads predict future latent state, selected sensor targets, task outcomes, and calibrated uncertainty. Pixel or waveform reconstruction is used only when the task requires that detail; normalized latent prediction follows the narrower evidence in C-006.

For target-event set $\mathcal{G}$, a probabilistic prediction loss is

$$\mathcal{L}_{\text{pred}} = -\frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} \log p_\theta(v_g \mid \mathcal{H}_{t_g}, \tilde{a}_{[t_g, t_g+\Delta_g)}),$$

where $|\mathcal{G}|$ is a target count, v_g is a declared target under a declared quantization or likelihood on dimensionless normalized coordinates, t_g is target context time in seconds, Δ_g is its prediction horizon in seconds, and $\mathcal{L}_{\text{pred}}$ is nats/target. A normalized latent-distance loss may be reported in dimensionless units, but it is not added to negative log likelihood without a declared conversion weight.

At a decision, the system may act on the world, request a modality, ask a question, consult sourced memory, or wait. Let $b \in \mathcal{B}_t$ denote one such acquisition or intervention choice. A conventional one-step value-of-information policy chooses

$$b_t^* = \arg\max_{b \in \mathcal{B}_t} \left[V(b \mid \mathcal{H}_t) - \lambda_E E(b) - \lambda_T T(b)\right],$$

where V is expected task-utility improvement in declared utility units, $E(b)$ is full sensor, communication, and compute energy in joules, $T(b)$ is added latency in seconds, λ_E has units utility/joule, and λ_T has units utility/second. `wait` is an explicit zero-acquisition option. This is the strongest conventional null for claims that prediction error should route sensing or compute, as described under P-007 and in the engineering analogue audit.

Controlled observability and plant binding

The observation operator is partly controlled. Sensor pose, locomotor microstructure, emission timing, sampling density, and contact can change which state directions are observable (C-589–C-592). At the same time, body, tool, attachment, payload, compliance, and contact change how commands become motion and sensory consequences (C-599, C-603–C-605).

Separate the immediate task controller from the sensing controller:

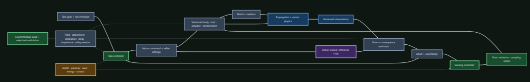
$$u_t = \pi_{\text{task}}(b_t, \Pi_v), \qquad a_t^{\text{sense}} = \pi_{\text{sense}}(b_t, \mathcal{O}_v, \Pi_v),$$

where b_t is a dimensionless belief state, task action u_t retains its physical units, sensing action a_t^{sense} retains pose, emission, sampling, or motion units, $\mathcal{O}_v$ is observation-contract version v, and Π_v is the controller–plant binding

$$\Pi_v = (B, T, A, C, S, J, Z, D, E),$$

with body/tool identity B, task T, attachment/payload A, contact model C, sensor/actuator calibration S, task Jacobian J, impedance/passivity envelope Z, delay model D, and safety/authority envelope E. Each field is typed, versioned, and linked to the controller, estimator, data, and tests that depend on it.

Wide diagram · fit to page



Editable source: controlled-observability-plant-binding.mmd.

Two held tests share this record:

1. **Controlled observability.** Keep the task policy fixed while a sensing policy maintains information through pose, emission, or sampling. It must beat fixed excitation, random acquisition, one-step expected value of information, active SLAM, observability MPC, and dual control at equal action, time, risk, compute, and energy (C-596).
2. **Counterfactual plant binding.** Swap one body, tool, attachment, payload, sensor, delay, contact, or impedance field at a time. Dependencies should predict which estimator/controller state is safe to reuse and which must be invalidated. It must beat reset/retraining, unversioned adaptation, model banks, gain scheduling, operational-space impedance control, and adaptive MPC (C-606).

This composition avoids two attribution errors. Passive mechanics and compliance are credited before a controller (C-597); an action-correlated internal signal is not assumed to be a complete accurate forward model (C-601).

Opportunity- and history-qualified adaptive action

Comparative cognition adds a sharper condition to the observation contract: the same apparatus does not create the same task when bodies, sensors, prior training, rewards, demonstrators, handlers, or feasible actions differ. The evidence in C-804–C-841 spans tool manufacture, causal transfer, future preparation, event memory, uncertainty control, imitation, teaching, exploration, negative transfer, and central-versus-peripheral control. Its durable contribution is not a species ranking. It is a requirement to record the opportunity that made an outcome possible.

Extend each episode with

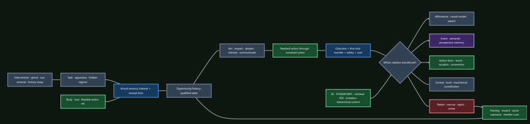
$$\Omega = (X, S, A, H, R, C, \tau, U),$$

where X is task and apparatus state, S is the sensory information actually available, A is the feasible realized-action set, H is prior training and social exposure, R is reward and stopping state, C is the intervention and control set, τ is elapsed time in seconds, and U is the sampling unit. The detailed definitions and units are maintained in the opportunity/history mathematics.

This record prevents several category errors:

1. **Terminal success is not transfer.** First-trial response after a material, geometry, cue, relation, body, or history change is retained separately from later test-time learning.
2. **One copied outcome is not one copied process.** Demonstrations are decomposed into action form, trajectory, end state, location, demonstrator, and functional relation. Ghost and result-only controls identify which component changed the learner.
3. **A future-directed action is not a unique memory mechanism.** Value tables, successor representations, planning, and retrieval remain nulls for delayed selection and resource reservation.
4. **Adaptive checking is not a unique uncertainty mechanism.** Public difficulty cues, response strength, calibrated confidence, selective prediction, and ordinary value of information must be beaten.
5. **Local control is not independent control.** Body mechanics, local sensing, peripheral feedback, central goal selection, override latency, and message traffic are measured separately.
6. **Repeated trials are not independent agents.** Episode, learner, dyad, group, site, rearing or training cohort, and population remain distinct sampling levels.

Wide diagram · fit to page



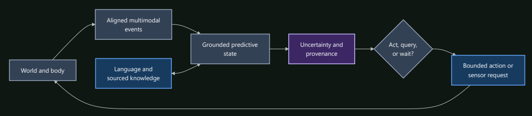
Editable source: opportunity-history-qualified-action.mmd.

Fixture F-003 turns the contract into eight separable tracks: manufacture transfer, causal inversion, future resource reservation, event memory, costed uncertainty, copy-target decomposition, social acquisition, and local/central/mechanical control. It compares the full composition with learned affordances and search, model-free and model-based RL, POMDP/MPC, semantic and episodic retrieval, selective prediction and value of information, imitation learning, central replay, and hierarchical control under one lifecycle boundary.

The fixture earns architectural credit only when opportunity qualification changes a conclusion, exposes a reproducible confound missed by the existing stack, or produces robust transfer beyond the complete nulls. Otherwise the episode contract remains useful instrumentation and the additional mechanism is retired.

Grounded learning loop

Wide diagram · fit to page



Editable source: `../assets/diagrams/grounded-learning-loop.mmd`.

Language attaches to grounded state

Language serves at least four different functions:

1. **reference:** name an object, relation, action, event, or trajectory span;
2. **instruction and query:** change the agent's task or request information;
3. **compression and communication:** summarize learned regularity for another time, module, or agent; and
4. **testimony:** introduce facts and abstractions not available through direct sensorimotor experience.

Reference is trained from the r_n links in language events, including negative and ambiguous referents. Instructions are available only from their receipt time and are treated as interventions that may change action. Testimony retains source, version, trust policy, and retrieval trace; it is not forced into a fictional physical observation. A concept can therefore combine direct experience, action–outcome evidence, and sourced symbolic knowledge without erasing which support came from where.

The staged design learns predictive physical state before or alongside a small language adapter, then compares late attachment, joint training, and language-first training under equal data and capacity budgets. Physical probes use paraphrases and unseen names; language probes include relations with no direct sensory referent. Demonstrated action and demonstrated outcome are separate targets, following the distinction motivated by C-064.

Uncertainty and provenance contract

Each prediction exposes a distribution or calibrated interval at the level at which action is chosen. Report negative log likelihood in nats/target, Brier score for declared categorical events, interval coverage in percent, and calibration error with binning fixed before evaluation. Calibration is broken out by modality status, prediction horizon, intervention type, environment, and linguistic versus direct evidence.

A single confidence scalar does not replace modality-specific uncertainty. The system records whether uncertainty arose from noisy observation, missing modality, disagreement among plausible dynamics, unseen composition, ambiguous language, or untrusted testimony when the estimator supports that distinction. If it does not, the output remains an undifferentiated predictive uncertainty.

Every training and evaluation event resolves through $p_{r,k}$, p_j, or p_n to:

- raw or generated source identifier and checksum;
- simulator, device, actor, and policy version;
- capture and receipt clock domains;

- calibration and transformation chain;
- missingness and corruption operations;
- split assignment and counterfactual parent; and
- license or access policy where applicable.

Answers and plans can then cite direct observation, intervened outcome, retrieved testimony, or model rollout separately. Provenance does not make a prediction correct; it makes its evidential path inspectable and reversible.

Strongest conventional baselines and nulls

ID	BASELINE	QUESTION IT ANSWERS
B0	Text-only model with matched language tokens and capacity	Does language alone explain the reported transfer?
B1	Passive multimodal encoder on the same recorded sensor events	Does temporal multimodality help without action conditioning?
B2	Synchronized masked/contrastive multimodal predictor	Does ordinary cross-modal prediction explain the gain?
B3	Standard action-conditioned latent state-space/world model	Does the proposed data and uncertainty contract add value beyond a conventional world model?
B4	Behavioral cloning and action-prediction model	Is copying the logged policy enough without predicting consequences?
B5	Calibrated linear/nonlinear system-identification model on tasks where its assumptions apply	Is learned grounding better than an established dynamics estimator?
B6	Random, entropy-seeking, and one-step EVSI acquisition policies	Does active sensing beat ordinary exploration and information purchasing?
B7	Oracle-state predictor/controller	How much error comes from partial observation rather than dynamics or control?
B8	Intact architecture with time, action, modality, or language links shuffled	Does performance depend on the claimed alignment and causal structure?

B7 is an upper bound, not a superiority baseline. Comparisons match trajectory seeds, raw sensor exposure where logically possible, model capacity, optimizer updates, action count, sensor acquisitions, context bytes, latency ceiling, and lifecycle energy. The active-data question and the action-conditioning question are evaluated separately: hiding action tokens in the same log is not the same experiment as allowing an agent to collect a different log.

Staged experiment

Stage 0 — Data and clock qualification

Build a deterministic replay harness before training a world model. It must reproduce event order, verify checksums and split isolation, expose capture and receipt time, reconstruct commanded versus realized actions, and audit every missingness code. Inject known offsets, jitter, packet delay, dropped intervals, and actuator lag. Reject a dataset version when these perturbations cannot be recovered or bounded from its metadata.

Stage 1 — Passive action-conditioned prediction

Use a controlled compositional environment with vision, audio, proprioception, touch/force, and optional depth. Objects vary independently in shape, color, mass, surface friction, and containment. Actions include look, approach, push, lift, rotate, drop, occlude, and tool contact. The held-out split contains:

- unseen combinations of known object properties;
- new visual textures and camera placement;
- changed mass or friction under familiar appearance;

- action delays and partial actuator failure; and
- one-modality and multi-modality outages.

Training, development, and test split by episode seed, object combination, and environment configuration—not by frames. Simulator state s_t^* is stored for evaluation and B7 only.

Train B1–B5 and the candidate predictive state on the same logged trajectories. Compare intact realized actions with actions masked, time-shifted, and permuted within episode. Primary outcomes are future-state and sensor NLL in nats/target, physical-property probe error in declared units, held-out intervention error, control success percent with a frozen small controller, calibration, bytes, and joules/qualified episode.

Stage 2 — Paired interventions and active acquisition

From a saved initial state, run paired branches differing in one action or sensor request while holding the environment seed fixed. Separately allow each active policy the same maximum world actions, sensor acquisitions, wall time, and energy. Compare random exploration, entropy seeking, one-step EVSI, and the candidate policy.

Measure prediction change in the correct physical direction, intervention effect error, successful disambiguations/action, downstream task utility, unsafe interventions, sensor byte, action energy, total joules, and latency. This stage tests the open issue in C-022 and the targeted-exploration translation from C-062.

Stage 3 — Language attachment

Attach language using trajectory-span and referent links. Compare late attachment to a frozen physical core, joint training from initialization, and language-first pretraining followed by grounded data. Equalize language tokens, grounded episodes, trainable parameter count, optimizer updates, and tuning budget.

Evaluate new names for known referents, paraphrased instructions, ambiguous reference, imitation versus outcome emulation, physical questions about unseen compositions, testimony-only facts with changed source versions, and removal of language at control time. Report whether language improves sample efficiency without becoming the only route to physical success.

Stage 4 — Asynchronous physical transfer

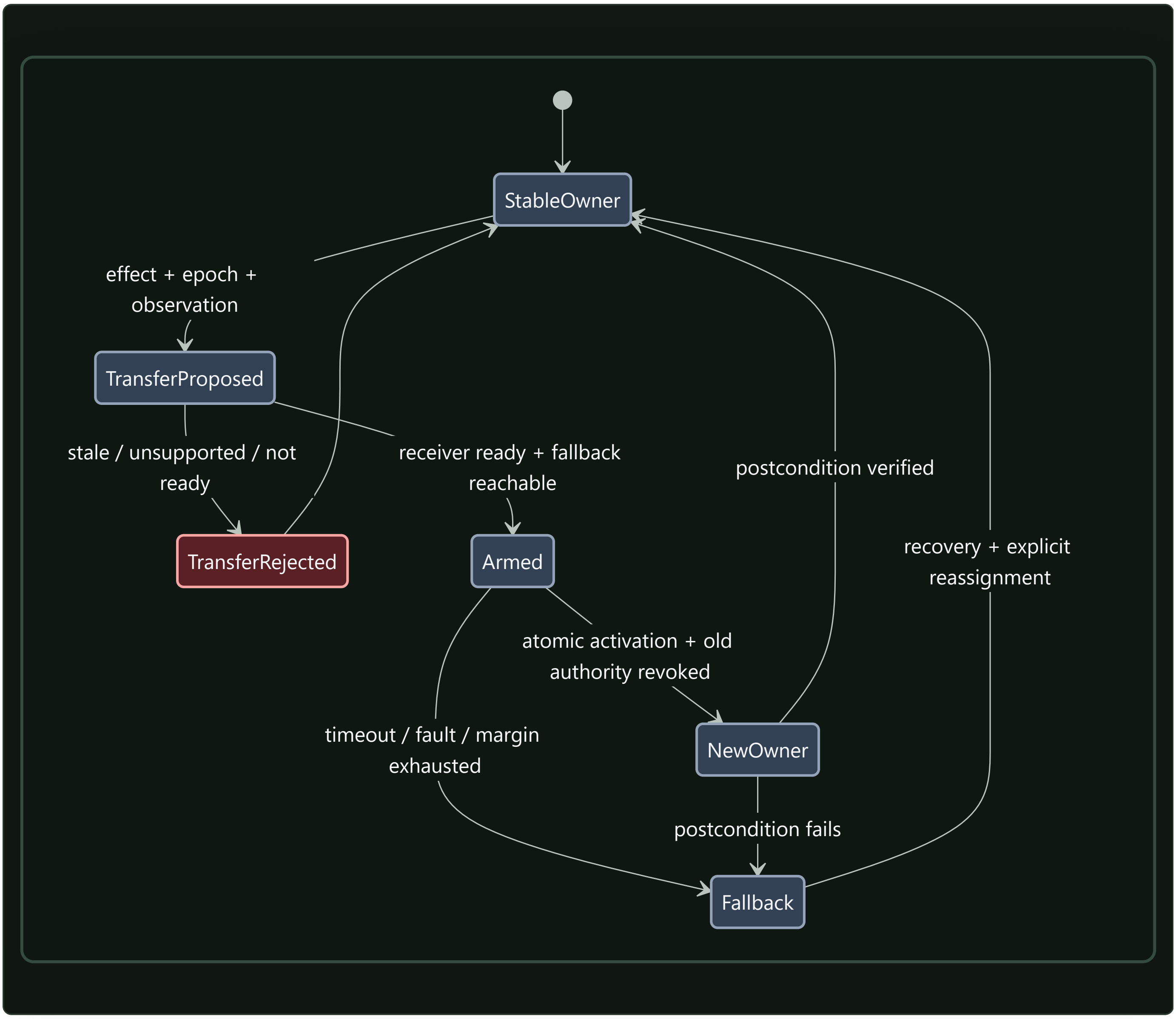
Move the frozen comparison to one named real sensorimotor platform. Preserve native clocks, packet delay, actuator mismatch, calibration changes, occlusion, and sensor failure. Recalibrate energy at device and node boundaries using the energy evaluation contract. Simulation-to-real transfer, recovery, and negative results are reported by intervention and missingness stratum rather than one average.

The real platform adds two contracts that simulation accuracy cannot supply. First, estimation is qualified by integrity. An illustrative position protection level is

$$PL = K_{\text{int}}\sigma_{\text{pos}}, \qquad \text{usable only when } PL \le AL,$$

where PL, position standard uncertainty σ_{pos}, and alert limit AL are in metres, while K_{int} is dimensionless and tied to a declared fault and risk allocation. A learned confidence score does not become integrity without bias bounds, dependence, coverage, and time-to-alert (C-446).

Second, a controller handoff must preserve exclusive effective authority while the world keeps moving. The transfer binds the controlled effect, epoch, outgoing and incoming controller, actual and pending mode, observation basis, headroom, time-to-boundary, faults, outstanding commands, fallback reachable set, expiry, acknowledgement, atomic activation, and observed postcondition.



Editable source: asynchronous-authority-transfer.mmd.

This is a refinement of Candidate 012, not a new autonomy layer. It must beat mode annunciation, interlocks, epoch leases, runtime assurance, explicit arbitration, Candidate 014, and the recoverable-initiative contract under split-brain, stale-command, no-owner, delayed-link, common-mode, and failed- postcondition injections (C-447–C-461).

Efficiency mechanism

Grounding adds sensors and interaction, so its efficiency claim is conditional: the learned state must avoid more future sensing, compute, data movement, or failed action than it costs to acquire and maintain.

The candidate levers are:

- **event-driven encoding:** update modality state on timestamped change rather than resampling every channel at the highest rate;
- **latent prediction:** predict task-relevant state without reconstructing every high-bandwidth detail;
- **local preprocessing:** reduce raw sensor traffic near the source while retaining calibration and uncertainty metadata;

- **conditional acquisition:** power, transmit, or process another modality only when its expected utility exceeds energy and latency cost;
- **temporal reuse:** carry forward stable state with an uncertainty increase instead of re-encoding unchanged observations; and
- **language compression:** communicate grounded referents and relations when sending the underlying sensor history is unnecessary.

Every experiment reports sensor-on time in seconds, sensor energy in joules, raw and processed bytes by boundary, acquisitions/episode, world actions/episode, encoder and world-model operations, latency in seconds, controller energy, maintenance/calibration energy, and total lifecycle joules/qualified episode. A lower model FLOP count is not a grounding-efficiency result when sensor collection or interaction dominates.

The relevant efficiency comparison is a frontier over task utility, risk, latency, energy, and physical interaction. One-step EVSI is the minimum null for conditional acquisition. A passive model trained on a larger recorded dataset is an additional null when interaction energy, not online autonomy, is the proposed benefit.

Evidence status

- Sparse natural-image representation under C-002: **established for the cited model and images**.
- Joint-embedding image prediction under C-006: **established for the cited image experiments**.
- Active sensor mechanics under C-022: **established for the measured whisker task**.
- Targeted exploration under C-062, pedagogical narrowing under C-063, and imitation/emulation difference under C-064: **established for the scoped behavioral tasks**.
- A generally useful predictive-coding abstraction under C-005: **plausible**, not a complete grounding theory.
- Robust general physical concepts from the integrated multimodal curriculum under C-007: **speculative**.
- Navigation integrity, protection-envelope scope, fault-stage separation, degraded/fallback distinctions, and assurance boundaries under C-445–C-460: **established or plausible within the audited vehicle domains**.
- Validated asynchronous authority transfer under C-461: **speculative**.
- Opportunity, history, causal-transfer, event-memory, social-learning, and distributed-control observations under C-804–C-840: **established, plausible, or disputed only within their declared tasks**; the integrated action contract under C-841 is **speculative**.

Speculative extensions

Controllability curriculum

Order early tasks by what an agent can reliably change and observe: persistence and contact, motion and occlusion, material response, tools and other agents, then linguistic abstraction. Curriculum order becomes an ablation rather than an assumed developmental law.

Learned morphology and sensor placement

Jointly adapt sensor placement, body parameters, and control only after the fixed-body grounding loop is understood. The simulation evidence under C-029 motivates the question, while lifecycle energy and reality-gap robustness remain required outcomes.

Socially supplied interventions

Treat demonstration, correction, question answering, and pedagogy as actions that change the learner's evidence policy. The system can preserve uncertainty after a demonstration and deliberately test affordances not shown, rather than assuming instruction is exhaustive.

Counterfactual branch memory

Store compact paired branches from matched initial states as high-value causal records. Promotion into long-term memory depends on reuse and provenance, not on intervention novelty alone.

Failure modes

SIGNATURE	INTERPRETATION AND REQUIRED RESPONSE
Action shuffling leaves held-out intervention error unchanged	the representation uses passive correlation; remove the action-grounding claim
Timestamp shuffling or large clock offsets cause no degradation	static identifiers or leakage dominate; audit split and alignment
Exact simulator timing wins but calibrated jitter collapses performance	temporal relation is brittle; retain time uncertainty or revise the encoder
Missingness code alone predicts task labels	acquisition policy or dataset artifacts leak the answer; rebalance and report the shortcut
One modality receives nearly all gradient or attention and its removal collapses all tasks	shared state has become single-modality state; enforce and test modality-specific competence
Commanded action predicts outcomes but realized action does not	actuator or logging shortcut; train and evaluate on realized intervention
B3 or B5 matches the candidate	conventional world modeling or system identification explains the result; use the simpler baseline
One-step EVSI matches active acquisition	no new sensing principle is supported; retain EVSI as the controller
Language removal destroys physical control learned before language	linguistic shortcut or catastrophic overwriting; separate adapters and replay physical probes
Fluent reports disagree with action-conditioned predictions	language head is not constrained by world state; expose uncertainty and provenance by source
Counterfactual performance is high only when oracle state leaks into input	the partial-observation problem remains unsolved; remove oracle fields and rerun
Active training wins only because it observed more frames or spent more action energy	data or resource exposure explains the result; compare at matched events, actions, and joules
Sensor and interaction energy erase compute savings	narrow the result to representation quality; do not claim system efficiency
Calibration fails specifically under missing modalities or novel interventions	uncertainty is not deployment-valid; block autonomous escalation in those strata
Low state-estimation error coexists with integrity alerts that are late or absent	average accuracy is not safe-to-use evidence; bind fault hypotheses, alert limit, and deadline
Handoff produces overlapping owners, a control gap, or stale queued effects	transfer is not atomic or command expiry is incomplete; revoke, fall back, and preserve the trace
Raw task success changes after body, cue, training, reward, handler, or site changes	the opportunity record is incomplete; stratify and rerun before attributing a mechanism
A socially exposed policy reaches the same end state but changes no novel action form	result, affordance, or location learning explains the effect; remove the imitation claim
A local controller wins until passive mechanics, local hardware, or communication are charged	the control-locus claim is misattributed; keep the cheaper complete baseline

Measurable predictions

H-G1 — Action-conditioned consequence

With identical logged sensor events, intact realized actions will improve held-out intervention NLL and physical-effect error over masked, shifted, and permuted actions. If only reconstruction or in-distribution probe accuracy improves, the result does not support action grounding.

H-G2 — Temporal alignment

Models trained with explicit capture/receipt time and calibrated clock uncertainty will degrade more gradually under held-out offset, jitter, and packet delay than fixed-step concatenation. The comparison reports error as a function of injected milliseconds and identifies the offset at which the quality envelope is crossed.

H-G3 — Missing modalities

Typed missingness plus modality-specific predictive heads will preserve a better quality–calibration frontier under contiguous sensor outages than zero imputation or independent encoders fused only at the output. Performance is reported separately for absence, failure, occlusion/corruption, and unknown status.

H-G4 — Active evidence collection

At the same action, acquisition, latency, and joule budgets, a learned active policy will require fewer interventions to identify held-out mass, friction, containment, or causal relations than random and entropy-seeking exploration. It supports a new mechanism only if it also improves beyond calibrated one-step EVSI; otherwise EVSI becomes the accepted implementation of P-007.

H-G5 — Compositional physical transfer

Action-conditioned predictive state will yield lower sample count to a fixed control-success threshold on unseen property combinations and dynamics than text-only, passive multimodal, and behavioral-cloning baselines of matched capacity. The threshold, maximum episodes, and unsuccessful runs are fixed before evaluation.

H-G6 — Language attachment

Late or joint language attachment with explicit referent and provenance links will improve new-name, paraphrase, and testimony-version tests without reducing pre-language physical probes beyond a preregistered equivalence margin. If language-first training alone matches intervention transfer at equal grounded episodes, the additional grounding curriculum has not earned its cost.

H-G7 — Calibrated abstention

Prediction uncertainty will increase under unseen interventions, multi-sensor failure, and ambiguous reference, and calibrated abstention or acquisition will reduce high-cost physical errors at a declared false-abstention rate. Average confidence without stratum calibration does not satisfy this prediction.

H-G8 — Lifecycle efficiency

Conditional sensing, event-driven updates, and latent prediction will reduce sensor byte, processed byte, and node joules/qualified episode at matched task utility, risk, and latency relative to always-on sensing and fixed-rate encoding. The claim is rejected if the confidence interval for lifecycle energy includes the preregistered equivalence margin after collection, calibration, language, maintenance, and failed-action costs are included.

H-G9 — Integrity-qualified authority transfer

Under delayed links, boundary misclassification, degraded sensors, outstanding commands, common-mode faults, and failed postconditions, a validated transfer record will reduce double-owner, no-owner, stale-command, and unsafe-boundary time beyond mode annunciation, interlocks, epoch leases, runtime assurance, ordinary arbitration, and the HCI/observation contracts at equal sensing, reserve, communication, training, review, latency, and joule budgets. A tie merges its useful fields into those conventional mechanisms.

H-G10 — Opportunity-qualified transfer

At equal episodes, interventions, action feasibility, search, storage, communication, human effort, and lifecycle joules, explicit opportunity and history state will improve first-trial functional transfer and calibration under held material, causal-relation, body/tool, acquisition-history, and social-channel changes relative to an otherwise identical learner. The claim is rejected if the gain disappears after actual sensor cues, feasible actions, prior training, reward schedule, handler/site, or later test-time learning are modeled, or if the complete affordance/search, planning, retrieval, VOI, imitation, replay, and hierarchical-control stack matches the result.

Promotion requires H-G1 plus at least one transfer result from H-G3–H-G6, valid uncertainty under H-G7, and no lifecycle contradiction under H-G8. Stage-4 authority-transfer claims additionally require H-G9. A comparative-cognition-derived transfer claim additionally requires H-G10. A positive result remains scoped to its trajectory schema, intervention family, environment shift, hardware boundary, and evaluated language role.

Representative adaptive performance

Scope

This chapter turns the sports expertise, adaptive performance, and team coordination audit into a system-wide performance contract. A result is interpretable only when it travels with the information that was available, the actions that were feasible, the history that produced the policy, the opponent and team it faced, the feedback it received, its current resource and damage state, and the selection process that determined which systems were observed.

The contract constrains:

1. sensing and observation provenance;
2. routing and allocation under deadlines;
3. action and closed-loop control;
4. memory of practice, opponents, damage, and recovery;
5. curriculum and online adaptation;
6. assurance, degradation, and staged return;
7. coordination and communication; and
8. population-level evaluation and lifecycle accounting.

It connects the sensorimotor grounding chapter, sparse predictive computation, memory and consolidation, system synthesis, and the energy model. Its executable specification is Fixture F-006, with notation and derivations in the mathematical contract.

Biological observation

Perception is coupled to a possible response

Experts can extract predictive information from early opponent kinematics, but the useful cue depends on the task, feature, opponent, prior, viewing window, and response mode (C-926–C-930). Predicting a label, moving a joystick, initiating a partial movement, and intercepting a physical event under its real deadline are therefore different experiments. Temporal or spatial occlusion can locate when or where information becomes useful; it does not identify a unique representation.

Training transfers more reliably when relevant information and action coupling are preserved, but “representative” is not synonymous with visually realistic (C-931–C-932). Timing, feasible action, opponent response, consequence, and resource state can matter even when surface appearance changes little.

Practice is not one outcome

Practice variability can improve learning when it explores a relevant task dimension at a useful challenge level; the effect is not universal, and larger perturbations do not automatically help (C-933–C-936). Accumulated practice correlates with expertise, yet amount, type, opportunity, selection, survival, access, and retrospective classification are entangled (C-937–C-939).

This creates three separate questions:

1. Does performance improve while repetition, instruction, and feedback are present?
2. Does the change remain after a declared delay without that scaffold?
3. Does it survive a change in cue, opponent, body, task, rule, deadline, or resource state?

The answers are practice performance, delayed retention, and transfer. They must not be substituted for one another.

Performance follows a frontier, not a single score

Movement speed and accuracy trade off under task-specific conditions, and throughput can appear stable while speed and error move in opposite directions (C-940–C-941). Risk, physical work, latency, and recovery add further axes. Optimizing only accuracy can hide slower action; optimizing only latency can hide errors or unsafe events; optimizing only completed work can hide depletion.

Fatigue changes the state from which an action is controlled. Opponent behavior, remaining work, feedback, recovery, and sleep can alter output, but effects remain task- and outcome-specific (C-942–C-948). Internal load, external load, readiness, damage, and injury risk are different constructs. A workload ratio or a single measurement cannot certify an individual system's safe capability (C-949–C-950).

Recovery and return are staged and reversible

Return to participation, return to the full task, and sustainable recovery of prior performance form a staged continuum. Time and functional evidence can carry nonredundant information, while an apparently high function score can be misleading after early return (C-951–C-953). Each promotion exposes the recovering system to more load and therefore creates new evidence. Deterioration must permit regression to a safer stage.

Teams coordinate through information and adjustment

Joint practice can increase predictive knowledge, and shared displays can change later team state. Yet synchrony is only an observable; similar movement can follow common input without reciprocal coordination (C-954–C-957). A coordination claim needs perturbation, lagged compensation, reduced task error, turnover, and cross-play. A shared-information claim additionally needs records of messages, observations, beliefs, and their discrepancies.

Deceptive behavior exploits an observer's cue policy. Expertise can reduce some susceptibility without removing it, and confidence may rise while the judgment becomes wrong (C-958–C-960). Feedback effects are likewise conditional on content, timing, task, autonomy, and whether the outcome is immediate performance or later learning (C-961–C-962).

Selection changes the population it measures

Relative age, maturity, present capability, coaching access, playing time, opponent quality, and later observation are coupled by selection. A talent system can amplify early differences and then mistake the resulting population for evidence that its original ranking was correct (C-963–C-965). Excluded systems have missing counterfactual careers precisely because the selection policy withheld the experience needed to produce them.

Finally, metabolic estimates and wearable estimates do not equal complete system energy. Human attention, facilities, sensors, equipment, computation, recovery, failed trials, maintenance, medical or safety work, and displaced opportunity alter the efficiency comparison (C-966–C-968). The transferable result is therefore an evaluation contract (C-969).

Proposed AI translation

Carry the state that makes a comparison valid

For agent i in episode e, preserve

$$\mathcal{K}_{i,e} = (X_e, O_{i,e}, A_{i,e}, H_{i,e}, M_e, F_{i,e}, R_{i,e}, D_{i,e}, G_{i,e}, C_e, U_e, B_e),$$

where:

- X_e is physical state, task, rules, deadline, consequence, and hidden regime;
- $O_{i,e}$ is information actually received, including source, support, latency in seconds, occlusion, noise, loss, and calibration;
- $A_{i,e}$ is the feasible action set under the current body or actuator, equipment, authority, rate, range, and safety limits;
- $H_{i,e}$ is timestamped practice, feedback, opponent, teammate, damage, exclusion, reward, and opportunity history;
- M_e is opponent and teammate composition, role, policy history, turnover, and communication topology;
- $F_{i,e}$ is feedback identity, information in bits, delay in seconds, and provider;
- $R_{i,e}$ is the resource and fatigue vector in its native units;
- $D_{i,e}$ is damage or fault state, diagnostic uncertainty, protected envelope, and return stage;
- $G_{i,e}$ is the selection and opportunity policy;
- C_e is the intervention, paired control, counterfactual seed, and stopping rule;
- U_e is the sampling unit, such as action, episode, agent, dyad, team, site, or cohort; and
- B_e is the complete ceiling in events, bytes, seconds, person-hours, joules, damage, unsafe events, replacements, and opportunity.

For method m and literal outcome k, the comparison target is

$$Q_{m,k}(\mathcal{K}) = \mathbb{E}[Y_k \mid do(m), \mathcal{K}]\,,$$

where Y_k uses the registered unit for outcome k. If a baseline and a new method receive different observations, actions, histories, opponents, feedback, resources, or selection opportunities, $Q_{m,k} - Q_{b,k}$ does not isolate method m from baseline b.

Treat representativeness as an inspectable vector

For training distribution P_{tr} and target distribution P_{te}, record

$$\mathbf{d}_{\text{rep}} = (d_O, d_A, d_T, d_M, d_F, d_R, d_D, d_G),$$

where the components compare actual observation O, feasible action A, deadline and consequence T, teammate/opponent state M, feedback F, resource state R, damage and return state D, and selection policy G. Each d_z is a declared divergence between the corresponding training and target distributions: dimensionless for a statistical divergence or in the declared ground-cost unit for optimal transport. The components remain visible; one weighted “realism” number would conceal which relation transferred.

Keep outcomes separate

OUTCOME	MINIMUM EVIDENCE	COMMON FALSE PROXY
anticipation	proper score by cue time and opponent; calibration; commitment latency	expert label or reaction time
physical interception	success, onset, trajectory, endpoint error, safety	video or joystick judgment
cue use	randomized removal, neutralization, conflict, or timing intervention	gaze or saliency
practice	acquisition curve by attempt and exposure time	end-of-practice score as retained learning
retention	delayed scaffold-free test	immediate post-practice score
transfer	first target trial and full source-to-target matrix	later target learning
exploration	action and outcome information, feasible coverage, later utility, cost	raw variance or entropy
adaptability	perturbation loss, recovery time, overshoot, recurrence, damage	stationary accuracy
pacing/readiness	output trajectory, state calibration, admissible actions, abstention	elapsed workload or one score
staged return	false promotion/withholding, dwell, recurrence, rollback, availability	calendar time or one test
coordination	perturbation-conditioned compensation, task stability, cross-play, repair	synchrony or proximity
shared information	messages, observations, predictive beliefs, discrepancies	similar behavior
deception	matched causal contrast, calibration, exploitability, regret, adaptation	confidence or surprise
talent prediction	prospective out-of-cohort calibration and counterfactual opportunity	current rank or selected-cohort accuracy
complete efficiency	protected outcomes plus all resource and harm axes	device energy or success/trial

Separate useful exploration from noise

For action variable A, reached-outcome variable Z, and method m, measure

$$\mathcal{X}_m = \left(H_m(A), H_m(Z), I_m(A;Z), K_m, Q_m^{\mathrm{tr}}, \mathbf{C}_m\right),$$

where both entropies H_m and mutual information I_m are in bits, K_m is the dimensionless fraction of the registered feasible region covered, Q_m^{tr} is later transfer in the task's literal unit, and $\mathbf{C}_m$ is the complete cost vector. More action entropy is useful only when it improves outcome information, transfer, or later task value at an acceptable cost.

An adaptive curriculum therefore needs the initial skill and history stratum, the perturbed task dimension, perturbation dose, retry cost, feedback bits, criterion exposure, and delayed tests. This directly constrains Candidate 004 and the memory boundary in the consolidation chapter.

Route and act through resource-qualified state

The runtime policy should expose the state on which pacing and authority depend:

$$a_t = \pi_m\left(o_{\le t}, \hat{r}_t, \hat{d}_t, s_t, \widehat{\pi}_{\mathrm{opp},t}, \widehat{b}_{\mathrm{team},t}, f_t\right),$$

where a_t is the commanded action or power target in its native unit, $o_{\le t}$ is causally received observation history, $\hat{r}_t$ is the estimated resource/fatigue vector, $\hat{d}_t$ the estimated damage state, s_t remaining work in metres, seconds, events, or joules, $\widehat{\pi}_{\mathrm{opp},t}$ the opponent-policy estimate, $\widehat{b}_{\mathrm{team},t}$ the teammate-state estimate, and f_t available feedback. Every estimate and channel receives a separate ablation.

This state conditions sensing, sparse routing, action authority, memory access, and recovery. It sharpens Candidate 002, Candidate 006, Candidate 007, and Candidate 012.

Compare a frontier, not a winner

For one task, retain

$$\mathcal{P}_m = (T_m, \epsilon_m, p_m^{\text{unsafe}}, E_m^{\text{life}}, H_m^{\text{human}}),$$

where T_m is latency in seconds, ϵ_m task error in its declared physical or task unit, p_m^{unsafe} dimensionless unsafe-event probability, E_m^{life} lifecycle energy in joules, and H_m^{human} role-stratified effort in person-hours. A method is more efficient only through a preregistered utility or a Pareto improvement with non-inferiority on protected outcomes.

Make readiness an action envelope

At decision time t, admissible actions are

$$\mathcal{A}_t^{\text{ready}}(\alpha) = \{a \in A_t : \Pr(Z_{t:t+h} \in \mathcal{Z}_{\text{safe}} \mid a, \mathcal{I}_t) \geq 1 - \alpha\}\,,$$

where A_t is the currently feasible action set, $Z_{t:t+h}$ the multidomain outcome vector over horizon h in hours, $\mathcal{Z}_{\text{safe}}$ the registered safe envelope, $\mathcal{I}_t$ information available at time t, and α the dimensionless tolerated risk. An empty envelope triggers abstention or escalation.

Let $g_t \in \{0, 1, 2, 3, 4\}$ denote protected, modified, controlled, full-load, and adversarial operation. Promotion requires the next stage's envelope; a violation requires that $g_{t+1} < g_t$ remain possible. This turns staged return into a concrete test for Candidate 009 rather than a one-time health classification.

Distinguish coordination from shared input

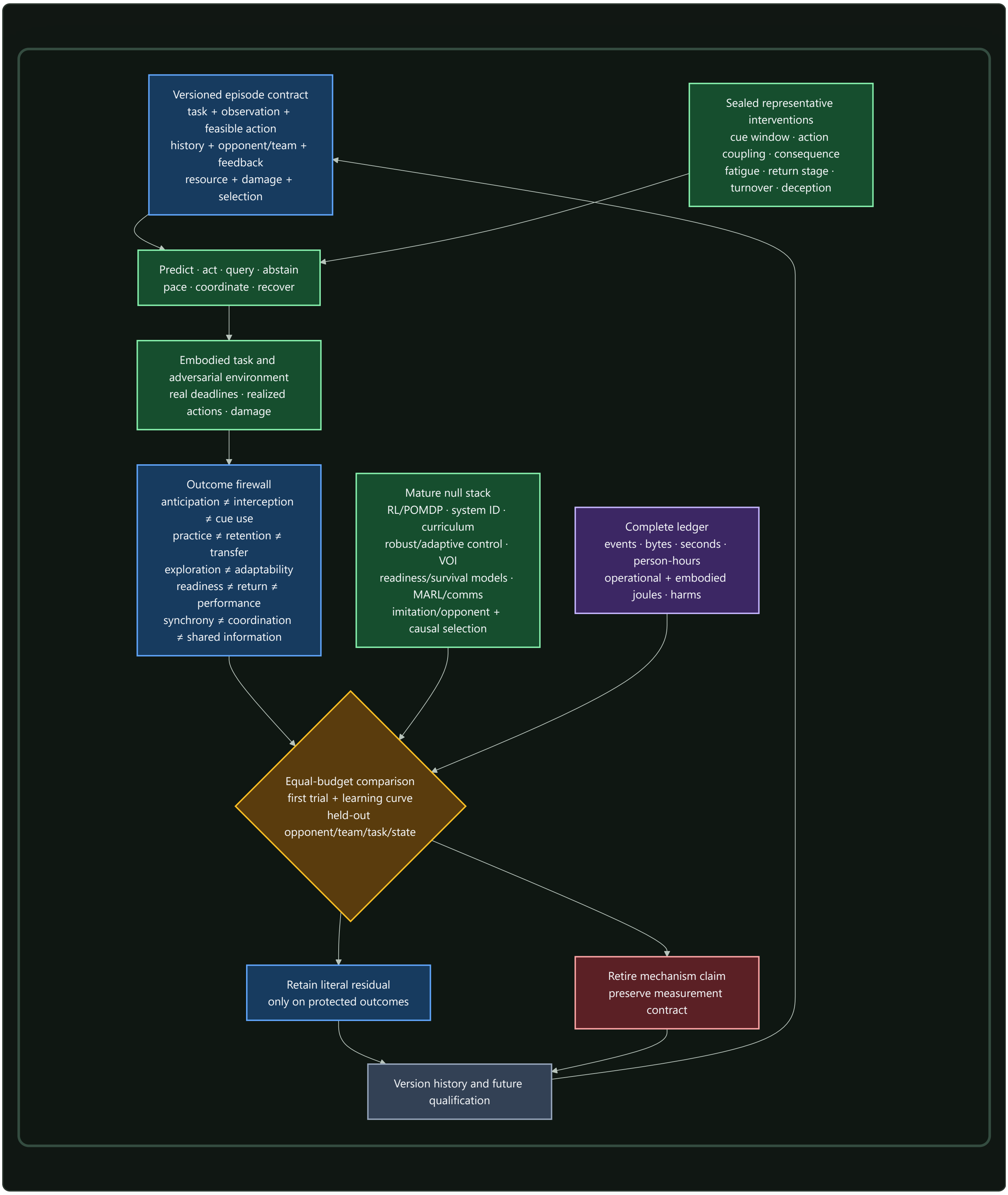
For agent contributions $u_i(t)$ and $u_j(t)$, apply perturbation $do(\eta_i)$ to agent i and estimate

$$\Gamma_{ij}(\ell) = \mathrm{Cov}(\Delta u_i(t), \Delta u_j(t+\ell) \mid do(\eta_i), X_t)\,,$$

where lag ℓ is in seconds, X_t is task state, u_i and u_j retain their native contribution units, η_i is a registered intervention in the unit of i's action or state, and Γ_{ij} has the product unit of the two contributions. A useful response must also reduce error in the protected task variable without increasing risk. Teammate turnover, role reassignment, message ablation, common-input controls, and never-co-trained cross-play separate reciprocal adjustment from synchrony.

Preserve selection and evaluator lineage

Population evaluation carries the policy that granted training, observation, feedback, compute, role, and survival opportunity. Selected and rejected systems remain in the analysis, with censoring and missing outcomes explicit. Additional opportunity near selection thresholds is randomized when admissible or handled with a declared causal design. This extends Candidate 019 and the observation lineage in Candidate 014.



Editable source: representative-resource-qualified-performance.mmd.

Efficiency mechanism

The contract can improve efficiency through five measurable effects:

1. **Less proxy optimization.** Coupled perception/action tests prevent spending training and inference resources on a symbolic score that does not improve the physical or operational task.
2. **Higher-value variation.** History-qualified curricula direct perturbations toward task-relevant uncertainty instead of buying undirected entropy.
3. **Resource-aware allocation.** Routing, sensing, and authority respond to remaining work, fatigue, damage, opponent state, and recovery rather than treating every episode as fresh.
4. **Reversible exposure.** Staged return seeks a better availability--recurrence frontier than either permanent exclusion or an immediate full-load restart.
5. **Complete comparison.** Population selection, human effort, failure, recovery, and embodied resources are charged before a claimed saving is accepted.

Lifecycle energy for method m is

$$E_m^{\text{life}} = E_m^{\text{train}} + E_m^{\text{infer}} + E_m^{\text{sense}} + E_m^{\text{act}} + E_m^{\text{comm}} + E_m^{\text{facility}} + E_m^{\text{recover}} + E_m^{\text{maint}} + E_m^{\text{emb}},$$

where every term is in joules under one declared service interval. The terms are training, inference, sensing, actuation, communication, facility, recovery, maintenance, and amortized embodied energy. Human design, demonstration, coaching, labeling, tuning, monitoring, repair, safety, and medical effort are reported separately in person-hours. A lower device-energy reading does not establish a lifecycle saving.

Evidence status

COMPONENT	EVIDENCE BOUNDARY	ARCHITECTURAL STATUS
early cue use and anticipation	task-, cue-, opponent-, and response-specific human studies; C-926–C-930	established scoped observations; transferable artificial mechanism unassigned
representative information/action coupling	transfer studies distinguish action coupling from visual similarity; C-931–C-932	plausible evaluation constraint
variability and practice	benefits depend on task dimension, dose, history, and outcome; C-933–C-939	established heterogeneity; adaptive curriculum residual speculative
speed--accuracy frontier	scoped task relations; C-940–C-941	established need for multi-axis reporting
fatigue, pacing, load, and readiness	outcome-specific interventions and measurement critiques; C-942–C-950	resource state is necessary metadata; learned controller untested
staged return	staged clinical framework and nonredundant criteria; C-951–C-953	plausible assurance translation; ordinary staged rollout remains the null
coordination and shared information	practice, shared display, synchrony, and coordination studies; C-954–C-957	perturbation and cross-play are evaluation requirements
deception and feedback	bounded effects under declared tasks; C-958–C-962	adversarial calibration requirement
selection bias and talent prediction	relative-age, maturity, and prospective-validity evidence; C-963–C-965	prospective causal evaluation requirement
complete efficiency	system-boundary and measurement evidence; C-966–C-969	required accounting contract; net advantage unknown

The proposed composition remains a benchmark target until it beats the complete ordinary stack in F-006. A complete stack includes calibrated prediction and retrieval; model-free and model-based RL; POMDP/MPC; system identification; domain randomization and curriculum; robust/adaptive control; value of information and selective prediction; workload, readiness, survival, canary, and rollback models; multi-agent control and explicit communication; imitation and opponent models; and causal selection models.

Speculative extensions

Resource-conditioned sparse routing

Routing could condition on expected remaining work, resource uncertainty, damage, and safe fallback rather than only token or task features. It must beat a conventional estimator plus constrained control at equal sensing, compute, reserve, and failure allowance.

Retention-aware curriculum control

A curriculum controller could value a perturbation by delayed retention and held-out transfer rather than practice loss. It must beat fixed augmentation, domain randomization, active learning, Bayesian experimental design, novelty, quality-diversity search, and automatic curriculum at equal attempt, perturbation, feedback, evaluator, time, and energy budgets.

Deception-calibrated opponent memory

Opponent memory could retain policy versions, cue conflicts, confidence, change points, and abstention value. It must improve calibration and adaptation on new deceptive opponents beyond Bayesian opponent models, fictitious play, self-play, recurrent policies, retrieval, and conformal abstention.

Turnover-resilient communication

Teams could compress communication after shared practice while retaining typed repair, acknowledgements, and belief discrepancy when membership changes. Lower message volume counts only if cross-play, repair latency, task quality, and safety survive.

Counterfactual development policies

Population management could compare selection with broad-development or threshold-lottery policies and explicitly model opportunity-mediated outcomes. It must predict later capability in new cohorts and recover false negatives without hiding attrition or development cost.

Failure modes

FAILURE	OBSERVABLE SIGNATURE	REJECTION OR CONTAINMENT RULE
symbolic proxy win	label accuracy rises while interception, deadline, or safety does not	require coupled action and protected physical outcomes
realism scalar	transfer is attributed to a surface label without factorial manipulation	publish the representative-distance vector and causal source--target matrix
practice/learning substitution	end-of-practice score is reported as retention or transfer	freeze delayed tests and first target trial before updates
useless variability	action entropy rises without outcome information or later utility	retire the schedule or use the simpler fixed/null curriculum
state-blind pacing	policy depends on elapsed work and fails unseen resource pathways	compare with calibrated state estimation and constrained control
single-score readiness	one sensor or aggregate certifies broad capability	use action-specific calibrated envelopes with abstention
irreversible promotion	a damaged system cannot regress after deterioration	require monitored stage rollback and recurrence accounting
synchrony attribution	common-input correlation is called coordination	perturb one agent; require reciprocal compensation and lower task error
brittle team convention	co-trained teams work but turnover and cross-play fail	retain typed protocol, repair, and never-co-trained evaluation
deception overconfidence	confidence rises as calibration, regret, or safety worsens	add genuine/deceptive causal controls and calibrated abstention
selection self-fulfillment	selected systems receive more opportunity and later validate the selector	follow rejected units and estimate opportunity-mediated effects prospectively
survivor-only evaluation	dropout, failure, exclusion, or missing follow-up disappears	retain every assigned unit, censoring event, and stopping decision
budget leakage	human work, failed trials, recovery, facility, or embodied energy is omitted	withhold efficiency claim until the complete ledger closes

FAILURE	OBSERVABLE SIGNATURE	REJECTION OR CONTAINMENT RULE
vocabulary-only residual	ordinary control, curriculum, or inference matches the result	keep the conventional mechanism and retire the added label

Measurable predictions

F-006 contains the full protocols and hard retirement rules. The chapter-level commitments are:

ID	INTERVENTION AND COMPARATOR	MEASUREMENTS	PREDICTION AND FAILURE BOUNDARY
RAP-01	cue-window/channel interventions versus calibrated sequence prediction, retrieval, Bayesian cue model, and POMDP	bits/event, calibration, commitment ms, endpoint m, unsafe events, J/event	residual must transfer to held-out deceptive opponents and full interception; a symbolic-only gain fails
RAP-02	factorial information/action/deadline/consequence/resource changes versus domain randomization, system ID, robust optimization, curriculum, and equal-sample fine-tuning	complete source--target matrix, regret, calibration, adaptation samples, safety, lifecycle cost	a named factor must predict held-out transfer beyond distribution distance; "more realistic" alone fails
RAP-03	history-qualified adaptive variability versus fixed schedules, augmentation, uncertainty-directed curriculum, and random search	practice curve, delayed retention, first-trial near/far transfer, information, failures, J/run	improve transfer at equal criterion exposure and perturbation dose; entropy without utility fails
RAP-04	resource-aware pacing versus state-space readiness, system ID, MPC, robust/adaptive control, risk-sensitive RL, and fixed reserve	task value, output trajectory, state error, admissibility, recovery, recurrence, unsafe events, J	improve the frontier under unseen resource pathways; equality retains the ordinary estimator/controller
RAP-05	reversible staged gate versus time-only, single-score, survival, canary, runtime-assurance, and hand-authored staged rollout	false promotion/withholding, dwell h, recurrence, rollback, availability, review person-hours, J	improve risk--availability on new fault classes and safely regress after deterioration
RAP-06	perturbation and turnover test versus centralized/decentralized control, shared display, protocol, MARL, and common-input controller	task error, lagged compensation, belief bits/event, messages, repair s, cross-play, J	require useful reciprocal compensation and never-co-trained cross-play; synchrony alone fails
RAP-07	deceptive-policy changes versus Bayesian opponent model, fictitious play, imitation, self-play, retrieval, robust and conformal prediction	opponent bits/action, calibration, exploitability, regret, abstention, adaptation s	improve calibrated adaptation at equal interactions, memory, search, and latency
RAP-08	prospective selection policy versus adjusted regression, causal selection, survival, threshold lottery, and broad development	new-cohort calibration, capability, false-negative recovery, opportunity h, attrition, harm, person-hours, J	improve sustainable capability without manufacturing validity through unequal opportunity
RAP-09	complete F-006 composition versus the strongest compatible conventional stack	all protected outcomes, events, steps, bytes, seconds, person-hours, lifecycle J, damage, withheld opportunity	require non-inferiority on protected outcomes and a preregistered Pareto resource gain across tasks, histories, opponents, states, sites, models, and hardware

Mechanism ablations are selective: removing actual-channel state should damage cue interventions; removing resource state should damage pacing and recovery; removing reversible stages should damage recurrence or availability; removing team-belief state should damage turnover and cross-play; removing selection lineage should damage prospective calibration. If every ablation merely reduces capacity and degrades every outcome, the proposed modules have not isolated their claimed roles.

Active acoustic inference

Scope

Sound is not a ready-made sequence of objects. A receiver obtains pressure over time through a medium, room, body, aperture, transducer, clock, calibration state, and preprocessing chain. Sources overlap; direct paths mix with delayed copies; motion changes the operator; an emitted probe changes both the evidence and the physical scene. The system must therefore infer sources and task state from a versioned, action-dependent measurement.

This chapter turns the acoustics, hearing, and auditory-scene-analysis audit into an engineering contract. The detailed equations live in operator-qualified acoustic inference, and Fixture F-009 tests the contract across nine hostile tracks. The editable systems diagram is kept in operator-qualified-active-acoustic-inference.mmd.

The chapter connects four existing parts of the project:

1. sensorimotor grounding, because head, body, receiver, and emitter actions change what can be known;
2. operator-qualified sensing, because acoustic inference is conditional on a physical forward operator;
3. sparse predictive compute, because multirate and event-driven representations can reduce traffic only under a complete task-and-energy test; and
4. the energy model, because sensing, emission, motion, communication, maintenance, and embodied hardware remain inside the service boundary.

The intended output is a calibrated decision, retained acoustic artifact, or safe action—not one universal “hearing” score. Detection, discrimination, intelligibility, grouping, separation, dereverberation, localization, ranging, uncertainty, and lifecycle efficiency remain distinct outcomes.

Biological observation

The ear is a level-dependent, multirate measurement system

Acoustic level has meaning only with a reference pressure, frequency and time weighting, integration interval, sensor position, calibration, and uncertainty (C-1054). Room quantities likewise depend on a declared procedure, frequency band, source-receiver geometry, and spatial and temporal support (C-1055). These constraints are part of the biological stimulus, not bookkeeping added after a model runs.

The cochlea implements frequency-dependent mechanical filtering whose response near characteristic frequency is compressively nonlinear. Outer-hair-cell motility contributes causally to that amplification, while psychophysical filter estimates depend on centre frequency and fitting assumptions (C-1056–C-1058). Temporal coding also has regimes: envelope sensitivity is band-limited, auditory-nerve phase locking degrades with frequency, and downstream circuitry can sharpen synchrony in scoped low-frequency conditions (C-1059–C-1061).

"Cochlear," "spiking," and "neuromorphic" are therefore insufficient mechanism descriptions. A transferable front end must state its filter shapes, bandwidths, sample rates, level dependence, compression history, saturation, thresholds, refractory periods, timestamp precision, and calibration state. FFT, wavelet, mel, ERB, gammatone, gammachirp, modulation, and learned filters remain competing implementations; cochlear-like filtering is already conventional signal processing (C-1095).

Spatial hearing combines bounded cues with movement

Coincidence-sensitive circuits can encode interaural time differences, but the useful cue changes with frequency content, head and pinna transfer functions, room response, and source spectrum. Interaural level differences and spectral cues contribute on different bands and geometries (C-1062–C-1063). Head motion can disambiguate otherwise similar observations (C-1064); it is an information-acquisition action, not nuisance augmentation.

Early-arriving spatial information can dominate later copies, while spatial separation can improve recognition under masking (C-1065–C-1066). That does not make reverberation irrelevant. Stable spectrotemporal context and prior room exposure can alter later judgments; monaural context can compensate for some reverberant speech distortion; direct-to-reverberant energy contributes to distance judgments (C-1070–C-1073). The useful state therefore includes direct, early, and late components plus their time-varying relation to source and receiver motion.

Auditory objects are inferred under masking

Energetic masking concerns overlap at the receiver. Informational masking also depends on whether target and masker are perceptually segregated, and selective listening changes strongly with ear and source configuration (C-1067–C-1068). Temporal coherence across frequency channels is a plausible grouping mechanism, not a settled universal decomposition rule (C-1069).

This makes four outcomes non-interchangeable:

- **detection:** whether task evidence is present;
- **grouping:** which events belong together and how uncertain that assignment is;
- **separation:** recovery of one or more waveforms under an explicit reference and permutation contract; and
- **causal identity:** which physical source produced a recovered component.

A separator can improve a waveform metric while switching identities, damaging localization, or changing perceptually important structure. Published separation metrics can hide severe waveform changes (C-1094), so no one metric is allowed to stand in for the scene.

Sparse activity and efferent control have narrow causal interpretations

Awake auditory cortex can represent sounds with low population activity, and sparse-coding objectives can learn cochlea-like kernels in some sound ensembles. The learned code depends on the training ensemble (C-1074–C-1076). Sparse activity is evidence about representation, not a direct measurement of system joules. Sensors, clocks, memory traffic, event routing, decoding, idle power, and training may dominate (C-1097).

Efferent activation can causally reduce cochlear gain and improve masked-tone auditory-nerve responses in the studied preparations (C-1077). A general human speech-in-noise benefit from medial olivocochlear strength remains disputed (C-1078). An AI analogue earns the name only when an intervention on task-conditioned front-end gain changes a registered external endpoint after distortion, recovery time, calibration, and energy are charged. An attention map or smaller hidden activation is not that evidence.

Echolocation is active observation with physical consequences

Echolocators vary call timing with spatial task difficulty, adapt emission level with distance, steer or reshape the acoustic field of view, and coordinate head or pinna motion with emission (C-1079–C-1082). Human experts can use self-generated echoes, but the demonstrated capability includes extensive experience and neural reorganization (C-1083). Spectral jamming avoidance in bats is context-dependent rather than universal (C-1084).

The transferable loop is emit, receive, infer, move or re-aim, and decide. Its benefit is conditional on waveform ambiguity, target reflectivity, range, clutter, association, other emitters, detectability, acoustic exposure, and the energy required to produce and receive the probe (C-1085, C-1096).

Proposed AI translation

Preserve the episode, operator, and action identity

For episode e, retain the contract

$$\mathcal{A}e = (Xe, Se, Ee, Re, He, Oe, Ce, Te, Ue, Be),$$

where:

SYMBOL	MEANING AND REQUIRED UNITS
X_e	geometry and boundaries [m], medium, temperature [K], relative humidity [1], flow [m/s], occupancy, and time-varying material state
S_e	target and interfering sources, waveforms [Pa], positions [m], velocities [m/s], directivity, identities, roles, and emission times [s]
E_e	commanded and realized active emissions: waveform, spectrum [Hz], duration [s], aim [degree or rad], acoustic energy [J], and exposure
R_e	receiver, body, and array pose; aperture [m]; morphology or manifold; transducer response; gain; health; saturation; and feasible motion
H_e	prior exposure, room and source history, adaptation, training, feedback, previous actions, and their timestamps [s]
O_e	versioned observation operator: impulse responses, sample rate [sample/s], precision [bit], clock, latency [s], preprocessing, support, and missingness
Ce	calibration identity, reference pressure and distance, covariance, traceability, method, interval, and data vintage
Te	literal target construct, decision deadline [s], and registered loss or utility in its native unit
Ue	independent waveform, event, source, room, receiver, body, array, device, site, or population unit
B_e	componentwise limits on evidence, time, memory, communication, action, exposure, risk, human work, and lifecycle energy

Calibration identity can become part of the learned task when a model depends on device, room, or preprocessing state (C-1098). Train-test splits must group clean sources, convolved variants, neighboring windows, impulse-response and renderer lineages, HRTF or array families, devices, speakers or listeners, and derived labels. Capture time and receipt time remain distinct, and only causally received observations may affect a decision.

Keep pressure, level, exposure, and energy separate

For pressure $p(t)$ in pascals over interval T in seconds,

$$p_{\mathrm{rms}} = \sqrt{\frac{1}{T}\int_0^T p^2(t)\,dt}, \qquad L_p = 20\log_{10}\left(\frac{p_{\mathrm{rms}}}{p_0}\right),$$

where p_{rms} is RMS pressure [Pa], reference pressure $p_0 = 20\,\mu\mathrm{Pa}$ in air, and L_p is sound-pressure level [dB re $20\,\mu\mathrm{Pa}$]. Frequency weighting, time weighting, band, position, orientation, and calibration must be explicit. Sound exposure $\int p^2(t)dt$ has units Pa^2 s; it is not acoustic emission energy [J], electrical energy [J], or a complete biological risk model.

For source pressure $x_s(t)$, received pressure $y_m(t)$, and noise $n_m(t)$, all in pascals, use the time-varying mixture

$$y_m(t) = \sum_{s=1}^{N_s} \int h_{m,s}(t,\tau) x_s(t-\tau)\, d\tau + n_m(t),$$

where m is receiver-channel index [1], s is source index [1], source count N_s is dimensionless, t and delay τ are in seconds, and propagation kernel $h_{m,s}(t,\tau)$ is in reciprocal seconds. A fixed convolution is only a special case; moving sources, receivers, boundaries, media, and emitters make the operator time-varying.

Build a qualified multirate front end

The front end may combine:

1. an FFT, wavelet, constant-Q, mel, ERB, gammatone, gammachirp, modulation, or learned analysis bank;
2. level- and history-dependent compression with declared attack, release, saturation, and distortion;
3. masking and source-count hypotheses rather than a single global noise scalar;
4. dense samples or sparse onset, offset, change, phase, or threshold events with timestamp and refractory identity; and
5. a decoder or downstream task that is charged for reconstructing information discarded by the encoder.

Band selection and update rate should follow the causal time scales needed by the task. Fine timing cannot be assumed above the supported band; weak sustained signals and rare hazards cannot be discarded merely because an event count falls.

Fuse spatial evidence without erasing ambiguity

For path-length difference Δr [m] and sound speed c_{snd} [m/s], interchannel delay is

$$\Delta t = \frac{\Delta r}{c_{\mathrm{snd}}},$$

where Δt is in seconds. For left and right RMS pressures p_L, p_R [Pa],

$$\mathrm{ILD} = 20 \log_{10}\left(\frac{p_L}{p_R}\right)$$

is interaural level difference [dB]. Both cues are band-, source-, room-, body-, and calibration-qualified. Spectral cues and head or receiver motion remain separate intervention axes. A correlation surface under reverberation may be multimodal; reducing it to one delay without calibrated uncertainty loses the ambiguity the controller needs.

The learned path must be compared with generalized cross-correlation, matched filtering, steered-response power, delay-and-sum, robust and minimum-variance beamforming, MUSIC or ESPRIT, Bayesian delay estimation, and source tracking. GCC, MVDR, and subspace localization are mature nulls, not discoveries created by a biological label (C-1086–C-1088).

Separate scene inference from metric substitution

Each output head declares one literal construct and uncertainty model. The minimum firewall is:

CONSTRUCT	REQUIRED MEASUREMENT	INVALID SUBSTITUTE
detection	hit and false-alarm rates, criterion, calibration, and dimensionless d'	recognition accuracy or mixture SNR
intelligibility or identification	confusion matrix and words, phonemes, or information units under a protocol	SI-SDR, loudness, or localization
grouping	event-to-source assignment, count, switch rate, ambiguity, and identity	separated waveform or attention weight
separation	SI-SDR plus waveform, perceptual, downstream-task, and identity measures	localization or intelligibility alone
dereverberation	direct, early, and late waveform/operator measures plus downstream effect	a “dry” label or one decay scalar
localization	azimuth and elevation error [degree], front-back error, coverage, and calibration	beam width, identity, or range
ranging	bias, absolute error, and interval coverage [m]	echo detection or direction
sparse timing	timestamp error [s], misses, false events, information, bytes, task value, and joules	event count or active fraction
active sensing	randomized emission or motion contrast, decision value, latency [s], detectability, exposure, risk, and energy [J]	action-success correlation
uncertainty	proper score, interval coverage, risk-coverage, and abstention value by held-out operator	confidence magnitude

Independent-component methods, time-frequency sparsity methods, deep clustering, permutation-invariant training, and time-domain separation are mature but assumption-bound baselines (C-1089–C-1093). The strongest compatible composition—not an isolated weak separator—is the decisive null.

Close the action loop

At decision time t, policy π_q for method q selects

$$(a_t^{\text{emit}}, a_t^{\text{body}}, a_t^{\text{sensor}}, a_t^{\text{gain}}, a_t^{\text{task}}) = \pi_q(\mathcal{H}_t, \widehat{X}_t, \widehat{O}_t, \widehat{U}_t, \mathcal{A}_t^{\text{safe}}, B_t),$$

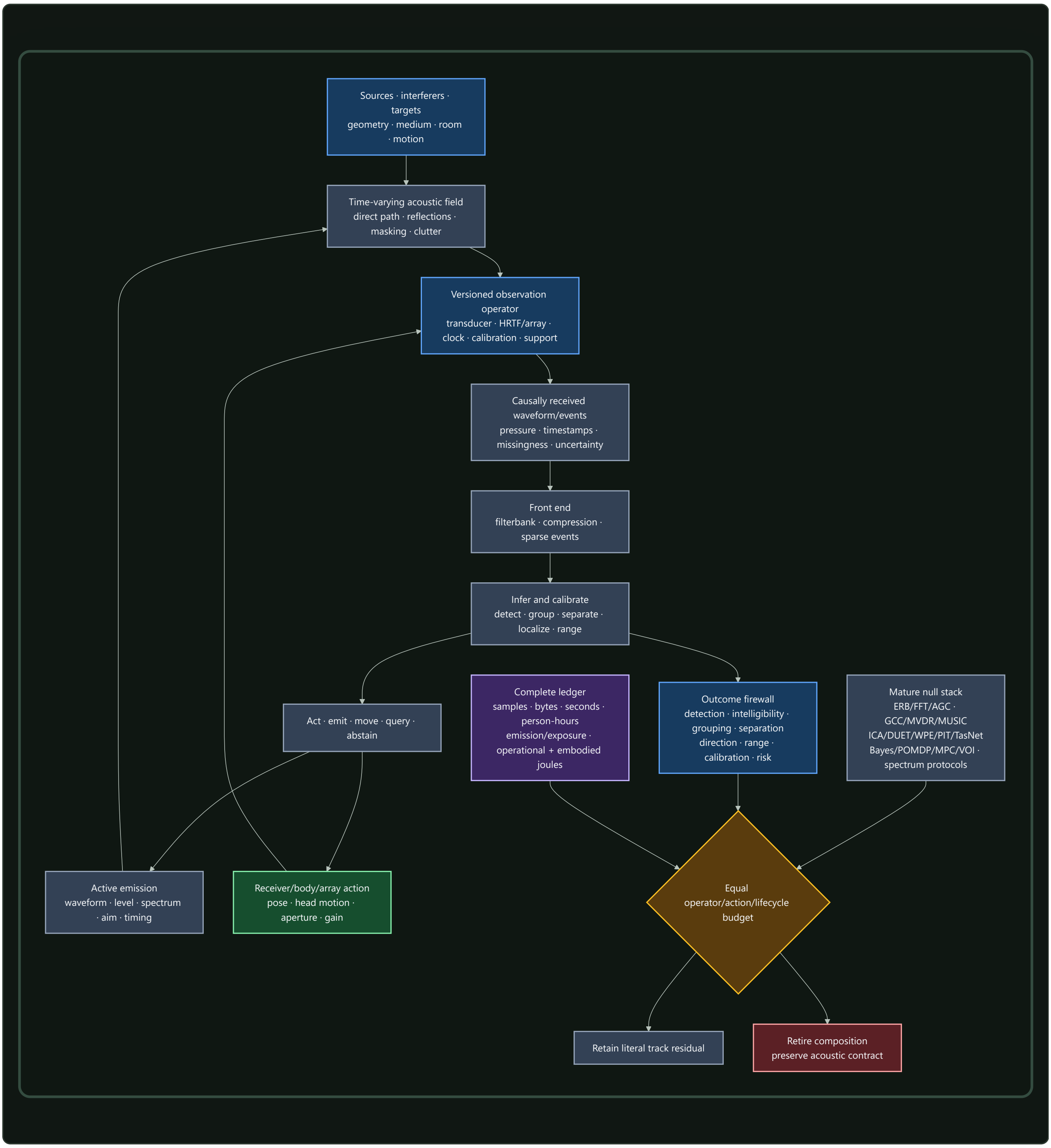
where $\mathcal{H}_t$ is causally received history, $\widehat{X}_t$ is estimated scene state, $\widehat{O}_t$ is operator and calibration state, $\widehat{U}_t$ is uncertainty, $\mathcal{A}_t^{\text{safe}}$ is the feasible action set, and B_t is remaining componentwise budget. Emission actions set waveform, spectrum [Hz], level [referenced dB], aim [degree or rad], and time [s]; body actions set pose and motion [m, rad, s]; sensor actions select aperture [m], channels, and sample rate [sample/s]; gain is dimensionless or declared in dB; the task action uses its native unit.

For monostatic echo emission and receipt times t_{emit} and t_{recv} [s], nominal range is

$$\hat{r} = \frac{c_{\text{snd}}}{2}(t_{\text{recv}} - t_{\text{emit}}),$$

where $\hat{r}$ is range [m]. The interval must propagate clock error, target motion, refraction, multipath, ringing, waveform ambiguity, association, and detector threshold. Multiple emitters add collision, authentication, fairness, adversarial imitation, and jamming state. The proposal must beat TDMA, FDMA, CDMA, random access, listen-before-talk, power control, coded probes, cancellation, authentication, and game-theoretic allocation at equal spectrum, power, messages, and exposure.

One system, tested against a complete null



Editable source: operator-qualified-active-acoustic-inference.mmd.

The contract is owned by existing components:

- Candidate 002 — Multiscale context broadcast for multirate context;
- Candidate 006 — Reversible physical skill for reusable analog or beamforming paths;
- Candidate 007 — Endogenous observation surveillance for action-dependent observation;
- Candidate 009 — Graded assurance envelopes for calibration, uncertainty, exposure, and fallback;

- Candidate 012 — Latency-qualified authority for travel time, evidence age, and decision authority; and
- Candidate 014 — Versioned observation contract for operator and calibration identity.

Efficiency mechanism

The proposed composition can improve efficiency through four literal pathways:

1. **Multirate analysis.** Slow envelopes and long context need not update at the rate required for fine timing. Savings must appear in samples, memory traffic, latency, and measured joules without degrading a protected band.
2. **Conditional events.** Onsets, offsets, changes, or uncertainty triggers can suppress redundant transport and downstream work. The decoder, timestamping, missed sustained signals, false events, and idle event hardware remain in the comparison.
3. **Selective physical action.** A head turn, receiver move, gain change, or acoustic probe can resolve an ambiguity more cheaply than processing another passive interval. Its motion, emission, exposure, delay, detectability, and interference costs must be lower than the decision value it creates.
4. **Reusable calibrated structure.** Stable filter, beamforming, correlation, or control paths can move to DSP, FPGA, ASIC, analog, or neuromorphic hardware when reuse amortizes conversion, characterization, drift monitoring, repair, and replacement.

For one registered service interval, lifecycle energy is

$$E_q^{\text{life}} = E_q^{\text{data}} + E_q^{\text{train}} + E_q^{\text{emit}} + E_q^{\text{sense}} + E_q^{\text{infer}} + E_q^{\text{move}} + E_q^{\text{comm}} + E_q^{\text{store}} + E_q^{\text{cal}} + E_q^{\text{maint}} + E_q^{\text{emb}},$$

where method q has data-acquisition, training, acoustic-emission, sensing, inference, motion, communication, storage, calibration, maintenance, and amortized embodied-energy terms, each measured in joules over the same service interval. The complete ledger also reports:

- pressure samples, sparse events, optimization steps, queries, and bytes;
- wall time [s], task latency [s], peak memory [byte], and network traffic [byte];
- acoustic exposure $[\mathrm{Pa}^2\ \mathrm{s}]$, unsafe-event count [1], and harm or detectability in a registered task-specific unit;
- emitted acoustic energy [J], electrical input [J], sensing [J], motion [J], compute [J], cooling [J], and facility allocation [J];
- recording, labeling, listening tests, calibration, tuning, monitoring, maintenance, and repair [person-hour]; and
- transducers, arrays, processors, actuators, replacements, wear, embodied hardware, and opportunity cost.

An arm that exceeds any binding budget is infeasible. Resources released by an ablation remain unused. Lower event count, lower FLOPs, quieter emission, or a faster accelerator is not an efficiency result unless accepted task service improves on the complete protected vector.

Evidence status

The stable claim ledger mirrors all 46 audit-local claims without promoting the audit itself into evidence:

TOPIC	STABLE CLAIMS	STATUS	CONSEQUENCE
calibrated acoustic and room measurement	C-1054–C-1055	established	reference, support, geometry, operator, and uncertainty are mandatory
cochlear mechanics, filters, timing, binaural cues, motion, precedence, masking, and selective listening	C-1056–C-1068	established	supplies scoped mechanisms and limits, not one front-end prescription
temporal coherence as a grouping mechanism	C-1069	plausible	test against spatial, statistical, and learned grouping nulls
context, room adaptation, distance, and low population activity	C-1070–C-1074	established	motivates explicit context and sparsity tests while keeping outcomes separate
sparse coding learns cochlea-like kernels	C-1075	plausible	representation result; no direct hardware-energy claim
ensemble dependence and causal efferent gain	C-1076–C-1077	established	qualify learned codes by data and test gain through intervention
general human speech-in-noise benefit from efferent strength	C-1078	disputed	do not use as a broad task-benefit premise
adaptive biosonar, field steering, coordinated motion, and expert human echo use	C-1079–C-1083	established	motivates costed emission-reception-action loops
universal spectral jamming avoidance	C-1084	disputed	use multi-emitter and adversarial controls rather than a universal law
waveform ambiguity and mature GCC, MVDR, MUSIC, ICA, sparse, deep, PIT, and time-domain separation nulls	C-1085–C-1093	established	compare with a complete competitive DSP and learned stack
metric failure, conventional cochlear-like DSP, and physical cost of active sensing	C-1094–C-1096	established	enforce the metric firewall and complete action ledger
sparse activity is not system energy; calibration identity can be learned task state	C-1097–C-1098	plausible	measure system energy and hold out operator versions
transferable residual is the operator- and action-qualified contract	C-1099	speculative	preserve the benchmark; retire the composition if the mature null matches it

No row establishes that the complete proposed system beats the conventional stack. The evidence supports the physical constraints, biological phenomena, and engineering nulls. The composition remains an experiment.

Speculative extensions

Operator-conditioned auditory state

Learn a compact state that jointly represents room response, body or array transfer, clock uncertainty, source hypotheses, and front-end state. It must predict held-out calibration probes and remain identifiable under scene changes. Compare it with explicit system identification, adaptive filtering, room banks, and Bayesian state-space models.

Counterfactual acoustic action

Train the policy to predict not only what a probe or head turn might reveal but also its exposure, detectability, collision, motion, and lifecycle cost. Use paired simulator seeds and physical replay so action value is estimated against the same latent scene, not easier episodes selected after acting.

Cross-timescale front-end control

Let fast protection and compression, intermediate event routing, and slower room or task adaptation share state without sharing unrestricted authority. Test whether the joint controller beats independently tuned AGC, PCEN, adaptive filters, selective prediction, and recurrent context after coordination cost.

Population-qualified acoustic hardware

Route recurring transforms among digital, analog, array, and event-driven front ends by measured device identity, calibration envelope, workload reuse, and remaining service life. Hold out hardware classes, lots, transducers, and array geometries. A nominal-device gain does not establish fleet-level benefit.

Strategic multi-emitter sensing

Model emitters as cooperative, selfish, deceptive, imitating, colluding, or turnover-prone agents. Joint waveform and scheduling policies must retain source authentication and calibrated uncertainty under never-coordinated population cross-play. Ordinary spectrum protocols remain the null.

Failure modes

SIGNATURE	INTERPRETATION AND REQUIRED RESPONSE
an unreferenced decibel value or unspecified integration interval drives the result	the stimulus and risk boundary are undefined; invalidate the comparison
train and test share a clean source, impulse response, renderer, HRTF, array family, device, or neighboring window	the result is an inverse crime or lineage leak; rebuild grouped splits
a fixed room convolution succeeds but moving-source, moving-receiver, or changing-boundary trials fail	the observation operator was misspecified; restrict the validity envelope
a cochlear label wins only against a weak mel front end	the mechanism was not isolated; restore FFT, wavelet, ERB, gammatone, learned, compression, and DSP nulls
sparse events lower active fraction but not bytes or joules	representation sparsity did not create system efficiency; retire the energy claim
event gating misses weak sustained signals, fine timing, or rare hazards	the threshold policy violates the protected outcome; retain a dense or hybrid path
localization collapses a multimodal GCC or beam surface to one confident point	estimator uncertainty is miscalibrated; preserve alternatives or abstain
apparent spatial super-resolution depends on source-count priors or training overlap outside aperture support	prior information is being reported as measured resolution; narrow the claim
SI-SDR rises while intelligibility, localization, identity, calibration, or downstream value worsens	metric optimization substituted for the task; fail the separation track
an efferent analogue changes gain or hidden activation without an external endpoint	no causal task benefit was demonstrated; remove the biological interpretation
active sensing wins only after extra samples, emission, exposure, motion, latency, or detectability	additional opportunity explains the result; match the full action ledger
echo range ignores clock drift, target motion, multipath, ringing, waveform ambiguity, or association	the interval is invalid; propagate those uncertainties or abstain
frequency shifting or timing changes fail against new emitters, collusion, authentication faults, or equal-power cross-play	the jamming policy is context-bound; retain conventional coordination
a learned path beats one beamformer or separator but not their strongest compatible composition	the baseline is incomplete; B7 in F-009 remains decisive
confidence fails in held-out rooms, bodies, arrays, clocks, calibration versions, sites, or hardware	the system learned operator identity without transferable calibration; restrict or retrain
an efficiency result omits recording, labels, calibration, failed trials, cooling, maintenance, replacement, or embodied hardware	the accounting boundary is incomplete; recompute the lifecycle result

Measurable predictions

The nine F-009 tracks convert the chapter into falsifiable predictions:

1. **Level-dependent front end.** A qualified nonlinear multirate front end improves registered detection or discrimination across held-out levels, bands, histories, sensors, and domains beyond the strongest fixed and learned filterbank-plus-compression stack at equal bytes, latency, and joules.
2. **Sparse timing.** Event encoding preserves protected timing, weak sustained sources, and rare hazards while reducing measured memory traffic and service energy beyond VAD, codecs, sparse convolution, and duty-cycled DSP.
3. **Active emission and reception.** Joint waveform, emission, and receiver action creates positive causal decision value after exposure, detectability, motion, interference, calibration, and lifecycle energy are charged.
4. **Localization under operator shift.** Calibrated direction and range remain better than GCC, Bayesian delay, SRP, MVDR, MUSIC, and tracking baselines on held-out HRTFs, arrays, bodies, apertures, rooms, spectra, faults, and clocks.
5. **Reverberation adaptation.** Context state improves literal waveform, intelligibility, event identity, distance, or calibration outcomes in new rooms beyond WPE, Wiener filtering, explicit operator estimation, and equal-sample adaptation without confusing compensation with recovery.

6. **Grouping and separation.** The model improves event-to-source grouping, source count, separation, causal identity, localization, uncertainty, and downstream value across new sources, rooms, counts, and cue conflicts beyond the complete ICA, IVA, DUET, NMF, WPE, PIT, deep-clustering, TasNet, and neural-beamforming stack.
7. **Efferent-like gain.** Randomized intervention on task-conditioned gain improves a protected external endpoint after distortion, false suppression, recovery, calibration, and energy are included; internal suppression alone predicts no pass.
8. **Multi-emitter interference.** Adaptive emission coordination improves ranging or detection, association, throughput, fairness, calibration, and exploitability under unseen cooperative and adversarial populations beyond mature spectrum-access and authentication protocols at equal power and messages.
9. **Complete lifecycle.** Any surviving benefit replicates across at least two source families, rooms or media, arrays or bodies, operator versions, scene types, model families, sites, and hardware classes after every acquisition, action, person-hour, risk, operational-energy, and embodied-energy term is charged.

Preregister protected outcomes, non-inferiority margins, material-improvement thresholds, 95% intervals, multiplicity control, tail-failure ceilings, and stopping rules. Randomize and analyze at the unit receiving the intervention; adjacent samples and overlapping windows are not independent sources.

The hard decision is simple: if the complete conventional composition matches the proposal at equal operator, action, calibration, uncertainty, exposure, human-work, and lifecycle budgets—or if no ablation isolates value beyond that stack—retire the architectural residual. Keep the measurement, operator, action, metric, and accounting contract.

Operator-qualified sensing and physical inference

Scope

This chapter defines what an adaptive system is allowed to claim from a physical measurement. A sensor does not deliver a scene, object, or fact. It delivers a finite observation produced by an aperture, illumination pattern, medium, detector, clock, calibration state, acquisition policy, and noise process. Inference adds assumptions and prior information.

The architectural consequence is precise: decoded tensors may be convenient runtime inputs, but they are not self-describing evidence. The system must keep enough of the measurement operator, uncertainty, validity envelope, and lineage to know what the observation could resolve, what it could not resolve, and when another measurement is worth buying.

This chapter is operationalized by:

1. the optics, photonics, and inverse-sensing audit;
2. the operator-qualified optical-inference mathematics; and
3. Fixture F-007, which tests the full contract against inverse-method, active-sensing, calibration, control, digital-accelerator, and passive-optics baselines.

Biological observation

Biological sensing is already physical inference. An eye has finite aperture, spectral sensitivity, sampling density, integration time, dynamic range, blind regions, motion, adaptation, and a body that can change viewpoint. A useful percept can therefore depend on both received evidence and prior structure. Movement can reveal a surface that one view leaves ambiguous; longer exposure can buy photons while losing temporal resolution; adaptation can extend useful operation while changing the response function.

The transferable observation is not a particular visual anatomy. It is the closed coupling among:

- a bounded physical measurement channel;
- an internal estimate that remains conditional on that channel;
- actions that change future observability;
- calibration and adaptation over multiple timescales; and
- task-specific decisions made before every latent detail is known.

Optics makes those constraints measurable. Finite apertures, null spaces, photon statistics, ambiguity classes, saturation, and drift are established properties of physical sensing (C-970–C-988). They prevent a fluent reconstruction from silently becoming stronger evidence than the acquisition supplied.

Proposed AI translation

Preserve the evidence-producing operator

For latent physical state x_e in episode e, let acquisition t produce

$$y_{e,t} = g_{e,t}(\mathcal{H}_{\nu_{e,t}}(a_{e,t}, c_{e,t})x_e) + n_{e,t},$$

where:

- $y_{e,t}$ is the raw observation in detector counts [count] or another declared sensor unit;
- $a_{e,t}$ is the acquisition action, such as viewpoint, exposure, wavelength, or illumination pattern;
- $c_{e,t}$ is the calibrated parameter vector in declared native units;
- $\nu_{e,t}$ is the immutable operator-version identifier;
- $\mathcal{H}_{\nu_{e,t}}$ is the physical forward operator;
- $g_{e,t}$ is detector conversion, clipping, and readout response; and
- $n_{e,t}$ is only the residual noise represented by an explicit likelihood.

The observation record travels with operator version, calibration covariance, saturation/dead-time mask, capture and receipt times, preprocessing lineage, and validity envelope. A compact representation can replace the raw record only for a registered family of future queries and only while reconstruction, uncertainty, and provenance obligations remain satisfied. This connects the versioned observation contract, semantic compaction, and value-aware retention.

Separate measured information from prior-supported reconstruction

For a linearized operator $H = U\Sigma V^*$, a component along v_j with singular value $\sigma_j = 0$ lies in the measurement null space. A decoder may still propose a plausible value for that component, but the value comes from a prior, another measurement, or a convention—not from this observation.

Every output therefore carries three distinguishable uncertainty sources:

SOURCE	WHAT VARIES	APPROPRIATE RESPONSE
measurement noise	photon arrivals, read noise, background, quantization	propagate the likelihood; change exposure or sensor when valuable
operator uncertainty	calibration, alignment, drift, response, timing	monitor residuals; recalibrate, downgrade, reroute, or abstain
prior or model uncertainty	training support, regularizer, latent family, task shift	expose support dependence; acquire discriminating evidence or retain alternatives

The separation matters under compressed sensing, phase retrieval, computational super-resolution, blind calibration, and learned reconstruction (C-972, C-976–C-981, C-985). Pixel count, sharpness, or confidence cannot substitute for newly identified physical information.

Buy another measurement only when it changes the decision frontier

Let b_t be the current belief, a a safe acquisition action, d a downstream decision, and $U(d, \theta)$ task utility for uncertain state θ. The expected value of information is

$$\mathrm{EVI}(a \mid b_t) = \mathbb{E}_y\left[\max_d \mathbb{E}[U(d,\theta) \mid b_t, a, y]\right] - \max_d \mathbb{E}[U(d,\theta) \mid b_t].$$

No scalar acquisition price is assumed. The controller compares EVI against a cost vector containing at least photons [count], energy [J], latency [s], dose or disturbance in its task-specific unit, actuator wear [cycles], and risk on a declared scale. An action is admissible only inside its safety and authority envelope.

This converts active perception from "collect more data" into a resource allocation problem. The active path must beat fixed acquisition, greedy value of information, Bayesian experiment design, POMDP planning, and model-predictive control at equal opportunity and cost (C-975).

Monitor validity instead of trusting calibration indefinitely

Calibration is versioned state, not a one-time property of a device. Reference channels and task-independent residuals monitor alignment, gain, timing, temperature, background, saturation, and component aging. A threshold crossing does not identify the cause; it changes what action is permitted.

A validity transition can trigger, in order:

1. a qualified reduction in confidence or supported query set;
2. a new reference or calibration acquisition;
3. rerouting to a different sensor or computational path;
4. a digital or conservative fallback;
5. reset, repair, or replacement; and
6. abstention when none of those paths restores the evidence contract.

Blind self-calibration is tested for identifiability, and task residuals are not allowed to conflate scene shift with device drift (C-984–C-989).

Route a transform to the substrate that actually makes it cheap

Passive and active optical hardware can execute physically matched linear transforms, sometimes before an observation becomes a large digital tensor. That is valuable when the input is already optical, the transform is reusable, conversion can be avoided, and required precision fits the device envelope.

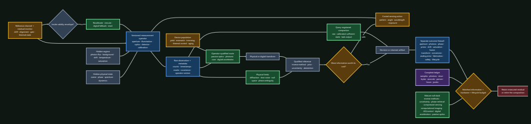
It is not a general preference for an optical path. The route record declares:

- input locality and format;
- transform identity, reuse count, sparsity, and required precision;
- source, modulator, detector, ADC/DAC, control, thermal, and host work;
- device-specific calibration, mismatch, yield, drift, and age;
- accepted-output latency and quality; and
- fallback and migration cost.

Routing then compares passive optics, a photonic core, a digital accelerator, and hybrid compositions on the same workload and service boundary. Optical propagation is credited only for work it actually displaces (C-989–C-999).

One closed contract

Wide diagram · fit to page



Editable source: operator-qualified-physical-inference.mmd.

Efficiency mechanism

The contract permits four efficiency gains, each with a matching way to fail:

1. **Acquire selectively.** Spend photons, time, and motion only where another observation changes an accepted decision. It fails when the acquisition controller costs more than fixed sensing or shifts risk outside the ledger.
2. **Transform before expansion.** Use a physical operator to filter, aggregate, or project local optical information before high-volume digital movement. It fails when conversion, source, control, or recalibration erases the saving.
3. **Retain the sufficient level.** Store raw evidence, calibrated sufficient state, or task output according to registered future queries and recovery obligations. It fails when later queries expose discarded information.
4. **Route by validity and reuse.** Amortize a stable transform on hardware that suits its precision and repetition. It fails under workload shift, fabrication spread, thermal control, low utilization, or short lifetime.

For each accepted service unit,

$$E_{\mathrm{service}} = E_{\mathrm{source}} + E_{\mathrm{mod}} + E_{\mathrm{prop}} + E_{\mathrm{detect}} + E_{\mathrm{ADC}} + E_{\mathrm{DAC}} + E_{\mathrm{control}} + E_{\mathrm{digital}} + E_{\mathrm{thermal}} + E_{\mathrm{facility}} + E_{\mathrm{embodied}},$$

where every E term is energy [J] measured over the same accepted-output boundary. Embodied energy includes fabrication, packaging, yield loss, replacement, and end-of-life treatment amortized over accepted lifetime service. The comparison also reports quality, calibration, latency, risk, photons, bytes moved, and human maintenance effort; joules alone cannot hide a worse sensing contract.

Evidence status

INGREDIENT	STABLE CLAIMS	STATUS AND ARCHITECTURAL USE
operator, aperture, null space, photon and precision limits	C-970–C-974	established physical constraints; mandatory measurement metadata
active illumination and structural priors	C-975–C-980	established scoped mechanisms; advantage remains task- and prior-qualified
multiplexing, coded acquisition, and adaptive correction	C-981–C-984	established tradeoffs; sensorless objective validity remains plausible
blind calibration, drift, saturation, and fusion	C-985–C-988	established constraints on identifiability and valid combination
physical transforms and avoided conversion	C-989–C-990	physical execution established; end-to-end benefit workload-dependent

INGREDIENT	STABLE CLAIMS	STATUS AND ARCHITECTURAL USE
system energy, conversion, and analog error	C-991–C-993	system-boundary constraints established; core-only efficiency claims disputed
fabrication, thermal control, and in-situ adaptation	C-994–C-996	variation and thermal cost established; recoverable mismatch is scoped
labels, routing, uncertainty, and lifecycle ranking	C-997–C-1001	neuromorphic-label inference disputed; routing and uncertainty composition plausible; scoped lifecycle reversal established

The sources support the constraints and component mechanisms. They do not yet show that their full composition improves this project's quality–risk–latency– energy frontier. F-007 is therefore a hostile fixture, not an architecture promotion.

Speculative extensions

Learned operator compaction

Learn the smallest operator state that preserves a registered family of likelihoods, counterfactual acquisitions, and calibration decisions. Compare it with explicit metadata, sufficient-statistic storage, low-rank calibration, and recomputation from raw evidence. A compact state that cannot answer a new registered query is rejected.

Joint query, sensor, and substrate routing

Let one controller decide whether to answer from retained state, acquire a new physical observation, or move the transform to another substrate. The claim is interesting only if joint control beats three separately optimized controllers after coordination and monitoring cost.

Population-calibrated physical modules

Treat fabrication variation as measured device identity rather than nominal noise. Assign workloads by the calibrated envelope of each device, then test whether characterization, placement, spares, and migration work less than trimming every device to one specification.

Future-query-aware sensing

Choose acquisitions that serve both the immediate decision and declared future queries. This could favor a slightly more expensive measurement now if it prevents reacquisition or unsafe inference later. The future-query distribution must be registered before results are inspected.

Failure modes

SIGNATURE	INTERPRETATION AND REQUIRED RESPONSE
sharper reconstructions appear without improved held-out physical decisions or calibration	the prior changed appearance, not measured information; narrow the claim
null-space pairs receive confident different answers from the same observation	the decoder hides prior selection; expose alternatives or abstain
active sensing wins only with more photons, time, dose, or actuator work	extra opportunity explains the result; match the acquisition ledger
a multiplex advantage disappears when the dominant noise source changes	the result is regime-specific; retain the crossover, not a universal rule
drift monitoring reacts to scene shift or misses reference-channel failure	the validity detector is not identifiable; add controls or conservative fallback
fused confidence improves while shared calibration error remains unmodelled	covariance was double-counted; use robust fusion or keep sensors separate
an optical path wins on core propagation but loses sensor-to-decision energy	conversion, control, or movement dominates; retain the digital baseline
nominal-device accuracy hides die, package, temperature, and age spread	the hardware claim is not population-valid; stratify devices and lifetime
in-situ adaptation consumes unreported training measurements or human tuning	calibration work is omitted; charge it to deployment
compact storage answers current tasks but prevents a registered later query	compaction violated the preservation contract; retain raw or richer state
one scalar efficiency score hides worse risk, calibration, or maintenance	report the Pareto vector; do not average protected outcomes away

Measurable predictions

1. **Null-space honesty.** On paired physical states that share an observation under the tested operator, an operator-aware system will retain ambiguity or abstain more accurately than a tensor-only decoder without reducing identifiable-task performance.
2. **Prior-shift qualification.** Under held-out scene structure, operator-aware uncertainty will predict super-resolution and compressed-recovery failure better than confidence from the reconstruction model alone.
3. **Costed active acquisition.** At equal photons, dose, latency, action count, risk, and joules, active selection will improve accepted task utility beyond fixed acquisition and one-step EVI—or the learned acquisition mechanism is retired.
4. **Noise-regime crossover.** Multiplexed and focused acquisition will exchange rank at a reproducible noise boundary predicted before the confirmatory run.
5. **Drift-aware recovery.** Versioned monitoring will reduce invalid confident outputs and recovery time under hidden alignment, gain, timing, and thermal changes beyond periodic calibration at equal reference and maintenance cost.
6. **Heterogeneous crossover.** A physical path will improve end-to-end accepted outputs per joule only in preregistered regions of transform reuse, precision, input locality, utilization, and device validity; digital routing will win outside them.
7. **Device-population validity.** Routing by measured device envelope will improve yield-adjusted lifetime service beyond nominal routing and uniform trimming after characterization, migration, spare, and control costs.
8. **Query-preserving compaction.** A query-registered retained state will use fewer stored bytes and lifecycle joules than raw retention while meeting every registered reconstruction, uncertainty, provenance, and recalibration tolerance.

All predictions are evaluated through F-007. A positive result remains bounded to its measurement operator, physical regime, query set, hardware population, workload, and lifecycle boundary.

Active chemical sensing

Scope

A chemical sensor does not receive an odor, analyte identity, source, or hazard. It receives a time-dependent response produced jointly by source release, transport, reaction, surfaces, the receiver's path, the sampling action, the inlet and chamber, sensor chemistry, temperature, humidity, calibration, adaptation, ageing, contamination, and previous exposure. In a turbulent plume, even the material reaching the receiver arrives as intermittent whiffs and blanks rather than a smooth pointer to its source.

This chapter turns the olfaction, chemical sensing, and plume-tracking audit into readable architecture. The detailed definitions live in operator-qualified chemical-sensing mathematics, and Fixture F-011 tests the architecture across fourteen hostile tracks. The editable diagram is kept in operator-qualified-active-chemical-sensing.mmd.

The chapter connects existing project components rather than adding another principle or candidate:

1. sensorimotor grounding, because sniffing, pumping, orientation, locomotion, and receiver geometry change the evidence;
2. operator-qualified sensing, because every chemical result remains conditional on a physical forward operator;
3. sparse predictive compute, because temporal events and sparse representations earn efficiency credit only through total task work and measured energy;
4. memory and consolidation, because fast adaptation, learned associations, slow calibration, drift, and maintenance occupy different state and update timescales;
5. reliability under mission profiles, because humidity, contamination, ageing, poisoning, replacement, and out-of-support operation change the device rather than merely the data; and
6. the energy model, because motion, pumps, heaters, preconcentration, chromatography, vacuum/ionization, calibration gases, consumables, maintenance, human work, and embodied devices remain inside the service boundary.

The intended output is a calibrated decision, qualified retained observation, safe action, or abstention. Presence, molecular identity, perceptual odor identity, concentration, mixture composition, direction, source position, source attribution, intensity, valence, hazard, exposure, and absorbed dose are different outcomes. One may help predict another; none may silently replace it.

The evidence range for this chapter is C-1152–C-1203: 50 claims are `established` within their stated experiments or authoritative methods, C-1188 is `plausible`, and C-1192 is `disputed`. Those statuses qualify the claim boundaries; they are not votes for the architecture.

Biological observation

Receptor populations provide coverage, not self-describing identities

Mammalian olfaction begins with a large receptor family. Within the receptor panel and odorants studied, individual receptors responded to multiple odorants, individual odorants recruited multiple receptors, and the population pattern changed with concentration (C-1152–C-1154). This supports a distributed measurement basis. It does not supply a universal odor code, open-world chemical coverage, or concentration-independent token.

Chemosensation itself is not one architecture. Manipulations of mammalian sweet and umami pathways provide a scoped dedicated-cell counterexample (C-1155). A receptor-like array must therefore state which chemicals and concentrations it can distinguish, where responses overlap, and which outcomes require another channel. “Combinatorial,” “labelled-line,” or “olfactory” is a description of evidence organization, not an implementation credit.

Bulb and cortex transform concentration, gain, and timing

Divisive normalization in the studied fly antennal lobe scaled projection-neuron responses with pooled receptor activity. Mouse olfactory-bulb and piriform measurements show transformations that can make identity representations more tolerant to concentration, while still preserving useful intensity or concentration-change information in other activity (C-1156–C-1162).

Three constraints follow:

1. concentration tolerance is an outcome to measure, not an invariant built into an anatomical label;
2. suppressing absolute level can damage leak, dose, or safety tasks even when identity improves; and
3. pooled inhibition and recurrence must compete with robust scaling, explicit gain-state estimation, concentration-conditioned inference, and ordinary recurrent models at equal latency and energy.

Receptor adaptation includes causal calcium-dependent feedback, receptor current and spike output can span different concentration ranges, and habituation depends on duration, interval, and odor similarity (C-1163–C-1166). Receptor adaptation, behavioral habituation, short-term sensor recovery, calibration drift, and irreversible poisoning must remain separate states.

Sniffing is part of the observation operator

Rats can alter sniff rate rapidly during discrimination; changing sniffing changes the peripheral-to-bulb filter; early inhalation-locked activity can carry information quickly; and mice can use millisecond-scale sniff-phase differences under scoped protocols (C-1167–C-1171). That timing cannot recover fluctuations already removed by tubing, a chamber, or a slow sensor. End-to-end bandwidth, capture and receipt time, and the realized sampling waveform set the usable support.

Spatial acquisition is also regime-dependent. Serial sampling sufficed in one mouse gradient task, while bilateral temporal correlation supplied odor-motion information in a fly preparation (C-1172–C-1173). Neither result establishes universal stereo or universal serial sampling. Body size, receptor spacing, movement, wind, range, plume intermittency, and sensor bandwidth decide which comparison is informative.

Turbulent plumes turn localization into inference under intermittent evidence

Theory tested against simulation, laboratory, and field observations describes structured whiff and blank statistics rather than a smooth instantaneous gradient. Fine plume structure requires fast ground truth; field gaps change with environment; and several intuitive burst summaries converge too slowly or vary too weakly to guide short-horizon search reliably (C-1174–C-1177).

Animals demonstrate multiple bounded strategies. Moths can surge after odor contact and cast after loss. Walking flies use distinct odor-ON, odor-OFF, and wind transforms, and in irregular plumes their stochastic turns and walk/stop decisions depend on encounter timing rather than continuous steering (C-1178–C-1180). Contact-correlated slowing in mice is an observation, not proof that slowing is optimal (C-1181).

Robotics already supplies strong nulls. Infotaxis, gas-plus-wind localization, transient processing for slow metal-oxide sensors, reactive off-zigzag search, geometry-aware modular policies, and particle-filter source belief have all been demonstrated in scoped settings (C-1182–C-1187). Compact temporal-memory reinforcement learning is only `plausible` for the studied simulated plumes until unchanged embodied transfer is shown (C-1188).

Sparse piriform activity is not a hardware or energy conclusion

Piriform odor responses can be sparse and distributed rather than neatly topographic; in one rat task, burst-count population information outperformed some precise-pattern accounts (C-1189–C-1191). The degree of sparsity is `disputed` as a universal characterization because it changes with concentration and protocol (C-1192). Longitudinal piriform ensembles can also drift while behavior remains stable (C-1193).

These observations motivate tests for sparse routing, event memory, and remappable readouts. They do not establish fixed semantic neuron addresses, low memory traffic, low organism energy, or a benefit over pruning, compression, low precision, sparse convolution, or dense execution on suitable hardware.

Association, valence, identity, and hazard are separable

Rapid reward-category coding appeared in olfactory tubercle within minutes in one task, while posterior piriform lacked the same explicit code even after overtraining. Arbitrary piriform ensembles could acquire opposite valence under different reinforcement, and innate aversion could be disrupted while learned detection or avoidance remained (C-1194–C-1196). Cortical-amygdala pathways causally contributed to scoped innate odor behavior (C-1197).

The architectural record must therefore preserve odor evidence, reinforcement, context, action, feedback, acquisition time, retention, transfer, reversal, and readout/module identity. Chemical identity, odor category, innate choice, learned choice, pleasantness, irritation, toxicity, external exposure, and hazard remain separately scored.

Mixtures and instruments expose the same ambiguity problem

Animals can learn a target in variable mixtures, but performance degrades with background count and overlap; chemically similar maskers can raise detection thresholds more in the tested regime (C-1198–C-1199). Cross-reactive artificial arrays are already an established sensing baseline, not a novel consequence of receptor analogy (C-1200).

Multi-year metal-oxide sensor data demonstrate drift. Humidity, stability, selectivity, and poisoning belong to the operator state, and validated analytical/safety workflows preserve sampling, calibration, recovery, identification support, exposure units, and uncertainty (C-1201–C-1203). A high closed-panel classifier score cannot turn an unsupported mixture into an identified chemical, a library match into source attribution, or odor detection into safety.

Proposed AI translation

Preserve the whole chemical episode

For episode e, preserve

$$\mathcal{C}_e = (S_e, X_e, A_e, R_e, O_e, K_e, H_e, T_e, U_e, B_e),$$

where:

- S_e records source identity, mixture, release in moles per second, temperature in kelvins, geometry in metres, phase, and motion in metres per second;
- X_e records domain, boundaries, surfaces, airflow in metres per second, pressure in pascals, relative humidity as a dimensionless fraction, temperature, turbulence, reaction, sorption, and chemical background;
- A_e records commanded and realized motion, orientation, sniff/pump flow in cubic metres per second, heater power in watts, valve, preconcentration, purge, query, confirmation, stopping, and abstention;
- R_e records receiver/body, bilateral or array geometry, pose, inlet, tubing, chamber, pump, heater, sensor, saturation, health, and feasible authority;
- O_e is the versioned observation operator: response/recovery, cross- sensitivity, nonlinearity, hysteresis, support, clock, latency, quantization, preprocessing, missingness, and selection;
- K_e records reference-gas composition and uncertainty, blanks, zero/span, flow, device and batch, compensation, age, drift, poisoning, maintenance, calibration validity, and traceability;
- H_e records prior exposure, adaptation, habituation, contamination, cleaning, training, reinforcement, feedback, previous actions, and readout remapping with timestamps;
- T_e declares the literal target, deadline in seconds, loss/utility, abstention policy, exposure rule, and safety constraint;
- U_e names the independent unit: sample, stock, injection, device, batch, day, source, plume realization, site, body, animal/subject, or model seed; and
- B_e is the componentwise ceiling in evidence, labels, standards, channels, actions, metres, seconds, bytes, searches, person-hours, joules, consumables, exposure, replacements, embodied devices, and opportunity.

This is the chemical instantiation of the versioned observation contract. The endogenous-observation candidate owns the coupling between acquisition action and future evidence; the latency-qualified authority envelope owns the action restriction when evidence is slow, stale, saturated, miscalibrated, or poisoned.

Model transport before interpreting the sensor

For analyte i, a minimum transport model is

$$\frac{\partial c_i}{\partial t} + \mathbf{u}\cdot\nabla c_i = \nabla\cdot(D_i\nabla c_i) + R_i(\mathbf{c}, T, P, H_r, \mathbf{x}, t) + q_i(\mathbf{x}, t),$$

where amount concentration c_i is in moles per cubic metre, position $\mathbf{x}$ is in metres, time t is in seconds, velocity $\mathbf{u}$ is in metres per second, diffusivity or declared effective dispersion D_i is in square metres per second, relative humidity H_r is dimensionless, and reaction/loss/phase-transfer R_i and volumetric source q_i are in moles per cubic metre per second. Every term has units of moles per cubic metre per second. Boundaries, buoyancy, droplets, thermal stratification, deposition, and unresolved turbulent fluxes remain explicit when they affect the task.

For channel m sampled at t_n, the measured trace is

$$y_{m,n} = g_{m,v}\left(\sum_i \int_0^\infty h_{m,i,v}(\tau;\mathbf{z}_n)c_i(\mathbf{x}_r(t_n-\tau), t_n-\tau)\,d\tau, \mathbf{z}_n\right) + \epsilon_{m,n},$$

where channel output $y_{m,n}$ and error $\epsilon_{m,n}$ use the calibrated sensor unit, causal response kernel $h_{m,i,v}$ is in reciprocal seconds, delay τ is in seconds, receiver path $\mathbf{x}_r$ is in metres, version v is dimensionless, and $\mathbf{z}_n$ contains flow, heater, temperature, humidity, interferents, adaptation, age, drift, saturation, and poisoning. A static feature vector is permitted only after this dynamic operator has been tested or shown irrelevant inside the declared support.

Represent non-identifiability instead of forcing a label

Let G_v be the calibrated mixture-to-sensor forward operator and $\mathcal{S}_c$ the supported set of nonnegative composition vectors in moles per cubic metre. For observation $\mathbf{y}$, retain

$$\mathcal{N}_v(\mathbf{y}) = \left\{\mathbf{c} \in \mathcal{S}_c : \|\mathbf{y} - G_v(\mathbf{c};\mathbf{z})\|_{\Sigma_y^{-1}} \le \varepsilon_y\right\},$$

where error covariance Σ_y is in squared sensor-output units, the Mahalanobis norm and tolerance ε_y are dimensionless, and $\mathcal{N}_v$ is the observation-equivalent composition set. If materially different identities, concentrations, exposures, or hazard states remain in that set, the result is ambiguous. The system can acquire another measurement, request analytical confirmation, retain alternatives, or abstain; it cannot convert a prior-selected label into new chemical evidence.

This state links to reset-coupled staged verification: an inexpensive array may screen, but escalation to GC--MS, PTR/SIFT--MS, IMS/FAIMS, or another qualified method must add conditionally useful evidence after sampling, standards, turnaround, analyst time, consumables, exposure, and energy are charged. The analytical method is itself an operator with blanks, recovery, retention, deconvolution, library support, calibration, and uncertainty—not an oracle.

Keep a literal outcome firewall

OUTPUT	NATIVE MEASUREMENT	MUST REMAIN SEPARATE FROM
presence	hits, false alarms, d', criterion, matrix and concentration	identity or recognition accuracy
chemical identity	confusion/unknown set, standards, retention/spectral support and calibrated probability	odor name, valence or source
concentration	`mol/mol`, `mol/m^3`, or `kg/m^3`, temperature, pressure, bias/error and support	raw sensor output or perceived intensity
mixture	component identity/concentration, recovery, censoring and non-identifiable set	dominant label
direction/position	angular error; position error and coverage in metres; source-off false declarations	contact or instantaneous gradient
source attribution	competing emitters, transport evidence, association and posterior calibration	chemical identity or location alone
association/valence	learning curve, context, reinforcement, retention, transfer, reversal; innate and learned choices separately	identity, toxicity or hazard
exposure/dose	external concentration-time in `kg s/m^3`; absorbed dose only with dosimetry	detection or sampling duty cycle
hazard/safety	chemical-, route-, population-, endpoint- and averaging-time-specific rule	odor threshold, intensity, preference or aversion
efficiency	protected outcomes plus evidence, time, bytes, actions, person-hours, consumables, exposure and lifecycle joules	event count, inference power or organism metabolism

Use action to change observability, not to obtain a free second dataset

At decision time t, choose

$$a_t = \pi_q(\mathcal{H}_t, \widehat{\mathbf{c}}_t, \widehat{\mathbf{s}}_t, \widehat{O}_t, \widehat{U}_t, \mathcal{A}_t^{\text{safe}}, \mathbf{B}_t),$$

where $\mathcal{H}_t$ is causally received observation/action history, $\widehat{\mathbf{c}}_t$ is concentration/mixture belief in moles per cubic metre, $\widehat{\mathbf{s}}_t$ is source belief with position in metres and release in moles per second, $\widehat{O}_t$ is operator/condition state, $\widehat{U}_t$ is uncertainty, $\mathcal{A}_t^{\text{safe}}$ is the independently constrained action set, and $\mathbf{B}_t$ is remaining budget in its component units.

The action may move or orient the body, change bilateral spacing, sniff or pump, change heater or valve state, purge, resample, request confirmation, stop, or abstain. Its causal value must be tested against fixed, random, replayed, and dose-matched acquisition. Equal wall time is insufficient if one method inhales or pumps more material, experiences more whiffs, travels farther, uses more energy, or accepts more exposure.

For plume search, the system retains a joint belief

$$p(\mathbf{s}, \mathbf{c}_{0:t}, O_t \mid y_{1:t}, a_{1:t}, \mathcal{C}_e),$$

not one gradient arrow. Surge--cast/off-zigzag rules, wind-only anemotaxis, particle-filter belief control, infotaxis, finite-state search, POMDP/MPC/value of information, and matched-memory reinforcement learning remain mandatory nulls. Success, false source declarations, location error, posterior coverage, path in metres, time in seconds, collisions, exposure, and joules are reported separately.

Maintain fast response and slow condition as different states

Use at least two state transitions:

$$\mathbf{r}_{n+1} = f_r(\mathbf{r}_n, \mathbf{c}_n, a_n) + \boldsymbol{\xi}_n, \qquad \mathbf{d}_{e+1} = f_d(\mathbf{d}_e, \mathcal{E}_e, m_e) + \boldsymbol{\omega}_e,$$

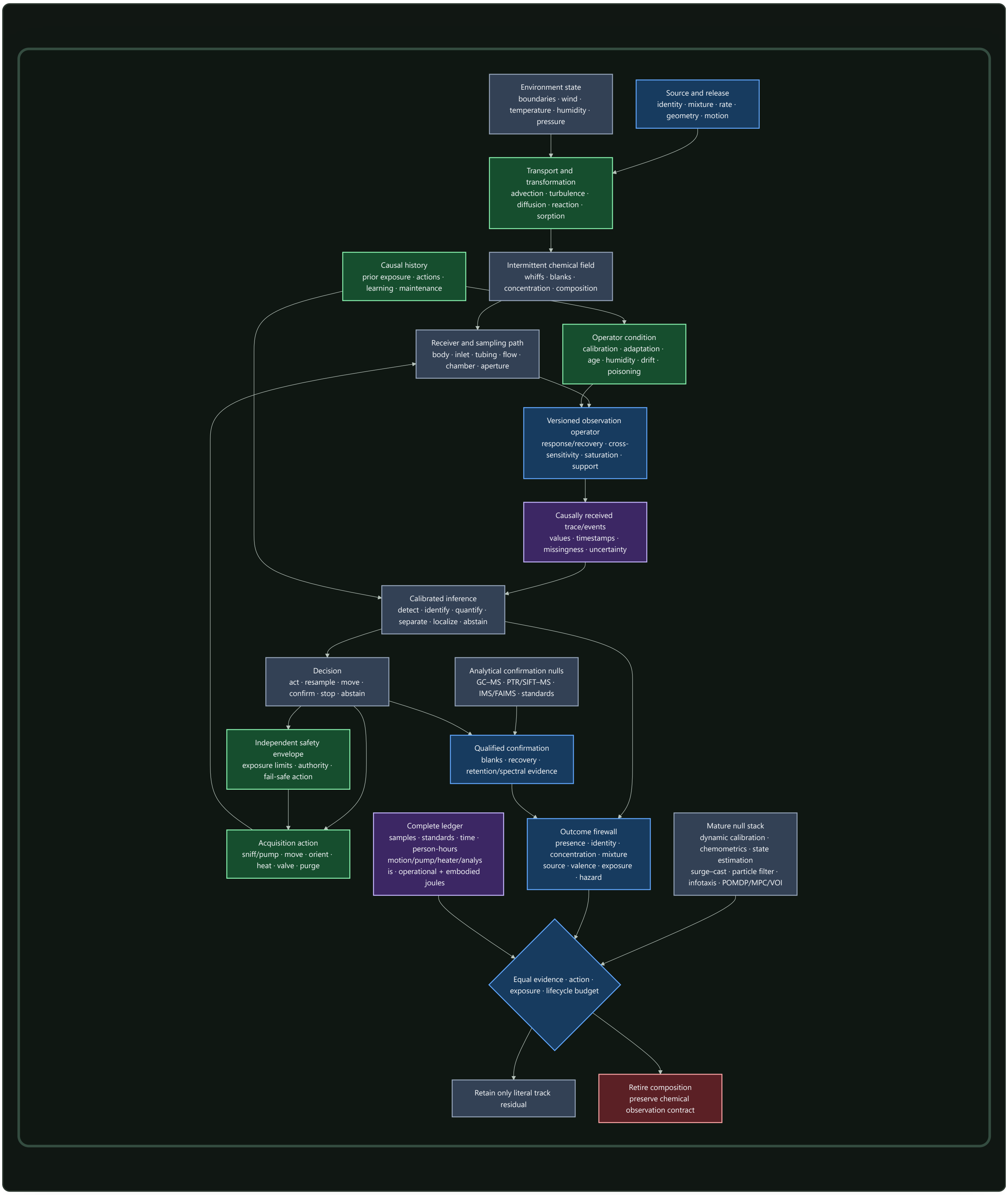
where within-episode state $\mathbf{r}_n$ includes response, adaptation, heater, and recovery; between-episode state $\mathbf{d}_e$ includes calibration, baseline/gain drift, contamination, ageing, and poisoning; concentration $\mathbf{c}_n$ is in moles per cubic metre; cumulative stress/exposure $\mathcal{E}_e$ retains its physical units; and maintenance action m_e records purge, cleaning, recalibration, repair, or replacement. A return to baseline does not prove restored selectivity or calibration. A task residual cannot by itself distinguish environmental change from device change.

The graded assurance envelope binds calibration and condition evidence to the exact operator version. The reversible physical-skill candidate receives credit for coatings, inlets, chambers, filters, heaters, or other physical transforms only after cross-sensitivity, reset, poisoning, replacement, fallback, and fabrication burden are measured.

Preserve evidence for recalibration and future interpretation

Raw traces, calibration/operator history, standards, sample lineage, analytical evidence, and retained physical samples have different reconstruction value. Contract-preserving compaction may replace them only for registered future queries; value- and reconstructability-aware tiering must survive hidden recalibration, changed-library, changed-exposure-rule, and poisoning-investigation queries without future-label leakage.

One closed sensing-and-action contract



Editable source: operator-qualified-active-chemical-sensing.mmd.

Efficiency mechanism

The architecture permits six distinct efficiency mechanisms. Each remains a hypothesis until it improves a protected outcome under F-011's matched budget.

MECHANISM	POSSIBLE SAVING	REQUIRED ACCOUNTING	IMMEDIATE RETIREMENT CONDITION
selective acquisition	avoid samples, motion, pumping, heating, or confirmation that cannot change the decision	sampled volume/mass, whiffs, actions, path, latency, exposure, wear and joules	fixed, random, replayed, or dose-matched acquisition reaches the same frontier
transient/event processing	act on causal onsets, offsets, whiffs and blanks without waiting for slow steady state	physical bandwidth, missed sustained signals, false events, bytes, memory traffic, decoder work and total energy	dynamic deconvolution, derivatives, matched filters, or a finite-state history matches it
cross-reactive population coverage	reuse partially selective channels across chemicals and mixtures	sensor chemistry/area, response support, calibration standards, interferents, unknowns, saturation and replacement	gain follows coverage, SNR, sampled material, or labels rather than the decoder
calibrated normalization and multiscale state	stabilize some identity information while retaining concentration, change, and device condition	absolute-signal error, rare targets, state updates, recurrence, calibration, latency and energy	robust scaling or explicit gain/state estimation matches it, or safety information is erased
staged analytical escalation	use a low-cost screen for easy cases and buy stronger separation/identification only when valuable	aliquots, standards, blanks, turnaround, analyst work, carrier gas, sorbents/columns, vacuum/ionization, exposure and joules	always-confirm, never-confirm, sequential tests, or an ordinary calibrated cascade matches it
qualified compaction and maintenance	retain only evidence needed for registered future queries; recalibrate, clean, remap, or replace only when justified	raw/sample retention, update writes, future-query loss, downtime, labels, maintenance, replacement, people and embodied burden	future recalibration, changed-library, exposure-rule, or poisoning queries cannot be reconstructed

Sparse or event-driven computation is not a seventh saving until its total physical cost is lower. A top-k or thresholded representation can reduce arithmetic while adding normalization, sorting, indices, irregular memory traffic, routing, remapping, idle hardware, missed-event risk, and maintenance. The relevant numerator is accepted task service, not active-unit count.

Lifecycle energy for method q over one accepted service interval is

$$E_q^{\mathrm{life}} = E_q^{\mathrm{data}} + E_q^{\mathrm{train}} + E_q^{\mathrm{move}} + E_q^{\mathrm{pump}} + E_q^{\mathrm{heat}} + E_q^{\mathrm{sense}} + E_q^{\mathrm{separate}} + E_q^{\mathrm{ionize}} + E_q^{\mathrm{infer}} + E_q^{\mathrm{comm}} + E_q^{\mathrm{store}} + E_q^{\mathrm{cal}} + E_q^{\mathrm{maint}} + E_q^{\mathrm{facility}} + E_q^{\mathrm{emb}},$$

where every term is energy in joules and covers data acquisition, training, receiver motion, pumping, heating, sensing, analytical separation, ionization/vacuum, inference, communication, storage, calibration, maintenance, facility overhead, and amortized embodied hardware. Carrier and calibration gases, sorbents, columns, dopants, filters, cleaning agents, samples, emissions, and disposal remain additionally reported in their native physical or lifecycle units.

Human work is

$$H_q^{\mathrm{human}} = H_q^{\mathrm{design}} + H_q^{\mathrm{sample}} + H_q^{\mathrm{label}} + H_q^{\mathrm{cal}} + H_q^{\mathrm{analyze}} + H_q^{\mathrm{tune}} + H_q^{\mathrm{safety}} + H_q^{\mathrm{monitor}} + H_q^{\mathrm{maint}},$$

where every term is in person-hours and roles are separated. An apparent energy gain is rejected if it moves work into sample preparation, calibration, chemical analysis, safety review, data curation, cleaning, or repair without counting it.

Evidence status

EVIDENCE BUNDLE	STABLE CLAIMS	STATUS	ARCHITECTURAL USE AND BOUNDARY
receptor family, combinatorial responses, concentration dependence, taste counterexample	C-1152–C-1155	4 established	justify population-coverage and dedicated-channel comparisons; no universal chemical code
normalization, bulb/piriform concentration transforms, intensity/change and sniff-phase state	C-1156–C-1162	7 established	test joint concentration--identity and explicit gain-state mechanisms; never erase safety-relevant level by default
receptor adaptation, transduction range and habituation	C-1163–C-1166	4 established	require separate response, adaptation, habituation, recovery and slow-condition state

EVIDENCE BUNDLE	STABLE CLAIMS	STATUS	ARCHITECTURAL USE AND BOUNDARY
active sniffing, response timing, serial and bilateral acquisition	C-1167–C-1173	7 established	justify causal sampling/body-action tests inside measured end-to-end bandwidth; no universal stereo/serial rule
plume intermittency, measurement bandwidth, environment and weak directional summaries	C-1174–C-1177	4 established	require measured transport, whiff/blank statistics and temporal controls rather than smooth-gradient assumptions
animal and robotic plume navigation, reactive/belief/search nulls	C-1178–C-1187	10 established	establish a regime-dependent policy library and strong robotics null stack; behavior is not optimality proof
compact temporal-memory RL in simulated plumes	C-1188	1 plausible	eligible only as a frozen simulation-to-embodiment hypothesis
sparse/distributed piriform codes and concentration-dependent sparsity	C-1189–C-1192	3 established; C-1192 disputed	motivate causal sparse/readout tests; no fixed sparseness, hardware, or energy conclusion
representational drift, rapid value learning, flexible and innate valence pathways	C-1193–C-1197	5 established	require remapping cost and separate association, region/readout, innate, learned and hazard outcomes
mixture foreground, masking, arrays, drift, condition and analytical/safety boundaries	C-1198–C-1203	6 established	require unknown/mixture tests, future-device splits, calibration/poisoning state, qualified analytical confirmation and exposure rules

The totals are exactly 50 established, one plausible, and one disputed claim. The established status applies only to the cited biological preparation, behavior, instrument, dataset, method, or authoritative standard. It does not establish that the project composition improves an engineering frontier.

The complete mature null is a composition, not a token baseline:

1. traceable sampling, standards, blanks, duplicates, recovery, flow and calibration;
2. GC--MS/GC--FID/PID and GC×GC where justified, retention indices, authentic standards, PTR/SIFT--MS, IMS/FAIMS, electrochemical/PID, and targeted spectroscopy under their support;
3. dynamic system identification, response/recovery modelling, deconvolution, filtering, robust scaling and temperature/humidity compensation;
4. PCA/PLS, LDA/QDA, calibrated regression/classification, SVMs, trees, ensembles, neural models, open-set detection, conformal/selective prediction, mixture models, domain adaptation and abstention;
5. measured/validated flow models, Kalman/particle filtering, Gaussian-process plume inference, observability and posterior calibration;
6. correlated random walk, gradient and wind baselines, surge--cast, off-zigzag, infotaxis, particle-belief control, finite-state search, POMDP/dual control/MPC/value of information and matched-memory RL; and
7. detector health, poisoning/out-of-support alarms, staged verification, exposure constraints, independent authority, fallback, maintenance, sample lineage, human work, consumables, and lifecycle accounting.

F-011 compares against that full stack. A weak static classifier, uncalibrated e-nose, single gradient controller, or instrument name is not the baseline.

Speculative extensions

The following are experiment-generating compositions. None is promoted by this chapter.

Action-conditioned identifiability

Use the current observation-equivalent set $\mathcal{N}_v(\mathbf{y})$ to select the cheapest safe action expected to separate decision-relevant alternatives. The action might change path, wind-relative orientation, bilateral geometry, flow, heater state, temporal support, or analytical method. This joins Candidate 007 with Candidate 014. It survives only if explicit Bayesian design, value of information, POMDP/dual control, and ordinary staged testing cannot reach the same calibrated decision frontier.

Dual identity--concentration state

Maintain shared evidence with separately protected readouts for chemical/odor identity, absolute concentration, concentration change, and operator condition. Normalization or recurrence may stabilize the identity readout while the other paths preserve dose and condition. A useful implementation must beat concentration-conditioned generative models and explicit gain-state estimators; it is rejected when one task improves by destroying another.

Remappable sparse population memory

Treat sensor/receptor channels and sparse learned units as replaceable evidence contributors rather than permanent semantic addresses. A readout-maintenance layer would detect drift, remap channels, preserve uncertainty, and request labels or calibration selectively. It is worth retaining only if future-time performance improves after update writes, labels, monitoring, downtime, memory traffic, replacement, and energy are charged. Standard recalibration, domain adaptation, ensemble remapping, pruning, compression, and dense low-precision execution remain the nulls.

Operator-matched event front end

Co-design inlet, chamber, sensor physics, heater/pump action, deconvolution, and event thresholds so the retained trace preserves task-bearing whiff/blank and transient information at lower traffic. This is a scoped extension of Candidate 006, not a claim that physical or event-driven sensing is intrinsically efficient. It must transfer across hardware, humidity, drift and plume timescale and must beat a calibrated dynamic model on the same device.

Qualified screen--confirm--retain loop

Compose an inexpensive cross-reactive screen, calibrated abstention, conditional analytical confirmation, and query-aware retention. The screen can provisionally act only inside the latency-qualified authority envelope; Candidate 010 owns escalation; Candidates 017 and 018 own retained evidence. The composition is rejected if an ordinary calibrated cascade or always-confirm policy matches protected risk, latency, and total cost.

Regime-switching plume search

Use calibrated transport, sensor-condition and source beliefs to switch among reactive ON/OFF behavior, wind-relative movement, local search, belief-driven exploration, confirmation, and safe withdrawal. The controller must expose the regime evidence that authorized the switch and must abstain when operator or wind evidence is invalid. It is rejected if one finite-state controller, particle-belief policy, infotaxis, POMDP/MPC, or matched-memory learner reaches the same held-out source-search frontier.

Failure modes

Physics and operator failures

1. **Smooth-gradient fiction:** instantaneous concentration is treated as a stable source direction despite intermittent transport.
2. **Static-vector fiction:** inlet, chamber, response, recovery, hysteresis, saturation, humidity, and prior exposure are discarded before inference.
3. **Bandwidth invention:** millisecond or event information is claimed after the physical transport or sensor has filtered it away.
4. **Mixture over-identification:** a single label is emitted while materially different compositions, concentrations, exposures, or hazards remain observation-equivalent.

5. **Conversion error:** parts per million are converted to mass concentration without molar mass, temperature, pressure, and fraction definition.
6. **Transport/operator confounding:** policy performance is credited to the learner although one arm received better wind, likelihood, sensor dynamics, field truth, calibration, or source prior.

Representation and learning failures

7. **Label-as-mechanism:** “receptor-like,” “bulb,” “piriform,” “sparse,” “temporal,” or “neuromorphic” replaces a causal ablation and literal endpoint.
8. **Coverage-as-decoder gain:** more sensor chemistry, area, concentration, standards, or SNR is attributed to architecture.
9. **Concentration erasure:** identity appears invariant because the system discarded information required for leak, exposure, or safety decisions.
10. **Stable-address assumption:** drifting or replaced sensor/representation units retain fixed semantic addresses without remapping cost.
11. **Event-count efficiency:** fewer active events are reported without bytes, memory traffic, routing, decoding, idle hardware, missed hazards, and lifecycle joules.
12. **Association collapse:** endpoint accuracy substitutes for acquisition curve, reinforcement/context, retention, transfer, reversal, and selective representation/readout intervention.
13. **Valence collapse:** innate choice, learned choice, pleasantness, irritation, toxicity, exposure, and hazard are merged into one score.

Evaluation, reliability, and safety failures

14. **Temporal/batch leakage:** random rows or adjacent windows share stock, dilution, sample, plume seed, sensor, device, batch, calibration, day, site, subject, or analytical run across splits.
15. **Simulation inverse crime:** training and evaluation share CFD mesh, response kernel, source schedule, random seed, or post-test retuning.
16. **No-source omission:** every episode contains a source, so unconditional declarations look successful and false reassurance stays invisible.
17. **Compensation without condition detection:** expected drift is corrected while humidity, contamination, poisoning, replacement, or out-of-support state remains undetected.
18. **Instrument-as-oracle:** GC--MS or another method is credited without sample lineage, blanks, recovery, breakthrough/carryover, separation, retention/spectral support, standards, library scope, and uncertainty.
19. **Odor-as-safety:** detection threshold, intensity, preference, or aversion substitutes for a chemical-, route-, population-, endpoint-, and averaging- time-specific exposure rule.
20. **Self-certified authority:** the same uncertain sensor/inference path defines its own safety envelope and fallback.
21. **Free active sensing:** an adaptive method samples more material, sees more whiffs, moves farther, waits longer, consumes more pump/heater energy, or accepts more exposure than its baseline.
22. **Incomplete lifecycle boundary:** analytical preparation, standards, calibration gases, consumables, motion, pumps, heaters, chromatography, vacuum/ionization, facility power, cleaning, replacement, human work, embodied devices, emissions, and disposal disappear from the ledger.

Any failure that creates the reported advantage retires the architectural claim for that track. The operator record can remain useful even when the proposed mechanism does not.

Measurable predictions

Fixture F-011 implements these predictions with frozen splits, common operator/action budgets, causal ablations, source-off trials, prospective device/time tests, and full resource accounting.

TRACK	TESTABLE PREDICTION	STRONGEST DECISIVE COMPARISON	RETIRE WHEN
T1 concentration--identity	a shared qualified state improves identity across held-out concentration while retaining calibrated absolute concentration	raw/dynamic calibrated models, concentration-conditioned generative inference, divisive normalization and recurrence	identity gain disappears when concentration is protected or relies on seen concentration/matrix
T2 coverage versus architecture	the proposed decoder extracts more task value from an equal channel basis, area, bandwidth and SNR	dense/sparse linear, kernel, tree, Bayesian and neural decoders on matched arrays	gain follows broader chemistry, more sampled material, labels or SNR
T3 normalization	state-qualified normalization improves the joint identity--concentration--rare-target--calibration frontier under saturation and interferents	no normalization, robust scaling, explicit gain-state estimation, divisive and recurrent alternatives	a conventional method matches, or absolute/safety information worsens
T4 temporal code	causal temporal order adds held-out chemical/plume information beyond the measured response operator	instantaneous/derivative features, matched filters, state-space deconvolution and event models	advantage vanishes under held-out inlet/sensor operators or marginal-preserving shuffle
T5 adaptive sniff/pump	closed-loop acquisition improves literal decision value per sampled material, exposure, time and joule	fixed-rate, random, replayed and dose-matched schedules with VOI/POMDP control	it receives more dose/opportunity or fixed acquisition reaches the frontier
T6 bilateral/serial/wind	cue value changes predictably with range, plume regularity, body spacing and bandwidth	instantaneous bilateral gradient, lag correlation, unilateral history, wind-only and calibrated fusion	one cue is claimed universally or gain fails held-out bodies/regimes
T7 plume statistics	whiff/blank history contains source-bearing information not captured by simpler causal summaries	mean, peak, slope, duration, frequency, time-since-hit, bilateral lag and full history at equal window	a simpler statistic matches, or prediction uses downstream distance/leakage rather than source evidence
T8 source search	the composition improves success, false declaration, calibrated position, path, time, risk, exposure and energy jointly	random walk, gradient, wind, surge--cast/off-zigzag, infotaxis, particle belief, finite-state, POMDP/MPC and matched-memory RL	B10 or any simpler policy matches on held-out sources/plumes with failures included
T9 embodiment transfer	a frozen simulated policy retains value under measured tubing, sensor, humidity, drift, saturation and action latency	finite-state, system-identified belief control, domain-randomized and recurrent/RL baselines	gain requires post-test retuning or disappears under the physical operator
T10 sparse total cost	sparse/event representation lowers complete accepted-service cost without losing rare, sustained, calibration or hazard signals	dense low precision, pruning, compression, top-k, threshold events, sparse convolution and indexed retrieval	only active count/FLOPs fall, or total bytes/joules and protected task do not improve
T11 drift/readout maintenance	condition-aware remapping improves prospective future-device performance at lower full maintenance cost	frozen readout, scheduled recalibration, state estimation, orthogonal correction, domain adaptation, ensembles and replacement	future labels leak, or conventional maintenance reaches the frontier
T12 mixture/masking	the system recovers or correctly abstains on unseen compositions while preserving target detection and component concentration	calibrated multivariate, nonnegative/generative mixture, open-set and selective-prediction baselines	closed schedules or dominant labels create the score, or non-identifiability is hidden
T13 humidity/poisoning	the system distinguishes reversible condition, drift, contamination, poisoning, replacement and unknown input early enough for safe degradation	no correction, dynamic calibration, supervised/unsupervised adaptation, condition diagnostics, redundancy and fallback	correction works only on expected drift or cannot constrain unsafe action prospectively
T14 tiered analysis	screen--abstain--confirm reduces protected risk/latency/cost across knowns, unknowns, mixtures, blanks and exposure boundaries	always-confirm, never-confirm, sequential probability tests, calibrated cascades and VOI escalation	false reassurance exceeds its ceiling or analytical/human/consumable/lifecycle cost removes the gain

For every track, report literal outcomes, calibrated uncertainty, independent unit, failures, abstentions, unused budget, and the complete cost vector. A residual must replicate across at least two target chemical families, interferent/matrix families, concentration and source/plume regimes, sensor chemistries, manufacture batches, operator/calibration versions, future times, sites, model families, and hardware classes. Active tracks additionally require unseen source positions, plume seeds, bodies, action limits, and paired counterfactual seeds.

If the complete mature stack matches the composition, if no selective ablation isolates value, or if the gain disappears after calibration, analytical work, exposure, maintenance, human effort, consumables, facility and lifecycle energy are charged, retire the architectural residual. Keep the chemical observation contract and the negative result; create no new principle or candidate.

Reliability under mission profiles

Scope

An AI system is not deployed on a nominal device. It is deployed on a physical population: lots, wafers, dies, blocks, packages, boards, power paths, cooling paths, memories, interconnects, converters, sensors, and controllers that begin different and continue to change. Reliability is therefore a relation among a required function, an acceptance criterion, a physical unit, its operating history, its environment, and a time horizon—not a scalar property of a hardware name (C-1002).

This chapter defines the architecture required to preserve accepted AI service under that reality. It separates:

1. time-zero variation;
2. reversible drift;
3. cumulative irreversible degradation;
4. abrupt permanent failure;
5. transient upset;
6. correction or compensation applied by the system; and
7. measurement, calibration, model-support, and classification error.

Those states are not interchangeable. Restored task score after recalibration does not prove physical recovery. A transient error does not establish wear. A monitor alarm does not identify a mechanism. A passed qualification does not cover every future workload or environment (C-1003, C-1022, C-1030).

The chapter joins four maintained artifacts:

1. the semiconductor device and circuit reliability audit, which establishes the evidence boundary;
2. the mission-profile-qualified reliability mathematics, which defines the observation, damage, service, and lifecycle ledgers;
3. Fixture F-008, which tests ten hostile experiment tracks against the complete conventional reliability stack; and
4. the editable degradation-and-recovery diagram.

The result composes existing candidates. It does not allocate a new principle or candidate (C-1053). Its purpose is narrower and more useful: make physical reliability a versioned runtime and lifecycle contract rather than an assumption hidden below the model.

Biological observation

Living systems maintain useful function while their physical components differ, accumulate stress, repair imperfectly, and turn over at different rates. They do not depend on every component remaining identical to an original specification. They combine local maintenance, selective replacement, redundancy, reserve, containment, feedback, and changes in activity. At larger scales, function can persist even while particular molecules, organelles, cells, or tissue regions are repaired, removed, replaced, or assigned less demanding work.

The useful abstraction is not a particular organ or biochemical mechanism. It is a substrate-independent maintenance contract:

- components have individual histories and unequal remaining capability;
- current output does not reveal all accumulated damage;
- stress and repair occur at several timescales;
- local defects are tolerated only while they remain contained;
- surveillance consumes resources and can itself fail;
- recovery may restore function without restoring the original substrate;
- reserve is finite; and
- replacement is part of operation, not evidence that degradation never occurred.

The translation must stop at that contract. Semiconductor bias-temperature instability, electromigration, radiation upset, solder fatigue, and memory-cell endurance do not become biological processes because they share words such as stress, repair, or recovery. Their causal models, units, evidence, and failure criteria remain physical-domain specific. The analogy contributes questions— what is damaged, what is observed, what can recover, what must be replaced, and what function remains acceptable—not answers.

This boundary is especially important because equal present performance can hide unequal remaining life. Some electrical effects partly recover after rest, while interconnect voids, consumed endurance, package fatigue, and other damage persist (C-1014, C-1017, C-1019, C-1022). The architectural lesson is to carry native margin, reversible state, irreversible damage, compensation, and reserve separately instead of naming every return of output quality "recovery."

Proposed AI translation

Make the physical population part of system identity

Every deployed unit receives an immutable identity chain for fabrication lot, wafer, wafer coordinates, die, block or array, package, board, power and clock path, cooling path, tester, calibration, hardware, firmware, compiler, model, repair map, and site. Replacement, remapping, recalibration, repair, or firmware change adds a versioned transition; it does not overwrite history.

The population is the denominator. Production yield, accepted yield, early-life failure, field reliability, and wear-out answer different questions (C-1007). Critical area, clustered defects, layout, repair, and edge structure affect yield (C-1008). Spare rows, blocks, experts, or tiles can raise accepted yield only when diagnosis and defect geometry permit repair, and they consume area, routing, test, latency, and residual common-mode reserve (C-1011).

For fabricated unit i and jointly required criterion q, let $A_{i,q} = 1$ when the unit passes that criterion and $A_{i,q} = 0$ otherwise. Joint accepted yield is

$$\widehat{Y}_{\mathrm{joint}} = \frac{1}{N_{\mathrm{fab}}} \sum_{i=1}^{N_{\mathrm{fab}}} \prod_{q=1}^{Q} A_{i,q},$$

where N_{fab} is the number of fabricated units [unit], Q is the number of required criteria [criterion], and $\widehat{Y}_{\mathrm{joint}}$ is dimensionless. Dead-on-arrival, untestable, unpackageable, screened-out, and discarded devices remain in N_{fab}. High marginal pass rates do not imply a high joint pass rate when timing, power, noise, memory, and analog constraints are correlated (C-1010).

Population evaluation groups data by lot, wafer, die, region, device, site, and future time. Randomly splitting transactions can leak shared fabrication and aging history (C-1013). Per-device control is credible only after it transfers across those grouped holdouts.

Carry typed physical state

For physical unit i at time t, maintain

$$z_i(t) = (\theta_i, D_i^{\mathrm{perm}}(t), R_i^{\mathrm{rev}}(t), W_i(t), \mathcal{F}_i(t), Q_i^{\mathrm{res}}(t)),$$

where:

- θ_i is time-zero physical state in declared native units;
- D_i^{perm} is cumulative irreversible damage [declared damage unit];
- R_i^{rev} is reversible degradation [same declared damage unit];
- W_i is consumed endurance [cycle, write, or normalized wear unit];
- $\mathcal{F}_i$ is abrupt or latent fault state [state]; and
- Q_i^{res} is remaining correction, timing, thermal, repair, and spare reserve [declared reserve unit].

Keep the evidence record separate:

$$O_i(t) = \left(o_i, v_i, \Sigma_i^{\mathrm{cal}}, m_i, c_i^L, c_i^U, a_i^{\mathrm{ev}}, \mathcal{V}_i\right),$$

where o_i is observed telemetry in sensor-native units, v_i is calibration and instrument version [version], Σ_i^{cal} is calibration covariance in squared native units, m_i is observation-availability mask [dimensionless], c_i^L and c_i^U are censoring bounds in native units, a_i^{ev} is evidence age [s], and $\mathcal{V}_i$ is the calibrated validity envelope.

This is the physical-device specialization of the versioned observation contract: telemetry may be compacted, but calibration, missingness, censoring, mission history, and version validity must remain available to the decisions they support.

This separation prevents five common category errors:

OBSERVATION	WHAT IT DOES NOT PROVE
a device began in a slow or leaky tail	that it has aged
a parameter moved and later returned	that irreversible damage was absent
a task output recovered after calibration	that native physical margin recovered
a fault disappeared after retry	that it was harmless or non-recurring
a monitor stayed quiet	that the protected path stayed inside margin

Time-zero mismatch and systematic gradients do not reduce to one global offset (C-1009). Device-specific trap activity can also produce stochastic tails that a deterministic shift misses; that stronger claim remains plausible rather than established (C-1012). Tester accuracy, calibration, sampling, guard bands, and the conformity rule determine false acceptance and rejection near limits (C-1006).

Drive degradation models from the actual mission profile

For episode e, record the measured mission profile

$$M_e = \left\{u_n, V_n, f_n, T_n, J_n, a_n, \phi_n, \dot{D}_n^{\text{ion}}, c_n, r_n, \Delta t_n\right\}_{n=0}^{N_e-1},$$

where u_n is workload class [class], V_n measured voltage [V], f_n operation or clock rate [Hz], T_n absolute temperature [K], J_n current density [A m^{-2}], a_n activity [dimensionless], ϕ_n particle flux [particle m^{-2} s^{-1}], $\dot{D}_n^{\text{ion}}$ ionizing-dose rate [Gy s^{-1}], c_n cooling state [state], r_n route and protection state [state], Δt_n interval duration [s], and N_e interval count [interval]. Commanded voltage, nominal temperature, calendar age, and mean utilization remain useful metadata but cannot replace the measured history.

The history matters because electric field and switching affect interface and oxide damage (C-1015); current density, waveform, geometry, material, and temperature affect electromigration (C-1017); local power changes temperature, which feeds back into delay, leakage, material transport, and degradation (C-1018); and thermal cycles can damage packages even when average temperature is similar (C-1019).

For physical mechanism k, a mission-qualified damage proxy is

$$D_k(t) = \int_0^t r_k(V(\tau), T(\tau), J(\tau), a(\tau), \phi(\tau), u(\tau))\, \mathrm{d}\tau,$$

where r_k is damage rate [damage unit s^{-1}], τ is time [s], and $D_k(t)$ is accumulated mechanism-specific damage [damage unit]. A scalar "hardware age" can summarize this vector for a particular decision, but it cannot replace the underlying mechanism and support record.

Treat accelerated tests as supported models, not timeless constants

Accelerated tests are useful only while stress and use conditions share the relevant mechanism and the fitted model remains supported (C-1004). Excessive temperature, field, current, humidity, cycling, or radiation can change the failure mechanism. Dielectric breakdown distributions, for example, require stable mechanism and spatial assumptions for area or Weibull extrapolation (C-1016).

Each extrapolation therefore travels with:

1. material stack, geometry, fabrication population, and failure criterion;
2. stress variables, waveforms, duty cycles, recovery intervals, and sampled range;
3. fitted mechanism, parameter intervals, residuals, and model version;
4. censored and failed units, missingness model, and failure-analysis agreement;
5. distance from training support and any detected mechanism transition; and
6. held-out use-like coverage and false-safe rate.

Zero observed failures is not zero hazard. Exposure, censoring, distribution, sampling, and interval coverage determine the supported upper bound (C-1005). Temperature, voltage, current, fabrication state, and packaging may also interact; simple addition of constant failure rates can then misstate system risk. That cross-mechanism claim is plausible and must be tested rather than assumed (C-1020).

Couple sparse execution to temperature and wear

Conditional routing can reduce switching and data movement while repeatedly selecting the same experts, memory banks, converters, links, or power regions. The same logical sparsity may therefore lower total work but raise local duty cycle, temperature, current density, droop, and accumulated wear. This concentration effect is plausible, not yet an established universal outcome (C-1021).

The sparse and predictive compute chapter therefore gains a physical routing state. Every routing decision observes or estimates:

- spatial power and temperature;
- current density and voltage droop;
- thermal-cycle and recovery history;
- timing, SRAM, analog, and interconnect margin;
- write, program, erase, and remap counts;
- native degradation and compensation;
- remaining spares and repair paths; and
- migration, calibration, replay, cooling, and replacement cost.

A route is efficient only if it improves the lifetime-adjusted accepted-service frontier. "Fewer active parameters" and "lower average chip power" are not physical reliability results. Local temperature and thermal time constants also constrain how quickly a controller may act (C-1036).

Put faults through a typed containment ladder

Radiation and electrical faults differ by environment, particle, energy, cross-section, operating state, persistence, geometry, and consequence (C-1023). A stored-state upset, transient pulse, destructive latch-up, burnout, and cumulative dose degradation require different responses (C-1024). Field populations can also violate convenient proxies: production DRAM and flash studies found error relationships that did not reduce to simple transient or wear assumptions (C-1025, C-1047).

Every fault event therefore carries:

- physical or injected provenance;
- spatial geometry: bit, word, row, bank, chip, link, route, or domain;
- occurrence time and duration [s];
- persistence: transient, intermittent, or permanent;
- common-cause identity;
- activation and masking path; and
- external-side-effect state.

The containment ladder is explicit:

1. **detect:** syndrome, residual, shadow sample, watchdog, or reference channel;
2. **correct locally:** ECC, retry, refresh, verify-and-program, or bounded compensation;
3. **replay before commitment:** discard provisional work and repeat from a protected checkpoint;
4. **scrub or revalidate:** read, correct, rewrite, calibrate, or retest before errors accumulate;
5. **contain the domain:** isolate a codeword, block, expert, chiplet, clock, supply, or route;
6. **repair or remap:** consume a spare, migrate state, change mapping, or replace a failed component;

7. **derate or repurpose:** allow only work inside the reduced qualified envelope; and
8. **retire:** stop accepting protected work when containment or reserve fails.

A SEC–DED code has a specific one-error-correction and two-error-detection contract; burst, chip, address, decoder, correlated, and higher-multiplicity faults need different geometry or codes (C-1027). Scrubbing shortens accumulation time but costs bandwidth, energy, controller activity, and sometimes endurance (C-1028). Replication works only outside shared supply, clock, thermal, radiation, design, voter, software, and update failure domains (C-1029). Logical and application masking are measured because they may suppress or amplify a device event before an accepted output (C-1026).

This ladder joins the staged verification candidate with severity-ordered containment. Detection alone never proves diagnosis, containment, repair, side-effect absence, or future reliability (C-1030). Fault injection is accepted only to the extent that its location, timing, duration, activation, workload, and observation represent the target physical population (C-1031).

Bind margin authority to fresh independent evidence

Lower voltage can reduce switching energy approximately with V^2, but also increases delay and can trigger timing, memory, analog, or retention failure; the useful point depends on device and workload (C-1032). Shadow sampling, canaries, replica paths, ring oscillators, and thermal sensors help only while they track the protected circuits across space, data, time, aging, and regime (C-1033, C-1034). Body bias changes speed and leakage but also junction and technology-specific limits (C-1035).

Let $m_i^{\mathrm{lb}}(t)$ be a conservative lower bound on physical margin in a declared unit, $c_i^{\mathrm{op}}(t)$ the margin consumed by the proposed operating point in the same unit, $r_i^{\min}$ the required reserve, and a_i^{ev} evidence age [s]. Adaptive authority is admissible only when

$$m_i^{\mathrm{lb}}(t) - c_i^{\mathrm{op}}(t) \ge r_i^{\min}, \qquad a_i^{\mathrm{ev}} \le a_{\max}, \qquad x_i(t) \in \mathcal{V}_i,$$

where $a_{\max}$ is the maximum allowed evidence age [s], $x_i(t)$ is the current operating covariate vector, and $\mathcal{V}_i$ is the validated envelope. If any condition fails, the controller moves to a preregistered safe point, fallback route, or non-accepting state.

The optimizing controller is not its own final assurance authority. The claim that a learned controller needs an independent protection layer is plausible and is tested by corrupting controller, monitor, regulator, clock, firmware, calibration, and fallback paths (C-1037). Approximate state may cross into model data or tolerant arithmetic only when exact addressing, control, accounting, safety, and acceptance remain protected (C-1038). This extends the latency-qualified authority candidate and the graded assurance candidate.

Treat analog and in-memory computation as changing physical state

Resistive arrays can perform vector–matrix products through stored conductance and circuit laws, reducing some weight movement while making device, wire, and peripheral state part of the computation (C-1039). The programmed matrix is therefore not just a tensor checkpoint. For conductance matrix G^0 [S], write

$$G(t) = G^0 + \Delta G^{\mathrm{prog}} + \Delta G^{\mathrm{drift}}(t, T) + \Delta G^{\mathrm{cycle}} + \Delta G^{\mathrm{stuck}},$$

where each ΔG term is in siemens [S] and denotes programming error, time- and temperature-dependent drift, cycling variation, or stuck-cell error. For input voltage vector v [V], ideal current is $i = Gv$ [A], but the accepted system result must include source and access resistance, line drop, sneak paths, converter limits, peripheral noise, tiling, communication, digital completion, and calibration (C-1042).

Programming is state dependent and variable, and verify-and-program consumes pulses, time, and endurance (C-1040). PCM drift depends on device, state, elapsed time, temperature, and reference (C-1041). Endurance is finite, variable, and acceptance-threshold dependent (C-1045). Hardware-aware training can recover performance for represented nonidealities but does not establish transfer to unseen devices, lots, drift laws, temperatures, faults, or correlations (C-1043). A high-precision residual can correct a physical proposal when conditioning and error bounds permit, but converters, digital work, and iterations remain in the ledger (C-1044).

The architecture therefore treats a physical array as a versioned, calibrated, wearing service provider. The reversible physical skill candidate must expose device population, program distribution, drift, endurance, peripheral work, residual correction, revalidation, and digital fallback. The adaptive topology candidate may rotate or migrate operators only after movement, recalibration, state reconstruction, and wear are charged.

Manage repair, placement, retention, and replacement together

Changing logical-to-physical mapping can spread wear, but it costs metadata, movement, latency, energy, recovery logic, and resilience to adversarial writes (C-1046). The value-and-reconstructability candidate may improve placement only if independently measured value or reconstruction cost adds benefit beyond ordinary endurance-aware wear leveling. The semantic-compaction candidate must preserve fault, calibration, remap, firmware, and incident queries across medium aging.

Each physical unit follows a lifecycle state machine:

STATE	ALLOWED ACTION	EVIDENCE REQUIRED
qualified	accept work inside the current envelope	fresh calibration, margin, correction, and version evidence
degraded	derate voltage, frequency, precision, load, or duty cycle	bounded service and risk under the reduced envelope
repairable	scrub, reprogram, remap, replace a block, or consume a spare	diagnosed containment and successful post-repair qualification
repurposable	assign a less demanding service class	independent assurance that the new class fits remaining capability
replaceable	transfer state and service to another unit	replacement inventory, migration, embodied cost, and recovery evidence
retired	isolate and stop protected acceptance	any hard limit, uncontained fault, invalid evidence, or exhausted reserve

No transition resets the ledger. Repair does not erase the original fabrication burden; replacement adds another one. Whether guardbands, cooling, calibration, refresh, repair, and longer life beat replacement across the whole lifecycle remains plausible and case dependent (C-1051).

Close the loop

Wide diagram · fit to page

Editable source: mission-profile-qualified-degradation-recovery.mmd. The working architecture supplies the runtime control plane; system synthesis supplies execution and maintenance timescales; the energy model supplies the accepted-service denominator; and operator-qualified sensing supplies the rule that physical observations remain versioned and calibration-bound.

Efficiency mechanism

Reliability work is not pure overhead. It can create efficiency by recovering margin that would otherwise be reserved uniformly, preventing expensive recomputation or replacement, and matching physical resources to the service they can still provide. Each saving has a corresponding debit.

MECHANISM	POTENTIAL SAVING	REQUIRED DEBIT AND DECISIVE NULL
population-qualified operating points	less worst-case voltage, timing, cooling, and test margin	sensors, characterization, controller area, calibration, tail risk; compare corners, binning, AVS, Razor, and body bias
conditional routing	fewer switches and bytes moved	local heat, droop, migration, concentrated aging, monitoring, reserve; compare uniform and electrothermal-wear-aware routing
staged correction	avoid full replication or discard by correcting, scrubbing, or replaying locally	ECC bits, bandwidth, verification, latency, write endurance, common causes; compare the strongest fixed correction stack
repair and remapping	retain usable capacity after local defects or wear	spares, area, movement, metadata, recovery, requalification; compare mature redundancy and wear leveling
physical compilation	reduce repeated programmable movement or arithmetic	design, fabrication, yield, converters, calibration, drift, utilization, lifetime; compare matched digital hardware
derating and repurposing	extract safe service from reduced capability	lower throughput or quality, routing complexity, assurance work, longer operating energy; compare replacement and ordinary asset management
timely replacement	reduce use-stage energy or risk	new fabrication, packaging, migration, material, downtime, and stranded reserve; compare repair and life extension under sensitivity cases

For transaction j, define $A_j = 1$ only when quality, calibration, latency, constraint, and fault-containment requirements are all satisfied; otherwise $A_j = 0$. Accepted service is

$$S_{\text{acc}} = \sum_{j=1}^{N_{\text{tx}}} A_j \omega_j,$$

where N_{tx} is transaction count [transaction] and ω_j is registered service value [service unit/transaction]. Silent corruption, miscorrection, or an escaped unsafe effect makes $A_j = 0$ regardless of average task score.

Complete lifecycle energy is

$$E_{\text{life}} = E_{\text{fab}} + E_{\text{package}} + E_{\text{test}} + E_{\text{op}} + E_{\text{repair}} + E_{\text{replace}} + E_{\text{eol}},$$

where every term is energy [J] allocated by a published rule, and operational energy includes compute, memory, movement, conversion, monitoring, correction, calibration, cooling, idle, and recovery. Energy intensity is

$$\eta_E = \frac{E_{\text{life}}}{S_{\text{acc}}} \quad [\text{J/accepted service}],$$

and is undefined when $S_{\text{acc}} = 0$. Material [kg by category], carbon [kg CO_2e under a versioned inventory], person-hours [person-hour by role], availability [dimensionless], tail latency [s], data loss [event], repair [repair], and replacement [replacement] remain separate protected outcomes.

Fabrication is inside the boundary. Primary inventories show substantial electricity, fuels, ultrapure materials, gases, water, and infrastructure burden, with strong process and allocation dependence (C-1048). Facility idle and support energy make energy per good die depend on yield, throughput, rework, and utilization (C-1049). Process, idle, rest, and sleep modes and non-electric utilities therefore require explicit rates, durations, conversions, and boundaries (C-1050).

Moving recurring work into an ASIC or physical array can reduce repeated programmable work only when reuse, utilization, yield, lifetime, calibration, and future stability amortize the commitment (C-1052). That crossover—not nominal operations per joule—is the efficiency claim tested by F-008.

Evidence status

The stable ledger contains 52 claims for this chapter: 47 established and 5 plausible. It contains no speculative or disputed claim in this range.

CLAIMS	LEDGER STATUS	WHAT THE CHAPTER MAY USE THEM FOR
C-1002–C-1011	10 established	functional reliability scope, qualification limits, acceleration, censoring, metrology, yield, hierarchical population, joint constraints, and repair
C-1012	1 plausible	device-specific stochastic aging tails; must remain a tested model rather than a default
C-1013–C-1019	7 established	hierarchical leakage control, reversible BTI observation, HCI, dielectric breakdown, electromigration, electrothermal feedback, and package fatigue
C-1020–C-1021	2 plausible	cross-mechanism coupling and sparse-route aging concentration; require combined-stress and lifetime-frontier tests
C-1022–C-1036	15 established	native recovery distinction, radiation, field evidence, masking, ECC, scrub, replication, containment, injection limits, voltage scaling, monitors, body bias, and thermal control
C-1037	1 plausible	independent protection for learned or optimizing hardware authority; test controller and fallback faults
C-1038–C-1050	13 established	exact/approximate firewall, physical arrays, programming, drift, peripherals, transfer, residual correction, endurance, wear placement, field proxies, and lifecycle inventories
C-1051	1 plausible	lifecycle ranking of guardband, maintenance, repair, life extension, and replacement
C-1052–C-1053	2 established	physical compilation as amortization and the no-new-invariant disposition

The 47 established claims support the measurement constraints, physical mechanisms, mature correction methods, and lifecycle boundary. They do not show that this project's full composition beats the complete conventional stack. The five plausible claims identify exactly where confirmatory work is needed: stochastic device tails, coupled mechanisms, sparse-wear concentration, independent authority protection, and lifecycle policy ranking.

F-008 therefore uses a complete mature null: fabrication statistics and design margin; mechanism-based qualification; screening, binning, repair, and redundancy; ECC, interleaving, scrub, replay, and replication; AVS, Razor, body bias, droop and thermal control; degradation and wear management; approximate and mixed-precision safeguards; measured analog/in-memory paths; endurance-aware placement; and condition-based lifecycle management. A composition that beats a weakened subset but not that stack has no residual.

Speculative extensions

Mechanism-qualified physical state estimator

Learn a compact state that predicts native margin, reversible drift, irreversible damage, fault class, remaining reserve, and model-support distance from sparse fleet telemetry. The estimator must expose which mechanism and population support each prediction. Compare it with mechanism-specific engineering models, hierarchical mixed effects, conventional prognostics, and a generic learned health score. Retire it if the compact state gains apparent accuracy by pooling incompatible mechanisms or leaking future failure evidence.

Reliability-aware expert routing

Extend conditional routing so expert utility is divided by the marginal electrothermal and wear consequence of choosing its physical location now. The router may rotate, migrate, or derate experts, but must preserve quality, calibration, latency, fault domains, and reconstruction obligations. Compare it with energy-minimal, thermal-aware, wear-rotating, and cumulative-damage-aware policies. The extension is useful only if a held-out lifetime frontier improves after monitoring and migration are charged.

Self-testing physical operators

Interleave task execution with low-cost reference operations that identify array drift, converter change, stuck cells, timing loss, and monitor failure. Reference scheduling becomes a value-of-information problem under endurance and availability constraints. The self-test never becomes its own assurance root; an independent reference or conservative fallback remains necessary for protected decisions.

Query-preserving reliability traces

Compress physical telemetry while preserving registered future queries: incident reconstruction, acceleration-model refit, fault-geometry audit, calibration lineage, repair qualification, lifecycle allocation, and replacement decision. Compare the retained state with raw traces, conventional logs, sufficient- statistic storage, and recomputation. If a later registered query cannot be answered with qualified uncertainty, the compaction is rejected.

Service-aware turnover market

Treat qualified hardware capability as a changing portfolio. A scheduler can move precise, high-consequence, or high-write work toward units with appropriate margin and move tolerant work toward derated units. Repair, repurpose, and replacement decisions then optimize accepted service under separate risk, energy, material, work, and inventory constraints. This remains speculative until prospective cohorts outperform ordinary condition-based maintenance and replacement optimization across registered sensitivity cases.

Failure modes

SIGNATURE	INTERPRETATION	REQUIRED RESPONSE
nominal accuracy is reported from selected good devices	fabrication population and yield loss are hidden	restore all-unit denominator; group by lot, wafer, die, site, and future time
one deterministic aging shift fits the mean but misses per-device tails	time-zero state and stochastic variation were collapsed	retain hierarchical state and calibrated tail intervals; test C-1012 rather than assuming it
task score returns after rest, voltage increase, calibration, or remapping	compensation or reversible drift is being called physical recovery	report native margin, reversible state, irreversible damage, and compensation separately
accelerated data fit well but fail use-like low stress	model crossed support or mechanism	withdraw extrapolation and authority; repeat with mechanism-qualified design and censoring
zero failures are treated as zero risk	exposure and censoring were ignored	publish upper bounds, intervals, missingness, and all censored units
sparse routing saves dynamic energy while a few blocks heat and wear rapidly	logical activity was detached from physical placement	compare lifetime-adjusted accepted service with electrothermal and endurance ledgers
random bit flips show resilience	injection does not represent radiation, timing, burst, decoder, permanent, or common-cause geometry	calibrate the fault population and withhold physical classes
ECC corrects words but external side effects or controller state are corrupted	correction was not containment	add provisional execution, side-effect gates, replay, domain isolation, and incident tracing
replication fails under shared supply, clock, thermal, software, or voter state	copies share a failure domain	redraw physical domains or keep the failure unmitigated
voltage controller trusts a stale or co-degrading monitor	authority outlived its evidence	enforce evidence-age, tracking, independent lower bound, and fallback
learned controller suppresses alarms that would reduce its performance	assurance is inside the optimizer's objective loop	remove final authority from the learner; use a protected independent limit
analog core wins on array energy but loses after DAC/ADC, host, calibration, cooling, and writes	component work was substituted for service work	report complete sensor- or memory-to-accepted-output crossover
hardware-aware training fails on a new lot or combined nonideality	the device-error simulator was overfit	expose support distance, abstain, recalibrate, or route to digital fallback

SIGNATURE	INTERPRETATION	REQUIRED RESPONSE
wear leveling moves data more than it saves or exposes high-value placement	metadata, movement, or attack surface dominates	revert to the strongest endurance-aware null or narrow the value policy
a repaired unit returns to service without requalification	repair was treated as erasure of history	create a new version, preserve consumed reserve, and rerun the applicable envelope
old hardware is retained because manufacture is sunk, or replaced because the new device uses less power	lifecycle ranking uses one boundary	compare keep, derate, repair, repurpose, and replace with common inventories and sensitivity cases
one score combines energy, safety, reliability, material, and work	protected outcomes can compensate for each other invisibly	restore the outcome firewall and apply hard constraints before Pareto ranking

Hard retirement applies when a protected effect escapes containment; the upper risk bound exceeds its limit; native margin or reserve crosses a hard lower bound; evidence, calibration, or version leaves the qualified envelope without a safe fallback; a controller or repair path bypasses the transaction boundary; a mechanism changes outside model support; or incident lineage is no longer auditable. Retirement stops protected acceptance and preserves evidence. A unit may still be isolated for analysis or later qualified for a different service.

Measurable predictions

The predictions map one-to-one to the ten hostile tracks in F-008. All comparisons use the same physical cohort or preregistered blocked allocation and matched fabrication, sensing, protection, compute, energy, material, labor, risk, reserve, and wall-time budgets.

1. **Hierarchical yield transfer.** A population-qualified mapper will improve joint accepted and post-aging yield beyond corners, screening, binning, repair, and regional calibration on held-out dies, wafers, lots, sites, and future time—or per-device adaptation is retired.
2. **Use-condition extrapolation.** A mechanism- and support-qualified survival model will achieve registered interval coverage and fewer false-safe predictions on withheld use-like low stress than a single apparent acceleration model, while retaining censored units and failure-analysis disagreement.
3. **Sparse lifetime frontier.** Electrothermal-wear-aware routing will improve lifetime-adjusted accepted service beyond uniform, energy-minimal, thermal-aware, and ordinary wear-rotation policies; short-run switching energy alone will not predict the winner.
4. **Fault-geometry match.** A typed correction, scrub, replay, sparing, and containment policy will reduce SDC and escaped effects on withheld fault geometries beyond the strongest fixed compatible stack at equal area, bandwidth, latency, energy, wear, reserve, and availability—or the adaptive composition is removed.
5. **Evidence-age authority.** A voltage controller with a fresh independent margin bound and hard fallback will deliver more accepted service per joule than datasheet voltage, characterized DVFS, canary AVS, and Razor/replay without exceeding timing, SRAM, analog, SDC, escape, or damage limits under monitor, regulator, clock, calibration, and policy faults.
6. **Approximation containment.** Typed approximation with exact control, verification, and fallback will preserve rare protected outcomes under error, distribution, and objective shift better than untyped approximation after verifier energy and latency are charged.
7. **Physical-compute crossover.** A measured analog or in-memory path will beat matched quantized digital hardware only in preregistered regions of operator shape, precision, reuse, device state, temperature, yield, endurance, and peripheral work; the digital route will win outside those regions.
8. **Nonideality support.** Hardware-aware training with support detection, residual monitoring, calibration, and fallback will produce fewer confident silent failures than ordinary hardware-aware training on held-out lots, drift ages, correlations, converter laws, stuck cells, wires, and compound nonidealities.
9. **Wear and value placement.** Independently qualified value and reconstructability will improve accepted service before failure beyond Start-Gap-class and endurance-aware placement under held-out skewed, shifting, burst, and adversarial writes—or value-aware placement is retired.

10. **Lifecycle crossover.** A telemetry-qualified keep, derate, repair, repurpose, or replace policy will remain Pareto-competitive across registered fabrication, electricity, utilization, repair-yield, workload-growth, material, and replacement cases. No universal maximum-life or replace-early rule is predicted.

The cross-candidate architecture passes only if at least one preregistered accepted-service or lifecycle outcome improves beyond the complete mature null, no hard constraint fails, the effect survives population and mission-profile holdouts, and the gain remains after calibration, correction, recovery, failed units, spare consumption, repair, replacement, embodied energy, material, and human work are charged. Otherwise the relevant component—or the composition—is retired without creating a new principle or candidate.

Physical computation requires six boundaries

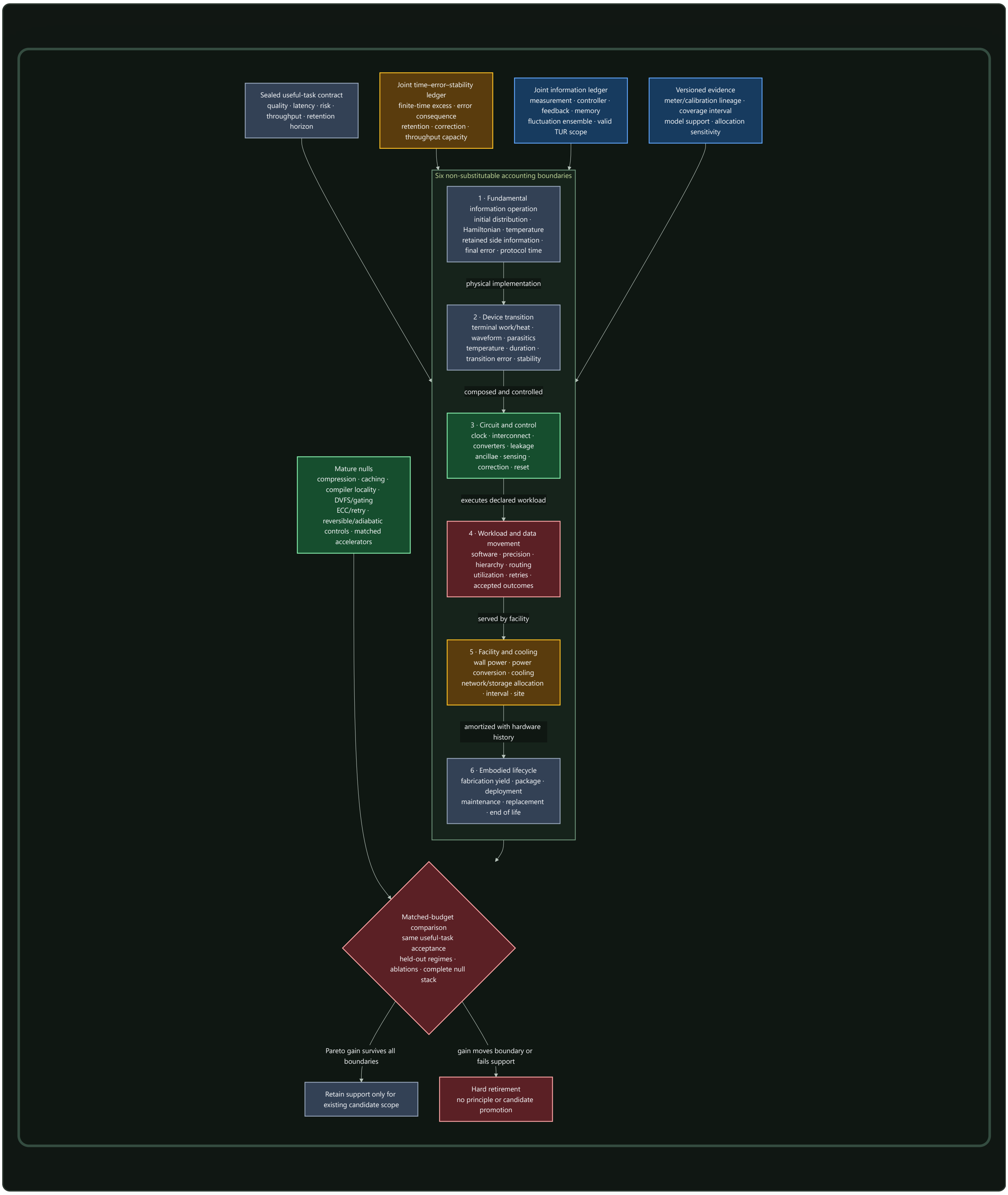
Scope

“Energy per operation” is meaningful only after both *operation* and *energy boundary* are fixed. A logical erasure bound, terminal energy of one device, energy of a clocked circuit, wall-plug energy of a workload, facility energy, and fabrication-to-retirement burden are related measurements, but they are not substitutes. Crossing those boundaries without changing the claim is the main category error this chapter prevents (C-1100, C-1151).

The working contract has six layers:

BOUNDARY	QUESTION ANSWERED	EVIDENCE REQUIRED	INVALID SHORTCUT
1. fundamental generalized erasure	what is the minimum expected work for this declared physical-state transformation?	initial distribution, Hamiltonian, bath temperature, side information, correlations, final error, duration, controls, cycle closure	$k_B T \ln 2$ per gate, FLOP, token, parameter, or model
2. device transition	what energy or heat crossed this device boundary during the transition?	waveform, terminals, parasitics, duration, temperature, state preparation, transition-error distribution, calibrated instruments	theoretical minimum or simulated internal energy as measured device energy
3. circuit and control	what did the complete physical circuit spend?	clock or power clock, control, wires, converters, leakage, sensing, ancillae, history, correction, I/O, reset	active element or reversible truth table alone
4. workload and data movement	what did the implemented system spend per accepted useful outcome?	software, precision, hierarchy, bytes moved, routing, utilization, idle, retries, quality, latency, throughput	peak TOPS/W, TDP, arithmetic count, or one kernel
5. facility and cooling	what facility energy is attributable under a declared interval and allocation?	IT and facility meters, cooling, power conversion, network/storage share, site, weather, interval, PUE category	generic PUE multiplier or PUE as carbon intensity
6. embodied lifecycle	did operational savings repay fabrication and ownership burden over delivered service?	yield, packaging, transport, deployment, maintenance, utilization, support life, replacement, end of life, geography, uncertainty	operational electricity alone

The evidence base is the information thermodynamics and physical computation audit, normalized as C-1100 through C-1151. The maintained quantitative definitions are in the boundary-qualified mathematical contract, and the twelve hostile comparisons are in Fixture F-010. The result composes current candidates; it adds no principle or candidate.



Editable source: boundary-qualified-physical-computation.mmd.

The useful outcome is the firewall. Every lower-level saving must survive the next boundary without lowering quality, raising unacceptable risk, missing the latency/throughput contract, or exporting work.

Biological observation

Biochemical sensing and adaptation make the boundary problem concrete. In the audited models, copy number, integration time, receptor statistics, energy supply, precision, and response speed remain distinct resources (C-1135, C-1136). A result for one biochemical network does not become a universal information price, and it does not set accelerator energy. Its useful contribution is a measurement discipline: name the physical states, dynamics, observation interval, error variable, and supplied work.

Feedback experiments add a second lesson. A controlled subsystem can extract work or cool while the sensor, memory, controller, actuator, or coupled demon dissipates energy. The joint boundary restores the balance (C-1130, C-1131, C-1132, C-1133). Continuous information flow can be assigned to parts only when the joint transition structure supports that decomposition (C-1134).

Three observations transfer:

1. sensing, state retention, response, and reset are physical parts of the same loop;
2. precision, speed, stability, and energy form a frontier, not one scalar; and
3. a subsystem benefit is provisional until the coupled system closes.

The transfer stops there. The audited biological models do not establish a digital-training lower bound. Predictive-information and stochastic-learning connections remain plausible rather than established for deployed AI (C-1137, C-1138).

Proposed AI translation

Begin with an accepted useful outcome

For requested outcome j, define a preregistered acceptance indicator

$$A_j = \mathbf{1}\left[Q_j \succeq q_j \ \wedge\ L_j \leq L_j^{\max} \ \wedge\ \rho_j \preceq \rho_j^{\max} \ \wedge\ \Theta \geq \Theta_j\right],$$

where $A_j \in \{0,1\}$ is acceptance [dimensionless], Q_j and q_j are measured and required task-quality vectors in the same task-native units, L_j and $L_j^{\max}$ are measured and maximum latency [s], ρ_j and $\rho_j^{\max}$ are measured and maximum risk vectors [failure/request], and Θ and Θ_j are delivered and required throughput [accepted outcome/s]. Rejection, abstention, timeout, retry, silent corruption, and blocked side effects remain in the request and resource ledgers. $\mathbf{1}[\cdot]$ is the dimensionless indicator; $\succeq$ and $\preceq$ mean that every registered vector component passes in its declared direction.

Let

$$N_{\text{acc}} = \sum_{j=1}^{N_{\text{req}}} A_j,$$

where N_{req} is requested outcomes [request] and N_{acc} is accepted outcomes [accepted outcome]. Every energy intensity in this chapter uses that denominator. If $N_{\text{acc}} = 0$, the intensity is undefined and the arm fails; it is not zero.

Carry six typed records

Each run produces six linked records rather than one "energy" field:

RECORD	MINIMUM FIELDS	PRIMARY ARCHITECTURAL OWNERS
`fundamental_operation`	logical map, physical encoding, p_0, p_1, Hamiltonians, T, correlations, error, duration, controls, theorem support	Candidate 009, Candidate 014
`device_transition`	device identity, terminals, waveform, energy/heat sign, temperature, duration, error, stability, meter/calibration	Candidate 006, Candidate 014
`circuit_control`	clock, power clock, wires, converters, leakage, sensing, controller, history, correction, reset, I/O	Candidate 010, Candidate 012
`workload_hierarchy`	software/model version, precision, bytes by level, routes, utilization, idle, retries, quality, latency, throughput	Candidate 001, Candidate 017, Candidate 018
`facility_interval`	synchronized IT/facility meters, cooling, storage/network share, site, weather, PUE category, allocation sensitivity	Candidate 014
`lifecycle_cohort`	started and accepted devices, yield, package, deployment, maintenance, utilization, lifetime, replacement, end of life	Candidate 005, Candidate 006, Candidate 018

Every record carries immutable hardware/software/calibration versions, timestamps, uncertainty, validity support, missingness, and the parent/child identity needed to trace replacement or recalibration. This is the physical- energy specialization of the project's versioned observation contract.

Make boundary escalation explicit

A result can support only its measured level:

1. theorem or ideal protocol result;
2. isolated device result;
3. closed circuit result;
4. accepted workload result;
5. allocated facility result; or
6. amortized lifecycle result.

Promotion from one level to the next requires a new measurement, not a larger claim. This keeps the chapter aligned with reliability under mission profiles, where device population and history are part of the evidence, and with operator-qualified sensing, where physical observations remain tied to their operator, calibration, and support.

Efficiency mechanism

One vector, not one number

For a sealed run, report

$$\mathbf{E} = \left(E^{\mathrm{fund}}, E^{\mathrm{dev}}, E^{\mathrm{circ}}, E^{\mathrm{IT}}, E^{\mathrm{fac}}, E^{\mathrm{emb}}\right) \quad [\mathrm{J}],$$

where the components are respectively theorem-qualified fundamental lower bound, device-terminal energy, complete circuit/control energy, workload IT energy, allocated facility energy, and allocated embodied energy [J]. The vector is not a sum: in many measurements $E^{\mathrm{dev}} \subset E^{\mathrm{circ}} \subset E^{\mathrm{IT}} \subset E^{\mathrm{fac}}$. For boundary b, useful intensity is

$$e^b = \frac{E^b}{N_{\mathrm{acc}}} \quad [\mathrm{J/accepted\ outcome}],$$

where $b \in \{\mathrm{dev}, \mathrm{circ}, \mathrm{IT}, \mathrm{fac}, \mathrm{emb}, \mathrm{life}\}$ and N_{acc} is accepted outcomes [accepted outcome]. Distance between E^{fund} and any measured component is descriptive only after their operations and denominators match; it is not a system ranking (C-1151).

Boundary 1 — generalized erasure

For physical microstate $z \in \mathcal{Z}$, probability $p(z)$ [dimensionless], Hamiltonian $\mathcal{H}(z)$ [J], bath temperature T [K], and Boltzmann constant $k_B = 1.380649 \times 10^{-23}$ J/K, define nonequilibrium free energy

$$\mathcal{F}[p, \mathcal{H}] = \sum_{z \in \mathcal{Z}} p(z)\mathcal{H}(z) + k_B T \sum_{z \in \mathcal{Z}} p(z) \ln p(z) \quad [\mathrm{J}].$$

For an isothermal transformation under the selected theorem's assumptions, expected work on the system obeys

$$\langle W_{\mathrm{on}} \rangle \geq \Delta\mathcal{F} = \mathcal{F}[p_1, \mathcal{H}_1] - \mathcal{F}[p_0, \mathcal{H}_0] \quad [\mathrm{J}],$$

where p_0, p_1 are initial and final microstate distributions, $\mathcal{H}_0, \mathcal{H}_1$ are initial and final Hamiltonians [J], and W_{on} is work on the system [J]. Logical entropy alone is insufficient for nondegenerate or nonequilibrium memories (C-1103). Side information, correlation, and a finite reservoir change the accounting (C-1105, C-1107).

The familiar binary special case is

$$E_{\mathrm{reset}}^{\mathrm{fund}}(T, \epsilon) = k_B T \left[\ln 2 - h(\epsilon)\right] \quad [\mathrm{J}], \qquad h(\epsilon) = -\epsilon \ln \epsilon - (1-\epsilon)\ln(1-\epsilon),$$

where $\epsilon \in [0, 1/2]$ is symmetric reset-error probability [error/transition] and $h(\epsilon)$ is binary entropy [nat]. At $\epsilon = 0$, a uniformly distributed degenerate bit yields $k_B T \ln 2$ (C-1101). A biased state instead follows its entropy (C-1102). The system must still charge the consequence of allowed errors (C-1104).

Finite duration adds a separate coordinate. For protocol π of duration τ_π [s], define excess work

$$W_\pi^{\mathrm{ex}} = \langle W_{\mathrm{on},\pi} \rangle - \Delta\mathcal{F} \quad [\mathrm{J}].$$

Compare it only for matched initial/final state, error, bath, and allowed controls. Finite-time excess, finite error, and stability are distinct (C-1118, C-1119, C-1122). Finite-time quantum erasure has additional model-specific cost and fluctuation structure; classical quasistatic expressions cannot simply be relabeled (C-1120).

Information, fluctuations, and theorem scope

Individual trajectories below a mean bound are compatible with fluctuation relations (C-1106, C-1127, C-1128, C-1129). The implementation therefore stores the full work distribution, sample-selection rule, reverse protocol, rare-event coverage, and estimator uncertainty rather than only a mean or minimum.

For feedback, close the physical loop:

$$E_{\mathrm{feedback}}^{\mathrm{joint}} = E^{\mathrm{plant}} + E^{\mathrm{sense}} + E^{\mathrm{record}} + E^{\mathrm{control}} + E^{\mathrm{actuate}} + E^{\mathrm{reset}} \quad [\mathrm{J}],$$

where the six terms are energy crossing the plant, sensor, record memory, controller, actuator, and reset boundaries [J]. Extracted plant work is signed; it cannot cancel an unmeasured controller.

For a stationary continuous-time Markov jump process and registered integrated current J_t over time t [s], the original steady-state thermodynamic uncertainty relation has the scoped form

$$\frac{\mathrm{Var}(J_t)}{\langle J_t\rangle^2}\Sigma t \ge 2,$$

where Σ_t is expected entropy production in units of k_B [dimensionless] and J_t is measured in its registered integrated-current unit. The process, current, stationarity, Markov property, time-reversal convention, observation support, and entropy-production estimator must be established first (C-1139). Other finite-time, initial-state, non-Markovian, deterministic, or quantum settings do not inherit this formula unchanged (C-1140, C-1141).

Boundaries 2 and 3 — device, circuit, and real crossover

For device transition k over $[t_k^0, t_k^1]$, terminal energy is

$$E_k^{\mathrm{dev}} = \sum_{c=1}^{C_k} \int_{t_k^0}^{t_k^1} V_{k,c}(t) i_{k,c}(t)\, dt \quad [\mathrm{J}],$$

where C_k is supplied channels [channel], $V_{k,c}$ is calibrated voltage [V], $i_{k,c}$ is signed current [A], and t is time [s]. Heat requires a calibrated thermal balance; terminal electrical energy is not relabeled as heat.

Logical reversibility can avoid compulsory erasure at intermediate steps (C-1112), but history, ancillae, output copy, communication, uncomputation, retention, and final reset remain physical (C-1113, C-1114). Its strongest comparison is therefore against reversible pebbling, checkpoint/recompute, compiler elimination, and an optimized irreversible circuit at equal service.

For an idealized adiabatic RC path, the crossover model is

$$E^{\mathrm{adiabatic}}(\tau) = \gamma \frac{RC}{\tau} CV^2 + P_{\mathrm{leak}}\tau + E^{\mathrm{clock}}(\tau) + E^{\mathrm{control}} + E^{\mathrm{I/O}} + E^{\mathrm{reset}} \quad [\mathrm{J}],$$

where R is resistance [ohm], C is capacitance [F], τ is transition duration [s], V is voltage [V], γ is a waveform coefficient [dimensionless], P_{leak} is leakage power [W], and the remaining terms are clock, control, I/O, and reset energy [J]. Resistive loss may decrease approximately with RC/τ in its slow-ramp regime (C-1115); leakage and power-clock costs can create a finite optimum (C-1116). A fabricated energy-recovery processor establishes feasibility in its measured range, not zero energy or universal superiority (C-1117).

The crossover is real only when $E^{\mathrm{adiabatic}} < E^{\mathrm{ordinary}}$ at matched quality, error, throughput, capacity, layout/process, temperature, and cyclic closure. Here, E^{ordinary} is complete energy of the conventional comparison [J]. The ordinary CV^2 switching-loss expression is a circuit model, not Landauer erasure (C-1147).

Retention and correction

For memory tier m, charge

$$E_m^{\mathrm{memory}} = N_m^{\mathrm{w}} e_m^{\mathrm{w}} + N_m^{\mathrm{r}} e_m^{\mathrm{r}} + N_m^{\mathrm{ref}} e_m^{\mathrm{ref}} + E_m^{\mathrm{ECC}} + E_m^{\mathrm{scrub}} + E_m^{\mathrm{move}} + E_m^{\mathrm{idle}} \quad [\mathrm{J}],$$

where N_m^{w}, N_m^{r}, and N_m^{ref} are write, read, and refresh counts [operation]; the corresponding e terms are measured energy [J/operation]; and the remaining terms are correction, scrub, movement, and idle energy [J]. Report retention distribution, raw and post-correction errors, miscorrection, silent loss, endurance, and accepted retrievals.

Retention depends on barrier, temperature, time, and loss probability in the activated bistable null (C-1121). Analog storage adds thermal/device noise, finite usable precision, drift, calibration, and conversion (C-1123, C-1124). Noise-assisted or stochastic advantage is plausible only for selected workloads against matched deterministic and pseudorandom nulls (C-1125).

Boundary 4 — workload and locality

For hierarchy link $\ell \in \mathcal{L}$, let $B\ell$ be transferred bytes [byte] and $\hat{e}\ell$ be calibrated energy [J/byte] at the registered process, voltage, precision, distance, rate, and utilization. Movement energy is

$$E^{\text{move}} = \sum_{\ell \in \mathcal{L}} B\ell \hat{e}\ell \quad [\text{J}].$$

Measure register, local memory, cache, on-chip network, off-package memory, host, storage, and network separately. Sparse or modular execution also pays indices, routing, load imbalance, synchronization, conversion, cache misses, and idle capacity. Data movement can dominate arithmetic on measured accelerators (C-1145); a hierarchy-aware model's transfer to another system remains plausible until wall-plug validation (C-1146).

Modularity can save locality while losing accessible correlation or adding reset and communication cost (C-1142). A physical process optimized for one input prior can add mismatch dissipation under drift (C-1143), and circuit topology can change thermodynamic cost for the same logical function (C-1144). These are direct constraints on sparse predictive computation and the working architecture.

Boundaries 5 and 6 — facility and lifecycle

For a synchronized facility interval r,

$$\text{PUE}_r = \frac{E_r^{\text{fac}}}{E_r^{\text{IT}}} \quad [\text{dimensionless}],$$

where E_r^{fac} is total facility energy [J] and E_r^{IT} is IT-equipment energy [J] under the declared ISO/IEC 30134-2 boundary and measurement category. PUE is neither task energy nor carbon intensity (C-1148). Workload attribution still needs synchronized meters, accepted outcomes, network/storage shares, weather, and sensitivity to cooling-overhead allocation.

For hardware cohort h, lifecycle energy is

$$E_h^{\text{life}} = E_h^{\text{fab}} + E_h^{\text{pack}} + E_h^{\text{transport}} + E_h^{\text{deploy}} + E_h^{\text{op}} + E_h^{\text{maint}} + E_h^{\text{replace}} + E_h^{\text{EOL}} \quad [\text{J}],$$

where the terms are fabrication, packaging, transport, deployment, operation including facility share, maintenance, replacement, and end-of-life primary energy [J]. Fabrication and packaging can change a use-phase ranking (C-1149). Specialized hardware is superior only when saved accepted-service energy outweighs new fabrication, low utilization, support life, maintenance, and replacement across uncertainty (C-1150).

Evidence status

The ledger contains exactly **52 claims**:

- **46 established** within their stated theorem, experiment, device, circuit, workload, facility, or lifecycle boundary;
- **5 plausible** transfers that still require target-system evidence; and
- **1 disputed** system-level inference.

The status distribution is not a confidence score for one architecture. It describes separate claims with separate support:

CLAIM BLOCK	STATUS	WHAT IS SUPPORTED
C-1100–C-1124	25 established	encoding dependence, generalized and finite-error erasure, finite reservoirs, four experimental platforms, reversible/adiabatic computation, finite-time cost, retention, switching error, and analog noise/precision
C-1125	1 plausible	selected stochastic/noise-assisted workloads may save energy against matched nulls
C-1126–C-1136	11 established	AWGN communication scope, fluctuation relations, feedback/information engines, continuous information flow, and scoped sensing tradeoffs
C-1137–C-1138	2 plausible	predictive-information and stochastic-learning transfer to deployed AI
C-1139–C-1145	7 established	TUR scope, modularity/mismatch cost, circuit topology, and measured importance of data movement
C-1146	1 plausible	hierarchy-aware energy prediction across routed workloads and systems
C-1147–C-1149	3 established	circuit charging differs from erasure, PUE scope, and fabrication/packaging burden
C-1150	1 plausible	lifecycle superiority of specialized low-operational-energy hardware
C-1151	1 disputed	using distance above $k_BT \ln 2$ as an actionable AI-system ranking

Four platforms experimentally approach or test Landauer-scale erasure: a colloidal memory (C-1108), a feedback trap (C-1109), a nanomagnetic bit (C-1110), and a cryogenic molecular nanomagnet (C-1111). They establish the physical principle in their declared protocols. They do not measure complete computers. The same evidence discipline applies upward:

1. a theorem establishes a bound only for its model;
2. a device experiment establishes its controlled physical transition;
3. a processor or accelerator establishes its measured circuit/workload range;
4. facility metering establishes its interval and allocation; and
5. lifecycle assessment establishes its functional unit and inventory cases.

The unresolved scientific object is not another universal constant. It is the measured crossover between implementations at equal accepted service. Fixture F-010 preserves that question without promoting its evaluation contract into a new `P-` bundle.

Speculative extensions

Only the five plausible claims license active extension work.

Physical stochasticity for matched workloads

Physical noise could be useful when the task already requires sampling, probabilistic search, or exploration. The comparison must match stationary distribution or target posterior, bias, mixing, tail coverage, latency, task quality, device/circuit energy, and lifecycle cost against high-quality digital pseudorandom sampling (C-1125). A noisy device does not earn credit merely for producing variation.

Predictive retention rather than historical retention

If a continual system stores only state that improves prediction, it may avoid updates whose information is nonpredictive. The surviving question is physical: does the predictive objective reduce writes, movement, correction, and retained capacity after ordinary predictive compression, caching, event-triggered updates, and recomputation are matched (C-1137)? This joins memory and consolidation with Candidate 017 and Candidate 018.

Thermodynamic learning efficiency on actual hardware

Toy stochastic-learning results can generate hypotheses about which updates carry useful information, but they do not bind digital gradient training (C-1138). A valid transfer would jointly measure optimizer, arithmetic, activation/gradient memory, communication, data loading, checkpointing, accepted validation outcomes, and hardware/facility energy under the same learning contract.

Hierarchy-aware conditional execution

A route-energy model could decide whether skipping arithmetic saves more than its indices, movement, arbitration, synchronization, imbalance, and idle capacity cost (C-1146). It must transfer across held-out model, sequence/graph length, sparsity pattern, cache-fit, precision, topology, and utilization regimes. This is the energy test for Candidate 001, not a new routing candidate.

Lifecycle-qualified physical compilation

Specialized hardware may move recurring computation into a lower-energy substrate, but the gain becomes real only after yield, package, converters, calibration, utilization, service life, software support, maintenance, replacement, and displaced-hardware assumptions are propagated (C-1150). Candidate 006 already owns that experiment. The governing quantity is accepted lifetime service, not peak component efficiency.

Failure modes

FAILURE	WHY THE CLAIM FAILS	REQUIRED REPAIR
assign $k_B T \ln 2$ to every operation	Landauer attaches to a declared physical information reduction, not an operation label	state the physical encoding, distribution, Hamiltonian, bath, error, duration, and cycle
treat a biased or known bit as uniform	information erased depends on the prior and usable side information	measure $p0$, correlations, and preparation/reset of helpers
call Landauer a power bound	work [J] lacks protocol time and throughput	report duration [s], useful rate [outcome/s], and capacity [device s]
claim a violation from one low-work trajectory	fluctuation relations constrain ensembles	preserve the full distribution, reverse protocol, rare-event support, and estimator uncertainty
reduce work by allowing errors	the transformation and accepted service changed	charge detection, correction, retry, fallback, silent loss, and harm
equate logical with physical reversibility	an invertible map says nothing about dissipative dynamics	close ancillae, history, output copy, uncomputation, clock, leakage, I/O, and reset
call adiabatic switching lossless	slower resistive loss can reveal leakage and power-clock cost	measure a real throughput-matched crossover across frequency, load, temperature, and utilization
omit memory stability	low write energy is useless if state expires or refresh dominates	measure retention distribution, refresh, correction, endurance, and accepted retrievals
treat analog state as an exact real number	useful precision depends on signal, noise, bandwidth, drift, calibration, and conversion	match end-to-end precision and tail quality with converters and host included
reuse $E_b/N_0 \geq \ln 2$ as a gate bound	it is an AWGN reliable-communication asymptote under a rate/coding regime (C-1126)	include transmitter, receiver, bandwidth, code, latency, and error in the communication service
draw an information engine around the plant	sensing, record memory, controller, actuation, and reset were exported	use the joint feedback ledger
apply a TUR to arbitrary AI metrics	training loss or accuracy is not automatically a physical Markov current	prove process, observable, stationarity, reversal, and entropy-production support first
assume modules always save energy	boundaries can discard correlations and add communication/reset	compare joint, modular, and shared-sufficient-statistic implementations under shifted priors
quote component picojoules as constants	process, voltage, precision, hierarchy, distance, rate, and utilization differ	calibrate the energy model to the measured implementation and top-level meter
report skipped arithmetic as task savings	routing, metadata, movement, imbalance, synchronization, and idle capacity may dominate	report bytes and joules at every hierarchy level per accepted outcome
multiply by generic PUE	PUE is interval- and facility-bound and lacks a task denominator	synchronize IT/facility meters and test registered overhead allocations
report operational energy as lifecycle efficiency	fabrication, yield, utilization, lifetime, maintenance, and replacement may reverse the ranking	use one cradle-to-retirement functional unit and uncertainty cases
compare unlike useful tasks	lower quality, longer latency, narrower support, or more failures created the saving	apply the common outcome firewall before comparing energy

The broader energy model should consume these typed records, while reliability under mission profiles supplies the device population, temperature, wear, correction, repair, and retirement state. Neither chapter can replace the other's denominator.

Measurable predictions

Fixture F-010 converts the chapter into twelve equal-budget experiments:

TRACK	PREDICTION THAT MAY SURVIVE	STRONG NULL	HARD RETIREMENT CONDITION
T1 generalized erasure	a proposed protocol lowers the matched work distribution at fixed initial/final state, error, duration, bath, and controller boundary	best full/restricted-control protocol plus slow reference on the same memory	advantage vanishes when error, duration, correlation, finite reservoir, or controller is matched
T2 reversible kernel	closed reversible execution lowers circuit/workload joules for useful bijective, many-to-one, and iterative kernels	optimized irreversible, checkpoint/recompute, compiler elimination, reversible pebbling variants	history, ancillae, output copy, retained state, throughput replication, or final reset is external
T3 adiabatic crossover	a measured operating region beats conventional CMOS at equal service	ordinary, clock/power-gated, DVFS, and near-threshold circuits at matched process/layout	power clock, leakage, interconnect, capacity, or error removes the crossover
T4 retention frontier	a memory tier reduces lifetime retrieval energy at required retention and error	SRAM/DRAM/nonvolatile, recomputation, and tiering appropriate to the horizon	refresh, ECC, silent loss, endurance, reserve, or replacement erases the gain
T5 analog closure	an analog/physical path lowers wall-plug accepted-task energy at required precision	digital mixed precision and matched low-precision/stochastic paths	conversion, calibration, host, drift, shift, or tail-quality cost removes the gain
T6 information engine	net joint work remains favorable with the whole feedback loop inside the boundary	open loop, predictive control, and randomized action at matched sensing/actuation	gain exists only around the plant or depends on uncharged records/control
T7 TUR scope	a registered physical current satisfies an applicable precision--dissipation bound and constrains task-relevant behavior	finite-time/transient variants, hidden-state/non-Markov models, predictive empirical null	process assumptions fail, entropy production is unidentifiable, or task relevance is absent
T8 modularity/mismatch	modules save energy after cross-boundary correlation, traffic, reset, calibration, and prior shift	joint implementation and module system with shared sufficient statistics	advantage assumes the deployment prior or disappears under correlation/shift
T9 locality	sparse/conditional execution lowers IT energy after all movement and capacity terms	dense optimized, structured sparsity, compiler tiling/cache/data reuse	saved arithmetic is offset by bytes, metadata, sync, imbalance, conversion, or idle
T10 facility	workload improvement reduces synchronized facility energy per accepted outcome	matched randomized facility blocks or calibrated side-by-side system	result comes from TDP, generic PUE, short interval, or unstable allocation
T11 lifecycle	operational savings repay incremental embodied burden within supported service life	deployed general hardware, software optimization, and shared specialized service	break-even requires unsupported yield, utilization, demand, lifetime, or displaced-hardware credit
T12 full stack	one candidate-backed composition Pareto-improves quality, latency, risk, capacity, energy, and lifecycle under uncertainty	complete ordinary stack plus boundary and mechanism ablations	no Pareto gain survives held-out regimes, support gates, and required sensitivities

All tracks use the same accepted-task contract and preserve failed devices, rejected requests, timeouts, retries, uncorrectable errors, silent corruption, abstention, idle capacity, maintenance, and replacement in their denominators. Confirmation groups withhold physical devices, fabrication cohorts, waveform and duration regimes, target errors, temperatures, retention horizons, workload families, hierarchy patterns, controller versions, sites, seasons, lifecycle cases, and future time.

The common result vector is

$$\mathbf{Y} = \left(f_{\mathrm{acc}}, Q, L_{0.50}, L_{0.99}, \rho, e^{\mathrm{IT}}, e^{\mathrm{fac}}, e^{\mathrm{life}}, E^{\mathrm{err}}, C^{\mathrm{cap}}, G, W, M\right),$$

where f_{acc} is accepted fraction [dimensionless], Q is task quality [task-native unit], $L_{0.50}$ and $L_{0.99}$ are median and 99th percentile latency [s], ρ is risk [failure/request], the three e terms are energy intensity [J/accepted outcome], E^{err} is error-consequence energy [J], C^{cap} is provisioned capacity [device s], G is greenhouse-gas inventory [kg CO_2e], W is water inventory [m^3], and M is a material/labor vector in declared native units.

A physical-efficiency claim survives only when its simultaneous uncertainty region is no worse on every hard-gated coordinate and strictly better on at least one preregistered primary coordinate against the strongest compatible null across required sensitivity cases. Passing supports only the existing candidate scope named in F-010. Failure identifies the boundary that produced the apparent saving and retires the wider claim.

Sparse prediction and adaptive compute

Scope

This chapter specifies the online control path that decides, for each event, whether to reuse an existing prediction, exit, invoke another module, read memory, acquire another observation, or escalate. Its objective is not minimum FLOPs. It is minimum measured resource use subject to explicit quality, calibration, deadline, and safety constraints.

The runtime has three coupled but distinct jobs:

1. estimate what changed and what remains uncertain;
2. choose the next computation or observation by expected decision value; and
3. allocate physical capacity without making the fast task path unstable.

Training curricula, long-term consolidation, structural pruning, and skill compilation are handled in other chapters. Their products may be invoked here, but this runtime does not silently rewrite slow model state.

Biological observation

The relevant evidence separates demand detection, distributed context, local response, and physical supply.

Activity is selective and regulated

Neural signaling operates under a strict metabolic budget, which constrains the fraction of cortex that can be strongly active at once (C-001). In the *Drosophila* mushroom body, feedback inhibition keeps odor codes sparse and decorrelated; disrupting that loop impairs discrimination between similar learned odors (C-025). These results support selective activity and active inhibition. They do not identify top-k routing, a pruning ratio, or a digital energy saving.

Prediction, uncertainty, and action are different signals

Hierarchical prediction with feed-forward residuals accounts for selected visual-cortical response properties (C-005). Active whisker sensing also shows why a passive residual is incomplete: self-generated sensor mechanics affect the observed neural code (C-022). A residual says that a prediction was wrong. It does not by itself say whether the error is irreducible noise, model ignorance, task-relevant novelty, or a reason to buy another observation.

Broad context is interpreted locally and temporally

Context-dependent disinhibition can change gain in a selected cortical population (C-020). More direct causal work on neuromodulation shows three constraints on a possible artificial abstraction:

- antagonistic signals can sustain different distributed behavioral modes (C-046);
- the frequency and duration of one broad signal can produce different and non-monotonic system responses (C-047); and
- receptor identity and location can turn the same source into transient or sustained local responses (C-048).

The engineering candidate is therefore a rate-limited context broadcast whose receivers own their gains, thresholds, and temporal filters. It is not a central command containing a route for every module. Its evidence, limits, and conventional analogues are developed in the neurodevelopment and global-control audit.

Demand and physical supply can be locally coupled

Presynaptic ATP production can be recruited by local activity (C-049); mitochondrial docking changes resource placement and local synaptic dynamics (C-050); and neural activity can recruit an adjacent vascular response in the studied rodent preparations (C-051). The transferable constraint is modest: measure demand near work, account for placement, and let an infrastructure layer adjust supply. The neurovascular paper's model-derived percentage is not an architecture constant, and none of these cellular results provides a conversion from ATP to accelerator joules.

Taken together, the observations motivate a layered controller: fast local prediction and gating, a narrow context path, and a slower resource plane. The architecture still has to beat standard estimation, decision, routing, and control methods.

Proposed AI translation

Separate the quantities before making a gate

Let y_t be an observation in its declared sensor units and $\hat{y}_{t|t-1}$ the prediction available before observing it. Define the raw residual

$$r_t = y_t - \hat{y}_{t|t-1}.$$

If the predictor supplies a calibrated residual covariance S_t, the normalized innovation is

$$\nu_t = r_t^\top S_t^{-1} r_t.$$

r_t has the units of y_t; ν_t is dimensionless. A large r_t with a large expected S_t may be unsurprising. Conversely, a small residual in a safety-critical variable may justify action. The runtime therefore carries separate fields for:

QUANTITY	MEANING	UNIT OR REPRESENTATION
residual r_t	observed minus predicted state	observation units
normalized innovation ν_t	residual relative to predicted covariance	dimensionless
aleatoric uncertainty	expected irreducible variability	distribution in output units
epistemic uncertainty	model uncertainty or support deficit	calibrated probability or ensemble statistic
task risk	cost of acting or exiting incorrectly	probability and consequence in task units
expected acquisition value $\Delta U_t(a)$	expected decision improvement from action a	declared utility units
estimated energy $\hat{E}_t(a)$	energy for acquisition and downstream work	joules
estimated latency $\hat{L}_t(a)$	completion time including queueing	seconds
traffic $\hat{B}_t(a)$	memory or network movement	bytes

QUANTITY	MEANING	UNIT OR REPRESENTATION
context rate R_b	transmitted global operating context	bits per second

No implementation may collapse all of these into “confidence” and retain the same claim. Residual magnitude, likelihood, expected value, and risk answer different questions.

Observation is affected by action

The acquisition model must also represent how sensing and response change the future observation channel. Operational telemetry can be delayed, incomplete, pooled, or policy-coupled: robust residual methods remain strong nulls (C-124); current counts can omit events that occurred but have not yet arrived (C-126); pooled signals can trade attribution for coverage (C-127); and a fast behavioral proxy can drift with platform, attention, or policy (C-128).

More importantly, an action can alter both hidden state and telemetry. Isolation may reduce propagation while also removing the isolated component's messages; throttling can suppress an error count by suppressing all traffic; a warning can change user behavior; and a new sensor policy changes which events can be observed. The controller therefore versions coverage, delay, ascertainment, observation-model state, and every sensing or response action. A drop in the post-action residual is not independent recovery evidence (C-131).

Candidate 007 tests joint process/observation estimation against residual CUSUM/GLR, nowcasting, maximum coverage, value-of-information sampling, and delay-aware POMDP or model-predictive control. It must use true data vintages, preserve subgroup coverage, estimate counterfactual observation paths, and charge investigation and action capacity as well as compute.

Price a menu of acquisitions

At event t, the system considers a finite action menu $\mathcal{A}_t$:

```
exit | reuse cached state | run local refinement | invoke expert
     | read episodic/factual memory | acquire sensor action | call tool
     | escalate to a larger model or human
```

For acquisition a, estimate the expected improvement in decision utility $\Delta U_t(a)$ and its full energy, latency, and traffic costs. A Lagrangian form of the selection rule is

$$a_t^* = \arg\max_{a \in \mathcal{A}_t} \left[\Delta U_t(a) - \lambda_E \hat{E}_t(a) - \lambda_L \hat{L}_t(a) - \lambda_B \hat{B}_t(a)\right],$$

subject to hard task constraints such as

$$\Pr(\text{unsafe outcome} \mid a, \mathcal{I}_t) \leq \delta, \qquad \hat{L}_t(a) \leq L_{\max}.$$

$\mathcal{I}_t$ is the information legally available when event t is decided, $\delta \in [0,1]$ is the preregistered maximum conditional probability of an unsafe outcome, and $L_{\max}$ is the latency ceiling in seconds. λ_E, λ_L, and λ_B have units utility/joule, utility/second, and utility/byte respectively, converting the three costs into the declared utility scale. When those conversions cannot be defended, use constrained multi-objective selection and report a Pareto surface instead of inventing a scalar score. An action with high information gain but no expected decision benefit receives no credit.

Three runtime timescales

LOOP	TYPICAL CADENCE	STATE OWNED	PERMITTED ACTIONS
event loop	per sensor event or token	prediction, residual, uncertainty, risk, route	no-op, exit, acquire, route, escalate
context loop	one or more explicitly rate-limited cadences	small broadcast plus receiver-local filtered state	change local gain, threshold, mode, or compute eligibility
resource loop	slower than task execution and bounded by actuation delay	queues, utilization, temperature, energy, bandwidth prices	adjust local capacity, apply backpressure, reserve or release routes

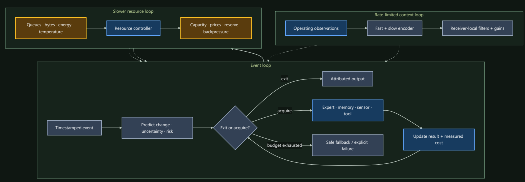
The event loop cannot raise its own power or bandwidth allocation. It requests capacity through the resource plane. The resource plane cannot change task semantics or fabricate confidence; it changes supply and prices. The context loop cannot transmit a per-module command table under the name “broadcast.”

Concrete per-event sequence

1. **Timestamp and classify the event.** Record source, modality, freshness, deadline, and risk class. Missing timestamps make prediction error and queue latency ambiguous.
2. **Run a local change test.** Compare the event with the last accepted local state using a sensor-noise model. Below-threshold change creates a no-op or cached-prediction candidate, not an automatic exit.
3. **Predict before incorporating the event.** Produce $\hat{y}_{t|t-1}$, residual covariance S_t, and the task outputs available at the cheapest depth.
4. **Decompose mismatch.** Compute r_t and ν_t; update a separate sequential change statistic for persistent drift; estimate epistemic uncertainty and task consequence.
5. **Test the cheapest valid exit.** Exit only if calibration for the event's risk stratum satisfies its error bound and no mandatory provenance, freshness, or tool check remains.
6. **Update the context receivers.** If a fast or slow broadcast update is due, each subscribed module applies its own causal filter, bounded gain, and local-state-dependent threshold. The received bitstream and local response are logged.
7. **Value the remaining acquisitions.** Estimate $\Delta U_t(a)$ and full costs for the available experts, memories, sensors, tools, and escalation paths. Reject actions whose expected value does not exceed their priced cost or whose completion misses the deadline.
8. **Route under capacity constraints.** Select the smallest useful set of modules. Include queue delay, expert load, communication, and state placement. Open reserve routes before tail latency collapses only if the measured congestion rule warrants it; the ant result in C-035 is a lead, not the routing algorithm.
9. **Execute with local backpressure.** A module consumes its local compute, memory, and bandwidth budget. Exhaustion triggers deferral, a cheaper approximation, or escalation; it never silently drops a safety check.
10. **Reconcile prediction and cost.** Update output, uncertainty, and the measured resource ledger. If the risk bound is unmet and deadline and budget remain, return to acquisition valuation. The loop has a fixed maximum iteration count.
11. **Emit an attributable result.** Return prediction or action together with confidence calibration version, routes used, evidence provenance, elapsed time, and measured or allocated energy. Runtime telemetry enters later maintenance analysis; it does not directly rewrite slow weights.

Control topology

Wide diagram · fit to page



Editable source: `../assets/diagrams/adaptive-compute-control.mmd`.

Authority follows information and remaining capability

Power-grid protection supplies a hard engineering version of local reflex and global escalation. Local digital relays can act quickly inside a declared zone (C-186), while wide-area schemes add broader but delayed and failure-prone evidence (C-190). Fast response is still constrained by current, headroom, energy, duration, and interacting controls (C-196).

The held translation assigns controller i an admissible action set

$$\mathcal{U}_i(t) = \mathcal{E}_i(\tau_i(t), q_i(t), m_i(t), b_i(t), c_i(t)), \qquad u_i(t) \in \mathcal{U}_i(t),$$

where τ_i is observation age in seconds, q_i is integrity state, m_i is operating-mode state, b_i is remaining physical or compute headroom, c_i is coordination availability, $\mathcal{E}_i$ is a certified set-valued map, and u_i is the proposed action. The vector components retain their own units, timestamps, uncertainty, and provenance; the notation does not make them interchangeable.

Stale evidence, lost integrity, exhausted reserve, or lost coordination should shrink authority toward an independently enforced fallback. A wider action requires validated handoff and a checked postcondition. The candidate loses if adaptive protection, gain scheduling, constrained control, barrier functions, or runtime assurance reproduce the same frontier. Candidate 012 tests this under saturation, hidden failure, attack, communication loss, partition, second events, and staged recovery.

Context is a constrained interface

The context candidate sends a small quantized code at declared fast and slow rates. Receivers expand it locally through distinct filters and bounded transfer functions. This is useful only when receiver-local structure reconstructs module-specific responses more cheaply than transmitting them.

Candidate 002 tests that proposition against instantaneous and shared-filter FiLM, memoryless and recurrent gates, bottleneck and standard global tokens, low-rank hypernetworks, fixed multirate receiver banks, and gain-scheduled supervisory control. Its primary channel is two 8-bit components at different cadences, with an average logical rate of $825\ \mathrm{bit\,s^{-1}}$. That number is an experiment setting, not a proposed universal bus rate.

Deficit travels; feasibility remains local

Plant nitrogen acquisition supplies a precise three-part allocation loop: nitrogen-starved roots send CEP deficit signals upward, shoot-derived CEPD signals return global context, and high-affinity uptake increases only where a root also encounters nitrate (C-207). This is not evidence for a new controller. Backpressure and primal–dual allocation are the strongest nulls.

For module i, let $d_i(t)$ be unmet task demand per second, $a_i(t) \in [0, 1]$ be local capability availability, $u_i(t)$ be allocated compute-seconds per second or watts, and $B(t)$ be the total budget in the same unit as u_i:

$$\sum_{i=1}^{n} u_i(t) \le B(t), \qquad u_i(t) = 0 \text{ when } a_i(t) = 0.$$

The possible advantage is communication structure: compress deficits upward, return a small scarcity code downward, and keep the expensive feasibility decision at the receiver. It should lose when complete state is fresh, capability is uniform, or backpressure already carries the required deficit and feasibility information more cheaply. Candidate 013 tests moving demand and resource patches, delay, topology churn, strategic over-reporting, reversible allocation, slow growth, and second events.

Engineering null models

Biological language is removed when an established method explains the result at the same cost.

PROPOSED ROLE	NULL MODEL THAT MUST BE INCLUDED	RESULT IF NULL MATCHES
detect unexpected input	calibrated Kalman innovation or likelihood residual	call it residual gating
distinguish a persistent change	Page/CUSUM accumulate–reset detector	call it change detection
decide whether to buy more information	one-step expected value of sample information	call it value-of-information control
skip depth on easy cases	calibrated intermediate classifier / early exit	call it adaptive depth
select capacity	top-k mixture-of-experts with load balancing	call it conditional routing
stabilize utilization or queue load	tuned PI and Kelly-style primal/dual allocation	call it feedback resource control
distribute operating context	FiLM, recurrent gates, learned global tokens, and hypernetworks	retain the simplest winning conditioner
separate fast and slow control	tuned multirate or gain-scheduled supervisory controller	call it multirate control
place recurring state near work	static placement, LRU/cache policy, and NUMA-aware scheduling	call it caching or placement

The equations and unit requirements for innovation, value of information, CUSUM, allocation, and energy are recorded in the engineering-analogue audit.

Adaptive resolution must be target- and regime-qualified

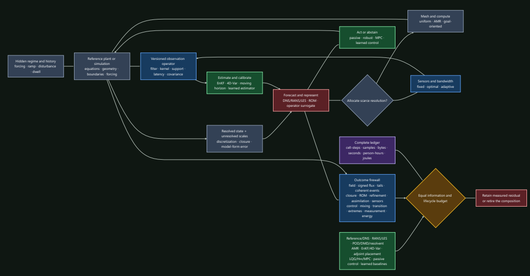
Fluid dynamics is a hostile test for “allocate more compute where prediction error is high.” The governing equations can be known while the useful state is still limited by unresolved scales, boundary and forcing uncertainty, discretization, closure error, partial observation, chaotic amplification, regime change, intermittency, and rare extremes. The audited evidence in C-881–C-925 therefore tightens the allocator contract.

An allocation signal is valid only relative to a declared target and measurement identity:

- **variable and support:** which physical or latent variable, spatial region, temporal window, and averaging kernel define the target;
- **filter and detector:** which scale, threshold, reference frame, event detector, and observation operator define a residual or coherent event;
- **signed relation:** whether energy, information, error, or another invariant should move toward larger or smaller scales under the current regime;
- **model boundary:** which part is resolved dynamics, closure, discretization, observation error, and out-of-support model-form discrepancy;
- **quantity of interest:** whether the reduced state must preserve average field reconstruction, control authority, transition timing, mixing, or an extreme tail; and
- **regime and history:** geometry, forcing, ramp direction, dwell, disturbance, prior occupancy, sensor drift, actuator condition, and current uncertainty.

A mean residual can be small while the signed scale flux is wrong, the tail is miscalibrated, or the rare event that matters is missed. A high-energy reduced basis can discard a weak direction that controls transition or actuation. A mesh can use fewer cells at one instant while spending more on regridding, subcycling, synchronization, load imbalance, data transfer, and failed solves. These are different failures and stay separate in the outcome vector.

Wide diagram · fit to page



Editable source: regime-qualified-flow-inference-control.mmd.

Fixture F-005 crosses ten adversarial tracks: signed multiscale transfer, closure portability, target-qualified reduced state, adaptive resolution, assimilation, sensor placement, closed-loop control, mixing, path-dependent transition, and extreme prediction. Its decisive comparator is a complete composition of reference simulation, classical RANS/LES closures, POD/DMD/resolvent or balanced reduction, goal-oriented AMR, EnKF/4D-Var and moving-horizon estimation, adjoint/Fisher/Gramian sensor placement, robust/MPC/passive control, and learned operators or policies.

The flow mathematics binds every result to geometry, equation, boundary, forcing, solver, grid, measurement, filter, detector, data lineage, actuator, target, regime, and resource identity. An adaptive mechanism earns credit only when it improves its preregistered target beyond that complete stack without reversing flux, losing stability, undercovering uncertainty, missing the sealed natural tail, or producing negative net lifecycle energy. Otherwise the ordinary method remains and the proposed composition is retired.

Efficiency mechanism

Sparse work has four independent levers

The runtime can save resources by:

1. suppressing unchanged events;
2. exiting at a cheaper validated depth;
3. activating fewer modules or memory paths; and
4. avoiding movement by keeping repeated work near its state.

These levers are not interchangeable. An early exit can reduce arithmetic while leaving input and cache traffic unchanged. Sparse experts can reduce active weights while increasing all-to-all communication. Local placement can reduce bytes moved without reducing operations.

For module gates $g_i(x) \in [0, 1]$ and declared operation costs $c_i(x)$, define the operation-weighted active fraction

$$\rho_{\text{ops}}(x) = \frac{\sum_i g_i(x) c_i(x)}{\sum_i c_i(x)}.$$

Also report byte-weighted activity ρ_{bytes} from measured memory and network traffic. Neither quantity is an energy estimate.

Charge every control path

Over the same measurement interval,

$$E_{\text{adaptive}} = E_{\text{predict}} + E_{\text{gate}} + E_{\text{route}} + E_{\text{active}} + E_{\text{memory}} + E_{\text{network}} + E_{\text{context}} + E_{\text{control}} + E_{\text{idle}}.$$

Every term is in joules. The runtime produces an efficiency gain only when

$$E_{\text{adaptive}} < E_{\text{dense}}$$

at matched quality, risk, latency class, batch opportunity, and hardware boundary. Report board, node, and facility energy separately when measured; do not infer one from another.

State movement and controller work must also amortize. For a placement, compilation, or cache migration completed N times, let E_{runtime} be incremental runtime energy in joules per completed use; let E_{discover}, E_{migrate}, and E_{validate} be one-time energies in joules; and let N be a dimensionless completed-use count. Then

$$\bar{E}_{\text{use}} = E_{\text{runtime}} + \frac{E_{\text{discover}} + E_{\text{migrate}} + E_{\text{validate}}}{N}.$$

$\bar{E}_{\text{use}}$ is joules per completed use. If N is not observed, report the break-even reuse count rather than claiming a saving.

Resource supply is a feedback problem

For one optimization instance, choose a single rate basis—requests per second, tokens per second, or bits per second—and let x_i be module i's allocation in that unit. Let $U_i(x_i)$ be a nondecreasing concave utility normalized to a common dimensionless scale, let R be a dimensionless resource-incidence matrix, and let each component of c be capacity in the same chosen rate unit. A conventional allocation null is

$$\max_{x \geq 0} \sum_i U_i(x_i) \quad \text{subject to} \quad Rx \leq c.$$

The slower resource plane observes queues, deadlines, utilization, energy, and temperature, then adjusts capacity or prices. Any learned controller must beat a tuned PI or primal/dual implementation on settling time, overshoot, constraint violations, tail latency, and joules per control update. “Local metabolism” is not a substitute for that comparison.

Evidence status

COMPONENT	CURRENT SUPPORT	STATUS FOR THIS ARCHITECTURE
strict biological activity budget	C-001	established biological constraint; digital magnitude unassigned
sparse conditional capacity	C-003	established AI mechanism; end-to-end benefit workload-dependent
input-dependent early exit	C-004	established on evaluated BERT tasks; risk-stratified generality open

COMPONENT	CURRENT SUPPORT	STATUS FOR THIS ARCHITECTURE
cortical residual hierarchy	C-005	plausible engineering lead, not a universal brain objective
event-driven on-chip learning	C-015	feasible on one published substrate; superiority not established
slower activity stabilization	C-018	established in cultured neurons; artificial set point open
context-dependent local gain	C-020	established for the studied mouse circuit; routing value open
active sensing mechanics	C-022	established for the whisker preparation; acquisition policy open
inhibitory sparse discrimination	C-025	established for the fly circuit and task; hardware benefit open
congestion-triggered reserve route	C-035	established in the ant experiment; conventional routing remains the null
multirate broadcast and local decoding	C-046–C-048	biological observations established; AI mechanism is Candidate 002
local demand, placement, and supply	C-049–C-051	cellular observations established; artificial resource plane untested
sparse, delayed, pooled, and policy-coupled surveillance	C-122–C-131	scoped epidemiological and statistical evidence; AI-system translation is Candidate 007
joint process/observation estimation with action provenance	C-132	speculative composition against POMDP, detection, nowcasting, and value-of-information nulls
regime-qualified closure, reduction, assimilation, refinement, and control	C-881–C-925	scoped fluid evidence and formal relations; integrated allocator remains Fixture F-005

The integrated runtime is speculative until its gates are tested separately and then recombined under one measurement boundary. A combined win cannot identify which control path caused it.

Speculative extensions

Residual queues instead of global lockstep

Each module could own a queue of unresolved residuals and wake only when local work, context, or a deadline changes its priority. The comparator is an efficient batched scheduler with the same queue and dispatch overhead. The extension is abandoned if asynchronous launches reduce utilization or increase p95 latency enough to erase skipped work.

Predictive placement

Frequently reused expert weights, key–value state, and memories could move toward their expected consumers. The hypothesis is about reuse-distance and migration amortization, not artificial mitochondria. Static placement, LRU, and NUMA-aware policies remain required baselines; every migration is charged in bytes, seconds, and joules.

Controlled recovery probes

A maintenance observer could inject small bounded perturbations and measure recovery time, overshoot, and restoring margin before ordinary quality metrics fail. This is specified in Candidate 003. Recovery diagnostics do not themselves stabilize the runtime, and their probe energy and task disturbance remain part of the cost.

Cross-modal acquisition brokerage

A common valuation layer could choose between deeper internal computation, a memory lookup, another sensor view, a physical action, a tool call, or a human query. Each option needs a calibrated outcome model and a common decision contract. Entropy reduction alone is not sufficient because an observation can be surprising yet irrelevant to the pending action.

Failure modes

FAILURE	OBSERVABLE SIGNATURE	REJECTION OR CONTAINMENT RULE
residual–uncertainty conflation	high-noise inputs always buy more compute without improving decisions	compare against normalized innovation and EVSI; reject residual-only gate
overconfident early exit	average accuracy holds while rare or shifted strata fail calibration	impose risk-stratified exit bounds and safe fallback
router collapse	a few experts saturate, queues and p95 latency rise, reserve capacity idles	compare load-balanced MoE and constrained allocation; reject unstable router
granularity below hardware break-even	active FLOPs fall but kernel count, bytes, latency, or joules rise	coarsen gates or return to dense fused execution
broadcast becomes a hidden router	context rate, token width, or receiver traffic scales with module count	enforce quantization, cadence, and logged bit/byte budgets
receiver saturation or mode lock	stronger context produces non-monotonic failure, unrecoverable hysteresis, or uniform module response	bound gains, test impulse responses, retain redundant fallback
positive feedback between routing and supply	busy modules receive more capacity, attract more traffic, and monopolize service	separate router and resource objectives; test step, burst, and delay stability
controller oscillation	periodic queue, power, or route changes with excess settling time	include anti-windup and delay-aware PI/primal-dual nulls; reject learned loop if dominated
resource-proxy gaming	predicted “useful work” rises while task utility per joule falls	reconcile estimates against measured outcome and energy
state-migration thrash	repeated placement changes exceed saved memory traffic	require hysteresis and observed break-even reuse count
shortcut no-op	unchanged-input gate suppresses slow but important drift	maintain persistent-change statistic and scheduled freshness checks
accounting boundary leak	reported savings omit host, network, idle, context, or control energy	withhold efficiency claim until the missing boundary is measured
mean-fit tail failure	average field error falls while signed flux, transition, or extreme-event calibration worsens	retain the full outcome firewall; reject pooled-score improvement
closure/numerics cancellation	a learned residual wins only on one solver/grid and degrades under refinement	separate closure, discretization, and model-form support; test hidden solver and grid lineages
adaptive-resolution bookkeeping leak	cell count falls while regrids, rejected steps, transfers, imbalance, or sensing dominate	compare complete work, wall time, bytes, and joules at equal target error
control saving without net saving	task drag or loss falls but actuation, sensing, compute, auxiliary, installation, or maintenance erase it	report service-interval net energy and reject non-positive benefit

Measurable predictions

These are experiment commitments. Directional improvements require paired uncertainty intervals and a predeclared practical margin; a lower theoretical operation count is not a pass.

ID	INTERVENTION AND COMPARATOR	PRIMARY MEASUREMENTS	PREDICTION AND FAILURE BOUNDARY
AC-01	calibrated early exit versus fixed depth and uncalibrated confidence threshold	error and calibration by risk stratum; layers/item; J/item; p95 ms	lower J/item at equivalent high-risk error and calibration; reject if gains come from degraded rare-event performance
AC-02	residual/uncertainty/EVSI acquisition policy versus raw residual threshold, Kalman innovation, CUSUM, and one-step EVSI	downstream utility; acquisitions/item; false escalations/item; bytes/item; J/item	candidate must move the utility–energy–latency frontier beyond the strongest composed null; otherwise use the null
AC-03	sparse expert and memory routing versus dense execution and load-balanced top-k MoE	active operations; memory and network bytes; queue depth; p95 ms; J/item	physical bytes and joules must fall with active operations at matched quality; a FLOP-only reduction fails
AC-04	Candidate 002 multiscale receiver versus FiLM, GRU gate, global token, hypernetwork, and multirate control	task error; impulse-kernel NRMSE; bit/s; bytes/step; J/step; p95 ms	require at least 5% task and 10% kernel improvement over the best equal-interface simple baseline, then task equivalence to the higher-bandwidth method with at least 25% fewer context bytes or 5% less energy and no more than 5% p95-latency increase
AC-05	local resource plane versus tuned PI, Kelly/primal-dual, and centralized fixed allocation	settling s; overshoot %; violations/s; utilization %; J/control update; J/item	learned/local control must improve a quality–tail-latency–energy frontier under burst and delay; equality means merge into conventional control
AC-06	predictive placement versus static placement, LRU, and NUMA-aware scheduling	migration bytes; cache misses/item; p95 ms; J/item; break-even reuse count	placement wins only after migration and validation amortize within observed reuse; otherwise retain the conventional policy
AC-07	difficulty-controlled input sets with equal length but varied ambiguity, relevance, and risk	acquired work/item; J/item; task utility; calibration	work should track expected decision value and risk, not length or irrelevant noise; extra work without utility improvement falsifies the allocator

ID	INTERVENTION AND COMPARATOR	PRIMARY MEASUREMENTS	PREDICTION AND FAILURE BOUNDARY
AC-08	distribution shift and rare-event stress with every adaptive gate enabled	worst-stratum error; expected calibration error; missed-hazard probability; safe-fallback rate; J/item	adaptive savings must survive the declared risk bounds; a favorable average with a worse safety tail fails
AC-09	Fixture F-005 complete regime-qualified composition versus its strongest classical/learned closure, ROM, AMR, assimilation, sensor, and controller stack	target-native field/flux/tail/event errors; calibration; stability; complete cell-steps, bytes, person-hours, and J/run	require a preregistered target improvement under hidden regime, solver, grid, observation, and hardware changes without degrading signed flux, sealed-tail calibration, stability, or net lifecycle energy; a tie retires the composition

Mechanism ablations are interpreted selectively. Removing the exit gate should increase depth without changing routing identity; removing sparse routing should increase active modules and traffic; removing the context fast path should selectively damage transient response; removing the slow path should selectively damage sustained response; freezing resource control should worsen burst recovery rather than semantic accuracy. If every ablation merely lowers capacity and hurts everything, the architecture has not isolated its claimed control loops.

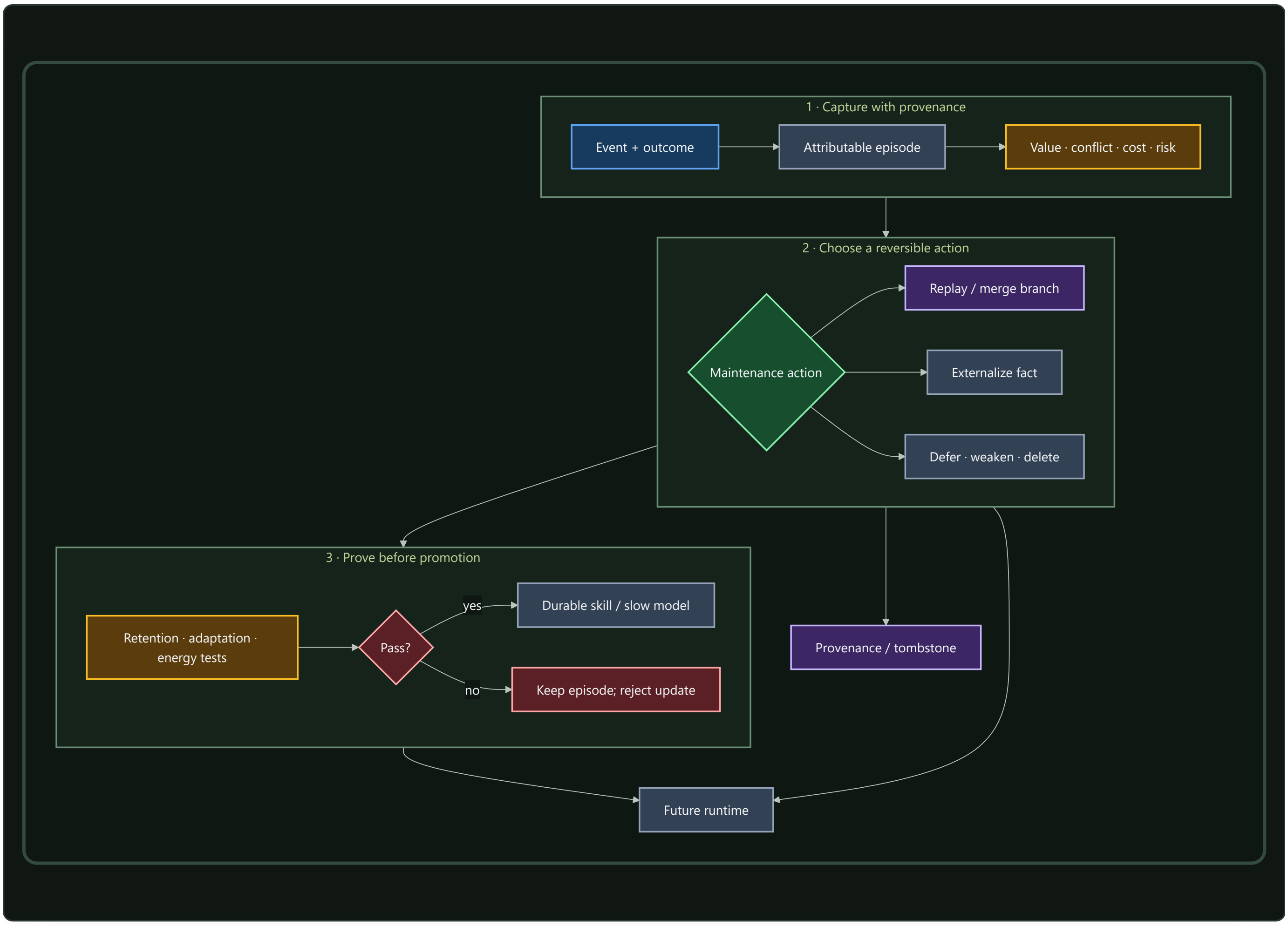
Fast memory, replay, and consolidation

> *Experience should change behavior immediately without receiving immediate permission to rewrite stable capability.*

Scope

Memory is a lifecycle: capture an event, preserve its origin, decide whether it deserves more work, test an integration, then retain, transform, externalize, weaken, or delete it. The objective is rapid adaptation without granting every surprising event permission to rewrite stable capability.

One memory lifecycle



Editable source: `../assets/diagrams/memory-lifecycle.mmd`.

The lifecycle separates two decisions often collapsed into "learning":

1. **What should be remembered now?** Capture is fast, attributable, and reversible.
2. **What should change the durable system?** Consolidation is selective, tested, and budgeted.

Biological observation

Complementary Learning Systems theory describes interacting fast hippocampal and slower cortical learning processes (C-008). The useful pattern is not only different storage speeds. It is controlled transfer between a rapidly changing record of experience and structure whose value depends on remaining stable.

The evidence also makes replay a selection problem:

- disrupting replay from a selected hippocampal assembly can selectively impair the associated rodent spatial memory (C-036);
- replay allocation varies with reward, learning, familiarity, and memory weakness rather than following one universal priority (C-037);
- existing relational structure can accelerate integration of compatible associations (C-038);
- retrieval can make an established memory temporarily update-sensitive (C-039), although the exact human prediction-error gate remains disputed (C-040); and
- forgetting can be actively regulated by neural and glial mechanisms in specific preparations (C-041, C-042).

Together they motivate a maintenance controller that allocates limited work across capture, replay, integration, protection, and forgetting.

Proposed AI translation

1. Separate stores by write authority

STORE	UPDATE RATE	CONTENT	NORMAL MUTATION
Working state	every event	active context, goals, predictions	overwritten freely
Episodic store	rapid	sourced trajectories, outcomes, errors	append, expire, redact
Slow model	controlled	reusable representations and skills	validated consolidation
Factual store	independent	mutable, attributable propositions	explicit versioned update

The separation is about authority, not hardware. A vector store, database, recurrent state, adapter, and model weights may share a device while obeying different write policies. One logical store may also span devices when locality or retention requires it.

2. Capture evidence before abstracting it

Each episode records enough context to explain a later update:

- observation and relevant prior state;
- action, answer, or intervention;
- outcome and uncertainty;
- data and tool provenance;

- active modules and retrieved memories;
- physical telemetry; and
- privacy, retention, and safety constraints.

The episode need not contain every hidden activation. It must preserve enough attributable evidence to reproduce, challenge, or reverse the lesson later derived from it.

3. Allocate the maintenance budget

At a maintenance window, the controller estimates what each candidate action could improve and what it would cost. Novelty, reward, uncertainty, familiarity, conflict, interference, and schema fit are features—not interchangeable definitions of importance.

The first controller should earn its complexity against ordinary policies:

1. uniform reservoir replay;
2. recency;
3. loss- or TD-error priority;
4. interference priority;
5. schema-fit priority; and
6. the proposed multi-signal lifecycle policy.

Every method receives the same episode bytes, replay examples, optimizer updates, wall time, and energy boundary. This makes scheduling policy—not extra maintenance—the independent variable.

4. Branch, test, then promote

Retrieval or replay opens a versioned candidate branch. It never makes the durable state writable by itself. The branch may modify a local adapter, module, route, representation, or factual record and is then tested for:

- retention of protected historical capability;
- acquisition of the proposed new capability;
- calibration and rare-case behavior;
- provenance and conflict handling;
- measured energy, bytes moved, and latency; and
- reversibility after rejection.

Replay and Elastic Weight Consolidation are distinct comparison mechanisms (C-009, C-010); neither is privileged as the final protection policy.

A passed branch can merge into the slow model, remain provisional, or enter the maturity and structural-consolidation lifecycle. A failed branch returns the episode with its negative result attached so the same invalid integration is not proposed indefinitely.

5. Forgetting is an action

Unbounded retention consumes storage, retrieval bandwidth, replay capacity, and attention. The controller therefore distinguishes:

ACTION	EFFECT	REQUIRED SAFEGUARD
Defer	preserve without more work	future reconsideration rule
Replay	spend work to test or strengthen	equal replay budget
Merge	compress compatible state	provenance survives compression

ACTION	EFFECT	REQUIRED SAFEGUARD
Externalize	move mutable information out of weights	source and version retained
Weaken	reduce retrieval or routing influence	rare-case regression probes
Delete	remove active state	reconstructable source or explicit retention exception

Weakening is separate from deletion because many obsolete or interfering memories should first lose influence while evidence accumulates. A tombstone or provenance record prevents deleted state from becoming an unexplained absence.

Storage contracts precede memory policy

Persistent state has several independent contracts:

CONTRACT	QUESTION	DOES NOT ESTABLISH
atomicity and recovery	which in-scope effects commit or abort after failure?	isolation, outside-effect reversal, or truth
isolation	which concurrent histories are visible?	real-time order or durability
durability	what survives acknowledged completion under the fault model?	correctness or indefinite retention
version visibility	which snapshot or temporal coordinate answers a read?	serializability or current-world truth
replication/coding	which exact bytes or log survive named faults?	independent judgment or semantic diversity
indexing/caching	where is a candidate copy found cheaply?	importance, authority, or source-of-truth status
retention/reclamation	which roots, readers, holds, and horizons keep state live?	future irrelevance or epistemic worth

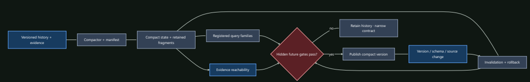
These boundaries are established by storage theory and systems evidence in C-325–C-338. Versioning alone does not make a history serializable; a durable statement can remain false; an agreed log can preserve a bad command; and event replay can change meaning when handlers, schemas, configuration, nondeterministic inputs, or outside effects are not version-compatible.

Valid time and system time remain separate. The first records when a proposition is asserted to hold in the modeled world; the second records when the store held that version (C-337). Capture, receipt, processing, correction, and supersession may add more clocks.

Semantic compaction has a finite preservation contract

Physical compaction preserves a declared storage view; it is not automatically semantic consolidation (C-334, C-341). A semantic compactor may replace history H with $Z = C_\phi(H)$ only under registered query, evidence, uncertainty, rollback, invalidation, deletion, and failure-degradation obligations.

Wide diagram · fit to page



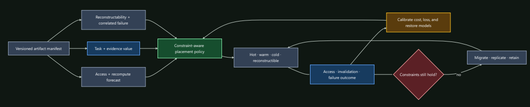
Editable source: semantic-compaction-contract.mmd.

Candidate 017 freezes the compactor before generating hidden future queries and invalidation requests. It compares with indexed history, snapshots plus suffix logs, materialized views, key compaction, lossless compression, extractive evidence summaries, and cold archives. Unsupported questions must be exposed, not filled with invented detail.

Placement separates access, value, and reconstructability

Recency and frequency are strong baselines, not definitions of importance (C-333). A rare cold artifact may still block a safety audit, source invalidation, rollback, or recovery. Conversely, a hot derived view may be cheap to reconstruct. The placement controller therefore keeps access forecast, recomputation cost, task loss, evidence value, staleness, and correlated-failure reconstructability as separately calibrated quantities.

Wide diagram · fit to page



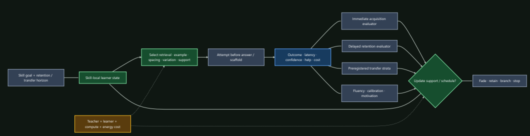
Editable source: value-aware-artifact-tiering.mmd.

Candidate 018 must beat LRU/LFU, ARC, W-TinyLFU, size- and miss-cost-aware caching, economic tiering, static optimization, and coded fault-domain placement without future labels. Physical bytes, movement, metadata, endurance, p99 latency, policy work, privacy, and failure costs remain inside the boundary.

Learning is a horizon-indexed outcome vector

Immediate task success does not identify durable learning. A curriculum event can improve acquisition while weakening delayed retention, novel transfer, fluency, calibration, motivation, or total efficiency. The evaluator therefore keeps those outputs separate and makes retention horizon and transfer distance part of the claim. The evidence boundaries are C-627–C-658.

Wide diagram · fit to page



Editable source: horizon-qualified-learning.mmd.

For each skill, retain current acquisition estimate, retention estimates at declared horizons, transfer by preregistered stratum, fluency/error frontier, calibration, support/scaffold state, confusability/context, intervention history, state version, and uncertainty. Difficulty is useful only when the attempt is processed successfully enough to generate information; failure is not beneficial merely because it was hard.

Candidate 004 must compare the adaptive sequence with tuned spacing, ordinary knowledge tracing, fixed fading, hard-example mining, and fixed curricula at equal attempt, feedback, time, storage, compute, and energy. Candidate 019 must compare interactive teaching with a versioned artifact plus tests, then measure retention and novel transfer after actual learner/model turnover while charging both teacher and learner effort. The learning-outcome mathematics defines the common evaluator.

6. Follow one event through the lifecycle

Suppose a tool returns a surprising result. Working state can use it immediately, while an attributable episode preserves the action, outcome, uncertainty, source, and active path. The slow model does not change yet. Related and conflicting outcomes accumulate until a maintenance window judges the episode worth replaying. Replay opens a local branch and tests it against protected history.

What happens next depends on the content. A reusable operation may become a skill or provisional module. A mutable proposition belongs in factual memory. Noise loses influence or expires under the retention policy. Later outcomes measure whether that choice was correct and recalibrate the scheduler. This is the feedback that turns a storage hierarchy into a lifecycle.

Efficiency mechanism

Online operation avoids immediate full-model gradient updates. Maintenance spends that work later and selectively, where expected future value justifies the physical cost.

For N_{served} events between maintenance windows, amortized energy per served event is

$$\bar{E}_{\text{event}} = E_{\text{online}} + \frac{E_{\text{maintenance}}}{N_{\text{served}}},$$

where E_{online} is runtime energy per event in joules and $E_{\text{maintenance}}$ includes scheduler, replay, memory movement, optimization, validation, and recovery energy in joules. A system saves energy only when the second term remains below the online work avoided by delayed and selective integration.

The full constrained action model—including joule, byte, second, and optimizer update budgets—is defined in `../math/memory-lifecycle.md`.

Evidence status

MECHANISM	EVIDENCE	CURRENT STATUS
Fast/slow learning split	C-008	established theory and supporting results; system translation incomplete
Interference protection	C-009	demonstrated in scoped sequential tasks
Replay for machine consolidation	C-010	plausible mechanism with task-specific evidence
Content-specific replay	C-036	established in the measured rodent intervention
Multi-signal replay allocation	C-037	established constituent observations; unified policy experimental
Schema-sensitive integration	C-038	established in scoped learning conditions
Retrieval-induced update window	C-039, C-040	lability established narrowly; exact human mismatch gate disputed
Active forgetting	C-041, C-042	established in scoped interventions; safe AI policy untested
Storage and temporal contracts	C-325–C-338	mature engineered mechanisms; mandatory nulls and vocabulary
Semantic compaction and value-aware tiering	C-339–C-342	held residual experiments under Candidates 017 and 018
Complete lifecycle controller	none	speculative synthesis

Speculative extensions

- Generate counterfactual variants around high-value episodes rather than only replaying recorded inputs.

- Learn expected knowledge gain per joule while retaining hard resource and safety bounds.
- Perform module-local consolidation first and synchronize globally only when a cross-module invariant changes.
- Use exact checkpoints to test several consolidation outcomes in parallel and retain the cheapest one that passes.
- Learn memory placement jointly with replay policy so frequently paired state becomes physically local.

Failure modes

- Replay amplifies biased, adversarial, or privacy-sensitive episodes.
- The scheduler starves quiet, rare, or safety-critical memories.
- A correlated shortcut is mistaken for schema compatibility.
- Generated replay drifts away from the environment.
- Retrieval becomes an adversarial write primitive.
- Weakening or deletion removes evidence later required for recovery or audit.
- A compact summary passes familiar queries but loses evidence, invalidation, rollback, deletion, or rare hidden-future obligations.
- A placement policy calls access frequency “importance,” leaks future value, or treats correlated replicas as independently reconstructible.
- Scheduler scans and telemetry consume the saved maintenance budget.
- The episodic store becomes an unbounded duplicate of the training corpus.

Measurable predictions

1. Fast attributable memory reduces adaptation latency without increasing protected slow-model regression.
2. A multi-signal scheduler improves the retention–adaptation–energy frontier beyond uniform, recency, loss-priority, and interference-priority baselines at equal replay count and bytes moved.
3. Schema-compatible episodes require fewer optimizer updates to integrate than violations while shortcut-controlled transfer remains unchanged or improves.
4. Explicit weakening reduces obsolete-memory intrusions without exceeding the declared rare-case deletion bound.
5. Separating mutable propositions from reusable skills reduces correction cost and unsupported factual carryover.
6. Maintenance energy amortized per served event remains below the online training work it replaces.
7. Registered semantic compaction reduces physical bytes without reducing hidden-query, evidence, rollback, or invalidation coverage below threshold.
8. Value/reconstructability features improve shifted-workload artifact placement beyond strong cache and storage policies after migration cost.

Maturity, grokking, and reversible structural consolidation

Scope

Define how a useful but plastic structure becomes protected, cheaper to run, and eventually eligible for pruning—without mistaking age, low training loss, or a grokking curve for proof of maturity.

Maturity is a lifecycle state, not a compliment. A mature module has earned a narrower update surface because its behavior is understood inside a declared validation envelope. That protection must remain reversible when the envelope changes or the module becomes brittle.

Biological observation

Development does not produce a final circuit by training every connection at a constant rate forever. In the studied mouse retinogeniculate preparation, relative activity and complement signaling affected microglial engulfment and retention of developing inputs (C-043). The useful abstraction is a slow maintenance process that helps refine structure under local evidence; it is not a universal deletion rule.

Mature circuits can also carry local constraints on further change. Targeted interventions on extracellular structure and a cholinergic brake reopened specific forms of adult visual-cortex plasticity (C-044, C-045). Protection and plasticity are therefore not opposite endpoints. They can be different operating states of the same structure, controlled locally and revisited under an explicit intervention.

Two results from ecological systems sharpen the control problem. Recovery from small perturbations slowed as one cyanobacterial microcosm approached a controlled tipping point (C-058); warning statistics also appeared during a manipulated whole-lake food-web transition (C-059). A system may look acceptable at rest while its restoring dynamics weaken. Maturity therefore cannot be certified by steady-state accuracy alone.

Nor is resilience one number. Greater species richness improved temporal stability but reduced resistance to warming in a large ciliate-microcosm experiment (C-060). In a separate microbiome analysis, functional redundancy was associated with resistance to newcomer engraftment (C-057). Stability, resistance, recovery, and capacity to admit a better replacement can move in different directions.

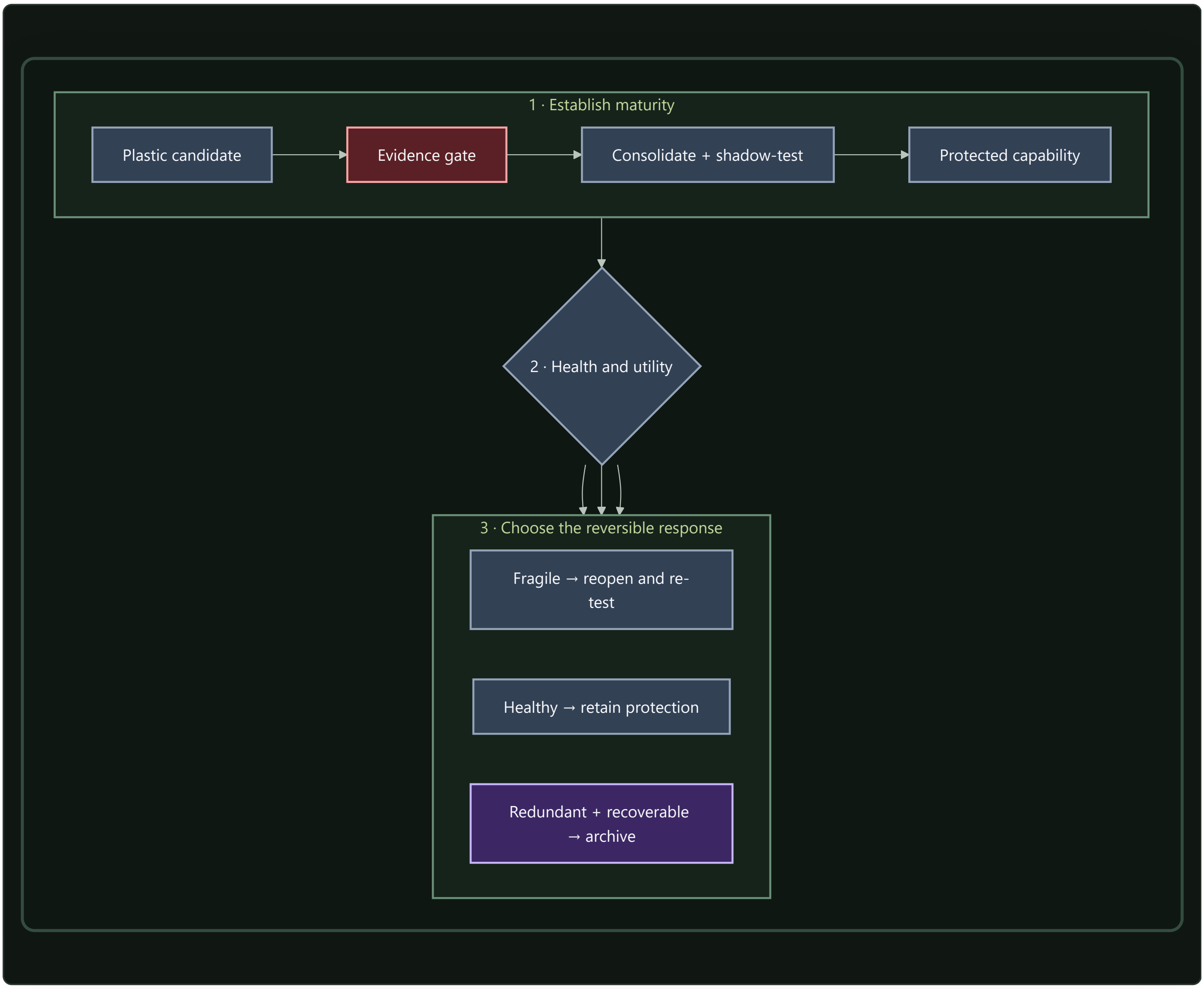
This makes grokking a useful phenomenon but a poor gate. Delayed generalization has been observed in bounded algorithmic settings, yet the claim that extended training reliably reveals the underlying rule and certifies readiness to prune is disputed (C-011). The system needs evidence about what a structure does, how uniquely it contributes, how it fails, and how it recovers.

Proposed AI translation

A reversible maturity lifecycle

Every structurally changeable module, route, memory transform, or compiled path has an explicit lifecycle state:

1. **Candidate:** highly plastic, attributable to its training episodes, and cheap to discard.
2. **Consolidating:** replayed against related, conflicting, rare, and intervention cases while its unique contribution is measured.
3. **Protected:** update rate and writable surface are reduced; structural changes require a versioned branch and shadow evaluation.
4. **Reopened:** a copy-on-write branch receives bounded adaptation while the protected version remains available for comparison and rollback.
5. **Retiring:** traffic is drained only after another path covers the required behavior and the physical execution graph can actually be compacted.
6. **Archived or removed:** provenance, validation envelope, and a reconstructable checkpoint remain for the declared retention period; hot execution state is released.



Editable source: `../assets/diagrams/maturity-fragility-cycle.mmd`.

The lifecycle separates three decisions often collapsed into “pruning”:

- **protect:** this path is useful and should stop drifting;
- **reopen:** this path no longer responds adequately inside its required envelope; and
- **retire:** this path is no longer uniquely useful and can be removed without making the system irrecoverable.

Protection is not retirement. A path may be mature precisely because it is important enough to preserve.

Maturity is a vector gate

For module i, maintain a maturity record

$$\mathbf{m}_i = (Q_i, U_i, S_i, F_i, A_i, C_i, P_i)\,,$$

where:

- Q_i is quality across the declared validation envelope;
- U_i is unique causal contribution under ablation and rerouting;
- S_i is behavioral and routing stability across time and environments;
- F_i is fragility measured from recovery and margin estimates;
- A_i is adaptation and newcomer-acceptance behavior under controlled shift;
- C_i is full-lifecycle physical cost; and
- P_i is provenance and rollback completeness.

The record remains a vector. A weighted score may rank candidates for review, but it cannot hide a failed safety, provenance, recovery, or rollback constraint. Thresholds are workload-specific and include uncertainty intervals; there is no universal maturity age, pruning percentage, or recovery constant.

Protection requires evidence that the module is useful, stable, attributable, and recoverable. Retirement reverses one condition: its *unique* contribution must be low because another tested path covers its role. Low weight magnitude or low average routing frequency is not enough.

Structural consolidation before deletion

When several routes repeatedly implement the same stable computation, the maintenance plane first tries to make their shared work explicit:

1. identify the recurrent subgraph and its boundary contract;
2. build a compact candidate through structured pruning, distillation, compilation, quantization, fusion, or relocation;
3. replay both common and conflicting cases through old and new graphs;
4. intervene on each source module to measure residual unique behavior;
5. shadow the compact path under live-like traffic;
6. drain old routes gradually while keeping rollback state; and
7. release tensors, optimizer state, routing entries, and communication only after the observation window passes.

Iterative pruning can reveal competitive sparse subnetworks in its tested settings (C-012). Here it is one operator inside the lifecycle, not the lifecycle policy itself. Magnitude pruning is a baseline; causal coverage and end-to-end physical savings decide promotion.

Reopening without overwriting the canonical path

A protected path is reopened only after a persistent signal, such as:

- calibrated error or intervention failure outside its historical variance;
- recurring novelty that existing candidates cannot absorb without interference;
- loss of recovery margin despite acceptable steady-state quality;
- repeated rollback or escalation around the same boundary; or
- evidence that protected redundancy prevents a superior newcomer from receiving a fair evaluation.

Reopening creates a branch. It does not make the canonical path globally writable. The branch receives a declared update, data, compute, and duration budget; the protected version continues on control traffic. The branch becomes canonical only after retention, calibration, intervention, cost, and recovery tests. Otherwise the branch is discarded and the trigger is retained as an unresolved event.

Recovery dynamics as a maturity signal

Snapshots answer whether a module is currently inside its envelope. Recovery dynamics ask how strongly it returns after a small displacement. The maintenance plane should begin with passive fluctuation analysis. If the state is a shadow or replica and the service budget permits it, a bounded probe can estimate return time or a local stability margin.

The proposed Stage-1 test is Candidate 003. It compares bounded recovery probes with SLO dashboards, queueing headroom, change detection, passive autoregressive estimates, and standard active system identification under equal budgets. If ordinary system identification performs as well, the system should use it; the design requirement is visibility into restoring dynamics, not a special biological estimator.

Recovery is only one axis. A maturity record should expose at least:

DIMENSION	QUESTION	EXAMPLE MEASURE
temporal stability	Does behavior fluctuate under a stationary regime?	quality variance per event
acute resistance	How far does quality fall during a bounded perturbation?	maximum quality loss, fraction
recovery	How quickly and completely does behavior return?	return time, s or events
adaptability	How much work is required to learn a valid new regime?	updates, examples, and J
newcomer acceptance	Can a better candidate receive traffic and prove itself?	time to useful routing share, s
rollback	Can the prior behavior be restored after a failed change?	success fraction and lost work
reserve	What capacity remains for unexpected demand or repair?	bytes, W, routing slots, or RE/s

These quantities must be reported separately before any aggregate resilience score is computed.

Structural health is a path-dependent capacity contract

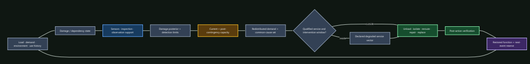
A module or route can continue producing output after pruning, damage, drift, or overload while its remaining margin and next-event tolerance collapse. "Healthy," "redundant," "gracefully degraded," and "reserve" are therefore not single scalar states.

For every structurally consolidated asset, record:

1. damage or dependency posterior and cumulative load/use history;
2. observation method, calibration, support, and detection limits;
3. present capacity and post-contingency capacity in native units;
4. surviving load/routing paths, common causes, and redistributed demand;
5. permitted degraded-service vector, affected strata, and exposure duration;

6. remaining time to a constraint boundary and available intervention;
7. unload, isolation, reroute, repair, replacement, or rollback action;
8. post-action verification; and
9. restored next-event reserve rather than merely returned output.

Wide diagram · fit to page



Editable source: residual-capacity-contract.mmd.

The mechanics and network evidence in C-481–C-499 establish mature nulls and measurement boundaries, not an AI effect size. The combined contract in C-500 remains an experimental schema shared by Candidates 005, 012, and 014. It is rejected if ordinary mechanics/reliability, asset management, fault handling, and network assignment match it.

Efficiency mechanism

Maturity can reduce recurring work in four places:

- protected modules need fewer parameter writes, optimizer states, and global synchronization events;
- stable routing can use smaller decision surfaces and better placement;
- recurring subgraphs can become fused, quantized, compiled, or cached paths; and
- retirement can release whole tensors, memory pages, routing entries, and network transfers.

The benefit is physical only when the runtime uses the new structure. Zero weights inside a dense kernel and dormant experts that are still loaded or synchronized do not count as structural consolidation.

For a proposed consolidation serving N future events, let $E_{\text{run,after}}$ and $E_{\text{run,before}}$ be steady runtime energy in joules per served event. Let $E_{\text{consolidate}}$, E_{validate}, E_{probe}, and E_{migrate} be one-time energies in joules, let $\mathbb{E}[E_{\text{recovery}}]$ be expected recovery energy in joules including failed branches, and let N be a dimensionless future-event count. Compare amortized energy per event:

$$\bar{E}_{\text{after}} = E_{\text{run,after}} + \frac{E_{\text{consolidate}} + E_{\text{validate}} + E_{\text{probe}} + E_{\text{migrate}} + \mathbb{E}[E_{\text{recovery}}]}{N},$$

against $\bar{E}_{\text{before}} = E_{\text{run,before}}$. The one-time numerator divided by N and both runtime terms are joules per event, so the comparison is dimensionally closed. Storage, data movement, tail latency, quality, calibration, and risk remain separate constraints rather than being silently converted into energy.

A consolidation advances only if its observation horizon is long enough that $\bar{E}_{\text{after}} < \bar{E}_{\text{before}}$ and it improves or preserves the declared quality–risk–latency–resilience frontier. The expected recovery term must include failed branches and rollback, not only successful releases.

Evidence status

ELEMENT	STATUS	WHAT IT SUPPORTS HERE
delayed generalization as universal maturity certificate (C-011)	disputed	grokking cannot be the gate
competitive sparse subnetworks (C-012)	established in tested settings	staged pruning is a valid operator, not a universal policy
activity-sensitive developmental refinement (C-043)	established in the cited preparation	a slower maintenance process can participate in structural contraction
reopening mature plasticity (C-044, C-045)	established in the cited preparations	protection can be local and reversible under intervention
redundancy and newcomer engraftment (C-057)	plausible association	stability may obstruct beneficial replacement
recovery warning signals (C-058, C-059)	established in scoped ecological systems	restoring dynamics are worth testing as a fragility signal
multidimensional stability tradeoff (C-060)	established in the cited microcosms	resilience dimensions must remain separate
mechanics, residual capacity, damage tolerance, and redistributed network load (C-481–C-499)	established or plausible in scoped engineering models	mature nulls for structural health, reserve, and recovery
path-dependent residual-capacity contract (C-500)	speculative synthesis	cross-layer fault-injection schema only
complete digital lifecycle controller	speculative synthesis	requires isolated and composed experiments

Speculative extensions

- Learn a lifecycle policy from logged promotion, rollback, and recovery outcomes while retaining hard provenance and safety constraints.
- Preserve cheap seed capacity outside the hot graph so retirement does not eliminate the ability to specialize under a future regime.
- Distill a coalition into a compact composite module, then retain the sources in a cold checkpoint until the composite survives a full recurrence cycle.
- Let recovery margin determine degrees of protection: lower update rate, narrower writable interfaces, or more stringent branch validation.
- Use capability-gap analysis to decide whether a failing mature module should reopen, be complemented by a newcomer, or retire.
- Reuse mature relational structure as a prior for faster consolidation while routing violations to a longer validation path.

Failure modes

- **False maturity:** a shortcut is stable on average and is protected before compositional, rare-event, or intervention tests expose it.
- **Brittle maturity:** quality remains inside its SLO while return time grows and the structure loses restoring margin.
- **Maturity monopoly:** protected redundant modules absorb all traffic and prevent a better newcomer from establishing evidence.
- **Rare-function erasure:** average routing and magnitude tests delete a path whose unique role appears only in a low-frequency or safety-critical regime.
- **Reopening thrash:** noisy triggers repeatedly create branches, retraining cost, and routing churn without a durable regime change.
- **Rollback rot:** checkpoints exist but dependencies, data schemas, or routing contracts have changed enough that restoration no longer works.
- **Cosmetic sparsity:** parameter count falls while bytes moved, kernel work, synchronization, and wall energy do not.
- **Maintenance inversion:** replay, probes, shadow traffic, migration, and regression testing consume more energy than mature execution saves.

- **Scalar resilience:** one score declares success while acute resistance, recovery, adaptability, or newcomer acceptance has deteriorated.
- **Causal misattribution:** a correlated low-usage path is pruned even though it stabilizes another module or handles a hidden confound.
- **Hidden reserve loss:** current output remains acceptable while damage, redistributed demand, common causes, or depleted intervention margin remove post-contingency capacity and next-event reserve.

Measurable predictions

1. A vector maturity gate using causal contribution and cross-context tests preserves rare and intervention performance better than fixed schedules, magnitude pruning, or training-loss thresholds at matched retained capacity.
2. Protected modules require fewer update joules and suffer less interference than continuously plastic modules, while branch-based reopening reaches a valid new regime with less regression than full unfreezing.
3. Recovery features identify some hidden loss of stability margin before steady-state SLOs. They remain in the architecture only if they add value beyond passive monitoring and standard system identification at matched probe and compute cost.
4. Structured consolidation lowers parameter bytes resident, bytes moved per event, communication, and measured joules together. Parameter reduction without those physical changes is rejected.
5. Increasing redundancy improves some stability dimensions while degrading adaptation or newcomer admission in at least one controlled regime; the raw resilience vector reveals the tradeoff that a scalar score hides.
6. Copy-on-write reopening plus rollback reduces catastrophic update loss relative to in-place adaptation after monitoring, checkpoint, and shadow costs are counted.
7. Consolidation produces a net lifecycle energy benefit only above a measurable reuse horizon N; below that horizon, leaving the computation plastic or interpreted is cheaper.
8. Residual-capacity fields predict unsafe second-event or redistributed-load failures beyond ordinary output, utilization, health, and redundancy scores; otherwise the cross-layer schema is removed.

Hardening, reflex paths, and factual memory

Repeated transformations, reusable skills, and mutable propositions require different execution, update, and recovery contracts.

Scope

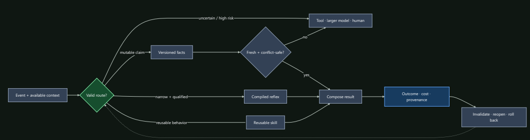
This chapter defines the boundary between five runtime outcomes:

1. a **compiled reflex path** executes a narrow, qualified transformation;
2. a **reusable skill** handles variable situations without embedding volatile propositions in its parameters;
3. **versioned factual memory** supplies mutable claims with provenance and validity state;
4. **escalation** sends an unresolved or high-risk event to a more capable model, tool, or human; and
5. **rollback** restores a last-known-good path or record after invalidation.

Hardening is a promotion decision, not a synonym for freezing. Every promoted artifact retains an applicability envelope, version, owner, regression set, physical cost record, invalidation policy, and recovery target.

Runtime paths and recovery

Wide diagram · fit to page



Editable source: `../assets/diagrams/hardening-memory-paths.mmd`.

The gate selects authority, not merely compute. A reflex may execute but cannot silently update a fact. A retrieved record may inform an answer but cannot rewrite a skill. An escalated result enters durable state only through the ordinary memory and consolidation lifecycle.

Biological observation

Biological and computational evidence supports separating rapid acquisition from slower integration (C-008). Retrieval can also make an established memory temporarily update-sensitive in a scoped preparation (C-039), while prediction error is disputed as a precise general trigger for that transition (C-040). Separate intervention studies show that mature constraints can be reopened under specific conditions (C-044, C-045).

The transferable requirements are multiple update timescales, guarded promotion, local reopening, and recoverable versions. The compiled path, factual schema, and escalation protocol below are engineering mechanisms tested against conventional systems.

Proposed AI translation

Five distinct contracts

PATH	STORES	MAY DO	MUST NOT DO
Compiled reflex	a narrow transformation, guard, and version	return a bounded result or action at low dispatch cost	answer outside its envelope or contain independently mutable facts
Reusable skill	a representation, policy, or parameterized transformation	generalize and compose across qualified contexts	present volatile propositions as current without retrieval
Factual memory	typed propositions and source relations	retrieve, supersede, dispute, revoke, or expire records	become true merely because retrieval ranked it highly
Escalation	no durable knowledge by itself	obtain more evidence or computation under a declared budget	bypass provenance, access, or promotion rules
Rollback	last-known-good versions, tombstones, and recovery metadata	restore routing and reconstruct prior state	erase the failed version or its affected-output trace

Compiled reflex paths

A compiled reflex is a versioned executable graph h with:

- an input and output schema;
- a validity predicate $G_h(x, \mathcal{I}_t) \in \{0, 1\}$;
- a declared applicability envelope $\mathcal{A}_h$;
- a deterministic or bounded-stochastic execution contract;
- a protected regression and adversarial set;
- a fallback route and last-known-good version; and
- a manifest for precision, kernels, placement, dependencies, and measured physical cost.

Here x is one event, $\mathcal{I}_t$ is the information available at decision time t, G_h is dimensionless, and $\mathcal{A}_h$ names the permitted input, environment, dependency, and risk strata. The runtime may dispatch h only when $G_h = 1$ and every dependency version remains valid. Guard evaluation is part of the path's latency, energy, and error budget.

Suitable candidates include parsing a fixed protocol, executing a stable local control law, applying a verified transform, or serving a repeatedly observed low-risk subgraph. Ordinary code, rules engines, memoization, and compiler optimization remain the first alternatives. A learned compiler is useful only if its generated guard and path outperform those alternatives under the same coverage and recovery requirements.

Compilation across physics

A mature reflex can also be stored in geometry, compliance, an analog transfer function, a physical reservoir, or a reprogrammable material state. Demonstrated components include passive task-specific dynamics (C-112), soft-body and physical-reservoir memory (C-113, C-

114), mechanical logic and physical learning (C-115, C-116), and local material repair (C-120). These observations extend the set of possible deployment substrates; they do not bypass the qualification gates.

A physical pattern is classified before it is credited with computation:

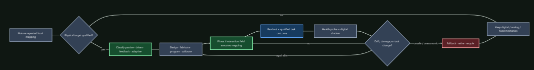
CLASS	WHAT CHANGES STATE	EXTERNAL BOUNDARY	LEARNING CLAIM
passive	relaxation or fluctuation under fixed energy landscape and boundaries	preparation, geometry, thermal/gravity/elastic conditions	none
driven fixed	continuous fuel, light, field, vibration, gravity, or flow under fixed interactions	all drive generation and dissipation	none
feedback controlled	explicit observation is converted into intervention	sensor, compute, communication, actuator, and drive	none unless the policy updates
adaptive	outcome evidence changes policy, interaction, morphology, or objective	all preceding costs plus training, validation, memory, and reset	testable

This separation follows C-463–C-479. Flocking, phase separation, defect motion, jamming, or assembly can be useful physical operations; visible order alone does not supply a task, evaluator, or policy update.

The held systems candidate is a rewritable physical path whose input and output remain locally coupled to the environment, while a versioned digital shadow preserves its specification, protected tests, calibration envelope, fallback, and output trace. The path is admitted only while health probes remain inside that envelope. Drift or damage returns authority to the digital path before a new substrate state is programmed and shadow-validated.

This path competes first with tuned passive mechanics, analog control, and FPGA/ASIC implementation, not only with an inefficient general model. Its lifecycle boundary includes design, fabrication, programming, drive, conversion, readout, reset, calibration, maintenance, fallback, repair, failed devices, and retirement. Candidate 006 tests whether a measured conversion, transport, recurrence, or command path is actually removed and whether break-even occurs before the qualified substrate lifetime ends.

Wide diagram · fit to page



Editable source: phase-field-compilation.mmd.

Candidate 006 now tests C-480 as one physical- phase specialization. Particle-scale power cannot win the comparison; the boundary includes fuel or field generation, sensing, control, transduction, readout, reset, fabrication yield, health probes, shadow, fallback, and retirement.

Reusable skills

A reusable skill is broader than a reflex. It accepts variable inputs, may consult current context or memory, and is evaluated for transfer outside the episodes that created it. It normally remains a slow-model module, adapter, tool policy, or callable subgraph.

Skill qualification asks whether the artifact preserves a reusable relation or operation. A proposition such as a price, office holder, software version, or medical recommendation is not a skill: its truth can change while the method used to retrieve, compare, or explain it remains valid. If a skill emits a mutable proposition, the output record must identify which factual version supplied it.

A skill may later receive structured pruning or quantization. Iterative pruning is evidence for competitive sparse subnetworks only in its tested settings (C-012). Ternary-weight language models are a plausible candidate under C-013; promotion still depends on end-to-end quality, risk, latency, bytes, and joules on the project workload.

Qualification gates

Promotion is conjunctive: a candidate fails when any hard gate fails.

GATE	REQUIRED RECORD	REJECT WHEN
Semantic class	reflex, skill, or factual record with one owner	the artifact mixes a stable transform with independently mutable claims
Applicability	schemas, $\mathcal{A}_h$, guard G_h, dependency versions	the guard cannot abstain before an out-of-envelope execution
Quality and risk	metrics by common, rare, safety, and shift stratum	average quality hides a stratum outside its tolerance
Causal contribution	ablation, reroute, and ordinary-code comparison	a cache, rule, smaller model, or router explains the gain
Physical accounting	guard, dispatch, execution, movement, idle, build, validation, and recovery costs	savings exist only in FLOPs or omit lifecycle work
Reversibility	immutable candidate version, atomic route switch, last-known-good target, rollback drill	prior behavior cannot be restored inside the recovery envelope
Provenance	source episodes, code/data versions, tests, approver, and artifact digest	the artifact or its qualification result cannot be reconstructed

For candidate path h and baseline b, let Q_h and Q_b be task quality in one declared unit, R_h and R_b be risk in one declared unit, and ϵ_Q and ϵ_R be preregistered tolerances in those respective units. Let $p_{\mathrm{FA},h}$ be the dimensionless fraction of out-of-envelope events incorrectly admitted by G_h, and let α_{FA} be its maximum permitted value. Qualification requires

$$Q_h \geq Q_b - \epsilon_Q, \qquad R_h \leq R_b + \epsilon_R, \qquad p_{\mathrm{FA},h} \leq \alpha_{\mathrm{FA}}.$$

These conditions are evaluated by stratum as well as in aggregate. A guard that rejects almost everything is exposed by reporting its coverage $c_h = N_{G_h=1}/N_{\mathrm{offered}}$, a dimensionless fraction. Coverage is a result, not a target inferred after testing; $N_{G_h=1}$ and N_{offered} are event counts.

Reversible verification before commitment

Kinetic proofreading shows that recognition and commitment can be separated by driven intermediate states with discriminatory rejection and reset (C-159). Its speed, error, and dissipation costs form a model-specific frontier rather than a universal accuracy multiplier (C-160). The systems translation therefore has four hard requirements:

1. temporary execution remains inside a declared rollback boundary;
2. the later verifier adds conditional information or a distinct detector;
3. rejected attempts, reset, delay, and provenance remain in the cost ledger;
4. irreversible authority is withheld until commitment.

For observations $z_1, \ldots, z_t$, the strongest statistical null conditions on the evidence already seen:

$$L_t = \sum_{i=1}^{t} \log \frac{p(z_i \mid R, z_{<i})}{p(z_i \mid W, z_{<i})}.$$

Here L_t is the dimensionless cumulative log-likelihood ratio, R and W denote correct/safe and wrong/unsafe hypotheses, and $z_{<i}$ is prior evidence. Ignoring that conditioning turns correlated rechecks into false confidence.

Candidate 010 tests reversible execution and risk-conditioned verification against this sequential test, calibrated cascades, abstention, retries, redundant verifiers, and error-detecting codes. It must tie or lose when the later stage is only a correlated copy or when reset leaks irreversible effects.

Graded assurance envelopes

Qualification records must state what kind of assurance each result provides. The classes are not interchangeable:

ASSURANCE CLASS	SUPPORTS	DOES NOT ESTABLISH
type, refinement, or proof	a named property under declared semantics and trusted base	termination, unspecified behavior, security, task quality, or truth
effect description	operations the model may perform under the analysis	authority to perform them or their correctness
capability grant	enforced authority inside a complete-mediation boundary	intent, competence, or safe outcome
empirical evaluation	behavior on declared data, environment, slices, and uncertainty	untested distributions or future versions
runtime monitor	a verdict over observed events under one temporal formula	unobserved channels or arbitrary future behavior
provenance	artifact identity and derivation path	source truth or claim entailment
transaction or compensation	recovery of participating state or a declared compensating action	reversal of time, disclosure, physical effects, or third-party actions

These boundaries are established in scoped programming-language and systems results: type soundness (C-145), effects versus capabilities (C-148), runtime-monitor scope (C-152), transactional rollback limits (C-154), and provenance without truth (C-156).

The held synthesis binds every assurance record to the same module version, artifact digest, dependency graph, state migration, authority policy, monitor schema, evidence set, and invalidation triggers. A dependency change rechecks only its affected cone, but stale assurance escaping to production and unnecessary rechecks are both measured. Candidate 009 compares this envelope against a complete conventional stack of typed APIs, sandbox/IAM, CI and static analysis, runtime policy monitoring, lineage, canaries, transactions, schema migration, and build-system invalidation.

Compromise-bounded authority and recovery

The security contract adds an adversary and trust boundary without collapsing distinct stages. Authentication establishes a scoped protocol property; authorization grants an action; detection classifies telemetry; containment blocks covered future use; and recovery re-establishes declared invariants from a tested root. None substitutes for the next (C-250, C-262, C-265).

For capability class j, let gj and rj be its grant and effective revocation times in seconds, and let wj be a declared dimensionless severity weight. Authority exposure is

$$XA = \sum_j wj \max(0, rj - gj),$$

with unit weighted-capability-seconds. The weights and individual intervals remain visible because a single broad destructive capability is not equivalent to many harmless reads. Sensitivity to plausible weights is reported.

Nominal credential lifetime is not the revocation result. If t_{comp} is the bounded compromise time and t_{last} is the last acceptance at every covered enforcement point, then

$$W_{\text{rev}} = \max(0, t_{\text{last}} - t_{\text{comp}})$$

is revocation exposure in seconds. Sessions, caches, delegation, offline verifiers, propagation delay, clock rollback, and missing acknowledgements are part of the measurement (C-260). The incident record keeps four clocks separately: compromise interval, detection, effective containment, and independently validated recovery.

The held profile binds principal and workload identity, capability scope, credential/key/attestation epoch, revocation freshness, approval-domain independence, observation age, adversary model, compromise horizon, and clean- root evidence to the same versioned artifact. It survives only if Candidates 009 and 012 reduce harm or secure recovery time beyond mature short-lived IAM and a rehearsed reimage–rotate–validate workflow at equal lifecycle cost.

Recovery is not one endpoint score

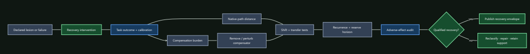
After a lesion or fault, the same task outcome can come from restored capability, a larger fallback, a different route, a tool, a cache, human intervention, or a more permissive environment. Rehabilitation evidence makes that underidentification explicit (C-316–C-324).

Let z be a declared post-recovery internal, routing, or behavioral state and z_{ref} a justified reference. Report

$$D_{\mathrm{native}} = d(z, z_{\mathrm{ref}}), \qquad \mathbf{B}_{\mathrm{comp}} = (E_{\mathrm{extra}}, M_{\mathrm{extra}}, L_{\mathrm{extra}}, H_{\mathrm{extra}}, P_{\mathrm{fragile}}),$$

where D_{native} has the declared metric's unit or is explicitly normalized; extra energy is joules; memory is bytes; latency and human support are seconds; and P_{fragile} is the dimensionless probability of failure under a named support-removal or shift test. These axes are not merged unless conversion weights are declared in advance.

Wide diagram · fit to page



Editable source: compensation-aware-recovery.mmd.

Compensation is not a failure when it is robust, affordable, and declared. The failure is to call it restoration, hide recurring support, or deploy outside the context that makes it work. Conversely, an ordinary cheap workaround may dominate native restoration. Candidate 005 therefore compares the complete accounting method with conventional multi-objective robustness, fault injection, failover, checkpoint restoration, and functional reconstruction.

Versioned factual memory

A factual record r contains at least

$$r = (k, v, u, s, v_s, t_{\mathrm{obs}}, [t_{\mathrm{from}}, t_{\mathrm{to}}), v_r, \pi, \sigma),$$

where:

- k is a typed key or subject–predicate identifier;
- v is the typed value and u is its declared unit, or `none` for a unitless value;
- s is the source identifier and v_s its source version;
- t_{obs} is the observation timestamp;
- $[t_{\mathrm{from}}, t_{\mathrm{to}})$ is the asserted validity interval;

- v_r is the immutable record version;
- π is the access, retention, and jurisdiction policy; and
- $\sigma \in \{\text{active}, \text{superseded}, \text{revoked}, \text{disputed}\}$ is record status.

Timestamps use UTC with declared resolution. Differences between timestamps are reported in seconds. A record may also carry source-supplied confidence or a calibrated probability, always dimensionless and never substituted for source identity or conflict handling.

Let t_{check} be the last successful source check and let τ_k be the maximum unchecked age for key class k in seconds. At query time t, the freshness gate is

$$F(r,t) = 1[\sigma = \text{active}]1[t_{\text{from}} \le t < t_{\text{to}}]1[0 \le t - t_{\text{check}} \le \tau_k].$$

$F(r,t)$ is dimensionless. The domain policy fixes τ_k before evaluation; an unbounded value is permitted only when the domain explicitly defines the record as non-expiring. Freshness means that the record passed its time and status policy, not that its proposition is correct.

For a key k, the conflict set $\mathcal{C}_k(t)$ contains active, policy- admissible records whose values cannot simultaneously hold at time t. Resolution may use an explicit source-authority rule, a time rule, or a domain-specific adjudicator. The system preserves losing records and the resolution trace. When no preregistered rule applies, the retrieval path abstains and escalates rather than averaging incompatible values.

Retrieval-augmented generation establishes that parametric generation can be combined with inspectable and replaceable non-parametric memory on evaluated knowledge-intensive tasks (C-014). This chapter adds version, freshness, conflict, and lifecycle accounting as requirements to test, not as evidence that retrieval is automatically correct.

Invalidation and rollback

Invalidation is triggered by any of the following observable events:

- an input-schema, dependency, tool, hardware, or source version changes;
- a validity interval or freshness allowance expires;
- a protected regression, shift probe, calibration check, or outcome fails;
- an authoritative source revokes or supersedes a record;
- a new admissible record creates an unresolved conflict; or
- guard false admissions, fallbacks, or escalations exceed their declared control limits.

The response depends on the artifact:

ARTIFACT	IMMEDIATE ACTION	DURABLE ACTION
Compiled reflex	atomically route new events to the last-known-good path	preserve failed binary, manifest, traces, and invalidation cause
Reusable skill	freeze the active version and open a copy-on-write branch	replay, validate, reconsolidate, or retire through maintenance
Factual record	remove the version from active retrieval and append a tombstone or dispute edge	retain prior values and rebuild an index version without destructive overwrite
Escalation policy	fall back to the conservative route and cap further delegated work	recalibrate on logged false admission, miss, cost, and outcome data

Every served output records the path, model, dependency, and factual versions that affected it. This makes the affected-output set enumerable after a defect. Report rollback time T_{rb} in seconds, rollback energy E_{rb} in joules at the declared boundary, lost or corrected events N_{loss} as a count, and restoration success as a dimensionless fraction.

Severity-ordered containment and triage

Rollback is one response to one fault contract. A stateful modular system also needs to keep the following actions distinct:

1. **Sense:** collect health evidence without treating a detector output as fault ground truth.
2. **Contain:** throttle inputs, revoke an interface, quarantine a route, or freeze a version to limit spread before diagnosis completes.
3. **Triage:** choose retry, local repair, selective reconstruction, restart, retirement, or replacement from the available evidence and declared cost.
4. **Verify:** test the affected behavior, protected rare behavior, provenance, and adjacent modules before restoring authority.
5. **Escalate:** move to a more destructive action only when verification fails or stronger evidence makes delay unsafe.
6. **Replenish:** restore validated capacity when a component is retired, rather than allowing maintenance to clean the system into capacity collapse.

The biological audit supplies scoped examples of fast load shedding (C-087), repair-versus-degradation triage (C-090), tag-dependent compartment routing (C-091), selective extraction (C-092), repair before removal (C-094), and removal coupled to replacement (C-095, C-096). The order is conditional: a rapidly spreading irrecoverable fault may require immediate replacement, while a local reversible fault should not trigger a global rebuild.

The engineering value of the composition remains unproven. Circuit breakers, taint tracking, checkpoints, replica failover, microreboots, scrubbing, rejuvenation, and Bayesian repair/replace policies are mandatory comparators. Candidate 005 tests the staged policy across locality, observability, repairability, and correlated detector error while charging sensing, reserve, copying, replacement, verification, downtime, and collateral loss.

Escalation is a budgeted route

Escalation activates when the local guard abstains, required facts are stale or conflicted, the event enters a protected risk stratum, or the available path cannot meet its quality contract. The escalation target may be a larger model, a deterministic tool, an authoritative data source, or a human.

The request carries the triggering uncertainty, attempted path and versions, relevant evidence, allowed data disclosure, deadline, and remaining energy or financial budget. Its result returns with provenance and observed cost. Escalation does not grant write authority; durable change still requires a versioned maintenance action.

Efficiency mechanism

Per-event and lifecycle accounting

For path version h serving N qualified events, define amortized energy

$$e_h(N) = e_{\text{guard}} + e_{\text{dispatch}} + e_{\text{execute},h} + \frac{E_{\text{build},h} + E_{\text{validate},h} + E_{\text{migrate},h} + \mathbb{E}[E_{\text{rollback},h}]}{N}.$$

Lowercase e terms are measured joules per qualified event. Capital E terms are one-time joules at the same device, node, cluster, or facility boundary; $\mathbb{E}[E_{\text{rollback},h}]$ includes failed promotions weighted by their observed or preregistered probability. N is a dimensionless event count, so $e_h(N)$ is joules per qualified event. If the deployment horizon is unknown, report the break-even event count rather than assuming amortization.

For event x, end-to-end latency is

$$\ell_h(x) = \ell_{\text{guard}}(x) + \ell_{\text{dispatch}}(x) + \ell_{\text{execute},h}(x),$$

with every ℓ term in seconds. Report the empirical end-to-end percentile $L_{h,p}$ for a declared percentile $p \in (0, 1)$ rather than adding separately measured component percentiles. Report physical traffic B_h in bytes per qualified event across each named boundary. Energy, latency, and bytes remain separate results.

The execution term expands by path:

PATH	COSTS THAT MUST BE VISIBLE
Compiled reflex	guard, dispatch, code and state loads, kernel execution, cache residency, precision conversion
Reusable skill	routing, parameter and activation movement, inference, memory/tool calls, synchronization
Factual memory	query encoding, index access, retrieval, reranking, freshness/conflict checks, evidence bytes, composition
Escalation	failed local attempt, serialization, network, remote or tool execution, waiting time; human time reported separately
Rollback reserve	retained versions, manifests, index generations, route switch, replay, correction, recovery validation

A candidate advances only when quality and risk remain qualified and it improves the preregistered energy–latency–traffic frontier after all listed costs are included. Skipped model operations alone are insufficient.

Provenance and freshness measurements

For N_f outputs that use factual memory and N_p of those outputs carrying a complete record-to-source trace, provenance coverage is

$$C_{\text{prov}} = \frac{N_p}{N_f}.$$

C_{prov} is dimensionless. Also report source-check age in seconds, stale-use and unresolved-conflict rates as fractions, index and evidence bytes, joules per lookup and update, and p50/p95 lookup latency in seconds.

If a source changes at t_{source} and the first correctly served version is available at t_{serve}, correction latency is

$$T_{\text{corr}} = t_{\text{serve}} - t_{\text{source}}$$

in seconds. Record source polling or event-delivery cost alongside this value; instant correction purchased by continuous high-cost polling is not free.

Strongest null models

All nulls receive the same task stream, factual sources, safety policy, hardware opportunity, and lifecycle horizon.

ID	NULL	REQUIRED COMPARISON
N0	full capable model on every event	tests whether any dispatch hierarchy beats unconditional execution
N1	calibrated early exit or static small/large cascade	tests ordinary adaptive depth under C-004
N2	conventional rules engine, compiler, or memoization cache with TTL	tests whether a compiled reflex adds more than established software practice

ID	NULL	REQUIRED COMPARISON
N3	separately trained smaller, distilled, or quantized model	tests whether path specialization beats a simpler fixed deployment
N4	parametric-only knowledge with scheduled fine-tuning	tests factual correction cost, carryover, and provenance
N5	strong hybrid-search RAG with reranking, citations, and ordinary freshness filters	tests whether version/conflict machinery improves the factual frontier
N6	in-place skill update with checkpoint restore	tests copy-on-write promotion and rollback overhead against standard recovery
N7	calibrated risk/confidence threshold using the same escalation target	tests whether the structured guard adds value beyond a scalar threshold
N8	encoded redundancy, replica/quorum recovery, integrity scrub, or self-stabilizing legitimate-set repair	tests whether established exact or rule-encoded reconstruction dominates learned recovery under the declared fault model
N9	circuit breaker, static isolation, microrestart, rejuvenation, or Bayesian repair/replace policy	tests whether ordered containment and triage add value beyond mature fault-management composition

If ordinary code or a standard data system matches the candidate, keep the conventional mechanism and retain only the qualification and accounting contract.

Evidence status

MECHANISM	EVIDENCE	STATUS FOR THIS CHAPTER
adaptive early exit	C-004	established on evaluated BERT tasks; shift and rare-risk gating open
fast acquisition versus slow integration	C-008	plausible architectural separation; exact tiers unvalidated
structured consolidation and pruning	C-012	established in scoped experiments; physical saving not automatic
ternary-weight models	C-013	plausible preprint evidence; project workload and full energy unresolved
parametric plus non-parametric memory	C-014	established on evaluated RAG tasks; freshness and conflict remain open
retrieval-sensitive updating	C-039, C-040	scoped observation; precise general mismatch gate disputed
reversible mature constraint	C-044, C-045	scoped biological intervention; digital rollback contract experimental
exact restore, encoded repair, failure detection, and integrity scrubbing	C-079–C-085	established engineering nulls under explicit fault and cost models
constraint-guided functional reconstruction	C-086	speculative residual candidate for semantic or capability loss without a clean exact state
load shedding, tagged routing, selective extraction, repair/removal ordering, and replacement feedback	C-087–C-096	scoped cellular mechanisms; the composed systems policy remains a held candidate
morphology, physical reservoirs, mechanical memory/logic, local assembly, and material healing	C-112–C-120	established in scoped substrates; end-to-end advantage over passive, analog, FPGA/ASIC, and digital nulls remains workload-specific
reversible physical skill compilation	C-121	speculative lifecycle systems hypothesis tested by Candidate 006
soft/active-matter physical order and external feedback	C-463–C-479	established or plausible in scoped physical systems; no audited task-success learning
phase/interaction-field skill compilation	C-480	speculative Candidate-006 specialization
types, contracts, effects, capabilities, proof checking, static analysis, runtime monitoring, transactions, hot update, and provenance	C-145–C-156	established scoped assurance classes with explicit trusted bases and invalidation boundaries
versioned graded assurance envelopes	C-157	speculative systems composition tested by Candidate 009
automatic reflex discovery and qualification	none	speculative until it beats the null models above

Speculative extensions

- Train a compiler to propose a small executable graph, applicability predicate, counterexample set, and rollback manifest as one candidate artifact.
- Learn source-check schedules from update hazard and consequence while keeping hard maximum ages for protected domains.

- Compile stable *relations* learned across factual versions while leaving the current proposition external and attributable.
- Use signed provenance graphs so a correction can enumerate dependent outputs, cached artifacts, and downstream derived records.
- Co-design path placement and precision with hardware only after the logical qualification gate passes on a conventional substrate.

Failure modes

FAILURE SIGNATURE	OBSERVABLE MEASURE	REQUIRED RESPONSE
guard admits shifted input	$p_{\mathrm{FA},h}$ rises by stratum	disable version; route to fallback; add counterexample
guard rejects most valid traffic	coverage c_h collapses while aggregate quality looks stable	report lost coverage; compare with N1/N7
shortcut becomes a reflex	protected counterfactual or subgroup regression	invalidate and reopen source skill
volatile fact leaks into parameters or compiled code	correction requires retraining or old value persists after record update	demote proposition to factual memory; trace affected outputs
fresh but wrong source record	outcome error despite $F(r,t) = 1$	preserve source trace; dispute/revoke; strengthen authority policy
unresolved contradiction is silently merged	non-empty $\mathcal{C}_k(t)$ without a resolution trace	abstain and escalate
index and model disagree on active version	served v_r differs from index-generation manifest	atomically roll back index and replay affected queries
promotion thrashes	repeated compile–invalidate cycles, migrations, or route flips	raise evidence horizon; charge churn energy; retain conventional path
quantization hides a rare regression	mean quality holds while protected-stratum R_h rises	reject precision change
rollback is nominal only	T_{rb}, N_{loss}, or recovery tests exceed envelope	block future promotion until recovery is repaired
retrieval savings vanish physically	operations fall while bytes, latency, or joules do not	reject efficiency claim; keep stronger null
escalation becomes an unpriced default	escalation fraction and remote cost rise without risk improvement	recalibrate guard and expose full route cost
ordered phase is credited as intelligence or learning	no independent task signal, evaluator, or outcome-driven update exists	relabel as passive/driven physics and retain it only as a substrate/null
particle-scale power looks excellent while wall energy rises	drive, sensing, compute, transduction, reset, or fabrication dominates	reject the system-efficiency claim and restore the full boundary

Measurable predictions

1. For recurrent, stable, low-risk transformations above a measurable break-even horizon, a qualified compiled path reduces joules per event or end-to-end latency relative to N0–N3 while preserving stratum-level quality, risk, and guard false-admission limits.
2. Reusable skills outperform memoization and rules on held-out compositions, while reflex paths outperform the skill only inside their narrower declared envelope.
3. Separating mutable propositions from skills reduces correction latency, retraining energy, and obsolete-value carryover relative to parametric-only N4 at matched task quality.
4. Version, freshness, and conflict gates reduce stale or contradictory factual use relative to strong N5; the result is rejected if lookup cost erases the quality–risk benefit.
5. A structured applicability guard reduces out-of-envelope false admissions under shift relative to N1 and N7 without achieving the result by rejecting nearly all traffic.
6. Complete provenance increases the fraction of factual outputs whose source and dependent artifacts can be enumerated after a correction, at a measured byte, latency, and energy cost.
7. Atomic route rollback and versioned factual tombstones reduce restoration time and lost events relative to in-place N6 after injected bad promotions and source revocations.

8. Quantized or compiled paths produce an end-to-end physical gain only when dispatch, memory movement, validation, retained rollback state, and failed promotions amortize within the observed deployment horizon.
9. A reversible phase or interaction field advances only if it removes a measured repeated digital/transport path and beats passive mechanics, analog control, FPGA/ASIC, and distributed digital control after complete drive, fabrication, sensing, reset, drift, fallback, and retirement cost.

System synthesis

> *Intelligence is the coordinated movement of state through perception, prediction, action, memory, adaptation, and maintenance under a finite physical budget.*

Scope

The working architecture defines the runtime, adaptation, and maintenance loops. This chapter follows information through those loops, identifies who may write each kind of state, and gives the order in which the integrated system should be assembled.

Biological observation

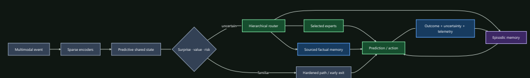
Living intelligence coordinates sensing, action, memory, plasticity, resource supply, and repair across different timescales. Local circuits handle repeated work, broader signals alter priorities, fast traces change behavior before slow structure moves, and maintenance continues during operation.

The transferable principle is **coordinated state ownership**. Reading a result does not grant authority to overwrite its source. Fast behavior, rapid learning, slow consolidation, and physical resource control can therefore interact without becoming one global update rule.

Proposed AI translation

Runtime path

Wide diagram · fit to page



Editable source: `../assets/diagrams/system-runtime.mmd`.

The path has six stages:

1. **Ground the event.** Encoders align perception with current action, timing, location, and tool state.

2. **Predict before expanding computation.** The shared state estimates what is likely, what is uncertain, and which goal currently matters.
3. **Choose the execution path.** A gate may dispatch a hardened skill, stop at an early exit, retrieve memory, activate selected experts, intervene through a tool, or escalate.
4. **Execute only admitted work.** Inactive capacity remains addressable without being loaded and operated on for the event.
5. **Measure the result.** Output is paired with observed consequence, calibration, latency, bytes moved, and energy.
6. **Capture an episode.** The adaptation loop receives an attributable record rather than an unexplained parameter change.

Authority is split across control planes

The system uses four control planes with different authority:

PLANE	CONTROLS	READS	CANNOT DO ALONE
Task	goals, quality, permitted actions	predictive state, user/tool feedback	hide physical or risk cost
Resource	energy, latency, communication, placement	estimates and measured telemetry	redefine task success
Adaptation	episodes, hypotheses, provisional modules	outcomes, conflict, uncertainty	promote a global slow-model change
Maintenance	replay, merge, weakening, deletion, topology	history, regressions, fragility, lifecycle cost	bypass provenance or release gates

The planes exchange compact declared state. A resource controller can deny an expensive route but cannot declare the cheaper result equally correct. A task controller can request deeper computation but cannot conceal the resulting traffic from the energy ledger. Adaptation can propose; only maintenance can promote a change into protected state.

Governance is activated by a real authority problem

Aggregation rules, delegation, vetoes, constitutions, and participation are not useful decorations for ordinary routing. They enter the architecture only when persistent actors have decision-relevant private information, can benefit from misleading reports, create spillovers across authority boundaries, hold externally authorized protected standing, or must repair higher-order rules without exposing them to ordinary updates.

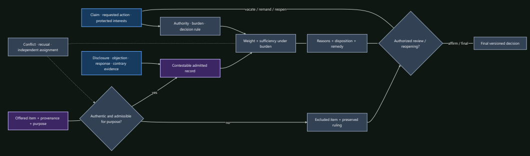
When that applicability gate passes, the control plane records more than a winning action:

1. the authorized objective and non-tradable invariants;
2. the actor's identity, scope, evidence, dependencies, and conflicts;
3. the proposal set, order, fallback, and omitted-alternative coverage;
4. every delegation edge, concentration measure, revocation, and cycle;
5. a veto's protected scope, severity, evidence, deadline, appeal, override, and expiry;
6. the decision and affected local and external interests; and
7. the rule version, amendment path, emergency authority, checked handoff, compatibility result, and rollback target.

A decision record is more than evidence plus a score

Where an action is contestable and authority-bearing, the system keeps normative authority, empirical evidence, formal inference, and procedural validity as separate predicates. Authenticity, admissibility for a stated purpose, weight, sufficiency under a burden, authorization, review, remedy, and finality are not ranks on one confidence scale.

Wide diagram · fit to page

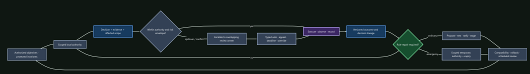


Editable source: burden-qualified-decision.mmd.

The record binds claim/action, authority, affected interests, proponent, burden, evidence purpose and provenance, disclosure/access, objections, contrary evidence, findings, reasons, decision, protected outcomes, review scope, remedy, successor version, and reopening conditions. It records excluded material so later review can test contamination without silently letting that material influence the decision.

The contestable-decision mathematics keeps error, delay, review, access, human work, energy, and protected procedure as raw axes. Its evidence boundaries are C-679–C-704. The composition must beat typed workflow, provenance, selective prediction, access control, rule graphs, independent review, red-team challenge, and full recomputation. It refines existing assurance, verification, operational learning, observation, convention, and governance candidates; it does not add a new control primitive.

Wide diagram · fit to page



Editable source: constitutional-control-plane.mmd.

The formal and empirical boundaries in C-368–C-395 forbid several shortcuts: an impossibility theorem does not select a moral objective; stability is not quality; participation is not legitimacy; popular or expert influence is not automatically capture; and an amendment count is not repair. Normative standing and non-tradable harms remain explicit authorized inputs.

Candidate 020 therefore has a deliberately hostile null: constrained optimization, typed IAM and interlocks, policy-as-code, independent evaluation, separation of duties, append-only lineage, runtime assurance, and mature incident/change management. If those match the result —or if every module shares one loss and directly verifiable state—the institutional composition is removed.

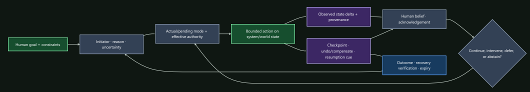
A person in the loop must have an executable control path

A visible approval step does not establish oversight. Effective mixed control requires a person to receive relevant state before the response deadline, understand the current and pending mode well enough for the task, hold actual authority over the consequential effect, execute an intervention, and observe whether the intended state changed. Explanation, confidence, trust, reliance, agreement, preference, and correctness stay separate.

Each machine-initiated or shared-control transition can therefore carry a recoverable initiative record:

1. initiator, reason, urgency, and calibrated uncertainty;
2. actual mode, pending mode, enabled effects, and response deadline;
3. effective human and machine authority, including silent arbitration;
4. intended action, observed state delta, and provenance;
5. acknowledgement or rejection and whether it changed execution;
6. checkpoint, undo, compensation, or safe-stop boundary;
7. resumption cue, pending goals, and assumptions that may have gone stale;
8. verified recovery result, expiry, and longitudinal-learning link; and
9. accessibility and assistive-technology conditions under which the path was tested.

Wide diagram · fit to page



Editable source: recoverable-initiative.mmd.

Human time is part of lifecycle cost. For interruptions $j = 1, \ldots, n$,

$$T_{\text{complete}} = T_{\text{active}} + \sum_{j=1}^{n} \left(T_{\text{interrupt},j} + T_{\text{resume},j} + T_{\text{rework},j}\right),$$

with every term measured in seconds under a declared coding rule. Error severity, stress, frustration, workload, training, and person-minutes remain separate outcomes; throughput alone cannot erase them.

The bounded evidence in C-396–C-416 supports specific interruption, mode, automation-bias, recovery, accessibility, and adaptive-interface constraints. The combined record is still speculative, so it refines Candidates 009, 011, 012, and 015 rather than becoming another registry principle.

State ownership

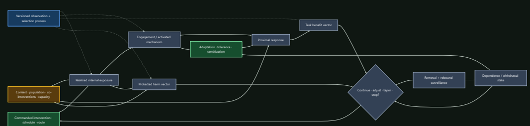
STATE	OWNER	NORMAL WRITE PATH	NORMAL READERS
Current predictive context	runtime	every event	selected runtime modules
Recent attributable experience	episodic memory	observed outcome	routing, adaptation, maintenance
Provisional hypotheses and modules	adaptation	bounded generation and trials	shadow runtime and evaluators
Reusable representation and skill	slow model	validated consolidation	encoders, predictors, experts
Mutable proposition	factual memory	sourced versioned update	retrieval router and task modules
Stable repeated transformation	hardened store	promotion pipeline	guarded low-cost dispatch
Resource and risk policy	budget controller	calibrated policy update	every gate and route
Low-dimensional operating context	context bus	rate-limited controller update	subscribed modules through receiver-local filters
Lifecycle and fragility state	maintenance	replay, probes, regressions	admission, reopening, pruning, rollback

The plane table says who controls an operation; the state table says who may write durable state. Each boundary becomes an ablation point: remove it, widen it, delay it, or replace it with a conventional baseline and measure the consequence.

Intervention is a stateful chain

A commanded change is not its realized effect. Training intensity, routing quota, memory injection, tool access, fallback support, pruning pressure, and human escalation can each be delivered incompletely, arrive late, alter later observations, produce different useful and harmful effects, and change the system's future response. The contract therefore keeps this chain explicit: (C-607–C-610, C-616, C-626).

Wide diagram · fit to page



Editable source: state-qualified-intervention.mmd.

The held system record includes commanded schedule; realized state; mechanism engagement; benefit and protected-harm vectors; adaptation, dependence, and withdrawal state; population/task/context support; observation provenance; uncertainty; reserve; and permitted continuation, adjustment, taper, or stop. Its scientific bounds are recorded in C-611–C-625. It rejects three shortcuts:

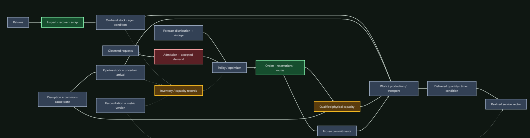
1. activation or occupancy is not downstream benefit or safety;
2. equal cumulative input does not imply equal peaks, spacing, state, or recovery; and
3. removing support is a new dynamical intervention, so rollback continues through rebound, recurrence, native-capability, and reserve checks.

The state-qualified intervention mathematics defines the unit-bearing state, authority, interaction, and withdrawal tests. The composition remains a domain track under Candidates 005/007/012/014 and must beat Bayesian state-space estimation, constrained MPC/POMDP, calibrated harm monitoring, and staged decommissioning at equal lifecycle cost.

A plan, a commitment, and service are different states

Resource-bearing execution has an informational path and a physical path. A forecast can change policy; policy can authorize an order, reservation, route, or allocation; none of those creates qualified capacity, moves material, finishes work, or proves service (C-627, C-659–C-678). The complete path is:

Wide diagram · fit to page



Editable source: material-commitment-service.mmd.

This distinction applies when modules allocate physical devices, accelerators, network paths, robot/tool capacity, spares, storage media, human attention, or external services with lead time. Every decision-bearing record preserves:

1. forecast vintage, observed request, admission, and accepted commitment;
2. on-hand, reserved, quarantined, pipeline, and available-to-promise state;
3. qualified capacity, queue, route, setup, common-cause, and lead-time state;
4. dispatched versus delivered work, condition, completeness, and timing;
5. the service metric's denominator, clock, substitutions, and version; and
6. backlog, lost demand, returns, recovery yield, reserve replenishment, and second-event readiness.

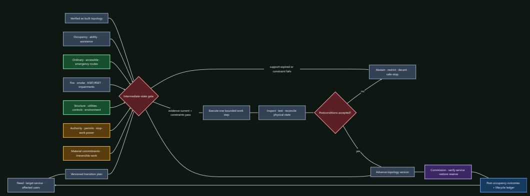
The material/service mathematics makes conservation, service, and recovery units explicit. The held composition must beat event sourcing plus inventory reconciliation, queueing, base-stock and multi-echelon control, stochastic/robust optimization, and receding-horizon planning with frozen commitments. It is an evaluation contract, not a new allocation algorithm.

Physical topology must remain valid while it changes

Logical routing can often switch within milliseconds and retry. Physical topology cannot assume that abstraction when a transition moves barriers, routes, utilities, controls, structure, people, or material. The system must represent the intermediate configurations—not only the before and after graphs.

The built-environment evidence in C-705–C-726 adds a demanding test case. A plan, BIM model, sensor dashboard, command, and verified physical state are separate records. Accessible use, egress, tenability, structure, utilities, environmental service, evidence validity, authority, and material commitment remain explicit through every work step.

Wide diagram · fit to page



Editable source: occupancy-qualified-spatial-transition.mmd.

This pattern is useful beyond buildings whenever reconfiguration has occupants, physical inventory, slow work, external authority, or irreversible steps. Its core rule is simple: a target state cannot authorize a path whose intermediate states are invalid. Hard constraints remain a conjunction rather than a scalar score; lost service stays visible by affected group; and "rollback" is claimed only when the predecessor is still physically reachable.

The full transition mathematics and Candidate 001 stress track compare the composition against ordinary design review, permits, impairment control, configuration management, commissioning, post-occupancy evaluation, and lifecycle asset management at equal budget. If that mature stack ties it, the extra composition is removed.

After an outcome: adaptation proposes, maintenance decides

An outcome can influence the next event through working state or episodic retrieval. Durable changes follow two paths with different authority:

PATH	IMMEDIATE WORK	POSSIBLE RESULT
Adaptation	capture an attributable episode; construct bounded proposals from existing fragments, relations, and abstractions	working-state change, hypothesis, or provisional module
Maintenance	score, replay, branch, compare, and account for lifecycle cost	retain, merge, externalize, weaken, delete, protect, reopen, relocate, prune, quantize, or compile

Variation creates candidates; observed outcomes and controlled interventions decide which survive. Bounded fragility probes remain on shadows or replicas and are charged to the maintenance budget.

The system may also create learning events: propose a structured hypothesis or latent rollout, choose a bounded intervention, and evaluate the outcome. This closed endogenous curriculum composes the evidence in C-061–C-066; random variation supplies candidates, not validation. Its operations are separated under equal budgets in Candidate 004.

The memory lifecycle governs evidence and retention. The maturity lifecycle governs protection, reopening, and structural consolidation. A proposal remains scoped and attributable until maintenance promotes it.

Every promoted structural action records:

1. a declared trigger;
2. a versioned candidate state;
3. protected historical and rare-case evaluations;
4. measured migration, validation, and recovery cost;
5. a promotion or rejection decision; and
6. a rollback target.

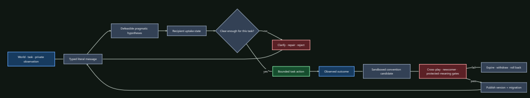
Communication is a versioned state transition

A fluent message is not one state variable. The runtime keeps at least six separate records:

1. typed literal payload and protocol version;
2. external referent, query, action, or constraint;
3. defeasible hypotheses about sender intention;
4. recipient-generated uptake state—received, parsed, understood-enough, accepted, rejected, or unresolved;
5. repair and supersession lineage; and
6. any resulting authority decision, which communication alone cannot grant.

This separation follows the evidence boundaries in C-268–C-281. Composition is relative to a grammar and interpretation; pragmatic inference depends on context and incentives; acknowledgement is not identical belief; channel capacity is not semantic value; and population convergence is not grounding, truth, or safety.

Wide diagram · fit to page



Editable source: versioned-repairable-conventions.mmd.

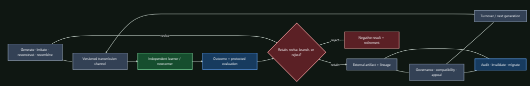
A local shorthand remains session-scoped until independent interpreters recover its declared denotation, older and newcomer agents pass cross-play, corrupted or version-mismatched messages trigger bounded repair, protected rare meanings survive, and migration plus rollback have been exercised. The full lifecycle is tested in Candidate 015 against typed protocols, schema registries, acknowledgements, replicated logs, calibrated inference, standard coding, and explicit migration.

Capability can cross learner turnover without staying inside one learner

A repository, demonstration, or popular practice is not yet cumulative inheritance. The system must expose six different operations:

1. **Generate:** produce a variant by invention, reconstruction, imitation, or recombination.
2. **Transmit:** identify exactly which actions, outcomes, explanations, artifacts, tests, and environmental state reach the next learner.
3. **Evaluate:** test the reconstructed capability independently rather than crediting popularity, prestige, or successful transmission as usefulness.
4. **Retain:** place accepted capability, evidence, failures, and lineage in a form that a newcomer can retrieve and interpret.
5. **Govern:** control compatibility, authority, appeal, migration, and retirement without treating a convention as truth.
6. **Survive turnover:** repeat the cycle with a genuinely new learner and a protected test suite.

Wide diagram · fit to page



Editable source: audited-cumulative-inheritance.mmd.

For generation g, a deliberately simple capability ledger is

$$K_{g+1} = q_g K_g + I_g + R_g - D_g,$$

where K_g is validated capability in a declared task-score unit, q_g is the dimensionless retained fraction, I_g is independently generated improvement, R_g is validated recombination gain, and D_g is degradation or compatibility loss in the same task unit. These terms are accounting fields, not a claim that capabilities are generally additive; protected outcomes stay visible beside the aggregate.

Raw population size also overstates diversity when every learner sees the same few lineages. If exposure weights are π_i and $\sum_i \pi_i = 1$, then

$$N_{\text{eff}} = \frac{1}{\sum_i \pi_i^2}$$

is the dimensionless effective number of equally exposed models. It measures attention concentration—not competence, independence, or complexity.

The evidence in C-343–C-367 supports bounded cultural accumulation, conditional social-information use, material scaffolding, and severe archaeological inference limits. It does not establish a population advantage for artificial learners. That residual is isolated in Candidate 019, which must beat centralized continual learning, replay, version control, retrieval, workflow engines, quality-diversity search, and fixed governance at equal cumulative effort.

Live response and longitudinal learning are separate loops

The runtime may need to contain a failure before its cause is known. Later analysis may identify a cause without improving the next live response. A usable operational-assurance plane therefore connects two loops without collapsing them:

LOOP	DEADLINE	REQUIRED STATE TRANSITION	FAILURE IF MEASURED ALONE
Live response	seconds to minutes	detect, acknowledge, assign scoped authority, contain, degrade, revoke, roll back, restore	fast containment can recur because no verified change enters future operation
Longitudinal learning	releases to months	retain competing traces, analyze, bind a finding to dependencies and an owner, test, deploy, verify, retrieve, retire	a complete report can arrive after preventable damage and may never change operation

Report volume is observation-biased. Over a declared exposure interval,

$$N_{\text{report}} = N_{\text{precursor}} p_{\text{detect}} p_{\text{report}} p_{\text{retain}},$$

where both N values are event counts and every p is a dimensionless conditional probability. A low count can mean low exposure, weak detection, reporting friction, fear, or deletion; a high count can mean hazard, improved coverage, duplication, or gaming (C-178).

Likewise, an incident archive becomes memory only when an applicable lesson is retrieved, used, evaluated, invalidated, and retired (C-184). The held composition binds live traces and precursor detections to scoped response, dependency-linked changes, protected tests, verified outcomes, retrieval triggers, and retirement. Candidate 011 compares it with a complete SRE stack and rejects it if role, interruption, reviewer, storage, stale-memory, or coordination cost erases the gain.

Build the system in dependency order

The full system should be assembled through progressively stronger contracts:

1. Establish dense and conventional modular baselines with one physical measurement boundary.
2. Align observation, action, outcome, location, and time before adding learned routing.
3. Introduce conditional execution and measure activation, data movement, and dispatch overhead.
4. Add attributable episodic memory without granting it slow-model write authority.
5. Add replay and consolidation under fixed maintenance budgets, then admit provisional modules only for measured capability gaps.

6. Test reversible protection, reopening, and structured pruning before promoting repeated behavior into cheaper paths.
7. Learn logical routing and physical placement together only after their separate baselines are stable.
8. Run continual operation with drift, conflict, newcomer admission, recovery, and complete lifecycle accounting.

Each stage keeps the strongest baseline from the previous one. A later mechanism cannot hide an earlier regression behind a higher aggregate score.

Efficiency mechanism

The architecture seeks three compounding reductions:

- **activation:** execute a small relevant subset of stored capacity;
- **movement:** keep repeated computation near the state it uses; and
- **reinterpretation:** convert stable repeated work into cheaper memory, precision, routing, or compiled paths.

Before execution, the resource plane prices the proposed route. Afterward, measured telemetry corrects that estimate. Hard risk floors remain constraints rather than terms that average efficiency can trade away.

Those reductions are useful only after control overhead is included. Lifecycle energy is therefore

$$E_{\text{life}} = E_{\text{runtime}} + E_{\text{memory}} + E_{\text{network}} + E_{\text{adaptation}} + E_{\text{maintenance}} + E_{\text{recovery}},$$

with every term measured in joules over the same workload and system boundary. The energy model defines the full comparison contract.

Evidence status

INGREDIENT	CLAIM RANGE	STATUS IN THIS SYNTHESIS
Conditional routing and early exit	C-003–C-004	available engineered mechanisms
Predictive representation and residual allocation	C-005–C-007	plausible runtime composition
Fast/slow memory and protection	C-008–C-010	scoped evidence; lifecycle policy experimental
Structural pruning and hardening	C-012–C-015	individual mechanisms available; promotion logic unvalidated
Sensorimotor control and local autonomy	C-017–C-024	biological and engineered constituents
Replay, reconsolidation, and forgetting	C-036–C-042	scoped observations; combined controller speculative
Maturation, reopening, and contextual control	C-043–C-051	scoped interventions; digital translations experimental
Collective coordination and resilience	C-052–C-060	scoped observations; quorum and fragility translations experimental
Endogenous generation and exploration	C-061–C-066	constituent observations; integrated curriculum speculative
Communication and convention lifecycle	C-268–C-281	constituent mechanisms established or scoped; versioned repairable composition speculative
Cumulative inheritance across turnover	C-343–C-367	bounded constituents established or scoped; population advantage unvalidated
Institutional authority and rule repair	C-368–C-395	formal constraints and bounded evidence; multi-level composition speculative
Human initiative, authority, and recovery	C-396–C-416	bounded HCI effects; combined record speculative
Complete integrated system	none	unvalidated project synthesis

Speculative extensions

- Modules bid for compute using expected task-value improvement per joule while hard risk and communication limits remain external constraints.
- Local clocks and queues replace a global synchronous step where causality and hardware permit it.
- Routing, memory placement, interconnect, and precision are learned jointly.
- Replicas run alternative maintenance decisions and compare later outcomes before one state becomes canonical.
- External tools and sensors become active experiments chosen for information gain, not passive input channels.

Failure modes

- A shared predictive state becomes a dense communication bottleneck.
- Task, resource, adaptation, and maintenance controllers oscillate or reward incompatible behavior.
- Routing overhead and scattered memory access erase activation savings.
- Episodic and factual stores disagree without an explicit conflict policy.
- Maintenance, regression, and recovery consume more energy than runtime saves.
- Premature hardening turns shortcuts or mutable facts into rigid behavior.
- Aggregate quality hides rare-case regression, fragility, or failed recovery.
- A scalar resilience score hides opposing resistance, recovery, adaptability, and newcomer-admission effects.
- Proposal generation collapses into high-temperature sampling without targeted intervention and outcome-based selection.
- Fluent exchange creates false common ground, silent semantic drift, or local conventions that newcomers and older versions cannot interpret or reject.
- A population receives credit for cumulative learning when it only performs more parallel search, preserves a headline score while losing rare skills, or copies one correlated lineage through every nominally independent agent.
- Governance vocabulary hides an ordinary optimizer, or veto, delegation, amendment, and participation add gridlock, capture, concentration, churn, or oversight cost without improving task-native or protected outcomes.
- A nominal human approval, explanation, or confidence display is credited as assurance even though state is stale, the mode is misunderstood, authority is ineffective, intervention arrives late, or recovery cannot change the world state.
- The architecture accumulates mechanisms faster than experiments can reject them.

Measurable predictions

1. Conditional routing reduces measured runtime data movement and energy at matched quality, calibration, and tail risk.
2. Attributable episodic memory improves adaptation latency without increasing protected slow-model regression.
3. Maintenance and structural consolidation reduce lifecycle cost after replay, validation, migration, and recovery are counted.

4. Separating factual propositions from reusable skills lowers correction cost and unsupported factual carryover.
5. Reversible maturation gates improve retention and relearning relative to fixed regularization, fixed pruning schedules, and naive fine-tuning.
6. Versioned uptake, repair, and cross-play gates reduce silent semantic failure under agent and protocol drift beyond a complete fixed-protocol stack.
7. Audited turnover retains and recombines more protected capability than a centralized continual learner at equal cumulative learning, coordination, storage, migration, and energy cost—or the population mechanism is retired.
8. Multi-level authority improves an applicable task-native or protected-harm frontier beyond the complete ordinary governance stack after gridlock, capture, concentration, human attention, and lifecycle cost—or it is merged.
9. A recoverable initiative record predicts failures beyond polished mode labels, previews, confirmations, undo/history, logs, runtime assurance, and user testing on consequential and accessibility-stratified tasks—or only its individually supported fields remain.
10. The integrated system occupies a better quality–risk–latency–energy frontier than every component ablation and the strongest ordinary controller, scheduler, cache, and router baselines.

Energy model and efficiency evaluation contract

Scope

Energy efficiency is not a property of a model in isolation. It is a measured relationship among a task, an input distribution, a quality and risk envelope, a latency or throughput requirement, a hardware–software system, a lifecycle horizon, and a physical measurement boundary.

This chapter defines the common contract for every efficiency result in the project. It replaces operation-count headlines with equal-budget comparisons, prices data movement and adaptation, and carries uncertainty through to an explicit reject, revise, or promote decision. The detailed notation and first models remain in the mathematical notes; experiment contracts instantiate this chapter rather than inventing new accounting rules.

An efficiency result is identified by the record

$$\mathcal{R} = (\mathcal{T}, \mathcal{D}, Q, R, L, \mathcal{B}, \mathcal{H}, \mathcal{S}, H, U),$$

where:

SYMBOL	MEANING	UNIT OR DECLARATION
$\mathcal{T}$	task and target behavior	named benchmark or environment
$\mathcal{D}$	evaluated input distribution	named dataset, stream, or generator
Q	primary task quality	task-specific score
R	declared risk or error measure	task-specific risk unit
L	latency requirement	seconds, with percentile
$\mathcal{B}$	physical accounting boundary	device, node, cluster, or facility
$\mathcal{H}$	hardware configuration	device type, count, clocks, memory, and interconnect
$\mathcal{S}$	software configuration	versions, precision, kernels, compiler, and runtime
H	lifecycle horizon	seconds and qualified-event count
U	uncertainty description	confidence interval and measurement model

Two energy numbers with different records are not directly comparable. A result may vary one field deliberately, but it must show the resulting curve instead of silently carrying the old conclusion across the change.

Biological observation

Neural signaling operates under metabolic constraints (C-001). Biological systems therefore provide examples of computation shaped by the cost of activation, communication, maintenance, and adaptation. Event-driven hardware also demonstrates that local sparse activity can be implemented outside biology (C-015).

The observation does not define a common operation between a brain and a digital accelerator. A spike, synaptic event, memory read, floating-point multiply, token, and successful decision are different functional units. The inherited brain-to-accelerator range remains disputed under C-016 because its numerator and denominator do not share a task, quality target, physical boundary, or operation definition.

Here the brain establishes the feasibility of severe resource allocation. The actionable translation is to make every proposed mechanism compete under a complete physical and statistical contract.

Proposed AI translation

The qualified event is the functional unit

Let x_j be deployment event j, and let $I_j \in \{0,1\}$ indicate that the event was served inside the preregistered quality, risk, and latency envelope. For a horizon containing N offered events, the qualified count is

$$N_q = \sum_{j=1}^{N} I_j.$$

N, N_q, and I_j are counts or dimensionless indicators. The envelope must define whether qualification is event-level, stratum-level, or run-level. Selectively dropping hard events cannot reduce the energy denominator: offered events, rejected events, failed events, and qualified events are all reported.

For tasks whose quality is defined only over a population, comparability is established at the run level. Candidate C and baseline B are inside the same envelope only if

$$Q_C - Q_B \geq -\varepsilon_Q, \qquad R_C - R_B \leq \varepsilon_R, \qquad L_{C,p} \leq L_{\max},$$

where ε_Q is the allowed quality loss in quality units, ε_R is the allowed risk increase in risk units, p is the declared latency percentile, $L_{C,p}$ is candidate latency at that percentile in seconds, and $L_{\max}$ is the latency ceiling in seconds. All margins are fixed before confirmatory runs. Energy superiority is tested only after this envelope is satisfied.

Measurement boundary and gross energy

For boundary b and measurement interval $[t_0, t_1]$, gross electrical energy is

$$E_b^{\text{gross}} = \int_{t_0}^{t_1} P_b(t)\,dt,$$

where $P_b(t)$ is measured electrical power in watts, t is time in seconds, and E_b^{gross} is energy in joules. The instrument, sample rate in hertz, clock alignment, integration method, and missing-sample policy are part of U.

Incremental energy may also be reported:

$$E_b^{\text{inc}} = \int_{t_0}^{t_1} \left[P_b(t) - P_b^{\text{idle}}(t)\right] dt,$$

where $P_b^{\mathrm{idle}}(t)$ is power in a separately measured, precisely defined idle state in watts. Gross energy remains primary. Incremental energy is meaningful only when candidate and baseline use the same idle definition and the idle subtraction does not hide reserved or provisioned capacity.

Every reported number carries one provenance label:

- **measured:** produced by a named instrument or counter during this run;
- **modeled:** calculated from measured counts and versioned coefficients;
- **cited:** copied with its original system, boundary, and date; or
- **hypothesized:** a preregistered value or direction awaiting measurement.

Mixed totals expose the provenance of every component. They are not labeled “measured energy” when any material term is modeled or cited.

The permitted boundaries are:

BOUNDARY	INCLUDED ENERGY
Device	accelerator package or named component only
Node	accelerator, CPU, memory, local storage, power conversion, and attributable node cooling
Cluster	participating nodes, fabric, shared storage, and attributable cluster infrastructure
Facility	cluster energy plus contemporaneous attributable facility overhead

If facility energy is derived from power usage effectiveness,

$$E_{\mathrm{facility}} = \mathrm{PUE}\ E_{\mathrm{IT}},$$

where E_{IT} and E_{facility} are joules and PUE is the dimensionless ratio of facility power to IT-equipment power for the same site and interval. A cited fleet average is not substituted for a measured node or cluster result. Facility, carbon, water, and financial cost are separate outcomes; none is used as a synonym for joules.

The energy number is a measurement result

The measurand must name the electrical boundary, object, state, interval, conditions, aggregation, workload, and intended decision. A counter reading is an indication, not yet a result (C-519, C-520). Each reported energy value therefore retains:

1. instrument, range, firmware, voltage/current/phase configuration, bandwidth, sampling, clock, environment, and raw trace identity;
2. measurement model, integration rule, preprocessing and software version, missing-sample policy, warm-up, retry, idle, and useful-output definitions;
3. calibration chain, stated reference, corrections, validity scope, checks, drift status, and every uncertainty contribution;
4. covariance from shared meters, clocks, coefficients, environments, and preprocessing, plus coverage method and reproducibility conditions; and
5. the decision rule, target uncertainty, permitted comparison, expiry, provenance, supersession, and invalidation dependencies.

Calibration does not mean validated, traceability does not mean accurate enough, and uncertainty is not unknown error (C-521–C-526). The measurement-contract note defines the record, dimensional checks, covariance propagation, guard bands, drift review, and invalidation graph.

For sampled power, the estimator

$$\widehat{E}=\sum_{m=1}^{M}P_m\Delta t_m$$

has units J because P_m is W and Δt_m is s. Its uncertainty model must retain correlations between samples and coefficients when they share a meter, calibration, clock, or correction. Repeated samples from one trace do not become independent experimental runs. An end-to-end calibrated boundary can rank systems differently from software estimates or device-only counters; the existence, sign, and frequency of such reversals are measured outcomes, not constants (C-536).

Paired meter blocks and seed-level inference

Fast work units can be shorter than a meter's sampling, clock-alignment, or resolution limits. Candidate 010 therefore freezes ordered opportunity blocks, counterbalances arm order within each scenario-seed cluster, and records warm-up and idle intervals separately. This reduces acquisition and review overhead without changing the inferential unit.

For seed s, arm a, and its set of measured blocks $\mathcal{B}_{s,a}$, define

$$\hat{e}_{s,a}=\frac{\sum_{b\in\mathcal{B}_{s,a}}E_b^{\mathrm{gross}}}{\sum_{b\in\mathcal{B}_{s,a}}C_b},$$

where E_b^{gross} is measured block energy in joules and C_b is the count of correct commits in that block. Thus $\hat{e}_{s,a}$ has units J/correct commit. Every repetition and scenario is aggregated inside the seed before a candidate-baseline contrast is formed:

$$d_s=\hat{e}_{s,C}-\hat{e}_{s,B}.$$

With n independently generated seeds, the paired mean and its two-sided Student-t interval are

$$\bar{d}=\frac{1}{n}\sum_{s=1}^{n}d_s,\qquad \bar{d}\pm t_{1-\alpha/2,n-1}\frac{s_d}{\sqrt{n}},$$

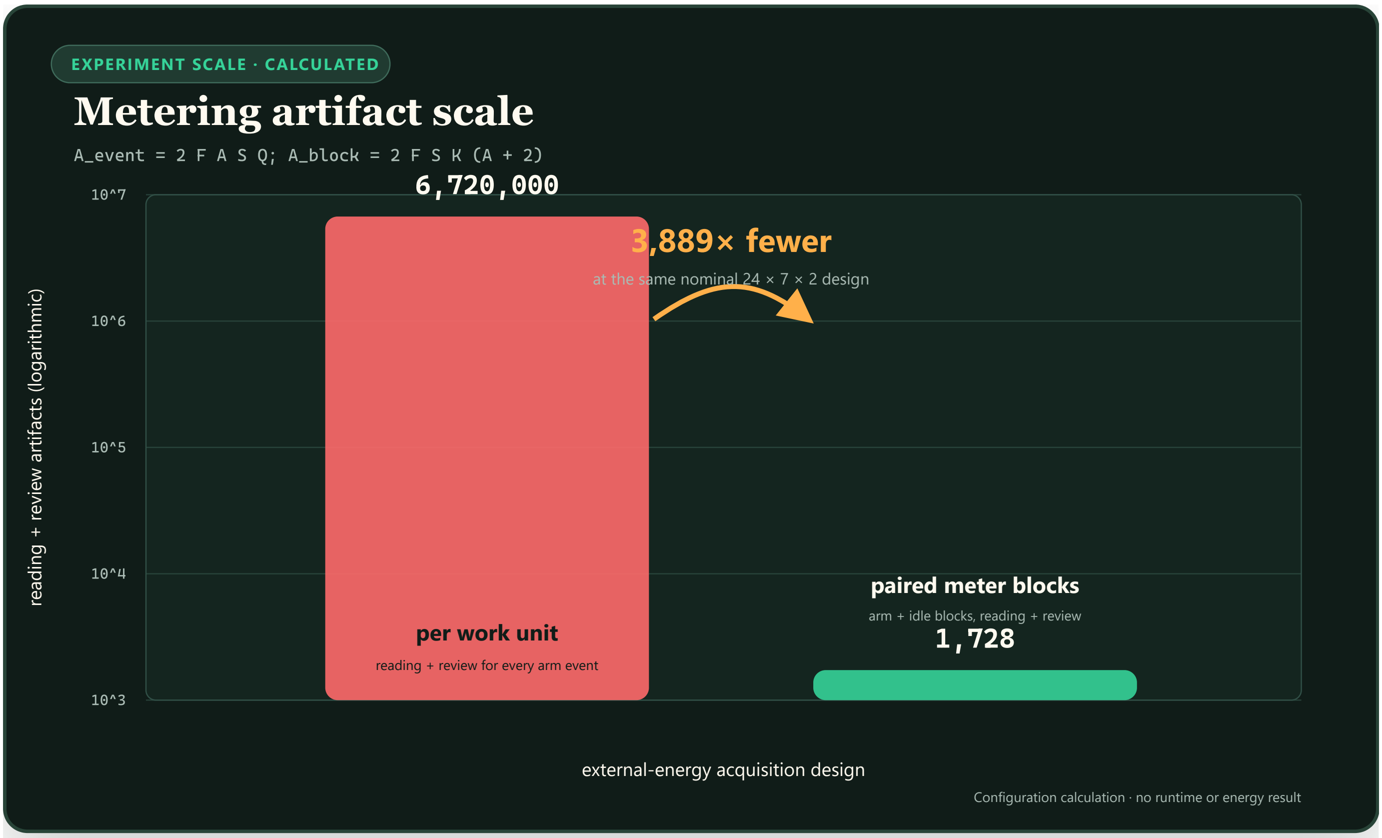
where s_d is the sample standard deviation of the seed contrasts in J/correct commit. Blocks, scenarios, power samples, and repeated measurements contribute precision and diagnostic information; none increases n.

Let $u_{s,a}$ be the declared conservative expanded measurement allowance for one seed-arm aggregate, including calibration contributions and half a meter resolution quantum per block, divided by its correct-commit count. A simple worst-direction reporting envelope widens the statistical interval by

$$\bar{u}=\frac{1}{n}\sum_{s=1}^{n}(u_{s,C}+u_{s,B}).$$

This envelope is deliberately conservative and does not replace a fuller covariance model. A zero correct-commit denominator, missing block, invalid review, expired calibration, excessive clock uncertainty, or fixture-shaped meter record makes the comparison undefined rather than silently dropping a seed.

The nominal two-seed design would create 6,720,000 reading-plus-review files under per-work-unit metering but 1,728 under the current paired-block design. That is an artifact-count calculation, not an energy result or a larger sample size.



Editable assumptions: `../assets/plots/core-models.json`.

Lifecycle energy

For candidate C over horizon H, define the disjoint lifecycle total

$$\begin{aligned} E_C^{\text{life}}(H) = E_C^{\text{search}} + E_C^{\text{train}} + E_C^{\text{consolidate}} + E_C^{\text{compile}} \\ + E_C^{\text{serve}}(H) + E_C^{\text{maint}}(H) + E_C^{\text{migrate}}(H) + E_C^{\text{recover}}(H) + E_C^{\text{idle}}(H). \end{aligned}$$

Every E term is energy in joules at the same boundary $\mathcal{B}$:

- E^{search} covers architecture search, hyperparameter tuning, and failed development runs attributable to the selected result;
- E^{train} covers final training and validation;
- $E^{\text{consolidate}}$ covers pruning, replay, merging, or structural stabilization before service;
- E^{compile} covers compilation, quantization, layout generation, and deployment preparation;
- E^{serve} covers event execution during H;
- E^{maint} covers monitoring, replay, repair, indexing, and lifecycle control during H;
- E^{migrate} covers state serialization, transfer, warm-up, and reconfiguration not already assigned elsewhere;
- E^{recover} covers additional recovery work after a declared failure or regime change; and
- E^{idle} covers provisioned but inactive devices, memory, communication links, and reserve capacity during H.

An event is assigned to exactly one term. A migration byte, for example, may appear in the movement ledger but its energy is not also charged as ordinary serving traffic.

Lifecycle energy per qualified event is

$$e_C^{\text{life}}(H) = \frac{E_C^{\text{life}}(H)}{N_{q,C}(H)},$$

where $N_{q,C}(H)$ is the candidate's qualified-event count during H and e_C^{life} is joules/qualified event. The same result is also reported per offered event so quality filtering remains visible.

Break-even horizon

Let ΔE_0 be candidate minus baseline one-time energy before service in joules, and let $\delta e = e_B^{\text{serve}} - e_C^{\text{serve}}$ be the measured steady serving saving in joules/qualified event. When $\Delta E_0 > 0$ and $\delta e > 0$, the event-count break-even point is

$$N^* = \frac{\Delta E_0}{\delta e}.$$

N^* is a count. At qualified service rate λ_q in events/second, the time break-even is $T^* = N^* / \lambda_q$ seconds. Maintenance, migration, recovery, and idle differences that grow with time must be included in the full numerical break-even calculation; the simple quotient is valid only when they are already represented in δe or are negligible over the stated horizon. If $\delta e \leq 0$, there is no energy break-even.

Data movement ledger

Executed arithmetic and moved data remain separate observables. For one run,

$$B_{\text{total}} = B_{\text{cache}} + B_{\text{device}} + B_{\text{host}} + B_{\text{fabric}} + B_{\text{storage}} + B_{\text{migration}},$$

where every B term is bytes crossing the named, disjoint boundary: on-chip cache levels, device memory, host–device interface, inter-device or inter-node fabric, persistent storage, and migration path. The ledger additionally reports byte-hops, defined as payload bytes multiplied by traversed logical or physical links, in byte-hops. Reads and writes are separated when their costs differ.

Operation and movement counts can support a calibrated model,

$$\widehat{E}_{\text{model}} = \sum_{o \in \mathcal{O}} n_o \epsilon_o + \sum_{\ell \in \mathcal{L}} B_\ell \epsilon_\ell + \int_{t0}^{t1} \widehat{P}_{\text{idle}}(t) dt,$$

where $\mathcal{O}$ is the declared set of operation classes, n_o is the executed count for class o, ϵ_o is calibrated joules/operation, $\mathcal{L}$ is the set of movement boundaries, B_ℓ is bytes crossing boundary ℓ, ϵ_ℓ is calibrated joules/byte, and $\widehat{P}_{\text{idle}}$ is modeled idle power in watts. A hat marks an estimate. The model is checked against gross measured energy; it never upgrades modeled joules into measured joules.

Numerical energy-per-operation tables are tied to process, device, precision, data locality, utilization, and year. The engineering audit therefore uses them as an accounting method, not constants (computer architecture analogue). The thermodynamic floor $k_B T \ln 2$ joules describes the minimum dissipation associated with erasing one bit under its physical assumptions; k_B is the Boltzmann constant in joules/kelvin and T is absolute temperature in kelvin. It is not an estimator for an inference, multiply, or memory transfer.

Equal-budget comparisons

Candidate and baseline receive matched opportunity to succeed. Each experiment freezes:

1. training and evaluation data, stream order, changes, failures, and seeds;

2. quality, risk, latency, and availability requirements;
3. input information and look-ahead—an oracle is labeled and never used as a superiority baseline;
4. tuning trials and tuning compute in device-hours or joules;
5. provisioned parameter, memory, module, edge, or replica capacity;
6. service compute, controller compute, telemetry, and actuation cadence;
7. migration, topology-edit, replay, and reserve ceilings;
8. software optimization effort appropriate to both methods; and
9. measurement boundary, duration, warm-up, repetitions, and instrumentation.

Some mechanisms intentionally exchange one resource for another. They are not forced into a single operation count; instead, all resource axes are recorded and the quality–risk–latency–energy–movement frontier is compared. A method that violates a hard budget is infeasible, not retroactively scaled into compliance.

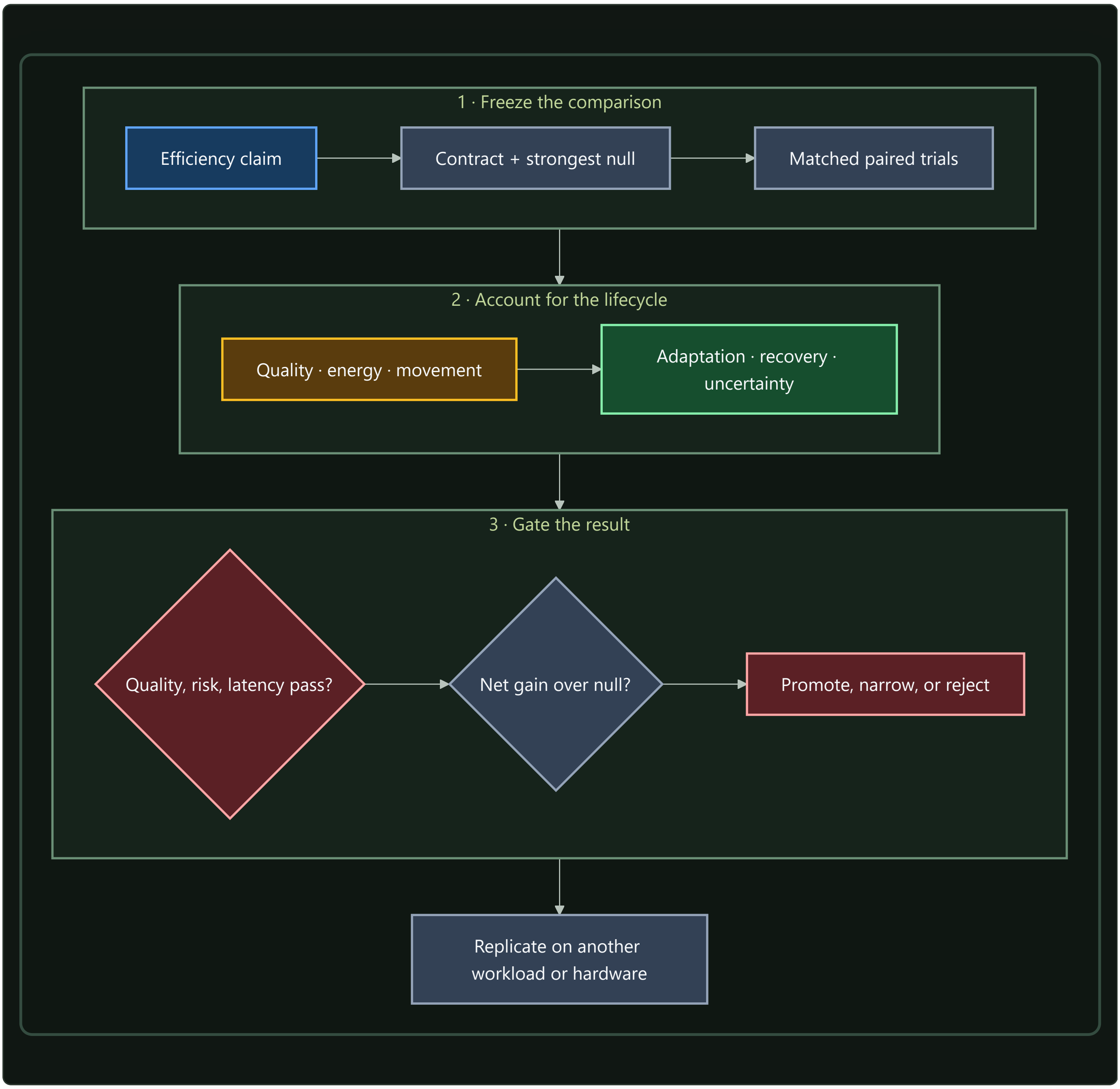
Engineering null models

The relevant null model is the strongest standard solution to the same constrained problem, not only a dense network:

PROPOSED MECHANISM	MINIMUM NULL MODEL
Conditional routing	tuned dense reference, fixed sparse model, and budgeted adaptive router
Prediction-error compute	calibrated residual/change detector plus value-of-information acquisition
Homeostatic allocation	tuned feedback or primal/dual resource controller
Adaptive topology	fixed topology with adaptive weights/routing and periodic global graph optimization
Temporal communication	fixed optimized schedule and work-conserving scheduler
Memory tiers	tuned cache/TTL/retrieval policy and, where possible, an oracle-lifetime upper bound
Maintenance plane	periodic/adaptive checkpoint, monitoring, and recovery controller
Structural specialization	profile-guided compilation, layout, quantization, and accelerator-aware kernel

The full mapping and formal reference points are in the engineering analogue audit. Component ablations determine whether the claimed mechanism causes a gain. An oracle supplies headroom, while a shuffled or random-action control detects benefit from adaptation without useful information.

Evaluation loop



Editable source: `../assets/diagrams/efficiency-evaluation-loop.mmd`.

Uncertainty and missing costs

The primary comparison is paired. Candidate and baseline run the same workload seed, event sequence, change schedule, and failure trace. For paired replicate k, define relative lifecycle effect

$$d_k = \frac{e^{\text{life}}_{C,k} - e^{\text{life}}_{B,k}}{e^{\text{life}}_{B,k}},$$

where each e is joules/qualified event and d_k is dimensionless. Report the paired point estimate and a confidence interval across independent full-run seeds. Hierarchical bootstrap resampling is used when events are nested within seeds or change episodes. Repeated samples from one power trace are not treated as independent runs.

For a modeled scalar energy $\widehat{E} = f(\theta)$, where f is the declared energy-model function and θ is its coefficient-and-count vector, local covariance propagation is

$$\mathrm{Var}(\widehat{E}) \approx J_f \Sigma_\theta J_f^\top,$$

where θ is the vector of calibrated coefficients and measured counts, Σ_θ is their covariance matrix in the corresponding squared mixed units, and J_f is the Jacobian row vector of partial derivatives of f with respect to θ. The result is variance in joules squared. Bootstrap or Monte Carlo propagation replaces this approximation for nonlinear, correlated, or non-Gaussian estimates.

If a material category cannot yet be measured, let its energy lie in declared interval $[E_u^-, E_u^+]$ joules. The conclusion must survive the least favorable assignment within that interval. A result whose sign depends on assuming the missing category is zero remains unresolved.

Uncertainty reporting includes:

- meter accuracy, resolution, sampling rate, clock error, and calibration date;
- run-to-run variation, warm-up state, thermal state, and background workload;
- coefficient covariance and extrapolation range for modeled components;
- seed, task-stratum, failure, and change-event variation; and
- censored failures to finish or recover.

The uncertainty interval accompanies the effect and the absolute values. A large sample size does not repair a mismatched boundary or missing lifecycle phase.

Efficiency mechanism

The project's mechanisms act on distinct terms of the lifecycle and movement ledger. They must earn their complexity against the appropriate null model.

LEVER	INTENDED PHYSICAL CHANGE	NECESSARY MEASUREMENTS	COMMON NULL EXPLANATION
Selective modules and early exit	fewer executed operations and avoided activation movement	operation classes, device/host bytes, gate energy, latency by difficulty	tuned smaller dense model or confidence threshold matches it
Compartmentalization and placement	fewer cross-boundary bytes and smaller blast radius	byte-hops, cut traffic, placement/migration energy, recovery	ordinary partitioning or cache-aware layout matches it
Adaptive logical topology	fewer idle links and shorter useful paths under drift	edge-seconds, byte-hops, controller work, migration, reserve, recovery	adaptive weights or periodic graph optimization matches it
Memory lifetime routing	fewer expensive writes, replays, and stale retrievals	tier bytes, reads/writes, migration, retention quality, maintenance energy	LRU/TTL/retrieval policy matches it
Quantization and compilation	less arithmetic and movement per stable path	executed precision, kernel mix, bytes, compile energy, break-even	standard profile-guided optimization matches it
Maintenance and consolidation	lower future update/recovery cost after paid background work	replay/checkpoint bytes, optimizer work, downtime, retained quality, lifecycle horizon	periodic maintenance matches it
Temporal communication	less broadcast while meeting deadlines	physical context bytes, synchronization power, jitter, schedule-update cost	optimized time-aware schedule matches it

The first candidate experiments instantiate the shared contract:

- adaptive topology prices edge updates, reserve, migration, and recovery against routing and graph-optimization baselines;
- multiscale context broadcast separates logical bandwidth from physical tensor movement and board energy; and
- recovery dynamics prices active probes, telemetry, latency, and false alarms against standard monitoring and system-identification baselines.

An operation saving becomes an energy mechanism only if it removes physical work at the chosen boundary. An arithmetic reduction that adds irregular dispatch, cache misses, synchronization, or migration may move cost rather than remove it.

Result hierarchy

Every experiment reports the narrowest supported level:

1. **Proxy reduction:** fewer operations, bytes, active edges, or updates.
2. **Component reduction:** lower measured energy at a named device or link.
3. **System reduction:** lower gross node or cluster energy inside the matched quality, risk, and latency envelope.
4. **Lifecycle reduction:** lower $e^{\text{life}}(H)$ after search, adaptation, maintenance, reserve, failure, and idle costs at a declared horizon.
5. **Transfer:** the lifecycle result replicates on another workload or hardware class without changing the claim after seeing the result.

Evidence at one level does not imply the next. This hierarchy makes a useful proxy result publishable without inflating it into an end-to-end claim.

Audit of the inherited comparison

The source discussion asserted that a 20-watt brain would correspond to hundreds of kilowatts or megawatts of accelerator power. The calculation mixed an assumed biological operation rate, peak accelerator arithmetic at varying precisions, and a facility multiplier. Its formal shape,

$$P_{\text{counterfactual}} = P_{\text{brain}} \frac{\eta_{\text{brain}}}{\eta_{\text{machine}}},$$

is dimensionally valid only when P_{brain} is brain power in watts, $P_{\text{counterfactual}}$ is machine power in watts, and both efficiencies η measure the same qualified functional output per joule under the same quality, risk, time, and boundary record $\mathcal{R}$. No such common output has been established. The inherited values are retained as provenance for C-016, not as constants, bounds, or project targets.

The title "20 Watts Was Enough" expresses the research constraint: useful adaptive intelligence exists under a small biological power budget. The project's numerical claims will come from the measurement loop above.

Evidence status

- Energy constrains biological signaling under the scoped evidence in C-001: **established**.
- Local sparse learning can be implemented on event-driven neuromorphic hardware under C-015: **established for the cited systems**, not proof of this architecture.
- The inherited biological-to-accelerator numerical range under C-016: **disputed**.
- Data movement, control, maintenance, and idle provision can dominate or erase nominal arithmetic savings: **engineering null hypothesis to measure for each system**, not a fixed proportion.
- Lower lifecycle energy from the integrated architecture at matched quality, risk, and latency: **speculative** until candidate experiments clear their preregistered gates and measured system boundaries.

Speculative extensions

Calibrated marginal-energy control

A controller could estimate the marginal joules of another layer, sensor, retrieval, replay, or route and compare it with expected decision improvement. The proxy would be trained from counters but recalibrated against gross energy measurements. Its calibration error and controller energy would be first-class outcomes.

Carbon- and scarcity-aware scheduling

Once joules are stable, execution time and placement could include marginal carbon intensity, water stress, or resource scarcity. These quantities retain their own units and uncertainty; they augment rather than replace the energy ledger.

Causal movement attribution

Hardware counters observe traffic but do not always identify which mechanism caused it. Controlled placement, cache-flush, route-freeze, and migration ablations could estimate the marginal bytes and joules caused by a gate, memory tier, or topology update.

Cross-substrate transfer model

A hierarchy of calibrated energy models could predict which mechanisms retain their advantage across GPUs, CPUs, accelerators, and distributed nodes. The transfer model would predict the sign and break-even horizon before the new hardware run, then be scored on calibration rather than refit after every result.

Failure modes

- **Boundary drift:** device energy is described as node, cluster, or facility energy without measuring the added components.
- **Quality leakage:** the candidate saves energy by rejecting hard inputs, lowering rare-event quality, or exceeding the latency envelope.
- **Weak null model:** a new controller is compared only with dense execution when tuned routing, caching, control, scheduling, or compilation solves the same problem.
- **Unpriced adaptation:** search, telemetry, replay, state movement, reserve, rollback, and failed updates disappear from the ledger.
- **Double counting:** migration or maintenance energy appears in both serving traffic and its own lifecycle category.
- **Proxy substitution:** FLOPs, parameters, active modules, logical messages, TDP, peak throughput, or a vendor operation table is reported as measured energy.
- **Idle erasure:** baseline subtraction removes capacity that the candidate must keep powered for latency or recovery.
- **Amortization without demand:** a one-time optimization is divided by an assumed event count beyond the measured deployment horizon.
- **Uncertainty collapse:** samples within one power trace are treated as independent replicates, or missing components are assigned zero uncertainty.
- **Optimization asymmetry:** the candidate receives more tuning trials, future information, custom kernels, or favorable batch/precision settings.
- **Hardware overgeneralization:** an irregular sparse win or loss on one substrate is stated as an architecture-wide property.
- **Thermodynamic rhetoric:** the Landauer floor or a 20-watt biological value is used to estimate attainable contemporary system energy.

Measurable predictions

H-E1 — Conditional execution

At matched quality, risk, and latency, a conditional path will reduce gross node joules/qualified event only after avoided module compute and physical data movement exceed gating, dispatch, and lost-utilization overhead. A tuned smaller dense model and fixed sparse model are required nulls. Reject the mechanism-level energy claim if the saving exists only in operation counts.

H-E2 — Data movement

Placement, compartmentalization, and structural consolidation will reduce host–device and inter-device byte/qualified event as well as byte-hops. The effect should predict a measured energy reduction after calibration. Reject the placement explanation if bytes do not fall or an ordinary partition/layout baseline matches it.

H-E3 — Lifecycle break-even

Pruning, compilation, consolidation, or specialization will have a finite N^* and T^* inside the measured deployment horizon. The result must remain positive after search, failed runs, maintenance, and rollback are included. Report "no break-even" when steady serving savings are non-positive or the observed horizon ends first.

H-E4 — Adaptive topology

Under recurrent changes and faults, use-dependent topology will reduce full lifecycle modeled and then measured joules/timely utility unit without violating quality or recovery margins. It must beat fixed topology with adaptive routing and budget-matched periodic graph optimization, and its frozen-topology ablation must lose the advantage. The Stage-1 thresholds are defined in Candidate 001.

H-E5 — Memory lifecycle

Lifetime-aware memory actions will reduce replay, write, retrieval, and migration energy at a declared retention and obsolete-intrusion envelope. A tuned cache/TTL/retrieval policy is the null. The scheduling and provenance overhead must be included in the break-even horizon described by the memory-lifecycle model.

H-E6 — Recovery-aware efficiency

A system with lower normal-operation energy but materially slower or less reliable recovery will not dominate. Energy, utility deficit in utility-seconds, and recovery time in seconds are reported jointly under paired faults and regime changes. Reserve capacity is charged in watt-seconds even when unused.

H-E7 — Substrate transfer

A mechanism promoted beyond one device will retain the direction of lifecycle effect on at least one second hardware class using the same task envelope and an independently calibrated movement model. If the sign changes, the result is reported as substrate-specific and the counters responsible for the reversal must be identified.

The project advances an efficiency claim only when the strongest engineering null model is outside the preregistered equivalence margin and the full lifecycle confidence interval clears the material-effect threshold. Otherwise the result narrows the design space and the biological principle remains a source of hypotheses rather than a performance claim.

CHAPTER 20

`concept/90-research-roadmap.md`

5,576 WORDS

Research roadmap

> *A publication checkpoint records progress. Completion is earned by closing a gate, retiring a hypothesis, or producing a reproducible result.*

Scope

This roadmap converts an open-world research ambition into bounded iterations. Breadth remains unlimited: any scientific field may contain a transferable operation. Work remains finite because each iteration must change a canonical claim, principle, chapter, equation, diagram, decision, or experiment—or record that the new evidence changes none of them.

The stages are dependency gates rather than dates. Evidence discovery, architecture writing, null-model comparison, and measurement design continue in parallel, but an integrated system does not inherit a mechanism merely because its audit is interesting.

Biological observation

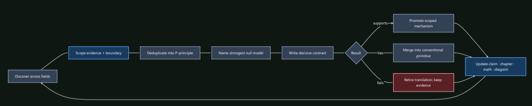
Development, adaptation, and maintenance operate on different timescales. Variation precedes selection; temporary states can precede durable change; stable organization remains repairable; and recurring behavior can migrate into structure. Relevant observations are separated across P-003, P-004, P-009, and P-010.

The engineering consequence is progressive commitment. Cheap reversible work comes first; expensive structural or hardware commitments require stronger evidence and longer reuse horizons.

Proposed AI translation

One research iteration

Wide diagram · fit to page



Editable source: `../assets/diagrams/research-iteration-loop.mmd`.

An iteration has three independently reviewable outputs:

OUTPUT	MINIMUM DURABLE ARTIFACT	MEANINGFUL COMPLETION
evidence delta	dated audit, primary reference, scoped `C-` claim proposal	the observation and its boundary can be checked without the architecture prose
synthesis delta	merge into a `P-` principle or a written discriminating reason to keep it separate	another field name does not create another mechanism by default
decision delta	experiment contract, rejection, or explicit no-change record	the result changes what the project would build or test

Parallel workstreams

Four workstreams move continuously, with different exit conditions:

1. **Discovery and evidence.** Sample under-covered fields by constrained problem, audit primary sources, extract causal operations, and preserve boundary conditions.
2. **Principle synthesis.** Deduplicate observations using the mechanism tuple in chapter 07, then update the shared primitive rather than accumulating themed components.
3. **Concept closure.** Turn retained principles into readable data contracts, control paths, state ownership, equations, null models, failure signatures, graphics, and measurable predictions.
4. **Decisive testing.** Compare one operation at a time against the strongest conventional method at equal interface, budget, hardware, and quality–risk envelope.

Stage 0 — Evidence, synthesis, and contracts

Stage 0 is complete for a mechanism only when:

- its observation has at least one scoped claim and primary source;
- its nearest existing AI and engineering analogues are recorded;
- its normalized operation maps to a `P-` principle or a held candidate;
- the affected chapter exposes state, information flow, timescale, physical boundary, and failure behavior;
- every equation defines symbols and units;
- a conventional null, ablation, measurement envelope, and rejection condition exist; and
- imported discussions remain provenance rather than evidence.

The repository structure satisfies this workflow. Individual mechanisms remain at different positions inside it; the domain inventory and adoption matrix expose those differences.

Historical evidence has a separate promotion gate

Longitudinal evidence enters the roadmap as one of four typed claims:

1. **Accounting identity:** the declared stocks and flows reconcile.
2. **Descriptive regularity:** a relation exists in a sampled period, place, coding system, and surviving archive.
3. **Causal estimate:** an intervention or design-specific contrast identifies an effect under explicit assumptions.
4. **Prospective prediction:** a frozen model succeeds on a period, place, lineage, or regime unavailable during development.

No row upgrades itself into the next. A cohort matrix can be arithmetically correct and forecast poorly under changing rates. An adoption curve can fit while influence, common cause, homophily, and command remain unidentified. A historical instrument can estimate a local effect without predicting a future system. A surviving record can support chronology while omitting destroyed or never-produced evidence.

For interval Δt seconds, the basic population identity is

$$N_{t+\Delta t} = N_t + B_t - D_t + I_t - O_t,$$

where N is qualified units at a timestamp and B, D, I, O are entry, retirement, immigration, and outmigration counts during the interval. The identity finds bookkeeping errors; it does not explain or forecast a flow. The full cohort, selection, diffusion, collapse-vector, and prospective-test contract is derived in population-observation math and folded into Candidate 014 rather than promoted as a new principle.

Stage 1 — Isolated mechanism experiments

The first tests isolate control operations before composing them.

CONTRACT	CANDIDATE OPERATION	STRONG NULL CLASSES	GATE
001 adaptive topology	reversible birth, routing, merge, and retirement	fixed sparse routing, periodic/global reoptimization, oracle bounds	structural changes improve a declared frontier after controller and migration cost
002 context broadcast	rate-limited context decoded by receiver-local temporal filters	FiLM, recurrent gates, global tokens, low-rank hypernetworks, gain scheduling	extra temporal decoding value survives an equal bit-rate and state interface
003 transition-class-aware recovery dynamics	bounded probes estimate hidden restoring margin and reject inapplicable transition classes	SLO/headroom, change detection, state-space ID, observability analysis, mechanism-specific indicators	useful warning survives probe and false-action cost while noise, rate, boundary, hidden-mode, and jump cases receive calibrated abstention
004 endogenous curriculum	generate, intervene, evaluate, and retain structured proposals	imitation, matched stochastic sampling, active learning, model-based planning, evolutionary search	compositional or causal gain survives equal proposal, interaction, evaluation, memory, and energy budgets
005 severity-ordered containment	contain spread, choose the least destructive qualified response, verify, escalate, and replenish	circuit breakers, static isolation, checkpoints, replicas, microrestart, scrubbing, rejuvenation, Bayesian repair/replace	collateral-loss and availability gains survive sensing, reserve, copying, replacement, and verification cost
006 reversible physical skill	compile a mature physical mapping into a rewritable local substrate with health probes and digital fallback	passive mechanics, analog control, FPGA/ASIC, fixed physical and matched digital reservoirs	a measured conversion, transport, recurrence, or command path is removed and lifecycle break-even precedes qualified retirement
007 endogenous-observation surveillance	jointly estimate hidden state and action-altered observation while preserving sensing/action provenance	residual CUSUM/GLR, nowcasting, maximum coverage, value-of-information sampling, delay-aware POMDP/MPC	decisions improve on true data vintages without mistaking policy-induced telemetry suppression for recovery
008 contestable modular allocation	withheld audits, protected entrants, real opportunity consequences, and replayable commitments under measured strategic private information	calibrated routers, constrained/primal–dual control, contextual bandits, randomized evaluation, applicable matching/scoring mechanisms	tie or lose under cooperative direct observability; improve protected outcomes under gaming, identity, collusion, and allocator threats at equal cost
009 graded assurance envelopes	bind distinct proof, effect, authority, evaluation, monitor, provenance, update, and recovery claims to a module version and invalidation graph	typed APIs, sandbox/IAM, CI/static analysis, runtime policy monitoring, lineage, canaries, transactions, schema/build invalidation	reduce unsafe admission, stale assurance, collateral rollback, and attribution time beyond the complete composed null at equal lifecycle budget
010 reset-coupled staged verification	create a reversible trace, invoke conditionally informative verification under commitment risk, then commit or reset with provenance	conditioned SPRT, calibrated cascade, abstention, retry plus rollback, redundant verifier, error-detecting code	improve false-commit, false-reject, latency, rollback, and energy frontiers only where later evidence is genuinely new and reset is cheaper than error
011 dual-loop operational assurance	connect bounded live command, containment, escalation, and restoration to dependency-linked verified learning, retrieval, and retirement	mature telemetry/tracing, SRE roles, IAM/interlocks, canaries, chaos drills, postmortems, action tracking, searchable runbooks	reduce tail containment and recurrence at equal authority, interruption, reviewer, storage, compute, human-time, and energy budgets
012 latency-qualified authority	derive admissible local action from evidence age, integrity, mode, headroom, and coordination, then abstain, fall back, or hand off	adaptive protection, gain scheduling, supervisory/constrained/robust control, barrier functions, runtime assurance	improve service and risk without self-certification, unsafe envelope transitions, extra authority, privileged state, or uncharged reserve
013 deficit–capability routing	send compressed unmet demand upward, return rate-limited scarcity context, and allocate or grow only where local capability is present	delayed centralized optimizer, backpressure, primal–dual allocation, learned routing, ordinary feasibility gates	improve fulfilled demand, tail starvation, regret, messages, oscillation, and energy without strategic capture or stale structural growth
014 versioned observation contracts	preserve response, exposure, spatial/temporal support, preservation, intervention, selection, association, uncertainty, vintage, supersession, and follow-up dependencies with each inferred claim	typed/event-time evidence schema, state-space and Gaussian-process models, calibrated likelihood, selection function, lineage, multiplicity control, predictive checks, simulation calibration, surveillance schema, graded assurance	reduce overconfidence, invalid mixed-clock fusion, stale claims, and wasted follow-up beyond the complete composed null at equal lifecycle cost
015 versioned repairable conventions	adapt local message forms while preserving typed literal meaning, uptake, repair lineage, protected meanings, compatibility, and rollback	typed protocol/DSL, schema registry, acknowledgement/retry, calibrated inference, replicated logs, standard coding, explicit migration and canary rollback	improve heterogeneous cross-play and drift adaptation without silent semantic failure or hidden decoder, repair, migration, and review cost

CONTRACT	CANDIDATE OPERATION	STRONG NULL CLASSES	GATE
016 conflict-bounded unit transition	define reproducible collective descendants, partition member and collective selection, and retain only conflict control with a net joint gain	ordinary modularity, typed interfaces, permissions/tests, clean versioning, external evaluation, routed experts, ensemble and population selection	demonstrate heritable collective capability and positive cost-adjusted between-collective selection under shortcuts, turnover, founder risk, and common-mode failure
017 contract-preserving semantic compaction	shrink histories under interpreter-, query-, representation-, evidence-, uncertainty-, rollback-, invalidation-, deletion-, access-, and corruption-degradation obligations	indexed history, snapshot plus suffix, views, key compaction, lossless compression, cold archive, extractive evidence summary, OAIS/PREMIS package, versioned schema/vocabulary, Candidate 014	pass frozen hidden future-query, migration, dependency-turnover, and invalidation suites across workloads and hardware after complete physical, curatorial, and maintenance cost
018 value-aware artifact tiering	place artifacts using access/recompute forecast, task/evidence loss, invalidation value, and reconstructability under correlated failure	LRU/LFU, ARC, W-TinyLFU, size/miss-cost caching, static/economic tiering, coded fault-domain placement	improve leakage-free out-of-sample placement under shift after metadata, migration, tail, privacy, and failure costs
019 audited cumulative inheritance	preserve and recombine validated capability across learner turnover through separately versioned generation, transmission, evaluation, lineage, external state, and governance	centralized continual learning, replay/distillation, version control, retrieval, repositories, workflow engines, quality-diversity search, fixed governance	improve protected capability, recombination, compatibility, and newcomer learning at equal cumulative work after coordination and lifecycle cost
020 constitutionalized multi-level control	scope local authority, escalate spillovers, type veto and appeal, revoke delegation, expose agenda, and repair higher-order rules under expiring emergency power	constrained optimization/MPC, IAM/interlocks, policy-as-code, separation of duties, independent evaluation, append-only lineage, runtime assurance, incident/change management	improve task-native or protected outcomes in two applicable task families after capture, gridlock, concentration, communication, oversight, and lifecycle cost
grounding loop	action-conditioned multimodal prediction and acquisition	passive multimodal learning, text-centric pretraining, standard world models	intervention and composition gains survive shortcut and missing-modality controls
memory lifecycle	capture, replay, externalize, weaken, or delete	reservoirs, prioritized replay, databases, caches, continual-learning regularizers	retention–adaptation improvement survives storage, maintenance, and deletion risk
mature hardening	compile, quantize, retrieve, or keep plastic through separate gates	compiler/autotuning, standard quantization, RAG, caching, distillation	each target produces a physical saving without freshness or recovery regression

Stage-1 exit gate: at least one isolated operation changes the measured quality–risk–latency–energy frontier relative to its strongest null. A tie merges the biological translation into the conventional primitive; a loss retires the translation while preserving the source evidence.

Stage 2 — Runtime plus adaptation

Combine the event controller from chapter 30 with attributable episodic capture from chapter 40. Keep slow structural state protected while a branch adapts.

Required composed ablations:

- runtime only;
- adaptation only;
- runtime plus adaptation without maintenance pricing;
- full two-loop system with storage, replay, rollback, and controller cost; and
- dense or monolithic continual-learning baselines at the same capacity and observation stream.

Stage-2 exit gate: sequential adaptation improves a declared task stream without a retention, calibration, rare-event, rollback, or lifecycle-energy regression hidden by the average score.

Stage 3 — Grounded developmental curriculum

Introduce temporally aligned perception, action, outcome, and language. Compare passive observation, action-conditioned prediction, targeted acquisition, and text-centric learning at matched capacity and interaction budget.

The environment must expose interventions, counterfactual structure, missing modalities, sensor noise, delayed outcomes, and distribution shifts. Language may describe, query, or compress the learned state; it cannot be the only path by which every evaluation variable enters the model.

Stage-3 exit gate: held-out intervention, temporal prediction, transfer, and composition gains survive leakage, memorization, simulator-state, and language-shortcut controls.

Stage 4 — Structural maturation

Apply the reversible maturity cycle from chapter 50 and the separated hardening targets from chapter 60. Candidate operations include structured pruning, module fusion, compilation, placement, quantization, retrieval externalization, write protection, and local reopening.

Every operation receives its own invalidation and recovery test. Parameter count is diagnostic; resident bytes, transferred bytes, wall energy, tail latency, maintenance, and rollback determine the system result.

Stage-4 exit gate: benefits remain positive after discovery, validation, migration, shadow traffic, failed promotions, recovery, and the observed reuse horizon are included.

Stage 5 — Substrate co-design

Compare accepted mechanisms on optimized conventional hardware and on event-driven, near-memory, neuromorphic, or other appropriate substrates. C-015 establishes feasibility for a scoped class of event-driven systems; it does not select a winner for this workload.

Algorithm, precision, memory layout, interconnect, batching, cooling boundary, and hardware control policy become joint experimental variables. The same qualified event and quality–risk envelope apply to every substrate.

Stage-5 exit gate: an independently reproducible end-to-end lifecycle advantage transfers beyond the workload or hardware used to tune it.

Breadth-wave decisions and next queue

The latest completed audits already changed the architecture rather than merely adding citations:

FIELD CLUSTER	DECISION NOW RECORDED
endocrine and circadian control	most pulse decoding maps to P-002/P-006/P-011; persistent receiver-local phase remains held against clocks, timers, PLLs, and schedules in Candidate 002
proteostasis and organelle quality control	sensing, containment, triage, repair, extraction, removal, replacement, and verification are separated; their ordered composition is Candidate 005
fault tolerance and reconstruction	exact restore, coding, replication, detectors, scrubbing, and self-stabilization are mandatory nulls; only underdetermined functional repair remains speculative
Earth-system transition signals	Candidate 003 is narrowed to gradual observable stability loss and now includes noise, rate, flickering, hysteresis, spatial, hidden-mode, boundary, jump, and benign-slowing controls
adaptive materials and self-assembly	demonstrated physical computation, memory, patterning, assembly, and healing map to P-006/P-009/P-010; reversible physical skill compilation becomes Candidate 006
epidemiology and surveillance control	placement, delay, ascertainment, pooled proxies, and intervention feedback map to P-001/P-007/P-009/P-013; endogenous observation becomes Candidate 007
economics, market design, and incentives	prices, auctions, matching, scoring, bandits, sanctions, and ledgers map to existing bundles; contestable allocation becomes Candidate 008 only for persistent strategic private information
programming languages and verification	proof, monitored behavior, authority, rollback, provenance, and empirical evidence remain distinct assurance classes; their versioned binding and invalidation becomes Candidate 009
chemistry and reaction networks	recognition, proofreading, nonequilibrium drive, autocatalysis, compartments, reaction–diffusion, transient assembly, molecular compute, oscillators, and integral control map to existing bundles; reset-coupled staged verification becomes Candidate 010 only where later evidence is conditionally new
high-reliability organizations and incident learning	field observations and doctrines do not transfer effect sizes; report denominators, topology, bounded escalation, incident command, and memory-in-use matter; the live/learning interface becomes Candidate 011
power grids and protection	local protection, adaptive settings, estimation, synchronization, reserve, islanding, cascade, and black-start restoration become mature nulls; evidence- and headroom-qualified authority becomes Candidate 012
plant distributed control	wound alarms, hydraulic/chemical drought response, patch foraging, tropisms, branching, source–sink flow, and mycorrhizal exchange map to existing bundles; organism, symbiosis, and community scales stay separate; deficit–capability routing becomes Candidate 013
astronomy and planetary remote inference	the latent-to-measurement-to-selection-to-association chain becomes explicit; non-detection, trials, model support, predictive checks, computation calibration, and causal identification remain distinct; Candidate 014 tests dependency-bearing observation contracts
geology and geomorphology	efficient-looking fracture and drainage networks become passive-physics nulls; connectivity is not conductance, erosion implies relocation, topology change can destroy service, and mixed supports extend Candidate 014 rather than creating a new principle

FIELD CLUSTER	DECISION NOW RECORDED
security and cryptography	authentication, authorization, information flow, detection, containment, and clean recovery remain separate; adversary, epoch, revocation, independence, compromise-horizon, and clean-root fields refine Candidates 009 and 012 rather than creating a new principle
linguistics and communication	formal composition, pragmatic inference, common ground, repair, turn timing, coding, iterated learning, grounding, and convention formation remain separate; Candidate 015 tests their versioned lifecycle composition against mature protocol engineering
paleobiology and major transitions	demographic and reproductive bottlenecks, extinction and recovery, acquisition and integration, aggregation and individuality, exaptation, constraint, and fossil observation remain distinct; Candidate 016 retains only reproducible collective heredity plus costed conflict control
pathology and rehabilitation	local selection can oppose host goals; sampled response is not eradication; tolerance is layered; output can hide compensation or depleted reserve; plasticity and dose have no monotonic sign; compensation-aware recovery becomes a cross-candidate evaluation method rather than a new principle
databases and storage	transactions, isolation, real-time ordering, replication, access paths, caches, compaction, reclamation, replay, temporal coordinates, coded repair, and tiering remain distinct; Candidates 017 and 018 retain only finite semantic preservation and task/evidence value beyond mature storage nulls
cultural evolution and archaeology	generation, transmission, evaluation, retention, governance, accessible model diversity, path dependence, and archaeological formation remain distinct; Candidate 019 tests only the residual population-level inheritance advantage across turnover
social choice and institutional governance	theorem assumptions, empirical effects, and authorized normative commitments remain separate; aggregation, strategy, delegation, veto, agenda, capture, amendment, participation, and path dependence refine Candidates 008/011/012/015/016; Candidate 020 tests only the applicable multi-level composition
human–computer interaction and human factors	nominal human presence, trust, transparency, explanation, preference, and adaptation are not assurance; effective initiative binds timely state, actual/pending mode, executable authority, state delta, recovery, resumption, expiry, accessibility, and full human cost into Candidates 009/011/012/015
quantitative history and demography	stock, flow, cohort, age, period, migration, replication, retirement, diffusion, archive selection, causal identification, collapse dimensions, and prospective validation remain distinct; their versioned composition strengthens Candidate 014 rather than creating another principle
aerospace, maritime, and safety-critical autonomy	stabilization, guidance, navigation, integrity, ODD, envelope protection, redundancy, fault stages, degraded service, fallback, collision avoidance, remote authority, certification, maintenance, and investigation remain distinct; validated asynchronous transfer refines Candidate 012 only
soft and active matter	passive relaxation, continuous fixed drive, external feedback, and adaptive policy remain separate; flocking, phase separation, defects, jamming, and assembly are physical mechanisms or nulls, while reversible phase-field compilation becomes only a Candidate-006 track
mechanical and civil resilience	compliant/passive mechanics, SHM, redundancy, robustness, graceful degradation, damage tolerance, fatigue/fracture, maintenance, max-flow/assignment, and transport recovery remain mature nulls; path-dependent residual capacity becomes a Candidates-005/012/014 test schema only
chemical and process engineering	conservation, nonlinear operation, separation, recycle, heat integration, MPC/RTO, fault stages, safety layers, operability, and plantwide control become mature nulls; only conservation-qualified flowsheet reconfiguration remains as a Candidate-001 stress track
metrology and measurement science	measurand, indication, calibration, adjustment, verification, validation, traceability, uncertainty, covariance, reproducibility, comparisons, decision rules, drift, software, and provenance remain distinct; their dependency-invalidating composition refines Candidates 009/014 and the energy model
developmental biology and morphogenesis	signal, competence, specification, implementation, commitment, sculpting, regeneration, metamorphosis, canalization, plastic response, current function, and evolutionary origin remain distinct; competence-gated structural transition becomes only a Candidates-002/006/009/010/014 fixture
microbial ecology, biofilms, and fungal networks	molecular signaling, receiver response, cell physiology, transport, lineage ecology, selection, organismal flow, symbiosis, and ecosystem consequence remain distinct; one transported-field fixture refines P-011/P-013 and Candidates 001/013
animal navigation, sensory ecology, biomechanics, and motor control	sensing policy, emitted signal, propagation, body mechanics, controller, calibration, environment, and task are bound explicitly; controlled observability and controller–plant binding refine Candidates 006/007/012/014 rather than becoming new principles
pharmacology and toxicology	commanded intervention, realized exposure, engagement, response, benefit, harm, adaptation, dependence, withdrawal, interaction null, and population support remain separate; a state-qualified intervention fixture refines Candidates 005/007/012/014
operations research, supply chains, and learning science	confidence and inventory records share one proxy-versus-state firewall, but instructional events and material commitments remain different transitions; horizon-qualified learning refines Candidates 004/019 while material service refines existing allocation/recovery candidates
legal evidence and procedure	normative authority, doctrine, empirical effect, formal inference, authentication, admissibility, weight, sufficiency, challenge, review, remedy, and finality remain distinct; a burden-qualified contestable-decision record refines Candidates 009/010/011/014/015/020
built environments and urban systems	accessibility, spatial configuration, passive physics, occupancy, controls, information models, commissioning, life safety, accessibility, infrastructure dependency, lifecycle burden, and recovery remain separate; no new principle survives, while an occupied-transition stress track narrowly refines Candidate 001
immune tolerance, adaptation, and memory	representation, recognition, deletion, ignorance, anergy, active suppression, exhaustion, expansion, contraction, affinity selection, trained state, context, memory maintenance, residency, and immunometabolism remain distinct; no new principle or candidate survives, while a typed lifecycle contract remains an evaluation discipline

FIELD CLUSTER	DECISION NOW RECORDED
music cognition, performance, and improvisation	statistical expectation, grouping, tonal state, phase correction, motif transformation, auditory–motor feedback, expressive plans, ensemble synchronization, improvisation, practice, far transfer, and enculturation map to existing bundles and ordinary sequence/control/retrieval/search/curriculum nulls; shared-clock-free partner- and phrase-specific co-adaptation remains a benchmark fixture only, not a principle or candidate
library, archival, and information science	provenance is not truth, fixity is not authenticity, retention is not useful memory, and retrieval rank is not epistemic confidence; appraisal, designated communities, representation dependencies, authority/vocabulary drift, migration, FAIR packaging, and institutional capability refine Candidates 014/015/017/018/019, while no new principle survives
comparative cognition, tool use, and flexible action	manufacture, causal transfer, future preparation, event memory, uncertainty control, social learning, teaching, recombination, exploration, inhibition, route recovery, and peripheral control are kept task-, opportunity-, morphology-, observation-, and history-qualified; species/task rankings are rejected, and Fixture F-003 refines evaluation for Candidates 004/006/007/014/018/019 without promoting a principle or candidate
visual art and design cognition	reconstructive generation, external workspaces, epistemic action, analogy, copying, fixation, proposal/selection separation, qualified aesthetic evaluators, material feedback, cultural accumulation, and provenance map to existing bundles; Fixture F-002 tests a versioned reconstruct–externalize–inspect–transform–evaluate composition across Candidates 004/014/017/018/019 without promoting a principle or candidate
mathematical practice and proof discovery	conjecture, analogy, abstraction, counterexample search, proof decomposition, libraries, representations, invariants, theorem proving, certificates, SAT/SMT, model finding, experimental mathematics, collaboration, and teaching map to mature methods; Fixture F-004 keeps attributable proposals separate from typed acceptance, reconstructs proof DAGs through small checkers, uses dependency-safe splits and reverse-dependency invalidation, and charges complete human/joule budgets across Candidates 004/009/010/011/014/017/019 without promoting a principle or candidate
fluid dynamics and turbulence	signed cascade/flux, intermittency and tails, detector-qualified coherent structures, closure/model-form support, objective-qualified ROMs, adaptive-resolution total work, observation/assimilation/observability, sensor-placement transport, stable net-benefit control, a mixing vector, path-dependent transition, calibrated extremes, measurement operators and uncertainty, and the full energy boundary remain separate; Fixture F-005 carries the outcome firewall across Candidates 002/003/006/007/012/014 without promoting a principle or candidate
sports expertise, adaptive performance, and team coordination	anticipation/interception, causal cue use, practice/retention/transfer, relevant variability, speed/accuracy/risk/energy, fatigue/readiness/damage/return, shared information/coordination, adversarial deception, feedback dependence, prospective selection, and full human/joule budgets remain separate; Fixture F-006 carries the representative history/resource-qualified performance contract across Candidates 002/004/006/007/009/012/014/019 without promoting a principle or candidate
optics, photonics, and inverse sensing	forward operators, finite information modes, null-space ambiguity, photon/noise budgets, structural priors, active/coded acquisition, adaptive correction, drift, saturation, fusion, optical transforms, conversion, analog error, fabrication, thermal control, future-query recovery, and full lifecycle energy remain separate; Fixture F-007 carries the operator-qualified measure–infer–intervene–monitor–route–retain contract across Candidates 001/006/007/010/014/017/018 without promoting a principle or candidate
semiconductor device and circuit reliability	qualification, hierarchical yield and variability, reversible drift, cumulative degradation, abrupt failure, transient upset, fault geometry, correction/recovery stages, adaptive margin, analog/in-memory physical state, endurance/wear, accepted field service, and fabrication-to-retirement burden remain separate; Fixture F-008 carries the mission-profile-qualified degradation/recovery contract across Candidates 001/005/006/009/010/012/014/017/018 without promoting a principle or candidate
acoustics, hearing, and auditory-scene analysis	calibrated pressure/level and room support, cochlear filtering/compression, temporal coding, spatial cues, masking/grouping/separation, reverberation/context, sparse activity, efferent control, echolocation and active emission/reception, array inference, uncertainty, calibration, and complete lifecycle cost remain separate; Fixture F-009 carries the operator- and action-qualified acoustic contract across Candidates 002/006/007/009/012/014 without promoting a principle or candidate
information thermodynamics and physical computation	46 established, 5 plausible, and 1 disputed claim keep logical transformation, device, circuit, computer/workload, facility, and lifecycle quantities separate; finite-time/error/reservoir/controller costs, reversible history, thermal noise, information engines, thermodynamic uncertainty, distribution mismatch, data movement, PUE, and embodied burden refine P-001/P-003/P-006/P-007/P-008/P-009/P-010/P-012/P-013; Fixture F-010 carries the boundary-qualified physical-computation contract across Candidates 001/005/006/009/010/012/014/017/018 without promoting a principle or candidate
olfaction, chemical sensing, and plume tracking	50 established, 1 plausible, and 1 disputed claim keep receptor/sensor coverage, identity, concentration, mixtures, normalization, sniff/pump dynamics, adaptation, turbulent transport, active search, sparse and drifting representations, valence, cross-reactive arrays, humidity, poisoning, analytical confirmation, exposure, safety, and lifecycle cost separate; Fixture F-011 carries the operator-qualified active chemical-sensing contract across Candidates 002/006/007/009/010/012/014/017/018 without promoting a principle or candidate

The music, library/archival, comparative-cognition, visual-design, mathematical-practice, fluid-dynamics, sports, optics, semiconductor, acoustics, information-thermodynamics, and olfaction/chemical-sensing queues are closed by explicit no-promotion decisions. The music Fixture F-001 retains E-MUSIC-07 as a cross-candidate benchmark; the archival refinement stays inside Candidate 017's query-registered preservation track; and comparative cognition contributes Fixture F-003 as an opportunity/history audit across Candidates 004/006/007/014/018/019; and visual design contributes Fixture F-002 as a versioned reconstruct–externalize–inspect–transform–evaluate audit across Candidates 004/014/017/018/019. It treats fixation and negative transfer as measured failure modes, qualifies evaluators and material evidence, and keeps lineage separate from truth, intent, and recoverability; and mathematical practice contributes Fixture F-004 across Candidates 004/009/010/011/014/017/019. It keeps proposal provenance separate from correctness, requires counterexamples and proof-DAG reconstruction, preserves typed lifecycle state, prevents dependency leakage, propagates reverse-dependency invalidation, and includes proof, checker, human, rebuild, and joule cost; and fluid dynamics contributes Fixture F-005 across Candidates 002/003/006/007/012/014. It requires signed scale transfer, tail fidelity, detector-qualified structures, explicit closure/model support, objective-qualified reduction, total adaptive work, actual observation and sensor operators, observability and posterior calibration, stable net-benefit control, vector mixing outcomes, transition history, extreme calibration, propagated uncertainty, and a full lifecycle energy boundary. Its null is the complete relevant CFD, reduced-model, estimation, sensing, control, rare-event, metrology, and accounting stack. Sports contributes Fixture F-006 across Candidates 002/004/006/007/009/012/014/019. It preserves actual observation/action coupling; separates anticipation from interception, practice from retention and transfer, relevant exploration from variance, and fatigue/readiness/damage from staged return; challenges coordination with turnover and perturbation, deceptive cues with calibration, selection with counterfactual opportunity, and apparent efficiency with full athlete, coach, facility, sensor, recovery, medical, failed-pathway, wall-time, and joule budgets. Its null is the complete calibrated prediction, retrieval, planning, control, curriculum, assurance, coordination, causal-inference, and lifecycle-accounting stack. Optics contributes Fixture F-007 across Candidates 001/006/007/010/014/017/018. It binds inferred state to the physical operator, acquisition action, photons/noise, calibration, ambiguity, prior support, saturation, typed uncertainty, route eligibility, retained-query obligation, and complete resource boundary. Its adversarial tracks test null- space honesty, prior mismatch, active photon allocation, multiplex crossover, drift attribution and fallback, optical/digital service curves, multi-device fabrication and thermal variation, and physical compaction under future queries. Its null is the complete inverse-method, experiment-design, control, fusion, digital/analog acceleration, calibration, and lifecycle-accounting stack. Semiconductor reliability contributes Fixture F-008 across Candidates 001/005/006/009/010/012/014/017/018. It binds accepted service to manufactured-unit hierarchy, mission profile, mechanism-specific acceleration, censoring, monitor age, electrothermal concentration, fault geometry, correction and recovery state, protected exact boundaries, physical- weight age, endurance, and fabrication-through-retirement resources. Its ten tracks test hierarchical transfer, accelerated-life support, sparse wear, correlated faults, voltage authority, approximation containment, analog/digital crossover, nonideality transfer, wear placement, and keep/repair/repurpose/ replace policy against the complete mature reliability stack. Acoustics contributes Fixture F-009 across Candidates 002/006/007/009/012/014. It binds each result to source and emission action, propagation/room state, receiver geometry and motion, calibrated operator support, context/history, literal target, uncertainty, independent unit, feasible action, exposure, and complete resource boundary. Its nine tracks test level-dependent front ends, sparse event timing, active emission/reception, localization under operator shift, reverberation transfer, grouping/separation, efferent-like gain, multi-emitter interference, and full lifecycle cost against the complete calibrated acoustics, inference, control, spectrum, uncertainty, and accounting stack. Information thermodynamics contributes Fixture F-010 across Candidates 001/005/006/009/010/012/014/017/018. It binds every physical- efficiency result to one accepted service while keeping logical, device, circuit, workload, facility, and lifecycle quantities distinct. Its tracks challenge lower-bound substitution, finite-time and finite-error operation, reversible-history closure, thermal reliability, measurement/controller reset, uncertainty-relation applicability, distribution mismatch, topology, data movement, PUE allocation, and fabrication-to-retirement ranking. Its null is the complete compression, reversible-computation, stochastic-thermodynamics, circuit/architecture, workload-metering, facility, and lifecycle-assessment stack. Olfaction and chemical sensing contribute Fixture F-011 across Candidates 002/006/007/009/010/012/014/017/018. It binds source and release, turbulent or reactive transport, realized sniff/pump/motion, receiver path, dynamic response and recovery, cross-sensitivity, calibration and exposure history, operator support, literal outcome, feasible action, safety, independent unit, and complete lifecycle cost. Its fourteen tracks test concentration–identity factorization, sensor coverage, normalization, deconvolution, active acquisition, bilateral/serial/wind cues, plume statistics, search policy, embodiment, sparse total cost, drifting representations, mixture masking, sensor poisoning, analytical confirmation, and safe tiered action against the complete calibrated sensing, analytical, estimation, control, standards, human-work, consumable, and lifecycle stack. The next breadth field is selected by expected ability to split, merge, reject, or re-baseline a current mechanism. Breadth alone does not close a row.

Efficiency mechanism

Stage gates reduce the cost of compounding weak mechanisms. Shared principle bundles reuse telemetry, baselines, and test regimes; isolated rejection avoids paying for full-system integration; exact provenance prevents repeated searches; and negative results retire duplicated work.

For stage k, let $E_{k,\text{research}}$ and $E_{k,\text{integration}}$ be joules spent within declared research and compute boundaries. Let $p_{k,\text{survive}}$ be the empirically estimated fraction of candidates that pass the isolated gate. The expected integration energy avoided by testing first is

$$E_{k,\text{avoided}} = (1 - p_{k,\text{survive}})E_{k,\text{integration}} - E_{k,\text{research}}.$$

This quantity is reported only when both energy boundaries are actually measured; its purpose is to expose the cost of evaluation, not to assume that research automatically saves energy.

Evidence status

- The ordering and gates are project decisions.
- Mechanism-specific evidence remains in the claims ledger; roadmap placement does not promote a claim.
- The three existing experiment contracts are specified but not executed.
- The five-stage composition and its lifecycle benefit remain speculative until the gates produce measured results.
- Sensorimotor transfer remains speculative under C-007.

Speculative extensions

- Maintain a machine-readable experiment ledger linked to commits, hardware records, raw telemetry, derived figures, and claim-status changes.
- Use active literature search to select the next field by expected impact on an unresolved principle, not by publication volume.
- Allocate compute to experiments by expected decision value and uncertainty reduction once the proxy is calibrated against actual outcomes.
- Run adversarial review agents whose only task is to find conventional explanations, hidden costs, or domain evidence that breaks a principle.
- Publish negative results and retired translations as reusable research outputs rather than silently deleting them.

Failure modes

- **Perpetual breadth:** new fields accumulate without changing principles or experiments.
- **Premature composition:** several plausible mechanisms are integrated before any one survives its null model.
- **Gate drift:** thresholds or comparison boundaries change after results are visible.
- **Mechanism creep:** a component enters the system without a claim, principle, ablation, or owner.

- **Proxy capture:** FLOPs, parameter count, or average accuracy substitutes for measured physical cost and qualified behavior.
- **Negative-result loss:** a failed translation disappears and later returns under another domain name.
- **Benchmark enclosure:** the roadmap optimizes a convenient task family that no longer tests the intended operation.
- **Documentation lag:** equations, diagrams, or claim status stop matching the executed artifact.
- **Hardware lock-in:** a mechanism is tuned to one substrate before its functional interface is stable.
- **Narrative immunity:** biological appeal preserves a mechanism after its measurable predictions fail.

Measurable predictions

1. Each breadth wave changes at least one principle boundary, null model, experiment regime, or explicit no-change decision; otherwise the sampling policy is revised.
2. The ratio of scoped claims to deduplicated principles increases as coverage grows, while duplicate organism-themed components decline.
3. At least one Stage-1 candidate ties a conventional baseline and is merged, and at least one loses and is retired; a roadmap in which every mechanism wins is not discriminating.
4. Shared telemetry and null-model infrastructure reduce the incremental cost of testing later principles relative to the first isolated contracts.
5. Mechanisms that pass isolated gates still sometimes fail composition, producing explicit interaction terms rather than post-hoc stories.
6. Lifecycle accounting reverses at least one apparent saving based on FLOPs, parameter count, or inference-only energy.
7. Every completed stage leaves reproducible baselines, scoped claim changes, negative results, and a smaller set of unresolved integration decisions.

Global field coverage

Census date: 2026-08-24

This is the repository's breadth control, not a claim that any discipline has been exhausted. It answers a narrower and mechanically checkable question: **does a durable field-centered audit exist, is the field present only through neighboring work, or has it not been reviewed at all?**

Editable data: `field-coverage.json`. The plot and this page are generated by `generate-field-coverage.mjs`.

Result

The OECD baseline contains **42 second-level fields**. The repository has a dedicated audit for **34 (81.0%)**, adjacent evidence without a field-centered audit for **8 (19.0%)**, and no durable review for **0 (0.0%)**.

The finer DFG probe contains **49 review boards and 214 subjects**. At review-board resolution, the repository has 36 dedicated, 13 adjacent, and 0 unreviewed areas. That higher apparent coverage is not greater depth: one audit can touch a large review board while leaving most of its constituent subjects untouched.

The independent ANZSRC census contains **23 divisions, 213 groups, and 1,967 fields**. At the deliberately coarse division level, 23 have a dedicated entry audit, 0 have adjacent evidence only, and 0 are wholly unreviewed. This is a disagreement detector, not evidence that the 213 groups or 1,967 fields have been audited.

No OECD second-level cell is wholly unreviewed at entry-audit resolution. That is an entry-census result, not near-complete science coverage: a dedicated label means one field-centered audit exists, while catch-all categories and most constituent subfields remain open. Large depth gaps remain in political science, public administration, stratification and broader communication studies; clinical medicine and medical biotechnology; agricultural biotechnology; analytical and food chemistry; inorganic and total-synthesis chemistry; water and ocean research; nanotechnology; production engineering; finance and management; comparative theology; and many named subfields inside every broad cell.

What the states mean

- **dedicated audit:** At least one durable audit is centered on evidence from this field. This does not mean the field is complete.
- **adjacent evidence only:** The field contributes evidence to another audit, but has no field-centered audit of its own.
- **unreviewed:** No durable field-centered evidence audit is present. Mentions and source leads do not count.

A dedicated audit is an entry ticket, not completion. Audit depth, evidence quality, principle deduplication, and executable-test readiness are separate axes.

Taxonomy contract

- The global backbone is OECD Fields of Research and Development (FORD), Frascati Manual 2015, Table 2.2. OECD notes that the classification evolves and does not map perfectly to education or department structures.
- The granularity check is the DFG subject classification for 2024-2028: 49 review boards, 214 subjects, and 4 research areas.
- The independent census and disagreement check is Australian and New Zealand Standard Research Classification (ANZSRC), 2020, official FoR workbook corrected 2025-10-24: 23 divisions, 213 groups, and 1,967 fields. All divisions are recorded; group- and field-level depth remains open. It is not a normative source for this EU/German project.
- The ERC whole-science panel structure is a routing sanity check only. The ERC explicitly says its panels are not a complete scientific classification and do not express research priorities.
- Catch-all categories remain open. Their presence cannot prove that unnamed or emerging disciplines have been sampled.

OECD FORD field-by-field record

1. Natural sciences

6 dedicated · 1 adjacent · 0 unreviewed

- **1.1 Mathematics — dedicated audit.**
  - Current route: mathematical practice proof discovery.
  - Missing depth: Mathematical practice and formal discovery are audited; most mathematical subfields remain unsampled.
  - Next discriminating question: Which structures from topology, category theory, geometry, and constructive mathematics change representation or verification rather than merely renaming it?
- **1.2 Computer and information sciences — dedicated audit.**
  - Current route: databases storage, programming languages verification, security cryptography.
  - Missing depth: Several systems fields are covered; networking, compilers, visualization, and software engineering remain uneven.
  - Next discriminating question: Which mature information-system mechanisms are still missing as null models for the proposed architecture?
- **1.3 Physical sciences — dedicated audit.**
  - Current route: information thermodynamics physical computation, optics photonics inverse sensing, soft active matter, particle nuclear high energy experimentation.
  - Missing depth: Statistical physics, optics, acoustics, soft matter, and particle/nuclear experimental methodology are represented; quantum information, much condensed-matter, atomic/molecular, plasma, accelerator, and nuclear-structure breadth remain partial or absent.
  - Next discriminating question: Which results survive explicit selection, detector response, nuisance, search-family, simulation, preservation, symmetry, and regime-breakdown tests?
- **1.4 Chemical sciences — dedicated audit.**
  - Current route: chemistry reaction networks proofreading, molecular chemistry synthesis systems, polymer research adaptive materials.
  - Missing depth: Reaction networks, proofreading, bounded molecular synthesis/control, polymer populations, viscoelastic spectra, phase/self-assembly paths, gels, dynamic networks, healing, sequence storage and circularity are represented; analytical, surface, electrochemical and theoretical chemistry plus inorganic, organometallic, total-synthesis and scale-up breadth remain uneven.
  - Next discriminating question: Which chemical control mechanisms survive explicit speciation, pathway, apparatus, operator, candidate-set, transport and lifecycle qualification?
- **1.5 Earth and related environmental sciences — dedicated audit.**
  - Current route: earth system transition signals, geology geomorphology, astronomy remote inference, mineralogy petrology geochemistry.
  - Missing depth: Geology, geomorphology, mineralogy/petrology/geochemistry, transition signals, and remote inference are represented; oceanography, hydrology, meteorology, geophysics, geodesy, and much experimental mineral physics remain uneven.

- Next discriminating question: How do sparse, delayed, transformed, and preservation-filtered observations distinguish local variation, material path, source mixture, and regime change across coupled scales?

- **1.6 Biological sciences — dedicated audit.**
  - Current route: cellular quality control, collective ecological resilience, developmental morphogenesis, microbial ecology biofilms, plant distributed control.
  - Missing depth: This is the most heavily sampled broad field, but genomics, marine biology, anatomy, physiology, parasitology, and many taxa remain incomplete.
  - Next discriminating question: Which mechanisms survive normalization across unrelated lineages and which are artifacts of shared ancestry or laboratory choice?

- **1.7 Other natural sciences — adjacent evidence only.**
  - Current route: soft active matter.
  - Missing depth: Hybrid natural-science material exists, but the residual category has never been explicitly searched for fields omitted by the named FORD classes.
  - Next discriminating question: Which emerging or hybrid natural sciences fall outside the named classes and expose a missing problem family?

2. Engineering and technology

9 dedicated · 2 adjacent · 0 unreviewed

- **2.1 Civil engineering — dedicated audit.**
  - Current route: mechanical civil resilience, built environment urban systems.
  - Missing depth: Resilience and occupied systems are represented; construction production, geotechnics, water engineering, and long-life asset governance remain partial.
  - Next discriminating question: Which inspection, reserve, staged repair, and graceful-service mechanisms transfer to long-lived adaptive systems?

- **2.2 Electrical engineering, electronic engineering, information engineering — dedicated audit.**
  - Current route: power grids protection and recovery, semiconductor device reliability, production maintenance nanomanufacturing communications material qualification.
  - Missing depth: Power, device reliability, adaptive link control, and interacting network-loop operations are represented; RF device and antenna design, signal processing, sensors, electromagnetic compatibility, and control hardware remain uneven.
  - Next discriminating question: Which protection, coding, state-age, and loop-coordination mechanisms preserve accepted service with less sensing, communication, and margin?

- **2.3 Mechanical engineering — dedicated audit.**
  - Current route: mechanical civil resilience, biomechanics motor control, production maintenance nanomanufacturing communications material qualification.
  - Missing depth: Mechanics and resilience plus bounded production-process, RCM, maintainability, support and restoration evidence are represented; tribology, machine design, thermal machines, factory systems and process-specific manufacturing remain shallow.
  - Next discriminating question: When can geometry, compliance, hysteresis, passive dynamics, maintainability, or mode-specific maintenance replace sensing, active control, and universal service schedules?

- **2.4 Chemical engineering — dedicated audit.**
  - Current route: process engineering.
  - Missing depth: Plantwide control and balances are represented; scale-up, separations, catalysis, fouling, and batch operation remain partial.
  - Next discriminating question: Which conservation-qualified decomposition and fault isolation methods constrain learned process modules?

- **2.5 Materials engineering — dedicated audit.**
  - Current route: adaptive materials and self assembly, semiconductor device reliability, polymer research adaptive materials, production maintenance nanomanufacturing communications material qualification.
  - Missing depth: Adaptive materials, polymer distributions/topology/rheology/healing, device degradation, and bounded material qualification/lineage are represented; metallurgy, ceramics, composites, joining, coatings, and most production physics remain shallow.
  - Next discriminating question: Which functions can be compiled into material response without losing distribution support, inspectability, repairability, remaining-life knowledge, or lifecycle advantage?

- **2.6 Medical engineering — adjacent evidence only.**
  - Current route: biomechanics motor control.
  - Missing depth: Biomechanical control touches the field, but devices, imaging, prosthetics, clinical instrumentation, and validation are not audited as engineering systems.
  - Next discriminating question: How do assistive and diagnostic systems share control, calibrate uncertainty, and fail safely across changing bodies?

- **2.7 Environmental engineering — adjacent evidence only.**
  - Current route: earth system transition signals, process engineering.
  - Missing depth: Environmental phenomena and industrial processes are separate audits; remediation, water treatment, waste systems, and environmental monitoring are not.
  - Next discriminating question: Which distributed treatment and monitoring strategies allocate intervention under uncertain transport and delayed feedback?

- **2.8 Environmental biotechnology — dedicated audit.**
  - Current route: biotechnology chemistry process systems.
  - Missing depth: Engineered groundwater remediation, intervention, transport, product/mass endpoints, rebound, and open-system evolutionary failure are audited; wastewater, gaseous emissions, mining, marine systems, containment, and environmental-release breadth remain unsampled.
  - Next discriminating question: Which intervention effects survive alternative transport paths, daughter products, oxygen ingress, invasion, succession, off-site export, monitoring designs, and post-treatment rebound?

- **2.9 Industrial biotechnology — dedicated audit.**
  - Current route: biotechnology chemistry process systems.
  - Missing depth: Strain burden/escape, fermentation, continuous culture, cybergenetic control, scale-up, separation, and lifecycle boundaries are audited; cell therapy, food fermentation, enzyme manufacture, product validation, contamination control, and manufacturing authorization remain unsampled.

- Next discriminating question: Which production-control results survive lineage drift, escape mutation, scale gradients, sensor distortion, transport limits, fouling, off-spec propagation, cleaning, and full seed-to-retirement cost?

- **2.10 Nano-technology — dedicated audit.**
  - Current route: adaptive materials and self assembly, soft active matter, production maintenance nanomanufacturing communications material qualification.
  - Missing depth: Nanoscale assembly mechanisms plus a bounded direct audit of population distributions, measurement operators, process history, and rare manufacturing defects are represented; nanodevice transport and sensing, toxicology, safety, fabrication breadth, and scale-up remain incomplete.
  - Next discriminating question: Which nanoscale assembly, sensing, transport, inspection, and defect-control mechanisms retain calibrated population evidence and lifecycle advantage after fabrication costs are included?

- **2.11 Other engineering and technologies — dedicated audit.**
  - Current route: aerospace maritime autonomy.
  - Missing depth: Safety-critical autonomy is represented; the residual engineering class is open-ended and cannot be considered exhausted.
  - Next discriminating question: Which engineering professions use mature operating constraints absent from the current null-model library?

3. Medical and health sciences

3 dedicated · 2 adjacent · 0 unreviewed

- **3.1 Basic medicine — dedicated audit.**
  - Current route: immune tolerance trained immunity, neurodevelopment global control, endocrine circadian control.
  - Missing depth: Several regulatory systems are represented; anatomy, general physiology, genetics, metabolism, and disease mechanisms remain uneven.
  - Next discriminating question: Which multi-organ control and maintenance mechanisms expose cross-module dependencies missing from neural analogies?

- **3.2 Clinical medicine — dedicated audit.**
  - Current route: pathology rehabilitation, pharmacology toxicology, clinical intervention pathways.
  - Missing depth: Pathology, rehabilitation, pharmacology, multimorbidity, diagnostic-to-action coupling, adaptive treatment pathways, antimicrobial stewardship, and intervention authorization now have bounded direct entries; surgery, most specialties, longitudinal disease management, and much diagnostic and therapeutic practice remain unsampled.
  - Next discriminating question: Which staged test, authorization, realized-treatment, monitoring, and stopping contracts improve person-level and population outcomes under incomplete observation, interactions, and bounded support?

- **3.3 Health sciences — dedicated audit.**
  - Current route: epidemiology and surveillance control, nursing care health services.
  - Missing depth: Surveillance plus a first nursing, care-continuity, and health-services audit are represented; occupational health, nutrition, implementation science, long-term care, community care, and most clinical service designs remain unsampled.
  - Next discriminating question: Which mechanisms preserve necessary service and legitimate person-level authority across community, long-term, acute, and digital care without hiding unfinished work or unpaid human burden?

- **3.4 Medical biotechnology — adjacent evidence only.**
  - Current route: cellular quality control.
  - Missing depth: Cellular mechanisms are audited as biology, not engineered diagnostics, therapeutics, tissue systems, or manufacturing.
  - Next discriminating question: Which engineered-cell and molecular-device control mechanisms remain stable, containable, and measurable in living hosts?

- **3.5 Other medical science — adjacent evidence only.**
  - Current route: sports expertise team coordination.
  - Missing depth: Exercise and performance science provide adjacent material, but the residual medical category has not been searched systematically.
  - Next discriminating question: Which clinical or care disciplines fall outside the named classes and contribute distinct coordination or evidence mechanisms?

4. Agricultural and veterinary sciences

4 dedicated · 1 adjacent · 0 unreviewed

- **4.1 Agriculture, forestry, and fisheries — dedicated audit.**
  - Current route: plant distributed control, soil crop multiresource colimitation, forestry fisheries aquatic food systems.
  - Missing depth: Soil/crop co-limitation, forestry, capture fisheries, aquaculture, and aquatic resource management have direct audits; silvicultural breadth, inland fisheries, pest management, livestock integration, farm economics, and many production systems remain unsampled.
  - Next discriminating question: Which policies remain effective across alternative population, spatial, climate, observation, implementation, rights, and market models without exporting harm or changing the denominator?

- **4.2 Animal and dairy science — dedicated audit.**
  - Current route: animal veterinary population health.
  - Missing depth: Resource partitioning, cattle feed restriction, welfare measurement, breeding diversity, rumen consortia, and population-event accounting are audited; species breadth, husbandry systems, reproduction, farm economics, and fisheries remain unsampled.
  - Next discriminating question: Which findings survive across species, production systems, lifetimes, protected welfare components, reproductive constraints, and start-cohort denominators?

- **4.3 Veterinary science — dedicated audit.**
  - Current route: animal veterinary population health.
  - Missing depth: Risk-based and One-Health surveillance, infection interference, evolving opponents, welfare, and dynamical identifiability are audited; comparative clinical care, surgery, companion and wildlife medicine, pharmacovigilance, and antimicrobial stewardship remain unsampled.
  - Next discriminating question: How do diagnosis, treatment, movement, surveillance, welfare, transmission, and resistance interact across species and changing target populations?

- **4.4 Agricultural biotechnology — adjacent evidence only.**
  - Current route: animal veterinary population health.

- Missing depth: Genomic selection, microbial intervention, vaccination, and evolutionary response provide adjacent evidence; gene editing, reproductive biotechnology, diagnostics, biological control, product platforms, and authorization remain unaudited.
- Next discriminating question: Which intervention effects remain after lineage concentration, off-target change, evolutionary response, containment, field transfer, and authorization are tested prospectively?

- **4.5 Other agricultural sciences — dedicated audit.**
  - Current route: forestry fisheries aquatic food systems.
  - Missing depth: Aquatic food systems and cross-resource governance provide a direct residual-field audit; rural studies, extension science, food logistics, agricultural education, and other residual specialties remain open.
  - Next discriminating question: Which residual agricultural disciplines change observation, authority, adoption, service, distribution, and long-horizon resilience beyond production output?

5. Social sciences

7 dedicated · 2 adjacent · 0 unreviewed

- **5.1 Psychology and cognitive sciences — dedicated audit.**
  - Current route: comparative cognition tool use, endogenous generation creativity, learning science skill acquisition.
  - Missing depth: Cognition, learning, and generation are represented; social, organizational, clinical, developmental, and differential psychology remain uneven.
  - Next discriminating question: Which cognitive effects survive representative tasks, prospective testing, and selection-bias controls?

- **5.2 Economics and business — dedicated audit.**
  - Current route: economics market design incentives, supply chain operations research, accounting audit actuarial insurance.
  - Missing depth: Markets, operations, accounting identities, audit and assurance boundaries, actuarial reserves, insurance selection and incentives, correlated tails, and solvency governance are represented; finance, management, marketing, taxation, organization design, and empirical institutional breadth remain shallow.
  - Next discriminating question: Which findings survive endogenous reporting, delayed obligations, correlated shocks, model misspecification, conflicting roles, and independently challengeable evidence?

- **5.3 Education — dedicated audit.**
  - Current route: learning science skill acquisition.
  - Missing depth: Learning and skill acquisition are represented; institutions, curriculum, inclusion, teacher coordination, and assessment systems remain partial.
  - Next discriminating question: How should an adaptive system sequence practice, feedback, assessment, support, and transfer under heterogeneous learning histories?

- **5.4 Sociology — dedicated audit.**
  - Current route: direct social research ethnography media, cultural evolution archaeology, high reliability organizations incident learning, social science depth institutions inequality digital media.
  - Missing depth: Direct social-research methods plus bounded stratification, household allocation, paid/unpaid work, migration measurement, collective-action, criminology, and digital-field entries are represented; education, urban/rural and

health sociology, science and technology studies, social work, historical sociology, population breadth, and most substantive theory remain incomplete.

  - Next discriminating question: Which substantive sociological results survive explicit construct, inequality operator, household allocation, migration denominator, participation cost, crime-observation, platform, positionality, and shared-error-root tests?

- **5.5 Law — dedicated audit.**
  - Current route: legal evidence procedure.
  - Missing depth: Evidence and procedure are represented; substantive EU/German fields, remedies, administration, competition, labour, and criminal law remain partial.
  - Next discriminating question: Which contestability and burden-of-proof mechanisms should govern automated decisions and scientific promotion?

- **5.6 Political science — dedicated audit.**
  - Current route: social choice institutions, social science depth institutions inequality digital media.
  - Missing depth: Formal governance plus bounded policy-implementation, administrative-burden, participation, collective-action, and repression mechanisms are represented; parties, elections, legislatures, executives, courts, comparative regimes, international relations, conflict, public finance, and policy breadth remain incomplete.
  - Next discriminating question: How do authority, veto, capture, administrative burden, participation cost, legitimacy, and implementation gaps change governance designs without treating visible action as private support?

- **5.7 Social and economic geography — adjacent evidence only.**
  - Current route: built environment urban systems, quantitative history demography.
  - Missing depth: Urban and demographic systems touch the field, but spatial inequality, mobility, place, scale, and regional change are not field-centered.
  - Next discriminating question: Which spatial aggregation and mobility mechanisms create misleading global state from locally unequal access and exposure?

- **5.8 Media and communications — dedicated audit.**
  - Current route: linguistics communication, social science depth institutions inequality digital media.
  - Missing depth: Communication mechanisms plus bounded platform-conditioned methods and comparative media-system operators are represented; journalism practice and history, media industries, interpersonal, intercultural, organizational, health and environmental communication, and audience breadth remain incomplete.
  - Next discriminating question: How do channels, provider-controlled interfaces, media institutions, platforms, gatekeepers, ownership, financing, incentives, and repeated exposure alter observation, shared state, and error propagation?

- **5.9 Other social sciences — adjacent evidence only.**
  - Current route: hci human factors, high reliability organizations incident learning.
  - Missing depth: Human factors and organizational work provide adjacent material, but the residual social-science category has not been searched systematically.
  - Next discriminating question: Which professions and hybrid social sciences contribute distinct observation, coordination, or accountability mechanisms?

6. Humanities and the arts

5 dedicated · 0 adjacent · 0 unreviewed

- **6.1 History and archaeology — dedicated audit.**
  - Current route: cultural evolution archaeology, quantitative history demography.
  - Missing depth: Material inference and quantitative history are represented; archival criticism, period-specific methods, oral history, and history of science remain partial.
  - Next discriminating question: How should path dependence, missing archives, provenance, survivorship, and changing categories constrain learned historical memory?
- **6.2 Languages and literature — dedicated audit.**
  - Current route: linguistics communication, textual criticism variant traditions.
  - Missing depth: Linguistics, textual criticism, variant traditions, scholarly editing, and translation qualification are represented; narrative theory, poetics, comparative literature, reception, performance, and most language traditions remain unsampled.
  - Next discriminating question: Which narrative and interpretive structures change counterfactual evaluation, perspective, compression, long-range coherence, or disagreement handling beyond the new witness-variant contract?
- **6.3 Philosophy, ethics and religion — dedicated audit.**
  - Current route: philosophy of science theory choice, theology religious practice ritual.
  - Missing depth: Philosophy of science, theory choice, theology, religious practice, and ritual now have field-centered audits; ethics, philosophy of mind and technology, comparative theology, and most religious traditions remain unsampled.
  - Next discriminating question: Which accounts of agency, moral status, value conflict, testimony, interpretation, conscience, and authority change system governance or evaluation contracts?
- **6.4 Arts (arts, history of arts, performing arts, music) — dedicated audit.**
  - Current route: visual art design cognition, music cognition improvisation.
  - Missing depth: Visual art and music are represented; theatre, dance, film, architecture as art, conservation, and art history remain uneven.
  - Next discriminating question: Which embodied, temporal, collaborative, and material practices produce useful variants and evaluations unavailable to static generation?
- **6.5 Other humanities — dedicated audit.**
  - Current route: residual humanities living heritage practice.
  - Missing depth: A deliberate residual-category audit now samples living heritage, rehearsal, conservation, archival absence, and digital preservation; the catch-all remains open and most direct humanities practices are still unsampled.
  - Next discriminating question: Which methods from oral history, museum and curatorial practice, film, theatre and dance traditions, folklore, interpreting, conservation science, and heritage conflict change a test rather than duplicate provenance, governance, or protocol versioning?

DFG granularity probe

The DFG layer catches gaps hidden by the broader OECD cells. It is shown at review-board level; its 214 individual subjects remain the next resolution step.

1. Geistes- und Sozialwissenschaften

11 dedicated · 2 adjacent · 0 unreviewed

- **1.11 Alte Kulturen — adjacent evidence only.**
  - Current route: cultural evolution archaeology.
  - Missing depth: Archaeological inference is present, but ancient-culture detail remains sparse.
- **1.12 Geschichtswissenschaften — dedicated audit.**
  - Current route: quantitative history demography.
  - Missing depth: Historical method breadth remains partial.
- **1.13 Kunst-, Musik-, Theater- und Medienwissenschaften — dedicated audit.**
  - Current route: visual art design cognition, music cognition improvisation.
  - Missing depth: Theatre and media studies are not directly audited.
- **1.14 Sprachwissenschaften — dedicated audit.**
  - Current route: linguistics communication.
  - Missing depth: Historical and typological breadth remains partial.
- **1.15 Literaturwissenschaft — dedicated audit.**
  - Current route: textual criticism variant traditions.
  - Missing depth: Textual criticism, scholarly editing, variant traditions, and translation qualification are audited; literary theory and most genres and traditions remain unsampled.
- **1.16 Sozial- und Kulturanthropologie, außereuropäische Kulturen, Judaistik und Religionswissenschaft — adjacent evidence only.**
  - Current route: direct social research ethnography media, theology religious practice ritual.
  - Missing depth: Religious practice and ritual have a field-centered audit, but ethnography, anthropology, Judaic studies, non-European cultures, and comparative religious studies remain unsampled.
- **1.17 Theologie — dedicated audit.**
  - Current route: theology religious practice ritual.
  - Missing depth: Practice, ritual, interpretation, canon, authority, conscience, and evidence boundaries are audited; doctrinal, historical, pastoral, comparative, and tradition-specific theology remain unsampled.
- **1.18 Philosophie — dedicated audit.**
  - Current route: philosophy of science theory choice.
  - Missing depth: Philosophy of science and theory choice are audited; most other philosophical subfields remain unsampled.
- **1.21 Erziehungswissenschaft und Bildungsforschung — dedicated audit.**
  - Current route: learning science skill acquisition.
  - Missing depth: Institutions and professional practice remain partial.
- **1.22 Psychologie — dedicated audit.**

- Current route: comparative cognition tool use, endogenous generation creativity, learning science skill acquisition.
- Missing depth: Clinical, organizational, and personality work remain uneven.

- **1.23 Sozialwissenschaften — dedicated audit.**
  - Current route: direct social research ethnography media, social choice institutions, social science depth institutions inequality digital media.
  - Missing depth: Empirical social research plus bounded policy implementation, administrative burden, inequality, household/work/care allocation, migration, collective action, criminology, platform methods and media-system operators are represented; parties, elections, comparative politics, international relations, journalism practice/history, media industries, education, urban/rural and health sociology, social work, and most substantive breadth remain incomplete.

- **1.24 Wirtschaftswissenschaften — dedicated audit.**
  - Current route: economics market design incentives, supply chain operations research, accounting audit actuarial insurance.
  - Missing depth: Market design, operations, accounting, audit, actuarial reserves, and insurance boundaries are audited; finance, management, marketing, taxation, organization studies, and much empirical institutional work remain unsampled.

- **1.25 Rechtswissenschaften — dedicated audit.**
  - Current route: legal evidence procedure.
  - Missing depth: Evidence and procedure dominate; substantive fields remain partial.

2. Lebenswissenschaften

6 dedicated · 1 adjacent · 0 unreviewed

- **2.11 Grundlagen der Biologie und Medizin — dedicated audit.**
  - Current route: cellular quality control, developmental morphogenesis.
  - Missing depth: Genomics and structural biology remain uneven.

- **2.12 Pflanzenwissenschaften — dedicated audit.**
  - Current route: plant distributed control, soil crop multiresource colimitation.
  - Missing depth: Field and crop contexts remain partial.

- **2.13 Zoologie — adjacent evidence only.**
  - Current route: animal navigation sensory ecology, comparative cognition tool use.
  - Missing depth: Behavior and navigation are present, but morphology, systematics, physiology, and evo-devo are not field-centered.

- **2.21 Mikrobiologie, Virologie und Immunologie — dedicated audit.**
  - Current route: microbial ecology biofilms, immune tolerance trained immunity.
  - Missing depth: Virology and parasitology are major gaps.

- **2.22 Medizin — dedicated audit.**
  - Current route: pathology rehabilitation, pharmacology toxicology, epidemiology and surveillance control, nursing care health services, clinical intervention pathways.
  - Missing depth: Pathology, rehabilitation, pharmacology, multimorbidity, diagnostic-to-action coupling, adaptive treatment pathways, antimicrobial stewardship, and intervention authorization have bounded entries; only a small fraction of the 33

listed medical subjects is directly sampled.

- **2.23 Neurowissenschaften — dedicated audit.**
  - Current route: neurodevelopment global control, memory replay forgetting.
  - Missing depth: Strongest life-science concentration; still not exhaustive.

- **2.31 Agrar-, Forstwissenschaften und Tiermedizin — dedicated audit.**
  - Current route: soil crop multiresource colimitation, animal veterinary population health, forestry fisheries aquatic food systems.
  - Missing depth: Soil/crop co-limitation, animal production, welfare, breeding, rumen ecology, veterinary population health, forestry, fisheries, aquaculture, and aquatic food systems are audited; many species, production systems, food science, farm economics, and clinical veterinary breadth remain unsampled.

3. Naturwissenschaften

9 dedicated · 10 adjacent · 0 unreviewed

- **3.11 Molekülchemie — dedicated audit.**
  - Current route: molecular chemistry synthesis systems.
  - Missing depth: Dynamic covalent assembly, self-sorting, context-qualified recognition, photochemical/electrochemical pathway control, structure elucidation and automated discovery nulls now have a bounded audit; inorganic/organometallic breadth, total synthesis, radical/pericyclic/C–H activation, flow/scale-up, impurity fate and reproducibility remain incomplete.

- **3.12 Chemische Festkörper- und Oberflächenforschung — adjacent evidence only.**
  - Current route: adaptive materials and self assembly, semiconductor device reliability.
  - Missing depth: No surface-chemistry audit.

- **3.13 Physikalische Chemie — adjacent evidence only.**
  - Current route: chemistry reaction networks proofreading, information thermodynamics physical computation.
  - Missing depth: Only neighboring reaction and thermodynamic material is present.

- **3.14 Analytische Chemie — adjacent evidence only.**
  - Current route: metrology measurement science, olfaction chemical sensing plume tracking, biotechnology chemistry process systems, mineralogy petrology geochemistry.
  - Missing depth: Sensor calibration, optical-density distortion, sampling, and mass closure provide adjacent constraints; chemical metrology, calibration hierarchy, detection/quantification limits, matrix effects, speciation, and interlaboratory comparison lack a field-centered audit.

- **3.15 Biologische Chemie und Lebensmittelchemie — adjacent evidence only.**
  - Current route: biotechnology chemistry process systems, animal veterinary population health, forestry fisheries aquatic food systems.
  - Missing depth: Metabolic pathways and edible nutrient/feed-flow accounting provide adjacent evidence; molecular biochemistry, food chemistry, processing, preservation, toxicology, and analytical methods lack a field-centered audit.

- **3.16 Polymerforschung — dedicated audit.**
  - Current route: polymer research adaptive materials.

- Missing depth: Polymerization mechanisms and populations, entanglement, viscoelastic spectra, conditional time–temperature superposition, phase/self-assembly paths, gel criteria, dynamic covalent networks, healing, response, ageing/fatigue, sequence storage, circularity and measurement operators now have a bounded audit; processing/manufacturing, biopolymers, membranes, conducting polymers, adhesion, composite interfaces, toxicology, fire and application qualification remain incomplete.

- **3.17 Theoretische Chemie — adjacent evidence only.**
  - Current route: chemistry reaction networks proofreading, biotechnology chemistry process systems.
  - Missing depth: Stoichiometric networks, microkinetics, scaling relations, and transport models provide adjacent formalisms; electronic structure, quantum chemistry, molecular dynamics, statistical mechanics, and method validation lack a field-centered audit.
- **3.21 Physik der kondensierten Materie — adjacent evidence only.**
  - Current route: soft active matter, semiconductor device reliability.
  - Missing depth: Soft matter and devices cover only a subset.
- **3.22 Statistische Physik, nichtlineare Dynamik, komplexe Systeme, weiche und fluide Materie, biologische Physik — dedicated audit.**
  - Current route: soft active matter, fluid dynamics turbulence.
  - Missing depth: Several components are represented; nonlinear dynamics remains broad.
- **3.23 Optik, Quantenoptik und Physik der Atome, Moleküle und Plasmen — dedicated audit.**
  - Current route: optics photonics inverse sensing.
  - Missing depth: Optics dominates; atomic and plasma physics remain shallow.
- **3.24 Teilchen, Kerne und Felder — dedicated audit.**
  - Current route: particle nuclear high energy experimentation.
  - Missing depth: Triggering, detector response, calibration, unfolding, nuisance uncertainty, rare-event search, blinding, simulation, preservation, evaluated nuclear data, symmetry, and effective-model breakdown are audited; accelerator, detector technology, nuclear structure/reactions, neutrino, astroparticle, phenomenology, and theory breadth remain unsampled.
- **3.25 Astrophysik und Astronomie — dedicated audit.**
  - Current route: astronomy remote inference.
  - Missing depth: Remote inference is represented; field breadth remains partial.
- **3.31 Mathematik — dedicated audit.**
  - Current route: mathematical practice proof discovery.
  - Missing depth: Practice and proof are represented; subfield breadth is not.
- **3.41 Atmosphären-, Meeres- und Klimaforschung — adjacent evidence only.**
  - Current route: earth system transition signals, forestry fisheries aquatic food systems, mineralogy petrology geochemistry.

- Missing depth: Earth-system work plus marine resource and farm-wild coupling are present; ocean physics, biogeochemistry, atmospheric process detail, and climate dynamics remain shallow.

- **3.42 Geologie und Paläontologie — dedicated audit.**
  - Current route: geology geomorphology, paleobiology major transitions.
  - Missing depth: Both have initial audits.

- **3.43 Geophysik und Geodäsie — adjacent evidence only.**
  - Current route: geology geomorphology, mineralogy petrology geochemistry.
  - Missing depth: No field-centered audit.

- **3.44 Mineralogie, Petrologie und Geochemie — dedicated audit.**
  - Current route: mineralogy petrology geochemistry.
  - Missing depth: Phase equilibria, kinetic trapping and replacement, P–T–t records, isotope and reservoir inverses, weathering/redox, reactive upscaling, detrital provenance, geochronology, preservation and metrology now have a bounded field audit; experimental mineral physics/crystallography, igneous and mantle processes, ore/economic geology, cosmochemistry, and analytical breadth remain incomplete.

- **3.45 Geographie — adjacent evidence only.**
  - Current route: built environment urban systems, geology geomorphology.
  - Missing depth: Physical and human geography are not audited directly.

- **3.46 Wasserforschung — adjacent evidence only.**
  - Current route: forestry fisheries aquatic food systems, mineralogy petrology geochemistry, biotechnology chemistry process systems.
  - Missing depth: Receiving-water capacity and aquatic flows provide adjacent evidence; hydrology, limnology, hydraulics, water treatment, groundwater, and water governance lack a field-centered audit.

4. Ingenieurwissenschaften

10 dedicated · 0 adjacent · 0 unreviewed

- **4.11 Produktionstechnik — dedicated audit.**
  - Current route: supply chain operations research, process engineering, production maintenance nanomanufacturing communications material qualification.
  - Missing depth: Statistical process state, qualified process windows, maintenance policy/support, manufacturing genealogy and bounded nanomanufacturing evidence have a direct audit; factory planning/layout, production logistics, machining, forming, joining, additive manufacture, human work and most process-specific production technologies remain incomplete.

- **4.12 Mechanik und konstruktiver Maschinenbau — dedicated audit.**
  - Current route: mechanical civil resilience, biomechanics motor control.
  - Missing depth: Design and product development remain shallow.

- **4.21 Verfahrenstechnik und Technische Chemie — dedicated audit.**
  - Current route: process engineering, biotechnology chemistry process systems.
  - Missing depth: Process engineering now includes a direct bioprocess/biotechnology audit; product-specific manufacturing, validation, downstream breadth, contamination control, and industrial authorization remain partial.

- **4.22 Strömungsmechanik, Technische Thermodynamik und Thermische Energietechnik — dedicated audit.**
  - Current route: fluid dynamics turbulence.
  - Missing depth: Energy machinery and applied thermal systems remain partial.
- **4.31 Werkstofftechnik — dedicated audit.**
  - Current route: adaptive materials and self assembly, polymer research adaptive materials, production maintenance nanomanufacturing communications material qualification.
  - Missing depth: Adaptive-material/polymer mechanisms plus bounded material qualification, inspection-document, lineage, substitution and requalification evidence are represented; metallurgy, ceramics, composites, joining, surface engineering, degradation and most production variation remain incomplete.
- **4.32 Materialwissenschaft — dedicated audit.**
  - Current route: adaptive materials and self assembly, polymer research adaptive materials, semiconductor device reliability.
  - Missing depth: Adaptive-material, polymer and device-reliability mechanisms are represented; biomaterials, electronic/quantum materials, composite interfaces and multiscale design remain partial.
- **4.41 Systemtechnik — dedicated audit.**
  - Current route: engineering analogues, aerospace maritime autonomy.
  - Missing depth: Transport and microsystems remain uneven.
- **4.42 Elektrotechnik und Informationstechnik — dedicated audit.**
  - Current route: power grids protection and recovery, semiconductor device reliability, production maintenance nanomanufacturing communications material qualification.
  - Missing depth: Power, device reliability, adaptive link control and interacting network-operations loops are represented; antenna/RF device design, electromagnetic compatibility, signal processing, sensors, control hardware and communications-system breadth remain incomplete.
- **4.43 Informatik — dedicated audit.**
  - Current route: databases storage, programming languages verification, security cryptography, hci human factors.
  - Missing depth: Several subfields are represented; visualization and software engineering remain partial.
- **4.51 Bauwesen und Architektur — dedicated audit.**
  - Current route: built environment urban systems, mechanical civil resilience.
  - Missing depth: Construction production and water engineering remain partial.

ANZSRC independent division census

This third lens records all 23 official divisions. Its much finer 213 groups and 1,967 fields remain an explicit resolution debt; a green division means one entry audit, not comprehensive coverage.

- **30 Agricultural, veterinary and food sciences — dedicated audit.**

- Current route: soil crop multiresource colimitation, animal veterinary population health, forestry fisheries aquatic food systems.
- Missing depth: Soil, crop, livestock, veterinary population health, forestry, fisheries and aquaculture now have direct audits; food science, farm systems, horticulture, reproductive biotechnology and many species remain unsampled.

- **31 Biological sciences — dedicated audit.**
  - Current route: cellular quality control, collective ecological resilience, developmental morphogenesis, microbial ecology biofilms, plant distributed control.
  - Missing depth: Cellular control, development, ecology, microbes and plants are represented; taxonomy, anatomy, marine biology, genomics, parasitology and most organismal breadth remain incomplete.

- **32 Biomedical and clinical sciences — dedicated audit.**
  - Current route: pathology rehabilitation, pharmacology toxicology, epidemiology and surveillance control, nursing care health services, clinical intervention pathways.
  - Missing depth: Pathology, rehabilitation, pharmacology, toxicology, epidemiology, nursing/care systems, multimorbidity, diagnostic-to-action coupling, adaptive treatment pathways, antimicrobial stewardship and intervention authorization are bounded direct entries; most clinical specialties, surgery, longitudinal disease management and medical biotechnology remain unsampled.

- **33 Built environment and design — dedicated audit.**
  - Current route: built environment urban systems, mechanical civil resilience, visual art design cognition.
  - Missing depth: Urban systems, structural resilience and design cognition are represented; architecture practice, construction production, landscape architecture, planning law and building operation remain partial.

- **34 Chemical sciences — dedicated audit.**
  - Current route: chemistry reaction networks proofreading, biotechnology chemistry process systems, molecular chemistry synthesis systems, polymer research adaptive materials.
  - Missing depth: Reaction networks, proofreading, molecular assembly/control, catalysis, process chemistry and polymer mechanisms are represented; analytical, surface, theoretical and much inorganic, organometallic, total-synthesis and scale-up chemistry remain uneven.

- **35 Commerce, management, tourism and services — dedicated audit.**
  - Current route: supply chain operations research, accounting audit actuarial insurance.
  - Missing depth: Operations, accounting, audit, actuarial reserves and insurance are represented; organization studies, management, marketing, finance, taxation, tourism and most service professions remain unsampled.

- **36 Creative arts and writing — dedicated audit.**
  - Current route: visual art design cognition, music cognition improvisation, residual humanities living heritage practice.
  - Missing depth: Visual art and music are direct entries, and rehearsal/conservation now have a narrow cross-field probe; writing practice, film, theatre and dance traditions, curating, and most performance research remain unsampled.

- **37 Earth sciences — dedicated audit.**

    - Current route: earth system transition signals, geology geomorphology, astronomy remote inference, mineralogy petrology geochemistry.
    - Missing depth: Geology, geomorphology, mineralogy/petrology/geochemistry, Earth-system transitions and remote inference are represented; geophysics, geodesy, ocean and atmospheric sciences, experimental mineral physics, ore geology and cosmochemistry remain uneven.

- **38 Economics — dedicated audit.**
    - Current route: economics market design incentives, social choice institutions.
    - Missing depth: Market design, incentives and social choice are represented; macroeconomics, development, labor, public finance, economic history and empirical identification breadth remain incomplete.

- **39 Education — dedicated audit.**
    - Current route: learning science skill acquisition.
    - Missing depth: Learning and skill acquisition are direct entries; institutions, curriculum, inclusion, assessment, vocational education and comparative education systems remain partial.

- **40 Engineering — dedicated audit.**
    - Current route: engineering analogues, mechanical civil resilience, process engineering, power grids protection and recovery, polymer research adaptive materials, production maintenance nanomanufacturing communications material qualification.
    - Missing depth: Several engineering families plus bounded production/process capability, maintenance/support, nanomanufacturing evidence, adaptive communications/network operations, and material qualification/lineage are represented; mining, transport, factory/layout and production-technology breadth, RF hardware, robotics hardware and many material systems remain uneven.

- **41 Environmental sciences — dedicated audit.**
    - Current route: earth system transition signals, collective ecological resilience, biotechnology chemistry process systems.
    - Missing depth: Earth systems, ecological resilience, remediation and environmental biotechnology are represented; water, ocean, atmospheric, contamination, conservation-management and environmental-policy breadth remain partial.

- **42 Health sciences — dedicated audit.**
    - Current route: epidemiology and surveillance control, pathology rehabilitation, nursing care health services.
    - Missing depth: Population surveillance, rehabilitation, nursing and health services are represented; public-health interventions, allied health, nutrition, health economics and implementation science remain uneven.

- **43 History, heritage and archaeology — dedicated audit.**
    - Current route: cultural evolution archaeology, quantitative history demography, textual criticism variant traditions, residual humanities living heritage practice.
    - Missing depth: Archaeological inference, quantitative history, textual traditions, living heritage, archival absence and conservation boundaries are represented; oral history, museum and curatorial practice, heritage conflict, and most periods and regions remain unsampled.

- **44 Human society — dedicated audit.**

    - Current route: direct social research ethnography media, social choice institutions, indigenous data knowledge governance, theology religious practice ritual, social science depth institutions inequality digital media.
    - Missing depth: Direct social-research methods plus bounded policy/administration, political science, inequality, household/work/care allocation, migration, collective action, criminology and digital-field depth are represented; human geography, social work, education, urban/rural and health sociology, historical sociology, comparative politics/international relations, and most substantive anthropology and sociology remain incomplete.

- **45 Indigenous studies — dedicated audit.**
    - Current route: indigenous data knowledge governance.
    - Missing depth: Indigenous data and knowledge governance has an Indigenous-led-source entry audit; languages, law, land, health, education, histories and community-specific knowledge require their own authorities and cannot be inferred from that entry.

- **46 Information and computing sciences — dedicated audit.**
    - Current route: databases storage, programming languages verification, security cryptography, hci human factors.
    - Missing depth: Storage, programming languages, verification, security and HCI are represented; networking, compilers, visualization, software engineering and many applied information systems remain uneven.

- **47 Language, communication and culture — dedicated audit.**
    - Current route: linguistics communication, textual criticism variant traditions, direct social research ethnography media, social science depth institutions inequality digital media.
    - Missing depth: Linguistics, textual criticism, translation qualification, bounded communication/media gatekeeping and effects, platform-conditioned methods, and comparative media-system operators are represented; journalism practice/history, media industries, interpersonal/intercultural/organizational communication, interpreting, cultural studies, rhetoric and most language traditions remain incomplete.

- **48 Law and legal studies — dedicated audit.**
    - Current route: legal evidence procedure.
    - Missing depth: Evidence and procedure are direct entries; substantive private, public, criminal, labor, environmental and international law remain partial and jurisdiction-specific.

- **49 Mathematical sciences — dedicated audit.**
    - Current route: mathematical practice proof discovery.
    - Missing depth: Formal discovery and proof practice are represented; most pure, applied, statistical and computational subfields remain unsampled.

- **50 Philosophy and religious studies — dedicated audit.**
    - Current route: philosophy of science theory choice, theology religious practice ritual.
    - Missing depth: Theory choice, theology, religious practice and ritual are direct entries; ethics, philosophy of mind and technology, comparative religion and most traditions remain unsampled.

- **51 Physical sciences — dedicated audit.**

- Current route: information thermodynamics physical computation, optics photonics inverse sensing, soft active matter, particle nuclear high energy experimentation.
- Missing depth: Thermodynamics, optics, soft matter and particle/nuclear experimental methods are represented; atomic, molecular, plasma, condensed-matter, accelerator and theoretical breadth remain partial.

- **52 Psychology — dedicated audit.**
  - Current route: comparative cognition tool use, endogenous generation creativity, learning science skill acquisition.
  - Missing depth: Cognition, creativity and learning are represented; clinical, personality, developmental, organizational, social and cross-cultural psychology remain uneven.

Taxonomy disagreement is a discovery signal

A field missing from one classification must not disappear from the research program. The first recorded disagreement is:

- **ANZSRC 2020 45 Indigenous Studies — dedicated audit.**
  - Current route: indigenous data knowledge governance.
  - Gap: Indigenous data and knowledge governance now has an Indigenous-led-source audit. Indigenous Studies is far broader: language, law, health, education, land, history, methods, and community-specific knowledge remain outside this narrow entry audit and require community-specific authority.
  - Next discriminating question: How can the breadth process include community-defined research questions and authorities without turning Indigenous Studies into extractable motifs or a universal policy template?
  - Handling rule: Treat this as a governance and knowledge-sovereignty audit led by Indigenous-authored sources; taxonomy coverage never authorizes extraction of biological or cultural motifs.

Breadth scheduler

Each breadth wave must contain four different selections:

1. one field from the least-covered OECD broad field;
2. one DFG subject hidden inside an OECD field already marked dedicated;
3. one field chosen for methodological distance from the current corpus, without requiring an obvious AI analogy; and
4. one depth item that can alter an existing claim, experiment, equation, or system boundary.

The first three selections collect leads. Only the fourth slot is allowed to consume substantial depth work before deduplication. This prevents familiar fields from monopolizing the research budget while still requiring useful residue to change a testable object.

Completed breadth waves

Wave 1 — taxonomy and method boundaries

1. **Soil, crop, and multi-resource co-limitation:** changed resource state from scalar availability to typed, transport- and service-window-qualified vectors; no new principle.
2. **Philosophy of science and theory choice:** changed discovery admission from model fit to an alternative-, auxiliary-, observation-, access-, and failure-root-qualified contract; no new principle.
3. **Indigenous data and knowledge governance:** added collective authority, purpose, benefit, refusal, and derivative-remedy obligations while prohibiting extraction of community knowledge; no new principle.
4. **Textual criticism and variant traditions:** separated attestation, transcription, collation, lineage, conjecture, edition, translation, and interpretation and made apparatus compaction query-qualified; no new principle.

Wave 2 — deliberate disciplinary distance

1. **Nursing, care science, and health services:** made unfinished necessary work, continuity type, receiver acceptance, response capacity, and verified service explicit; five synthetic protocols, no new principle.
2. **Animal production and veterinary population health:** protected multidimensional welfare from productivity proxies and added cohort, lineage, interference, consortium-flux, breeding-diversity, and changing-opponent tests; no new principle.
3. **Environmental and industrial biotechnology:** added dilution/selection, evolutionary escape, scale-gradient, transport, remediation-rebound, measurement, and lifecycle falsifiers while leaving chemistry boards conservatively open; no new principle.
4. **Accounting, audit, actuarial science, and insurance:** separated balance from truth, materiality from detection, reserves from headroom, dependence from pooling, and governance from model validity; nine protocols, no new principle.
5. **Theology, religious practice, and ritual:** separated instrumental effect, conventional form, protected commitment, authority, interpretation, and belief data; five synthetic protocols, no new principle.
6. **Forestry, fisheries, aquaculture, and aquatic food systems:** added stock–flow–cohort–space–rights, carbon-pool, management-loop, externality, nutrient-recovery, and edible-endpoint tests; six protocols, no new principle.
7. **Particle, nuclear, and high-energy experimentation:** added trigger support, detector response, unfolding, nuisance, blinding, search-family, simulation, preservation, covariance, and effective-model breakdown tests; eight protocols, no new principle.

Wave 3 — independent taxonomy and remaining empty cells

1. **Residual humanities and living heritage:** separated archival selection from event absence, contextual authenticity from a universal score, living practice from frozen tokens, rehearsal history from free compression, re-treatability from byte rollback, and representation from identity; six protocols, no new principle.
2. **Molecular chemistry and synthesis systems:** bounded dynamic covalent assembly, self-sorting, context-qualified recognition, temporal pathway control, structure elucidation, and automated discovery against mature chemical nulls; eight protocols, no new principle.
3. **Polymer research:** made distributions, mechanism, topology, relaxation spectra, phase path, gel criteria, healing, ageing, sequence storage, circularity, and measurement operators explicit; twelve protocols, no new principle.
4. **Mineralogy, petrology, and geochemistry:** separated equilibrium, kinetic trapping, material replacement, P–T–t records, tracer inverses, weathering, redox, upscaling, provenance, dates, and preservation-filtered deep time; eight protocols, no new principle.

5. **Direct social research, ethnography, and media:** separated instrumented response, nonresponse, category construction, measurement invariance, situated observation, reflexivity, talk/action, network interference, formal routine, media selection/exposure, mixed-method failure roots, and distinct authority bases; eight protocols, no new principle.
6. **Taxonomy control itself:** added all 23 ANZSRC divisions, 213 groups, and 1,967 fields as an independent disagreement probe beside OECD and DFG; division coverage never substitutes for group- or field-level review.

Wave 4 — intervention, production, and institutional depth

1. **Clinical intervention pathways:** separated multimorbidity composition, diagnostic performance from pathway utility, prognostic from predictive markers, sequential-policy support, antimicrobial patient/population outcomes, and recommendation from execution authority; nine claims and six synthetic protocols, no new principle.
2. **Production, maintenance, nano, communications, and material qualification:** separated statistical control from specification and qualification, prognosis from decision value, images from nano-population evidence, physical-layer rate from accepted service, interacting automation loops, and lineage from qualification; nine claims and nine CPU-only protocols, no new principle.
3. **Institutions, inequality, crime, and digital media:** separated formal policy from delivered service, inequality operators, household totals from allocation, visible action from private support, recorded crime from incidence, platform traces from populations, and media-system transport; seven claims and eight CPU-only protocols, no new principle.

Next gap wave

1. **Measurement-heavy adjacent fields:** audit analytical and food chemistry, water research, ocean/atmospheric science, geophysics/geodesy, and chemical speciation with explicit operator, calibration, transport, and interlaboratory boundaries.
2. **Remaining clinical specialties and biotechnology:** sample surgery, imaging, paediatrics, psychiatry, obstetrics, dentistry, gene and cell therapies, medical and agricultural biotechnology, gene editing, biological control, field heterogeneity, and biosafety without treating the new pathway audit as specialty completion.
3. **Computing and engineering remainder:** deepen networking, compilers, visualization, software engineering, nano-process yield, hardware ageing, lifecycle qualification, and interacting operational control beyond the new production audit.
4. **Institutions, economy, and education:** deepen public finance, management and organization, labour and employment, education systems, migration, comparative law, and community-specific methods beyond the new social-system audit.
5. **Subfield resolution plus execution:** sample the 214 DFG subjects and 213 ANZSRC groups explicitly, then convert the most discriminating protocols into versioned workstation artifacts instead of accumulating prose-only readiness.

No item is promoted because it sounds novel. Every retained mechanism still passes the open-world extraction record, mature-null comparison, deduplication, and equal-budget rejection gate in the discovery policy.

Maintenance rule

After each audit, update the machine record first, regenerate this page and plot, and record whether the new material produced: a duplicate, a sharper boundary, a new claim, a changed experiment, or no durable residue. A field remains incomplete even after its first dedicated audit.